



in28minutes

Google Cloud

Professional Cloud Architect



What This Course Is All About

Google Cloud: A broad ecosystem of services

- **Our Focus:** The core services and architecture skills needed for the Professional Cloud Architect exam
- **Real Challenge:** Choosing suitable services and architecture patterns for each scenario

Course Design: Learn key services, compare options, and learn architectural trade-offs

Our Goal: Help you architect cloud solutions and prepare for certification with confidence



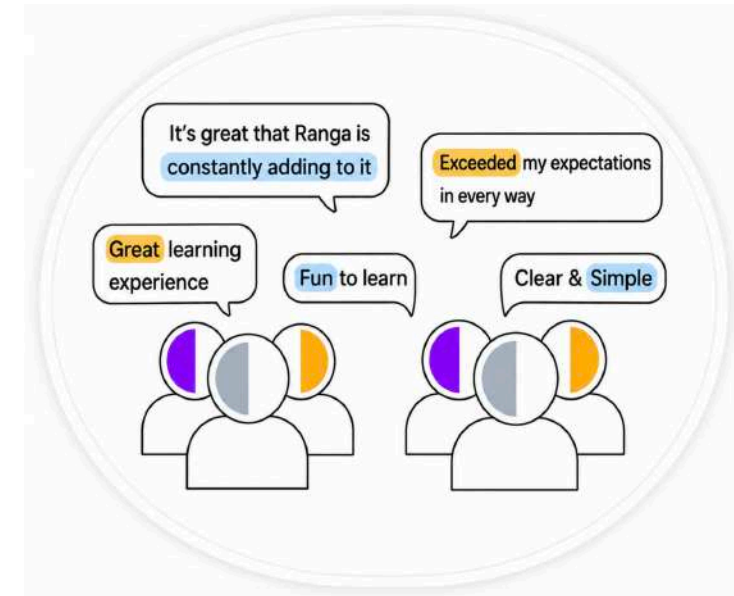


What Learners Are Saying

in28minutes: Trusted by millions of learners worldwide

What Learners Say:

- ★★★★★ - I passed my Google Cloud Certified Professional Cloud Architect exam. I must say that this course is excellent.
- ★★★★★ - Without this course I would have been lost on my journey to cloud architecture.
- ★★★★★ - Thank you Ranga, I passed my Google Cloud Certified - Professional Cloud Architect on my first attempt.





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Getting Started with Cloud Fundamentals



The Journey: From Own Data Centers to Cloud

Goal: Understand the enterprise journey from managing data centers to adopting the cloud

The Starting Point: Challenges with managing data centers

- **Servers:** What a server really is
- **Data Centers:** Where servers live
- **The Pain:** Why managing data centers is difficult

The Solution – Cloud

- **The Benefits** – Why the world is moving to the Cloud





How does in28minutes.com work?

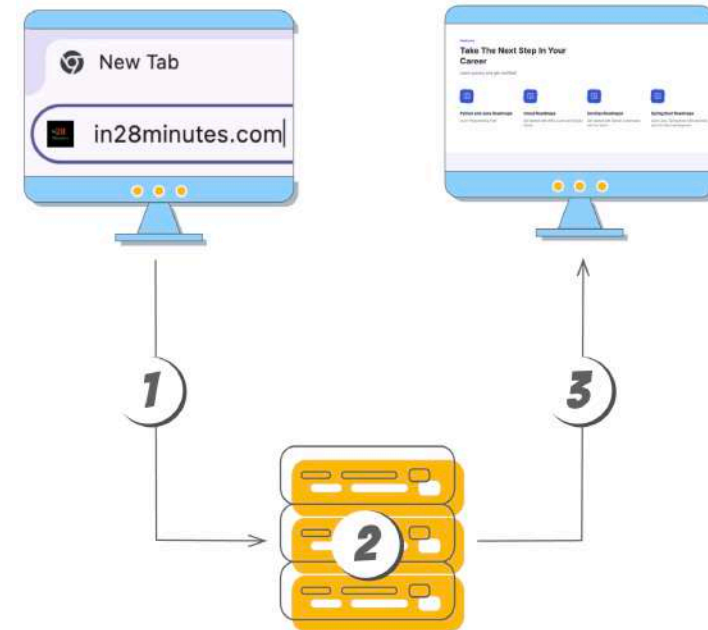
1: Browser - User types URL in the browser

- **Send Request** - Browser finds the server and sends a request to the server

2: Process Request - Server processes the request and prepares the response

3: Send Response - Server sends response back to the browser

- **User Display:** Browser shows the response to the user





What Is a Server?

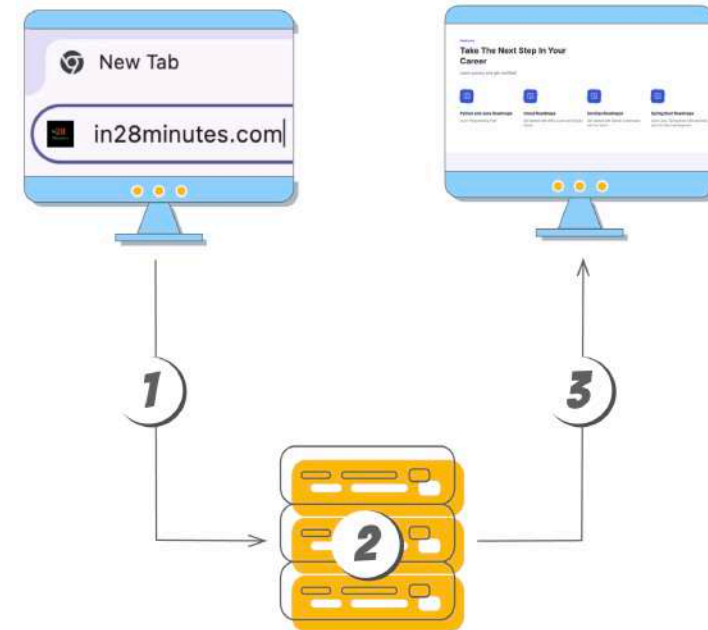
The Core Concept: A server is a powerful computer designed to provide services

- **Server:** The provider (e.g., in28minutes web server)
- **Client:** The requester (e.g., browser)

What Does a Server Serve?:

- **Websites:** HTML pages, images, videos
- **Files:** Documents, photos, backups
- **Emails:** Sending and receiving messages

Summary: Servers provide services, and clients consume those services





Can One Server Handle Everything?

Scenario: in28minutes.com goes viral

- **Traffic Spike:** Millions of users try to access it

Challenge 1: One server has limits

- **Capacity:** Can only handle a limited number of users

Challenge 2: What if the hardware fails?

- **Single Point of Failure !** : If the server fails, the application goes offline

Conclusion: Relying on a single server is risky for most applications





The Enterprise Reality – 1000s of Servers

Scenario: Consider a large bank or a financial institution

- **Lots of Apps:** Websites, mobile applications, internal applications and others
- **Thousands of Servers:** Needed to run all kinds of applications reliably

The Challenge: Not recommended to manage thousands of servers in an office

- **Requirements:** You need massive space and power
- **Question:** Where to house this infrastructure?





What is a Data Center?

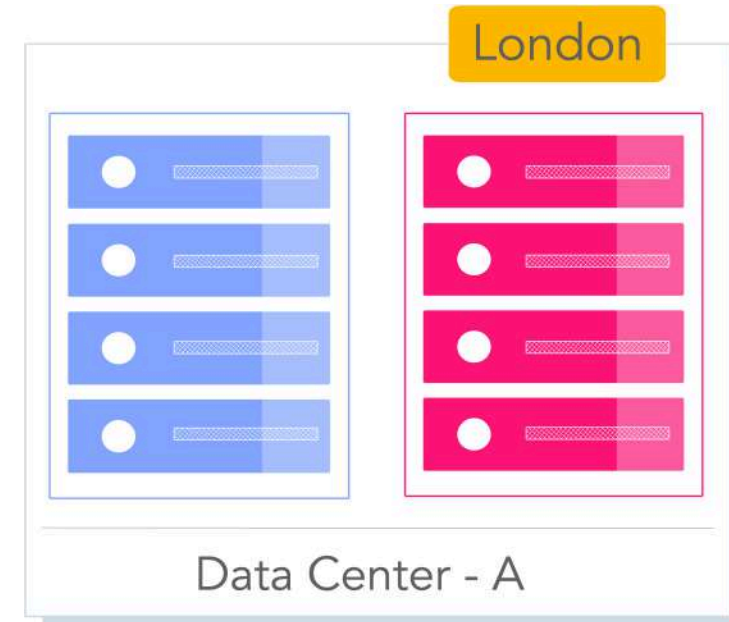
Basic Idea: A secure place for hosting servers

- **Massive Scale:** Houses hundreds, thousands, or even millions of servers

It is Not Just a Building: Needs specialized infrastructure

- **Power:** Uninterrupted power supply
- **Cooling:** Prevent servers from overheating
- **Networking:** High speed connectivity
- **Strong Security Needs:** Hosts key business systems

The Challenge: Building and managing data centers is expensive and complex





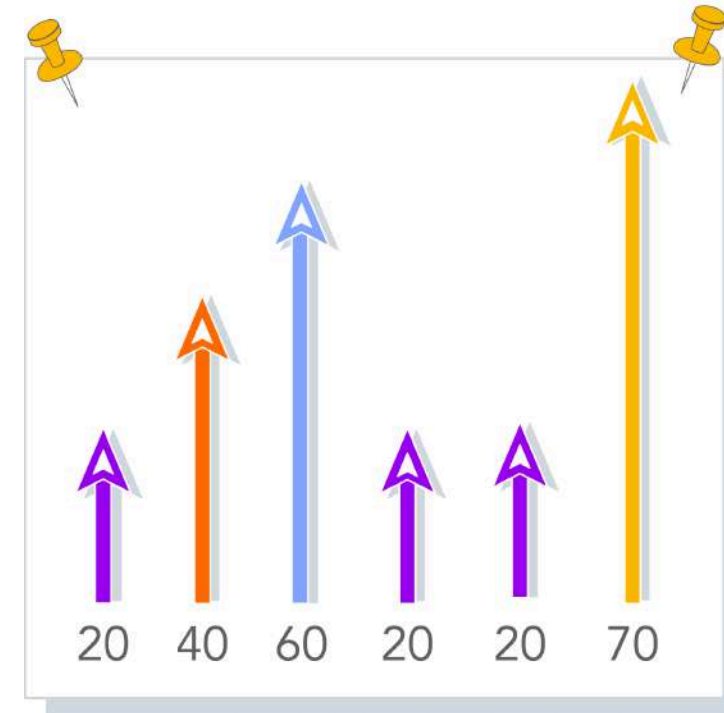
Example 1: Shopping Website

What Happens?: Online shopping demand is not constant

- **Peak Times:** Holidays, sales, weekends
- **Low demand:** During other times

Traditional Approach: Provision For Peak

- **Peak Load Provisioning:** Buy enough servers for peak (maximum) usage
- **Why?:** If you don't provision for peak, the site crashes when you need it most
- **Under Utilized Resources:** Most servers remain idle during periods of low demand





Example 2 : A Startup Succeeds

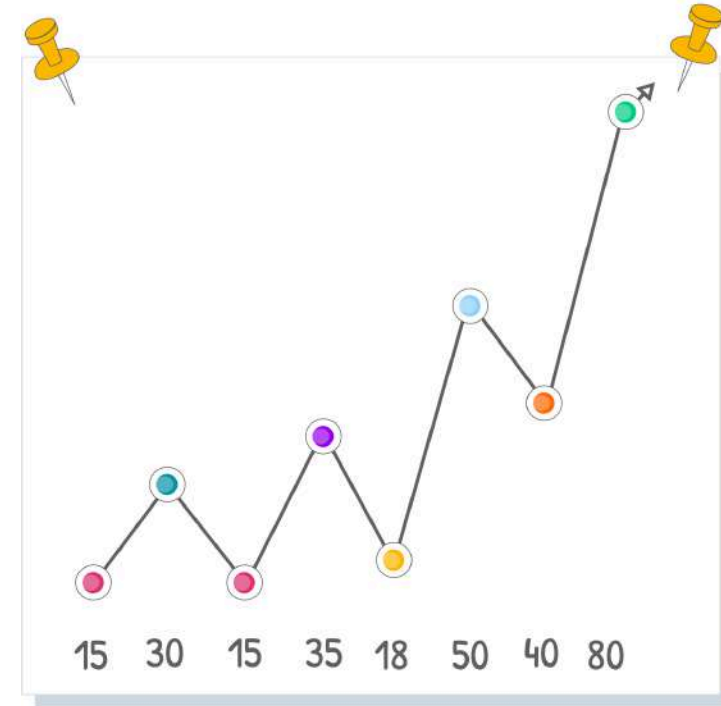
Scenario: Launches a new app that goes viral

The Problem: Unpredictable growth

- **Surge:** Users grow from 1000 to 1 million overnight
- **The Constraint:** Servers take time to buy and install

Traditional Approach: Buy servers in advance

- **Over-Provisioning:** Buy infrastructure in advance for future growth
- **Risk:** What if growth does not happen?
 - **Outcome:** Expensive hardware remains unused





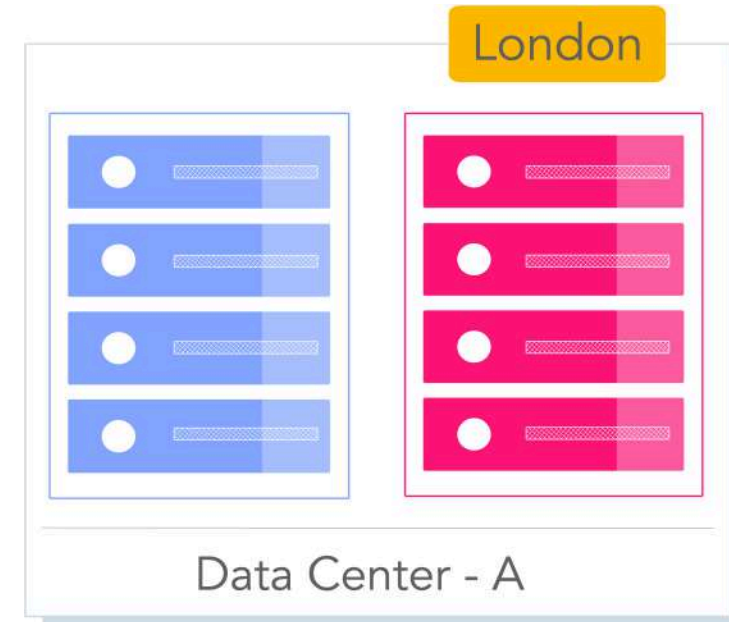
Example 3 : Enterprise Expansion

Scenario: A European company wants to serve Indian customers from a closer location

- **Current Infrastructure:** A Data Center in London
- **Goal:** Enable faster access for Indian users
 - Decrease Latency !

Traditional Approach: Set up a New Data Center in India

- **Time:** Takes months or years to set up
- **Cost:** Massive upfront investment





Challenges In Setting Up Own Data Centers

High Upfront Cost: To Buy Servers

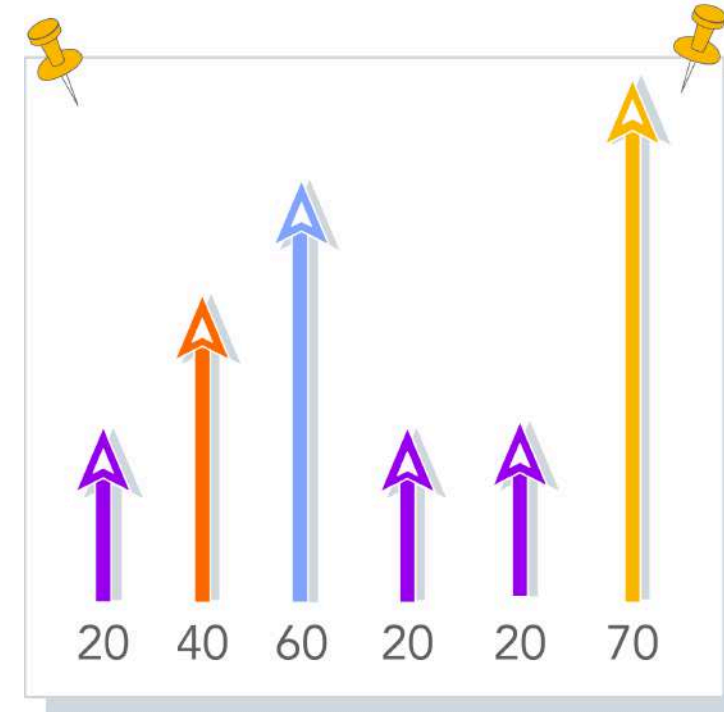
- **Slow Speed:** Takes weeks or months to set up

Needs a Dedicated Team: To maintain infrastructure

- **Think of a startup:** Can a startup afford it?

Guessing Game: You must predict traffic months in advance

Geographic Limitations: Difficult to expand into new locations quickly





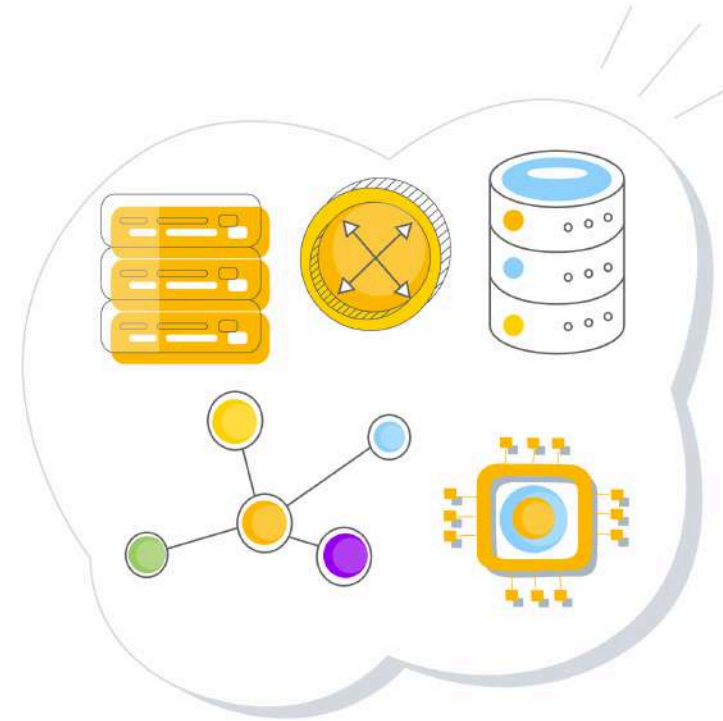
The Cloud: A Silver Lining

Key Question: Why buy infrastructure when you can rent it?

- **Cloud:** Rent servers when you need them!
- **Remember:** Cloud is NOT magic - it's just someone else's infrastructure (servers) that you rent

Core Concepts:

- **On-Demand Access:** Rent resources (servers, storage, databases) instantly
- **Elasticity !** : Increase or decrease the number of resources (servers) based on demand
- **Pay-As-You-Go !** : Pay only for what you use





Cloud Advantage – Pay for Use

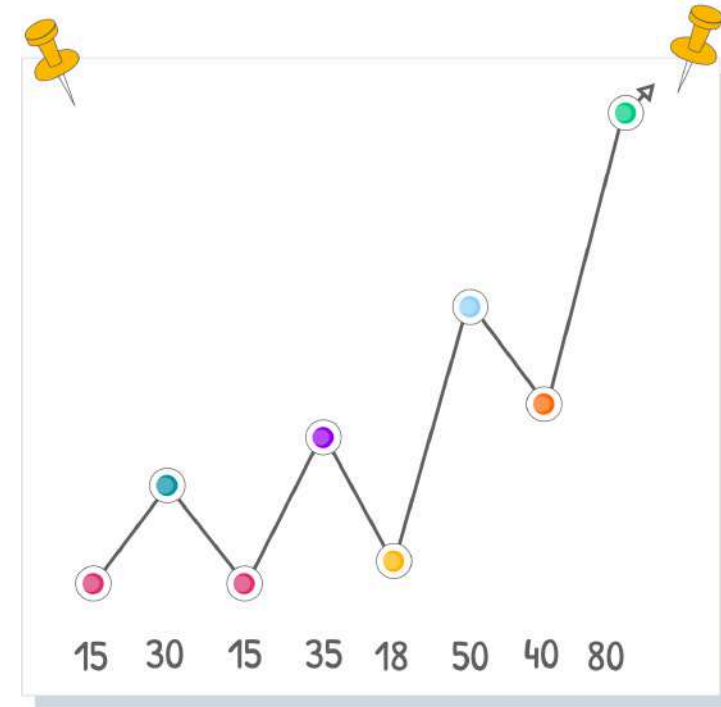
Scenario: A startup launching a new app

Traditional Way: Needs investment to set up a data center before launching

- **CapEx !** : Capital Expenditure or Capital Expense

The Cloud Solution: Rent and start small

- **Rent Servers:** From cloud provider
- **Pay for Use:** Costs only grow if the app grows
- **Lower Risk:** Fail without heavy investment
- **OpEx !** : Operational Expenditure or Variable Expense





Cloud Advantage – Economies of Scale

Scenario: Who gets a better deal?

- **You:** Buying 10 hard disks OR
- **Your Cloud Provider:** Buying 10 million hard disks

Economies of Scale ! : Buying infrastructure in large quantities typically lowers the cost per unit

The Cloud Reality: Cloud providers purchase infrastructure in massive volumes

- **Result:** Lower cost per unit
- **Customer Benefit:** Lower prices passed on to users





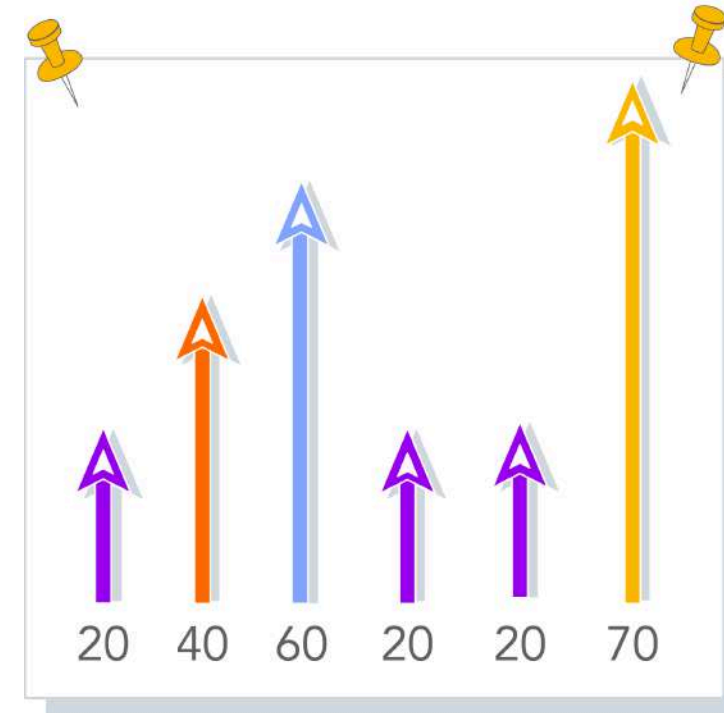
Cloud Advantage – Stop Guessing Capacity

Scenario: Applications and websites need to predict traffic a few weeks ahead of time

- **Not easy:** Difficult to predict accurately
- **Consequence:** Over-provisioning or Under-provisioning

The Cloud Solution: Elasticity !

- **Scale on Demand:** Add more servers instantly when traffic peaks
- **Release Servers:** Release servers when traffic drops
- **Impact:** You do NOT need to pay for idle resources





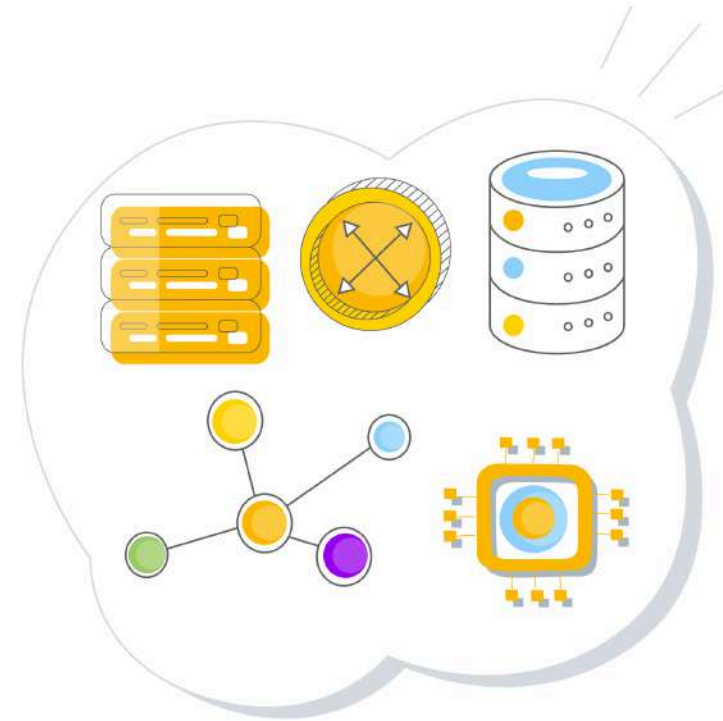
Cloud Advantage – Increased Speed and Agility

Scenario: Your team has a great idea for a new application

- **Traditional Approach:** "Wait 6 weeks for us to buy new servers"
- **Result:** Innovation slows down!

The Cloud Solution: "Click a button, get a server in minutes"

- **Agility !** : The ability to move fast, and respond to changes quickly
- **Faster Time to Market:** Launch features faster





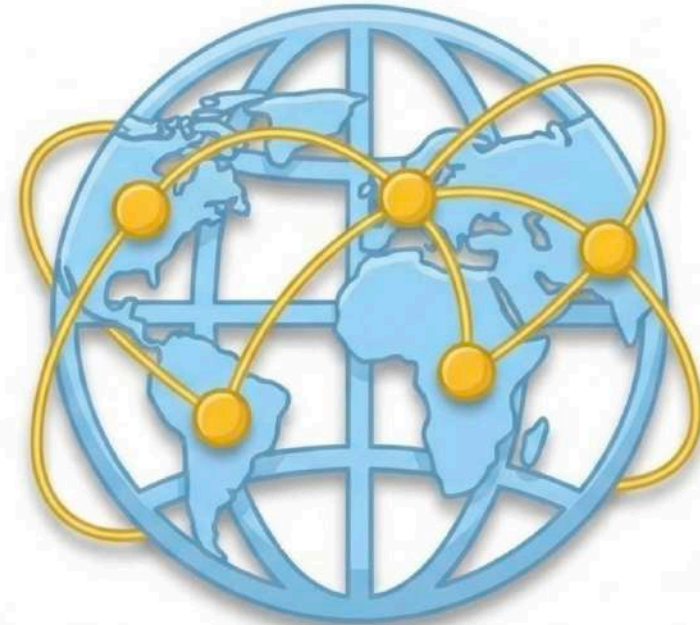
Cloud Advantage – Go Global

Scenario: You have users around the world!

The Challenge: High latency for global users if your servers are only in one location

Cloud Solution: Deploy to multiple regions in minutes

- **Global Reach !** : Expand to new markets around the world quickly
- **Better Experience:** Provide low latency (faster access) to users worldwide
- **Lower Cost:** No need to build data centers and manage infrastructure in multiple locations





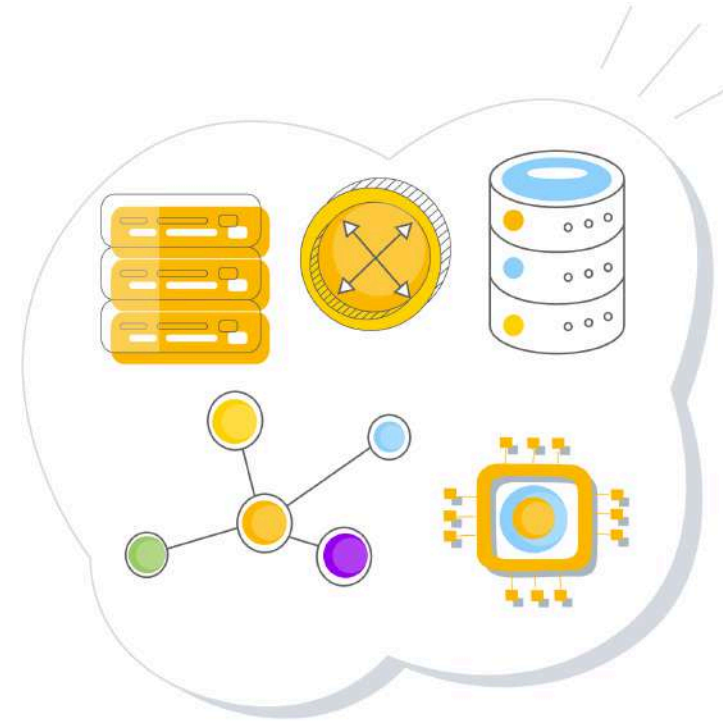
Cloud Advantage – Avoid Heavy Lifting

Scenario: Insurance Company

- 30% on infrastructure, 70% on core business
- **Undifferentiated Heavy Lifting !** : Tasks that are hard but don't make you unique

The Cloud Solution: Cloud reduces infrastructure burden

- **Avoid Undifferentiated Heavy Lifting:** Spend more time on what makes your business unique
- **Example – Insurance Company:** Get better at calculating premiums or detecting fraud (using AI)





Cloud – Key Terminology to Remember (1 of 2)

Term	Description
Elasticity	Ability to automatically scale resources up or down based on demand
Agility	Ability to move fast and respond quickly to business needs Quickly deliver software Quickly provision infrastructure
Global Reach and Geo-Distribution	The ability to expand to new markets and deploy applications to data centers around the world quickly
Availability	Are applications available when users need them
Latency	Time taken for a request to travel between user and server Reduced by serving users from nearby locations



Cloud – Key Terminology to Remember (2 of 2)

Term	Description
Capital Expenditure (CapEx)	Large upfront investment - For example, in infrastructure like servers and data centers
Operational Expenditure (OpEx)	Ongoing cost for renting resources (or services) as needed
Pay As You Go	Trade capital expense (CapEx) for variable expense(OpEx). Pay only for the resources you actually use.
Economies of Scale	Buying resources in large quantities lowers the cost per unit
Avoid Undifferentiated Heavy Lifting	Spend more time on what makes your business unique Example: Insurance company focusing on calculating better premiums.



Cloud Concepts – A Few Scenarios (1 of 2)

Scenario	Cloud Concept
Netflix adds thousands of servers instantly on Friday nights to stream movies to millions, then releases them on Monday morning.	Elasticity - increase or decrease no of servers based on demand
A startup launches a new app with minimal cost and increases spending only as user adoption grows	Pay As You Go or Operational Expenditure
A product team provisions infrastructure in minutes to test new features and releases updates faster	Agility
A healthcare application ensures patients can access services 24x7, even during maintenance or failures	High Availability



Cloud Concepts – A Few Scenarios (2 of 2)

Scenario	Cloud Concept
A SaaS company deploys its application in multiple regions to serve customers worldwide with low latency	Global Reach and Geo-Distribution
A media streaming platform serves videos from locations closest to users for faster playback	Low Latency
A cloud provider buys hardware in massive quantities and offers lower prices to customers	Economies of Scale
An insurance company focuses on fraud detection and customer experience while the cloud provider manages infrastructure	Avoid Undifferentiated Heavy Lifting



Google Cloud - Google's Cloud Platform

Built by Google: Runs powerful infrastructure driving products serving over billion users - Google Search, Gmail, YouTube among others

- **The Best Network:** Google owns one of the world's largest private fiber-optic networks
- **Speed:** Your data travels on Google's own private "internet" for faster, more secure connections
- **Data & AI DNA:** Widely recognized as the leader in Data Analytics and AI

Enterprise Ready: Used by startups and massive enterprises

- **Massive Service Catalog:** Hundreds of cloud services





We Worked Out the Best Path to Learn Cloud

So Many Services!: Compute, Databases, Storage, Networking, Analytics, and more

Overwhelming Choices – You could learn them one by one

- **Risk** – Easy to get lost in the details

The Big Question: *Where should you start?*

Our Solution – Structured, step-by-step learning path

Goal – Make learning cloud easy, practical, and fun





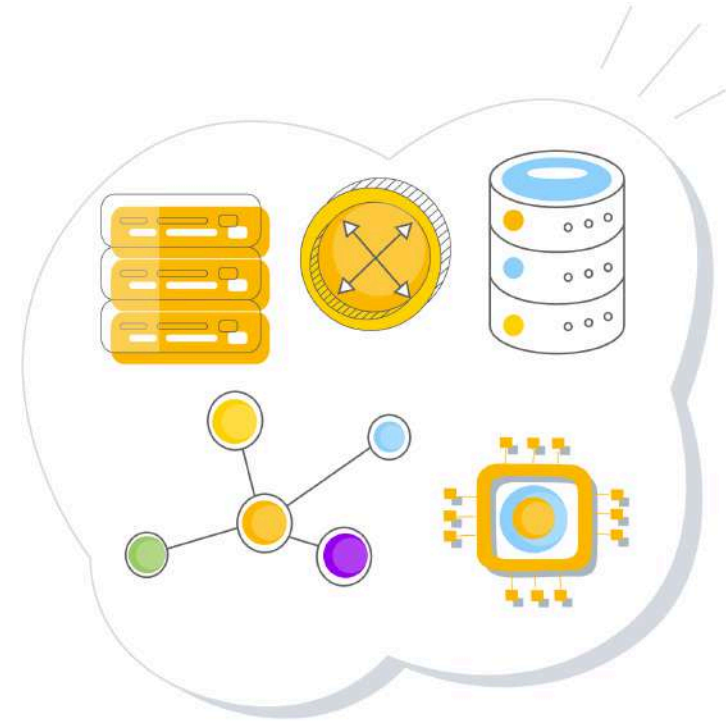
Setting Up Your Cloud Account

Gateway to Cloud – You need an account to access services

- **Global Access** – Deploy resources in regions around the world
- **Free Tier** – Most cloud providers offer free credits for beginners
- **Benefit** – Learn and experiment without immediate cost

First Step – Create your account to play with the cloud

- **I'm Excited!:** To Get You Started





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Getting Started with Regions and Zones

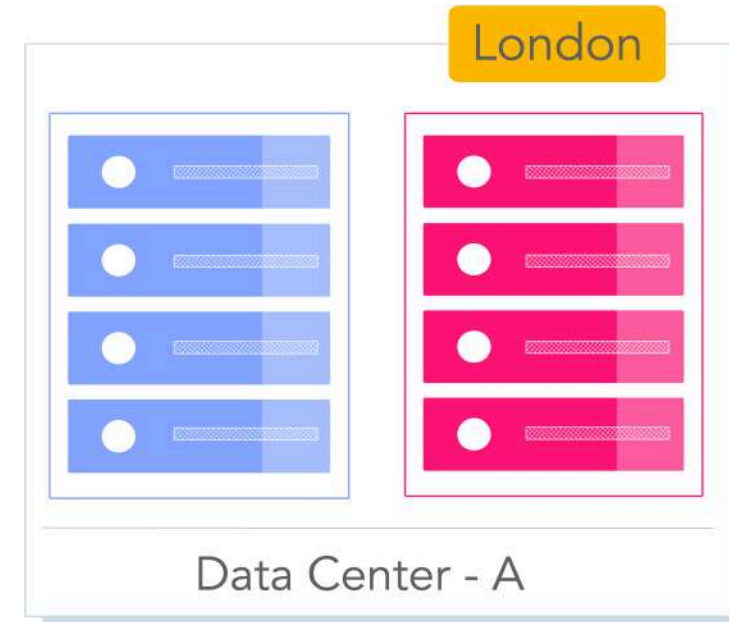


Scenario - Application in One Data Center

Scenario: Imagine that your application is deployed in a data center in London

Challenges:

- **Challenge 1 :** Slow access for users from other parts of the world (**High Latency !**)
- **Challenge 2 :** What if the data center crashes?
 - Your application goes down (**Low Availability !**)





Scenario - Application in Multiple data centers

Scenario: Let's add in one more data center in London

Challenges:

- **Challenge 1** : Slow access for users from other parts of the world
- **Challenge 2 (SOLVED)** : What if one data center crashes?
 - Your application is **still available** from the other data center
- **Challenge 3** : What if **entire region** of London is unavailable?
 - Your application goes down





Scenario - Add a New Region

Scenario: Let's add a new region - Mumbai

Challenges:

- **Challenge 1 (PARTLY SOLVED)** : Slow access for users from other parts of the world
 - You can solve this by adding deployments for your applications in other regions
- **Challenge 2 (SOLVED)** : What if one data center crashes?
 - Your application is still live from the other data centers
- **Challenge 3 (SOLVED)** : What if entire region of London is unavailable?
 - Your application is served from Mumbai





Regions

The Challenge: Imagine a startup in India setting up data centers in different regions around the world

- Would that be easy?

Cloud Makes It Easy: Cloud Providers provide Regions all around the world

Region : A specific geographical location to host your resources

- **Examples:** New York, London, Mumbai, Sydney





Choosing The Right Region - Factors

1 – Compliance ! : Be aware about regulations

- **Why:** Some countries require citizens data to stay within borders

2 – Latency: Be close to your users

- **Goal:** Choose a region closest to the majority of your users

3 - Services Available: Does the region support the services you need?

- **Note** – New services often launch in specific regions first

4 – Pricing: Check pricing in different regions

- **Why:** Costs can vary between regions for same resources





Why Use Multiple Regions?

Low Latency – Serve users faster by placing resources closer to them

Global Footprint – Expand services and reach customers worldwide

Data Residency – Store data in specific geographic locations to meet regulations

High Availability – Reduce downtime by running apps in multiple regions

Disaster Recovery – Protect against regional failures (e.g., natural disasters)





Zone(s) or AZ(s) - Availability Zones

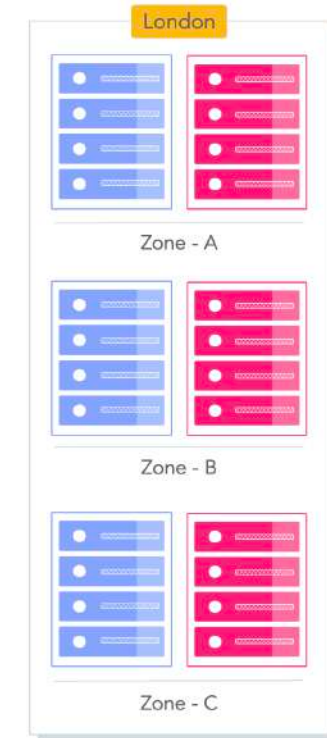
Scenario: How do you achieve high availability within a single Region?

Solution: Use Zones (Availability Zones)

- **Zones:** Isolated locations within a Region
- **Independent:** Each Zone has independent power, n/w & connectivity
- **High Speed Link:** Zones in a region are connected through low-latency links

Advantage: Increased availability and fault tolerance within same region

- Survive the failure of a complete data center





Regions vs. Zones - Quick Summary

Feature	Region	Zone/AZ
Definition	A separate geographic area (e.g., London, Mumbai)	An isolated location <i>within</i> a Region
Distance	Regions are separated by large geographic distances	Zones are connected by ultra-low latency links within the same region
Failure Scope	Deploying in multiple regions protects against massive disasters (earthquakes, floods) affecting a whole region	Deploying in multiple zones in a region protects against local failures (power outage, fire) in a specific building
Primary Goal	Disaster Recovery, Data Residency, Low Latency for global users	High Availability within the same region



Common Misconceptions – Regions and Zones

Misconception	Reality
Regions are only for large enterprises	Cloud makes regions accessible even to startups
All regions cost the same	Costs vary from region to region - based on local taxes and costs of running a data center in that specific region
Using one zone is enough for high availability	Zone failure can still bring apps down
All regions have same number of zones	Number of zones can vary by region
All zones have just one data center	Nope - Some zones can have more than one data centers



Regions and Zones – Scenarios

Scenario	Best Practice
Design highly available applications within a Region	Multi-Zone Deployment
Global SaaS application with users worldwide	Deploy across multiple regions
Survive data center failure	Distribute across Zones
Survive entire city or regional outage	Distribute across Regions
Meet data residency laws	Select specific compliant Region



Google Cloud - Regions and Zones examples

New Regions and Zones are constantly added

Region Code	Region	Zones	Zones List
us-west1	The Dalles, Oregon, North America	3	us-west1-a us-west1-b us-west1-c
eu-north1	Helsinki, Finland, Europe	3	eu-north1-a eu-north1-b eu-north1-c
asia-south1	Mumbai, India APAC	3	asia-south1-a asia-south1-b asia-south1-c



Getting Started with Virtual Machines



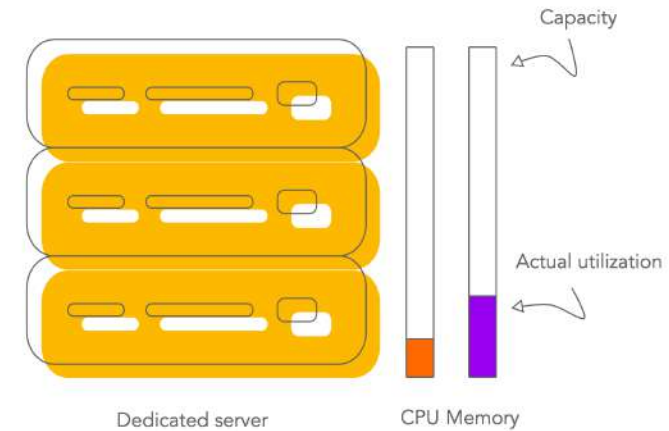
Getting Started – What is Virtualization

Goal: Take you on journey with virtual machines (VMs)

The Journey:

- **Problem:** Why dedicated servers are inefficient
- **Solution:** How virtualization solves the problem (enables creation of multiple VMs on a single physical server)
- **Evolution:** How VMs moved from data centers to cloud

Ready?: Let's get started on this fun journey





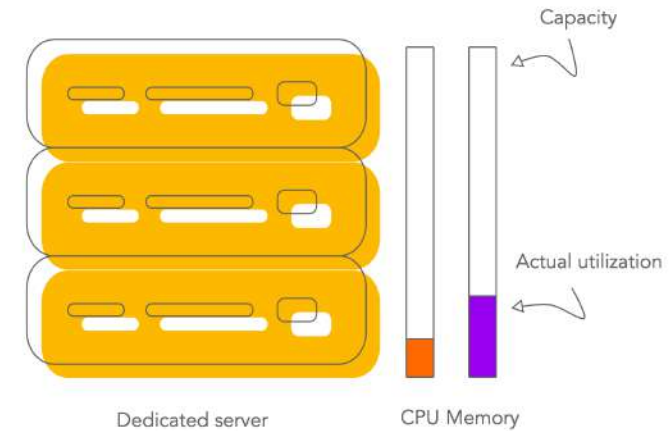
The Dedicated Server Dilemma

Scenario: You deploy an app on a dedicated physical server

Issues:

- **Resource Wastage:** Your app uses only 20% CPU (resources are under utilized)
- **Takes Time:** Setting up a new physical server for each app takes time
- **High Cost:** Buying a new physical server is expensive

Question: How can we do better?





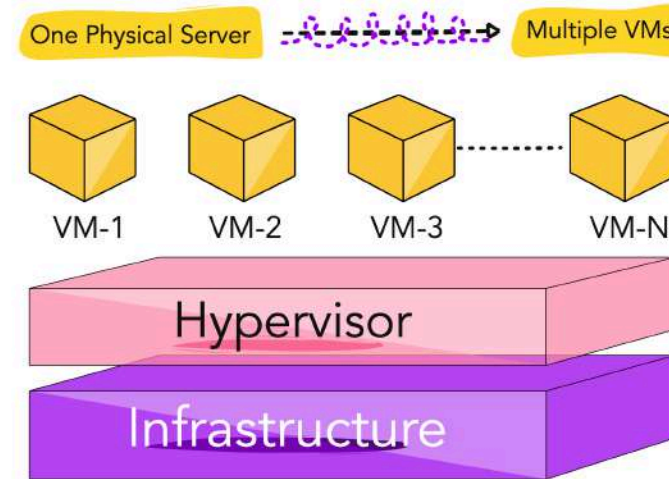
Enter Virtualization – The Solution

Goal: Run 10 apps on 1 server safely and efficiently

How?: Divide one physical server into multiple "virtual" servers

Requirement: A magic software layer

- **Role:** Sits on top of a physical server, creates & manages VMs
- **Resource sharing:** Resources (of the physical server) are shared between VMs
- **Isolation:** A VM should not impact other VMs





VMWare - Driving Virtualization in the Data Center

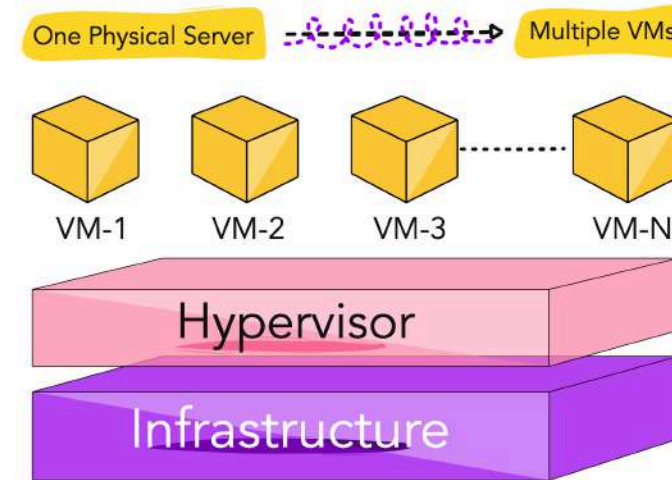
Question: How can multiple virtual machines run on a single physical server?

- **VMware:** Provides virtualization software (hypervisor)

Key Capabilities:

- **Flexibility:** Create, resize, and delete VMs via hypervisor
- **Isolation:** Failure in one VM does not impact others
- **Centralized Management:** Manage servers and VMs at one place

Challenge: Enterprises still need to maintain physical data center infrastructure



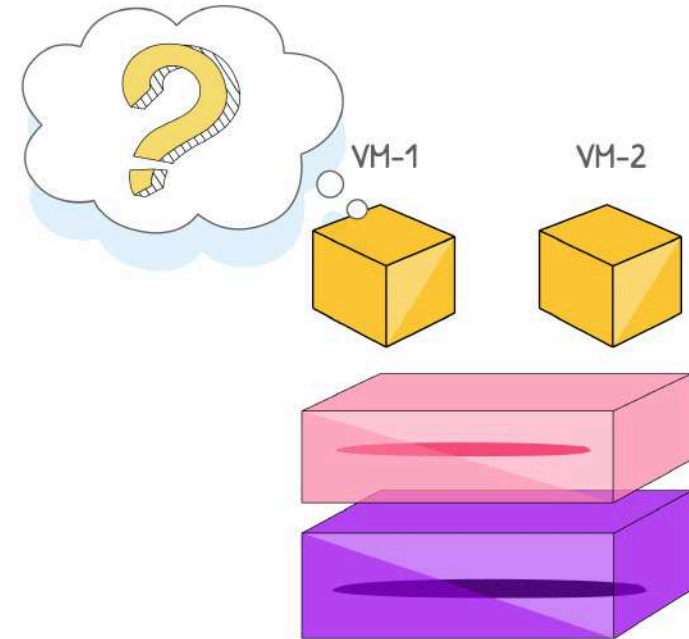


The Cloud Shift – Virtualization on Demand

The Next Step: Moving VMs from "Your Data Center" to "Cloud"

- **AWS:** Amazon EC2 (Elastic Compute Cloud)
- **Azure:** Azure Virtual Machines
- **Google Cloud:** Google Compute Engine
- **You Manage:** The VM configuration, OS, runtime and your app
- **Cloud Provider Manages:** Infrastructure (hardware, power, cooling, maintenance etc.) and Hypervisor

Freedom From Infrastructure: Focus on your VM and your app, not on infrastructure





How To Create Virtual Machines in the Cloud?

1: Choose the Right Service: EC2 or Azure VMs or GCE or ..

2: Choose Your VM Configuration:

- **Pick a Region:** Decide where to run your VM
- **Select an OS:** Choose from Linux, Windows, or ..
- **Pick the Hardware:** Pick CPU, memory, storage, ..

Launch in Minutes: Create a VM in minutes

- **Pay as you go:** Pay only for usage
- **Complete flexibility:** For example, create VMs in London from Hyderabad (India)





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Getting Started with Compute Engine



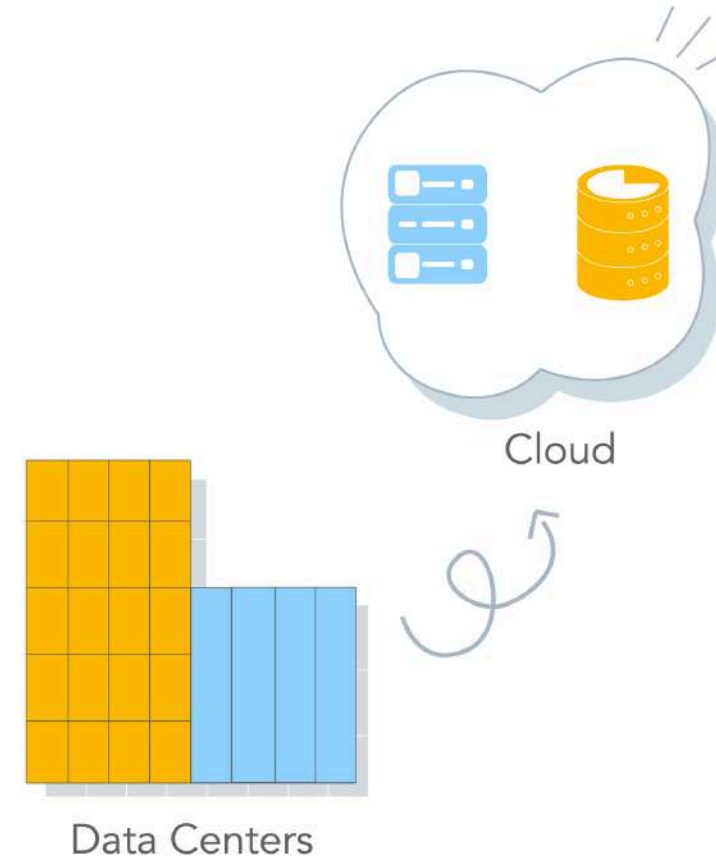
From Data Centers to the Cloud

VMware: Revolutionized data center virtualization

- Run multiple isolated VMs on a single physical server
- **Limitation:** Enterprises still responsible for infrastructure (hardware, networking, power, cooling, security)

Question: What if you could provision VMs without managing physical infrastructure?

- **Compute Engine:** Google Cloud Infrastructure-as-a-Service for creating and managing VMs

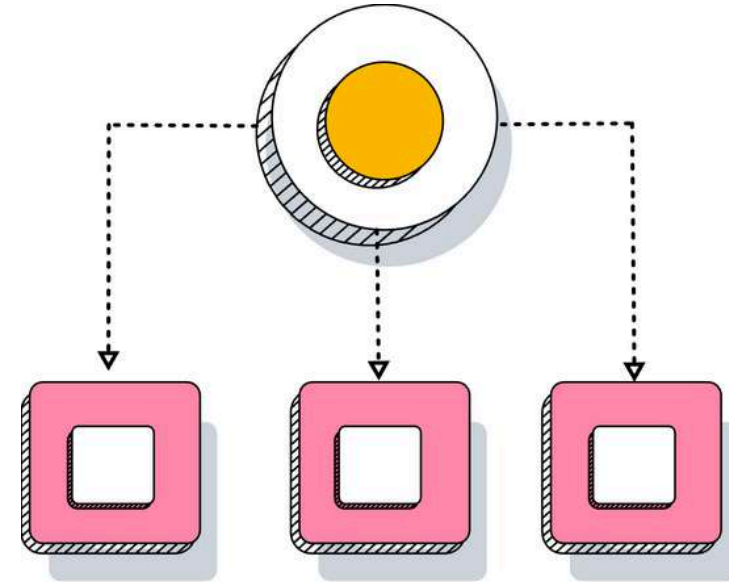




Compute Engine - Create VMs in Google Cloud

Compute Engine: Google Cloud managed service to create & manage VMs

- **Flexible Configuration:** Choose CPU, memory, disk, and OS to match your needs
- **VM Lifecycle Management:** Create, start, stop, resize, or delete virtual machines
- **Global Reach:** Deploy VMs close to your users
- **Pay-As-You-Go:** Pay only for the resources you use
- **Easily Scalable:** Add or remove VMs as needed
- **Group of VMs:** Create multiple VMs
 - **Load Balancing & Auto Scaling:** Handle more traffic automatically





Compute Engine - Key Concepts

Feature	Explanation
Compute Engine instance (VM)	A virtual machine running in Cloud
Image	Choose Operating System and Software
Machine Family	Choose the right family of hardware (General purpose or Compute/Storage/Memory optimized or GPU)
Machine Type (e2-standard-4, e2-standard-8, etc.)	Defines CPU and memory configuration (e.g., 4 vCPUs, 8 vCPUs)
Hyperdisk & Persistent Disk	Attach Virtual Disks to VMs (Block Storage)

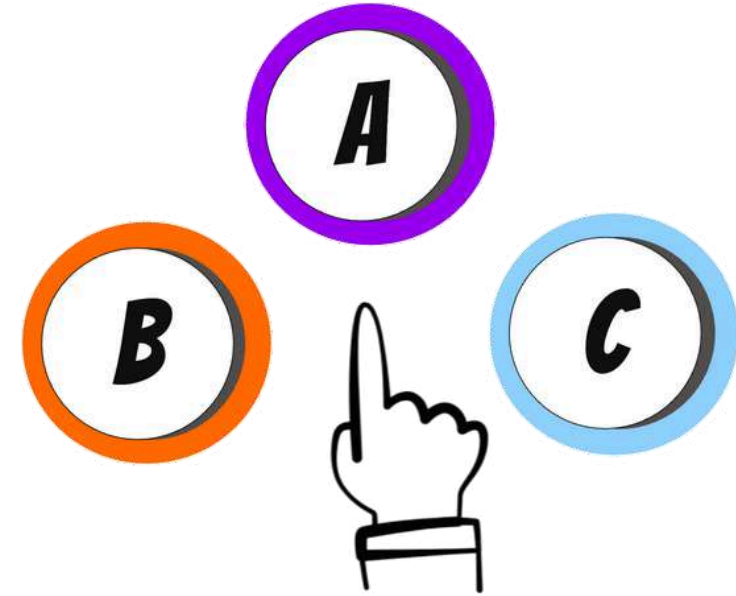


Choosing VM Hardware - Different Apps Have Different Needs

Goal: Achieve the best performance-to-cost ratio for your application

Different Applications: Different needs

- **Web applications:** Balance of CPU and memory
- **Analytics:** Lots of memory to perform calculations
- **Gaming:** High CPU and/or GPU for real-time processing
- **AI/ML or graphics:** Specific hardware - GPUs/TPUs
- **Big data or DW:** Large & fast storage - large data sets



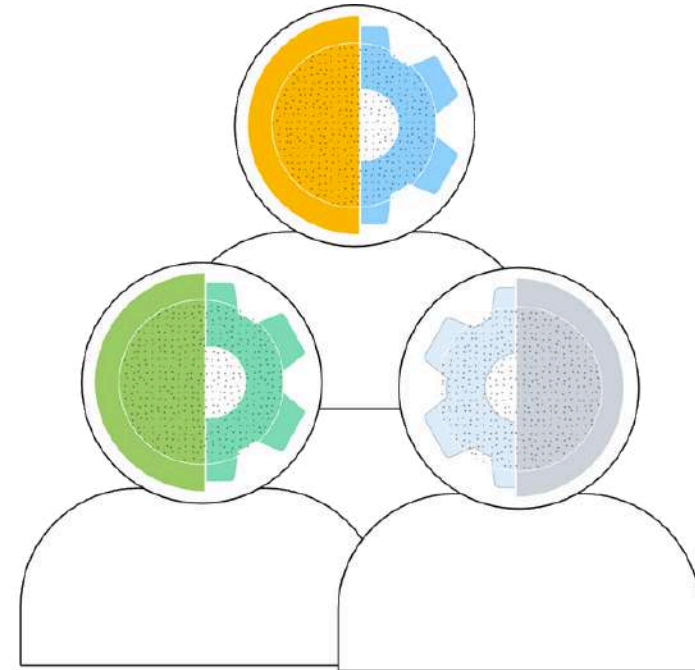
Question: How to make the right choice?



Choosing VM Hardware - Machine Family

Google Cloud: Offers multiple machine families for different workloads

- **General purpose (E, N series):** Balance of CPU and memory (web apps, dev env., small databases)
- **Memory optimized (M, X series):** High-memory workloads such as in-memory databases and analytics
- **Compute optimized (C, H series):** CPU-intensive workloads like batch processing and simulations
- **Storage optimized (Z series):** High disk I/O for big data and data warehouse workloads
- **Accelerator optimized (A, G series):** GPU or TPU workloads for AI, ML, and graphics





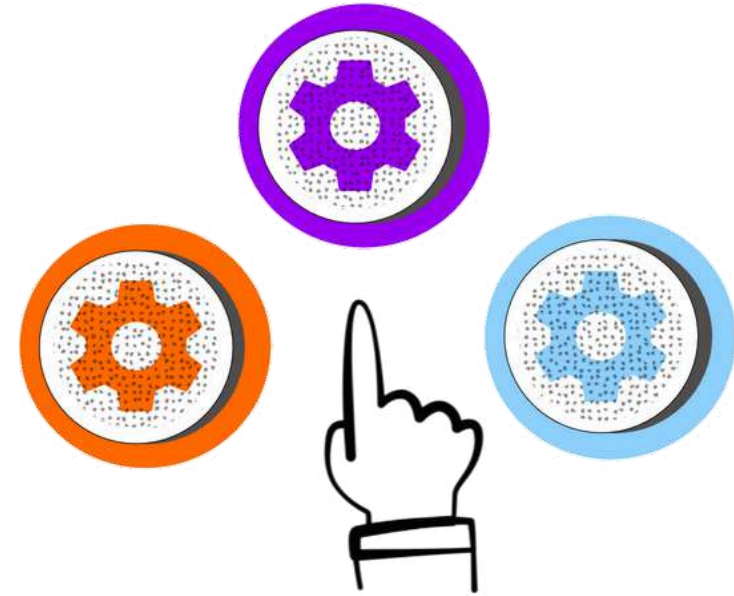
Choosing VM Hardware - Machine Type

Each machine family includes multiple machine types

- **Examples:** Options from E2 family: e2-micro, e2-standard-4, e2-highmem-8, e2-highcpu-16, etc.

Naming example: e2-standard-4

- **e2:** Machine family
- **4:** Number of vCPUs
- **standard:** Configuration type
 - **System memory per vCPU (e2 family):** standard (4 GB), highmem (8 GB), highcpu (1 GB)





Choosing Your OS and Software - Google Cloud - Image

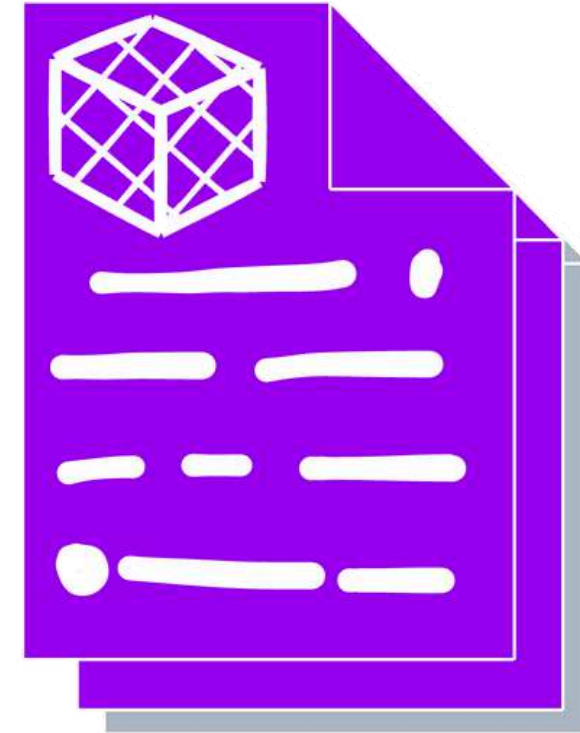
Image: Defines the OS and software your VM will run

Public Images: Ready-to-use images

- **Provided by:** Google, open source communities,...
- **Launch quickly:** With popular OS and software
- **Ex:** Debian, Ubuntu, Windows Server, SQL Server

Custom Images: Create your own image with exactly what your project needs

- **Full Control:** Include your preferred OS, pre-installed apps, configuration, and security settings





Compute Engine Hands-on : Setting up a HTTP server

```
#!/bin/bash
sudo su
apt update
apt -y install apache2
sudo service apache2 start
sudo update-rc.d apache2 enable
echo "Hello World" > /var/www/html/index.html
echo "Hello world from $(hostname) $(hostname -I)" > /var/www/html/index.html
```

Commands Explained:

- `sudo su`: Switch to the root user to run administrative commands
- `apt update`: Refresh package index from APT repositories
- `apt -y install apache2`: Install Apache2 web server
- `sudo service apache2 start`: Start the Apache2 web server
- `sudo update-rc.d apache2 enable`: Enable Apache2 to start automatically on reboot
- `echo "Hello world from $(hostname) $(hostname -I)" > /var/www/html/index.html`: Write a test message with hostname and internal IP to the web page



Google Compute Engine - Scenarios

Scenario	Solution
You want to select hardware for your VM based on workload	Machine Family and Machine Type
What does 8 in e2-standard-8 indicate?	No of vCPUs
You want to choose the operating system and pre-configured software for your VM	Image
You want to attach storage volumes to your VM for data persistence	Persistent Disk / Hyperdisk
You need to ensure that your VM runs a specific operating system and software stack for your application	Custom Image



IP Addresses - Virtual Machines

IP Address	Description
Internal IP Address	Permanent Internal IP Address (restricted to the specific Google Cloud private VPC network) that does not change during the lifetime of an instance [no charge]
External Ephemeral IP Address	Temporary External IP Address (accessible from internet) that might change when an instance is stopped or restarted
External Static IP Address	Permanent External IP Address that can be attached to a VM ["If you reserve a static external IP address and do not assign it to a resource such as a VM instance, you are charged at a higher rate than for static and ephemeral external IP addresses that are in use"]



Compute Engine IP Addresses - Scenarios

Scenario	Solution
You want your VM External IP address to remain the same across stops, restarts, and/or reassignments	Static External IP Address
You do not want your VM to be directly reachable from the internet	Assign Only Internal Ip Address (No External IP Address)
You want to talk to a VM from another resource inside the same private network. Which IP address should you use?	Use Internal IP Address of the VM (Traffic will stay within the network - it will not go over internet)
True or False: If you reserve a static external IP address and do not assign it to a resource such as a VM instance, you are charged at a higher rate than when it is assigned to a VM instance	True



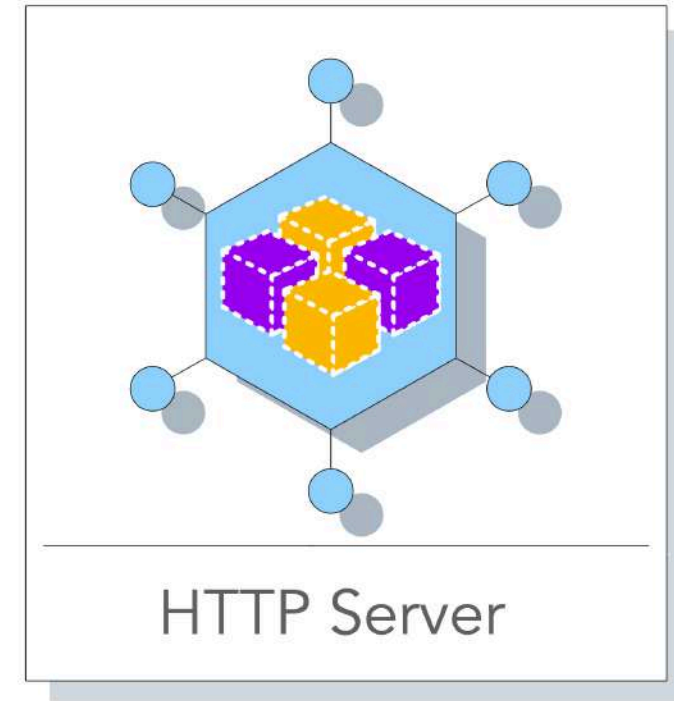
Simplify VM HTTP Server Setup

Goal: Reduce the number of steps in setting up a VM and HTTP server

Startup Script: Run setup automatically when a VM starts

Custom Image: Create ready-to-launch VM Disk Image with everything pre-installed

Instance Template: Reuse predefined VM configuration





Bootstrapping with Startup Script

Requirement: Install software on 10 VMs

- **Problem:** Each time you launch a VM, you need to connect to VM and install software
- **Solution:** Bootstrapping

Bootstrapping: Automatically install software when VM first starts up

- Install web server, patches, .. automatically

Startup Script: VM configuration to perform automatic bootstrapping

- Your VM sets itself up automatically, saving time and effort

```
#!/bin/bash
apt update
apt -y install apache2
echo "Hello world from $(hostname)"
$(hostname -i)" > /var/www/html/index.html
```



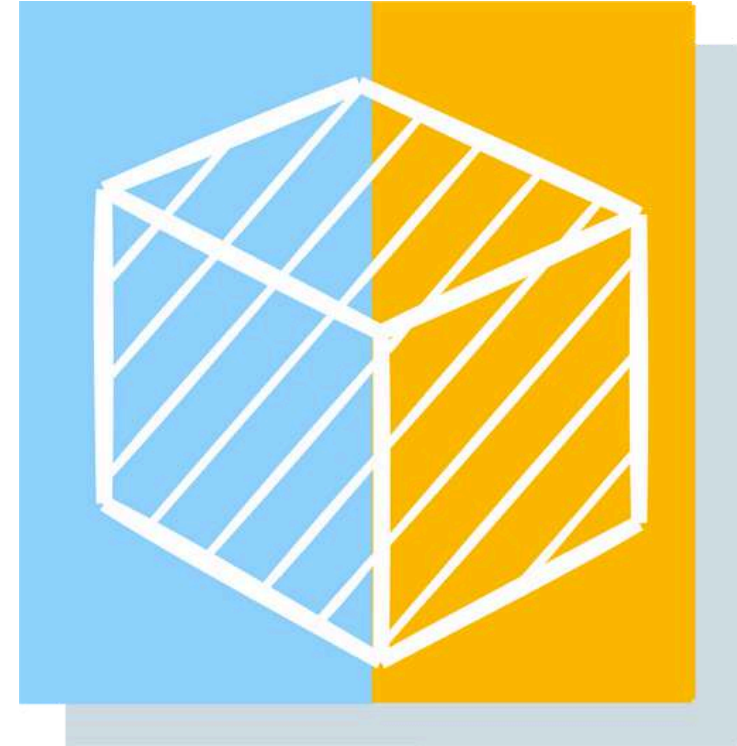
Reducing Launch Time with Custom Image [1/2]

Scenario: Every time you launch a new VM, it installs updates and software from scratch

Problem: Increases your VM startup time

Solution: Create a **Custom Image** with everything pre-installed

- Built from an existing VM, disk, snapshot, or Cloud Storage file
- Can be shared across multiple projects for consistency



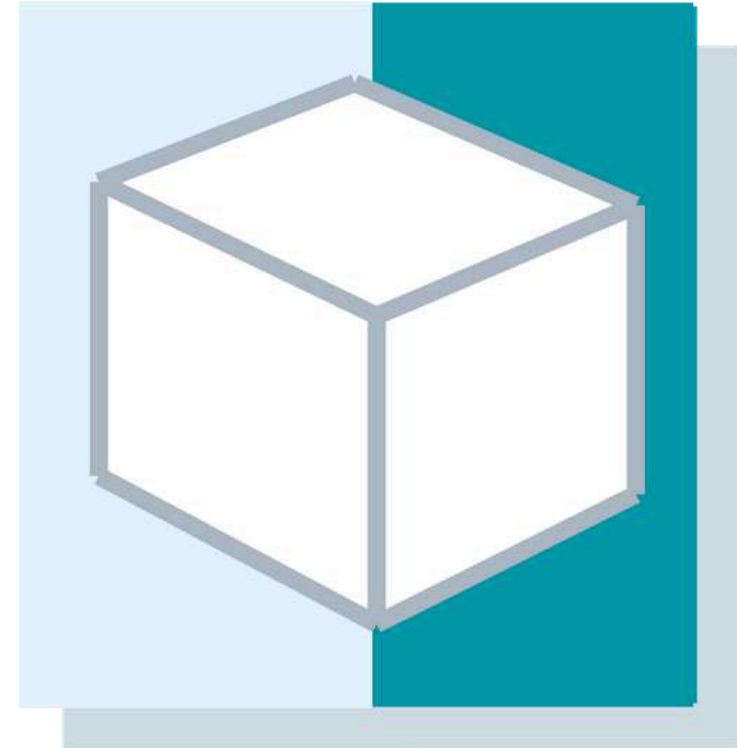


Reducing Launch Time with Custom Image [2/2]

Best Practices:

- Deprecate old images and specify replacements
- **Harden Images** to meet your organization's security standards
- **Prefer using Custom Image to Startup script**

Why It Matters: Custom Images make VM creation faster while being secure and reliable





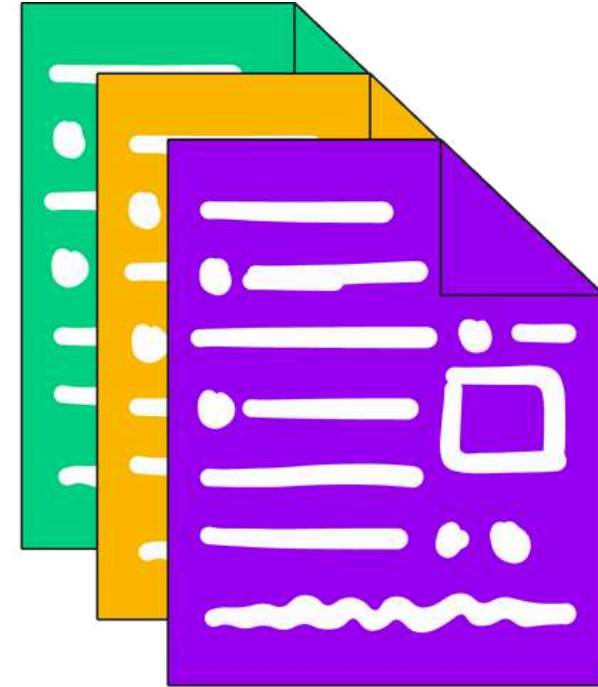
Instance Templates

Scenario: You've just set up (configured) a perfect VM - right image, right machine type, startup script, enabled http access ..

Problem: Each time you create a new VM, you need to configure it all over again

Solution: Use an **Instance Template**

- Configure machine type, image, labels, startup script, ..
- Quickly launch new VMs





Instance Templates - Remember

Multiple Use Cases: Create individual VMs or managed instance groups from same template

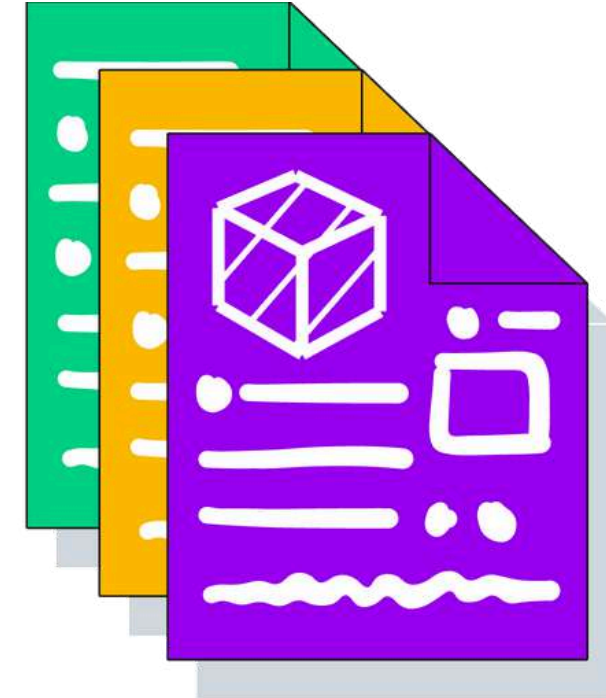
- **Very Convenient:** Launch similar instances quickly without configuring again

Immutable Templates: Cannot modify existing templates once created

- **Change Process:** Copy existing template and modify the copy to make changes

Optional: Specify Image Family(ex:*debian-13*)

- Automatically uses latest image from image family





Compute Engine - Launching VM - Scenarios

Scenario	Solution
You want to run software installation and setup every time a VM starts	Startup Script
You want to launch multiple VMs with the same machine type, image, startup script and other configuration	Instance Template
You want to create Managed Instance Groups using a reusable VM blueprint	Instance Template
You want to reduce VM launch time by pre-installing OS updates and software	Custom Image
You want to create secure, hardened VM images that follow organization standards	Custom Image



Reducing Costs - Compute Engine Virtual Machines

Feature	Description
Sustained use discounts	Automatic discounts for using resources for long periods of time Example: If you use certain machine types for more than 25% of a month, you get a discount on every incremental minute
Committed use discounts	Reserve compute instances ahead of time Commit for 1 year or 3 years Discount varies based on machine type
Spot VMs	Cheaper, temporary instances for non critical workloads (Cheapest but availability not guaranteed!)

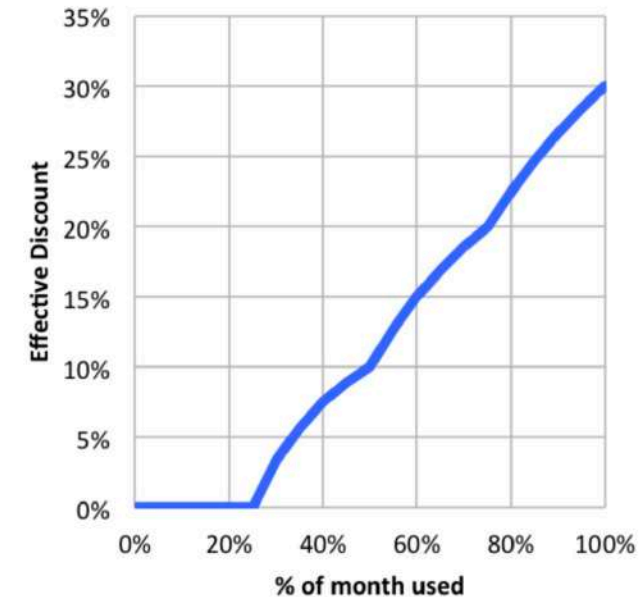


Sustained Use Discounts

Scenario: What if your VM runs for most of the month?

Concept: Google Cloud provides automatic discounts for sustained usage

- **Constraints:** 1) Used for more than 25% of a billing month 2) NOT receiving any other discounts
- Discounts increase gradually with usage over the billing month
- No upfront commitment required (Applied automatically to eligible resources)
- **Applicable To:** VMs of certain machine types in GCE and GKE





Committed Use Discounts

Scenario: What if you know your workload will run continuously for a long time?

Concept: Get a **discount** by committing resources for 1 or 3 years

- Google Cloud offers CUDs in return for purchasing committed use contracts (also known as commitments)
- Commit either to a minimum amount of resource usage or to a minimum spend amount for a duration of one or three years
- Can be a good option if your workloads have predictable resource needs



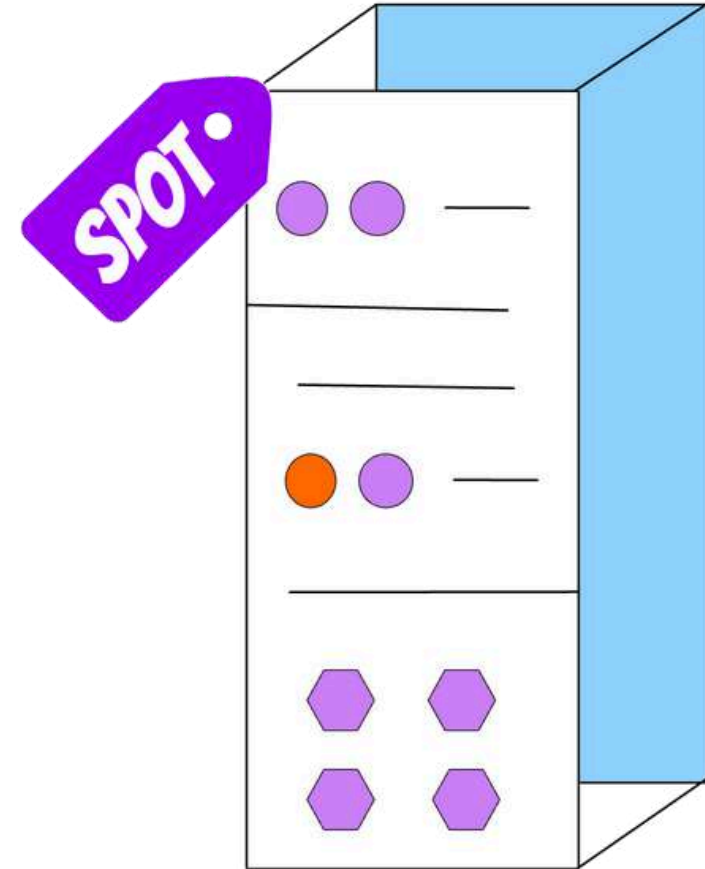


Spot VMs

Scenario: How can you run non-urgent workloads at the lowest possible cost?

Spot VMs: Use spare Google Cloud capacity at deep discounts (sometimes more than 80%)

- **Dynamic Pricing:** Changes based on available capacity
- **30-Second Warning:** You get a short warning to save work before shutdown
- **Use When:** Your apps are fault tolerant, cost-sensitive, and workload is not urgent (e.g., batch jobs)
- **Restrictions:** Not always available, no SLA, can't migrate to regular VMs, free tier credits do not apply





Spot VMs - Step by Step

Step 1: Create a Request – Launch a VM with Provisioning model - Spot

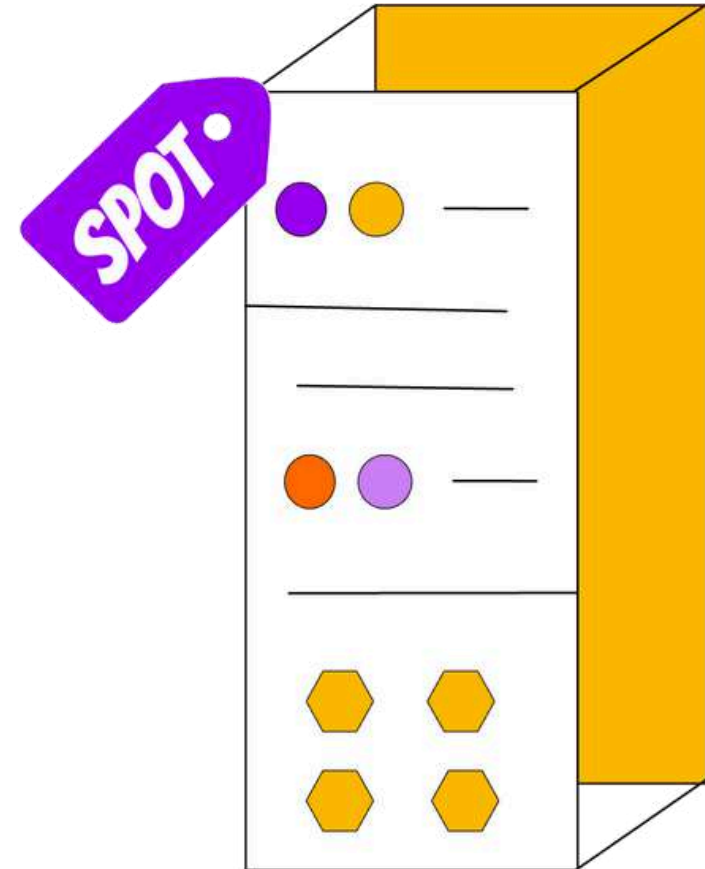
Step 2: Allocation – VM is provisioned if spare capacity is available

Step 3: Signal – Google Cloud sends a termination signal

- (or you can choose to stop the VM yourself)

Step 4: Termination – VM is stopped and resources are released

Note: You get a **30-second warning** before termination to save your work



Google Compute Engine - Understanding Billing and Costs



Per-Second Billing: Charged per second after the first minute

- **Stopped VMs:** No compute charges for stopped VMs
 - BUT you need to pay for storage (disks, etc.)

Best Practices:

- **Budget Alerts:** Always set budget alerts to avoid surprises
- **Right-size resources:** To match your workloads
- **Understand Discounts:** Understand Sustained Use, Committed Use, and Spot VM discounts and use them as applicable





Cost Optimization - Scenarios

Scenario	Solution
You have predictable resource needs and can commit for 1 or 3 years	Use Committed Use Discounts
You want automatic discounts for workloads running continuously over time	Use Sustained Use Discounts (applied automatically)
You have fault-tolerant, short-lived workloads that can handle interruptions	Use Spot VMs
You want the lowest cost for non-critical, restartable batch jobs	Use Spot VMs
Can you use Spot VMs? - You are running time-sensitive or critical backend APIs	Do NOT use Spot VMs



What is Observability?

Scenario: Your app is running slowly or returning errors

- But you do not know what the problem is or why it is occurring

Observability: The ability to understand what is happening inside your system

- Just by looking at **external outputs** like
 - logs
 - metrics
 - traces



Compute Engine - Getting Started with Observability



Metrics: Numeric values tracked over time

- **Default Basic Monitoring:** Enabled by default
 - CPU utilization, Network traffic, Disk throughput & I/O metrics
- **Monitor memory/disk space:** Install Google Cloud Ops Agent
- **Cloud Monitoring dashboards:** Provide visualization

Logs: Text records of events (errors, warnings)

- Needs Google Cloud Ops Agent
- View in Cloud Logging

Traces: The journey of a request through multiple services (Out of scope for now!)





Exploring Scope of Compute Engine Resources

Machine types: Availability and pricing can vary between regions

- **Change Machine Type:** Stop the VM first (cannot change while running)

Image: Multi-regional (ex: multiple regions in us) or regional (ex: us-central1)

Instance templates: Regional (ex: us-central1) or global

VM instances: Belong a specific zone (ex: us-central1-a) in a region (ex: us-central1)

- **Project Scoped:** VM instances belong to a project





GCE VMs - What is Internal DNS Name?

How can VMs in a VPC talk without IP Addr?

- Use **Internal DNS name**: Assigned when VM is created
- **Resolves to**: Internal IP address of the VM instance
- **Zonal DNS** recommended -
`VM_NAME.ZONE.c.PROJECT_ID.internal`
 - **VM_NAME** - What is your VM instance name?
 - **ZONE** - Which zone is it located in?
 - **PROJECT_ID** - Which project is it created in?
 - **Example**: `my-first-vm.us-central1-a.c.my-project-id.internal`

VM instances in the same VPC: Can use internal DNS names to communicate





Licensing Costs for OS

License: Usage agreement between you and a software vendor

- **Free licenses:** No charge for usage
- **On-demand/Pay-as-you-go (PAYG):** Google Cloud charges for usage. Cost varies depending on the OS
- **Bring your own license (BYOL)/Bring your own subscription (BYOS):**
 - You have a previous agreement with OS vendor
 - You can reuse it in Google Cloud (Remember: There are a lot of constraints)
 - **Recommendation:** Use your existing license while it is valid
 - After that shift to Pay-as-you-go model (PAYG)

Boot disk

Select an image or snapshot to create a boot disk; or attach an existing disk. Can't find what you're looking for? Explore hundreds of VM solutions in [Marketplace](#)

[Public images](#)

[Custom images](#)

[Snapshots](#)

[Archive Snapshots](#)

Operating system

SUSE Linux Enterprise Server

Version *

SUSE Linux Enterprise Server 12 SP5 x86_64

x86_64, built on 20260506

License type ⓘ *

Select the type of license to use with this OS image.



Pay-as-you-go (PAYG)

Google Cloud charges a license fee for usage of this image. [Pricing](#) varies based on the operating system.



Bring your own subscription or license (BYOS or BYOL)

Google Cloud doesn't charge you a license fee to use this OS image. With this option, you use the license subscription from your OS vendor.



Compute Engine Features - Custom Machine Types

Scenario: Predefined VM options are NOT appropriate for your workload

- **Custom Machine Type:** A machine type customized to your needs

Adjust: vCPUs, memory & GPUs

- **Limited Machine Families:** Among G, N and E machine series
- **Wide OS Support:** Use popular operating systems like Ubuntu, Debian, Red Hat, Windows, and more
- **Key Cost Factors:** Number of vCPUs and amount of memory provisioned

General purpose Compute optimized Memory optimized Storage optimized **GPUs**

Graphics processing units (GPUs) accelerate specific workloads on your instances such as machine learning and data processing. [Learn More](#)

GPU type: NVIDIA T4 Number of GPUs: 1

☐ Enable Virtual Workstation (NVIDIA GRID)

Series ?	Description	vCPUs ?	Memory ?	CPU Platform
☒ N1	Balanced price & performance	1 - 96	1.8 - 624 GB	Intel Haswell

Machine type

Choose a machine type with preset amounts of vCPUs and memory that suit most workloads. Or, you can create a custom machine for your workload's particular needs. [Learn more](#)

Preset Custom

Creating a custom machine incurs additional costs

Cores: 1 96 1 vCPU

Memory: 1 6.5 1 GB



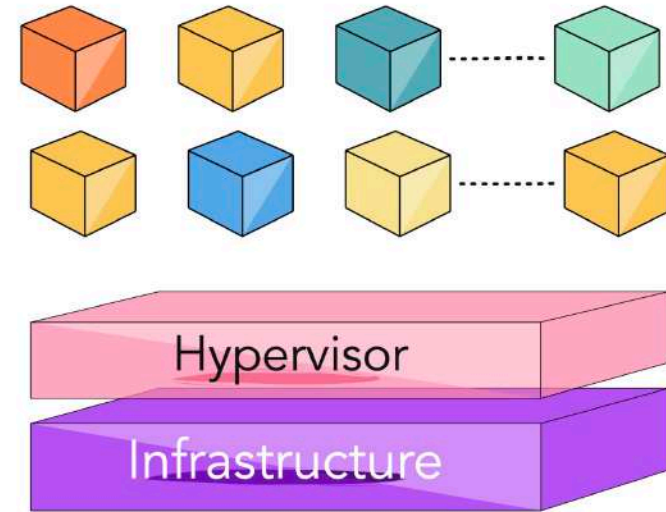
Need For Sole-Tenant Nodes

Do You Know?: Regular VMs share physical hosts with other customers

Sensitive Workloads: Some workloads require host level isolation for security, compliance, or licensing

- **Licensing Needs:** Some software (for example, Windows Server or SQL Server) is licensed based on the number of physical cores or processors on a host
- **Organizational Policies:** Internal policies may require dedicated hardware

Solution: Use **Sole-Tenant Nodes** to run VMs





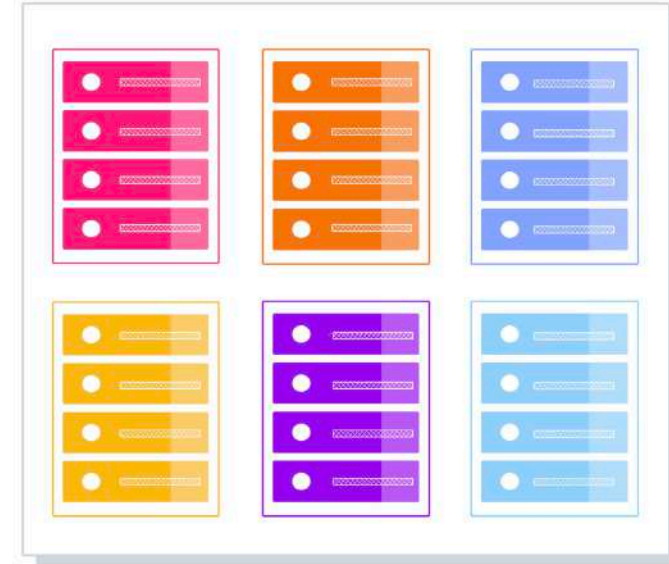
What Are Sole-tenant Nodes?

Dedicated Physical Servers: The entire physical host is dedicated to one customer

Fully Integrated with Compute Engine: Use the same machine types, images and other features

How?:

- **1: Create a Node Group:** Create one or more nodes with affinity labels (`group:a`, `group:b`, `group:c`)
- **2: Create VMs:** Use affinity rules (`group:IN:a`, `group:IN:a,b`, `group:NOT_IN:a`) to control the placement of VMs on nodes





When Should You Use Sole-tenant Nodes?

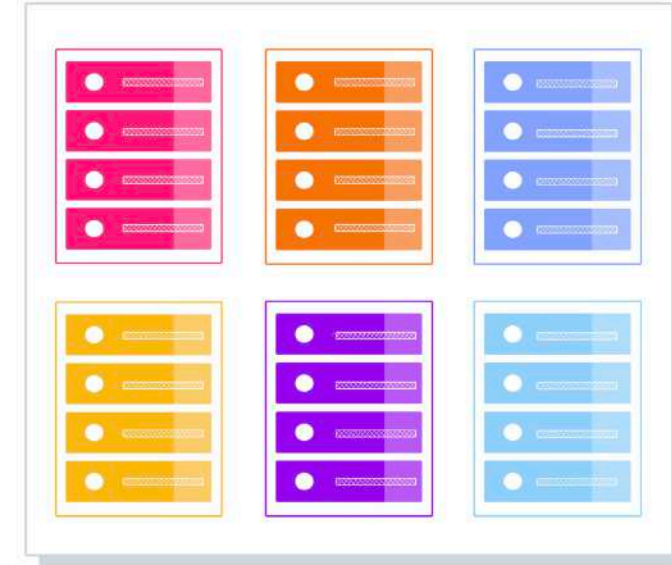
Strict Isolation Requirements: Compliance frameworks like PCI-DSS, HIPAA, FedRAMP

BYOL Licensing: Use an existing per-core or per-processor license

- *Examples:* Windows Server, SQL Server

Need Flexibility: With VM placement

- **1:** Place related workloads on the same host (All VMs on one Node Group)
- **2:** Spread workloads across hosts for High Availability (Some VMs on one node group and others on another node group)





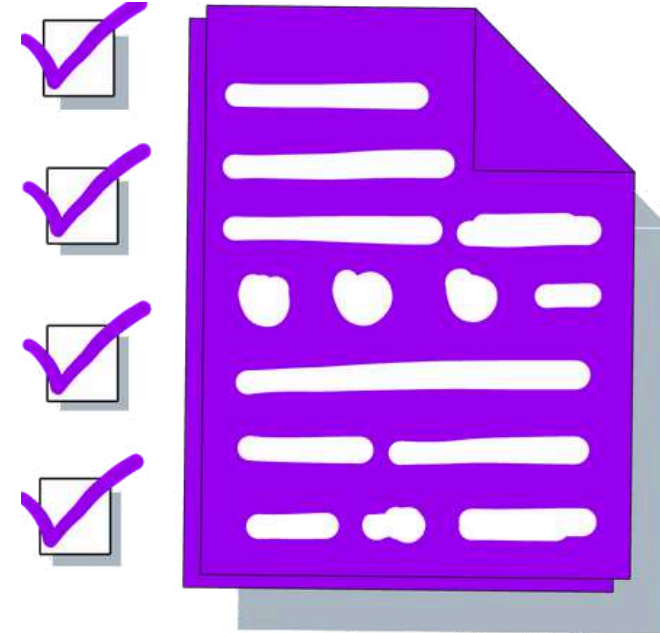
Compute Engine - Enabling Live Migration For a VM

Scenario: How to keep your VM instances running when a host server needs to be down for a planned maintenance event?

- **Live Migration:** VM instance is migrated to another host server in the same zone
- Does NOT change attributes or properties of the VM

Trigger Event Examples: Infrastructure maintenance, Hardware failures, etc.

Remember: Live Migration is NOT always supported (Examples: Not supported on GPU attached instances, Spot VMs)





Compute Engine - How to Enable Live Migration?

Key Configuration: Availability Policy

1: On host maintenance: What to do during periodic infra. maintenance?

- *Migrate (default when supported)*: Migrate VM instance to other hardware
- *Terminate*: Stop the VM instance

2: Automatic restart: Restart VM instances if terminated due to non-user-initiated reasons

- Maintenance event, hardware failure etc.

Provisioning model

VM provisioning model
Standard

Determines how long your VMs will run. Choice is permanent

☐ Set a time limit for the VM ?

☐ Gracefully shut down the VM ? [Preview](#)

On VM termination

Choose what happens to your VM when it's preempted or reaches its time limit

On host maintenance

Migrate VM instance (Recommended)

Terminate VM instance

Host error timeout
Unspecified (default)

When the VM becomes unresponsive, it will terminate after this amount of time

Automatic restart
On (recommended)

Compute Engine can automatically restart VM instances if they are terminated for non-user-initiated reasons (maintenance event, hardware failure, software failure and so on)

[^ VM provisioning model advanced settings](#)



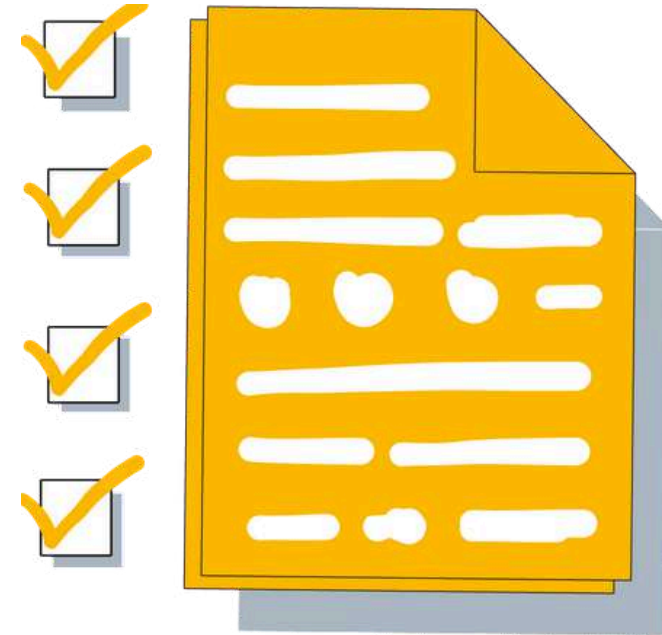
Cloud Best Practice - Use Labels

Labels: Key-value pairs attached with cloud resources

- Examples: `environment:production`, `team:devops`, `business:finance`
- Added during resource creation (or edit later)

Key Benefits: Of VM Labels

- **Cost Management:** Break down costs by project, team, or environment in billing reports
- **Resource Organization:** Group related resources (of a specific env. or team) making management easier
- **Bulk Operations:** Run scripts or automate tasks across multiple resources using labels





Compute Engine - Quick Review

Topic	Details
Machine Types	Optimized combination of compute (vCPUs) and memory for your workloads. Create Custom Machine Types when needed.
Image	What OS and software to use? Reduce boot time and improve security by creating custom hardened images. Images can be shared with other projects.
Instance Templates	Pre-configured templates simplifying the creation of VM instances
Sustained Use Discounts	Automatic discounts for running VM instances for significant portion of the billing month
Committed use discounts	1 year or 3 year reservations: For workloads with predictable resource needs
Spot VMs	Short-lived cheaper compute instances for non-time-critical fault-tolerant workloads



Virtual Machine - Best Practices

Best Practice	Description
Choose Zone and Region Carefully	Consider cost, legal regulations, availability requirements, latency, and machine type needs
Distribute VM Instances	Spread your VMs across multiple zones and regions to ensure high availability and fault tolerance
Pick the Right Machine Type	Experiment with different machine types to find what fits your workload best
Consider GPUs	For math-heavy or graphics-intensive applications like AI/ML to boost performance
Consider Committed Use Discounts	For steady long running workloads
Consider Spot VMs	For fault-tolerant, non-urgent, cost-sensitive workloads



Compute Engine Scenarios

Scenario	Solution
What are the pre-requisites to be able to create a VM instance?	1. Project, 2. Billing Account, 3. Compute Engine APIs should be enabled
You want to login to your VM instance to install software	You can SSH into it
You want to allow HTTP or HTTPS traffic to your VM	Configure Firewall Rules
You have predictable resource needs that will remain constant for a few years. How can you reduce your VM costs?	Committed Use Discounts
You want to create an image with your preferred OS, a few pre-installed apps, custom configuration, and security settings	Use a Custom Image
You want to launch multiple VMs with the same configuration - machine type, image, startup script, etc.	Create an Instance Template



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Getting Started with Instance Groups



Instance Groups

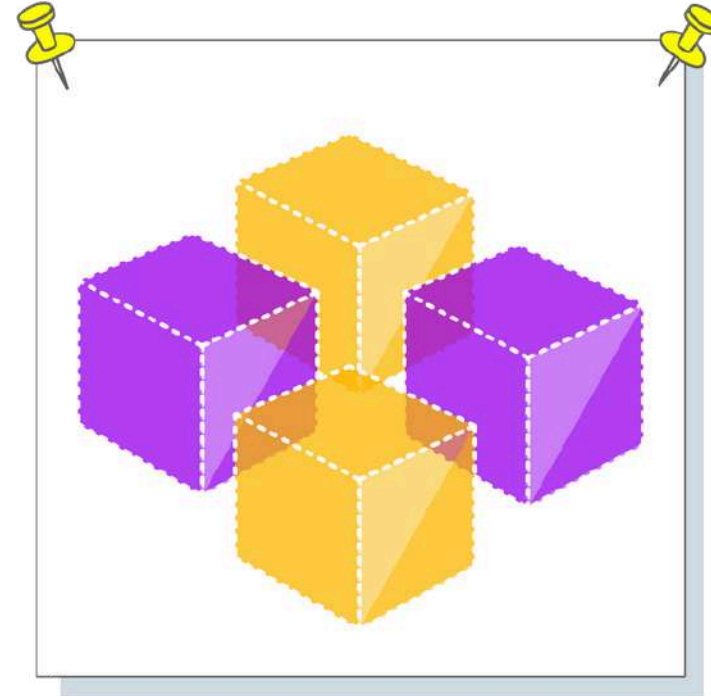
Scenario: What if you need to manage multiple VMs belonging to same app?

- **Instance groups:** Collection of VM instances that can be managed as a single entity

Two Kinds:

- **Managed** – Identical VMs created from an instance template (with auto-scaling, auto-healing, etc.)
- **Unmanaged** – VMs with different configurations

Zonal or Regional: **Zonal** (all VMs in one zone) or **Regional** (spread across zones in one region for higher availability)



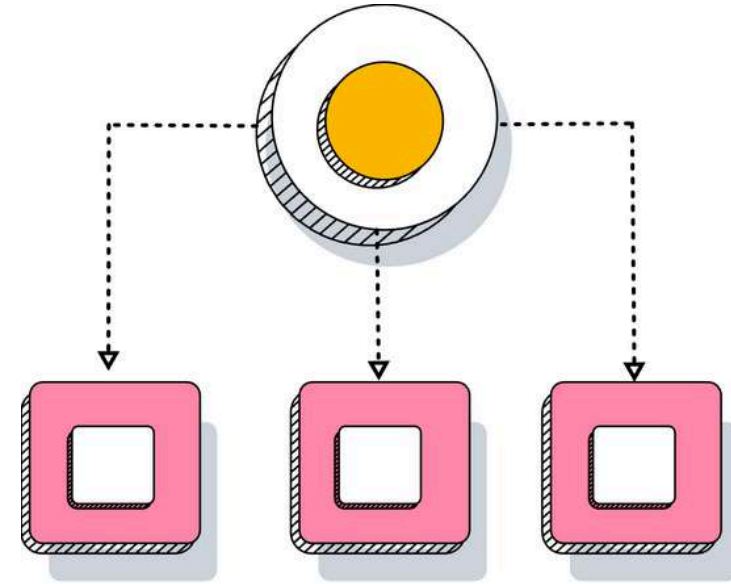


What is a Managed Instance Group (MIG)? [1/2]

Managed Instance Group (MIG): Group of VMs created using an instance template

Key Features:

- **Maintain Instance Count:** Keep a constant number of VMs running all the time (If a VM fails, MIG replaces it)
- **Self Healing:** Uses health checks to spot and fix app failures
- **Auto Scaling:** Automatically add or remove VMs depending on the load
- **Load Balancing:** Distribute traffic across healthy VM instances in MIG



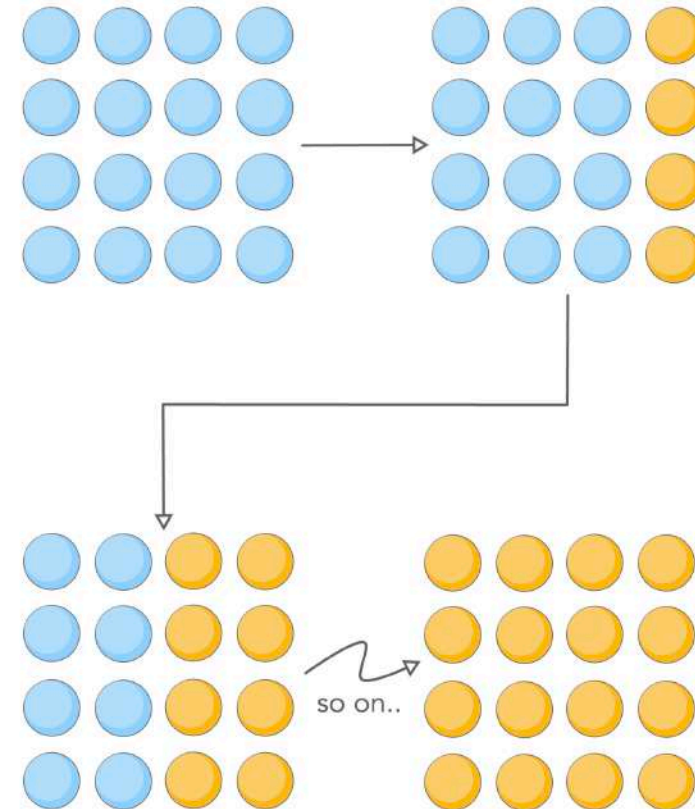


What is a Managed Instance Group (MIG)? [2/2]

Regional MIGs: Create instances in multiple zones in a single region (higher availability compared to zonal MIGs)

Release New Versions: Without downtime

- **Rolling updates:** Release new version step by step (gradually). Update a percentage of instances to the new version at a time. [Multiple Intermediate Steps]
- **Canary Deployment:** Test new version with a group of instances before releasing it across all instances. [Typically One Intermediate Step]





Instance Group Scenarios

Scenario	Solution
You want to create and manage a group of identical VMs in an easy way	Use a Managed Instance Group with an instance template
You need simple grouping of already existing VMs without autoscaling	Use an Unmanaged Instance Group
Your application must remain available even if an entire zone goes down. You want to create an Instance Group. Which type (regional or zonal) would you recommend?	Create a regional MIG (or multi zone MIG) to distribute instances across zones
You want failed or unhealthy VMs to be automatically detected and replaced	Configure health checks in MIG to enable autohealing



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Getting Deeper with Instance Groups

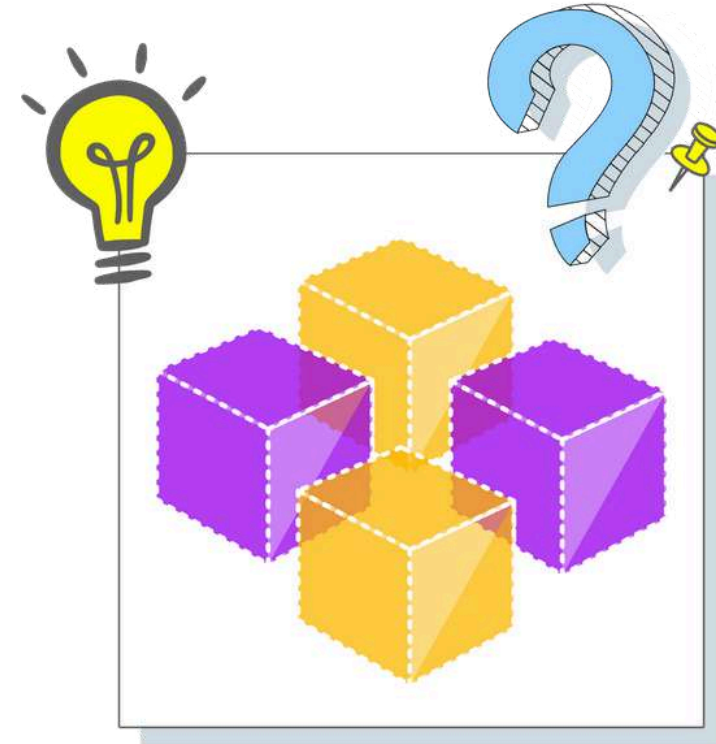


Creating a Managed Instance Group - Key Configuration

Instance Template: Blueprint for VM configuration (Mandatory)

Auto-Scaling: Automatically adjust number of VMs based on load

- **Minimum & Maximum** number of instances
- **Metrics:** CPU utilization, load balancing utilization, Cloud Monitoring metrics
- **Initialization period (Cool-down period):** How long does your application need to initialize?
- **Scale In Controls:** Avoid sudden reduction in instances
 - **Example:** Limit scale-in to 10% or 3 instances in 5 minutes





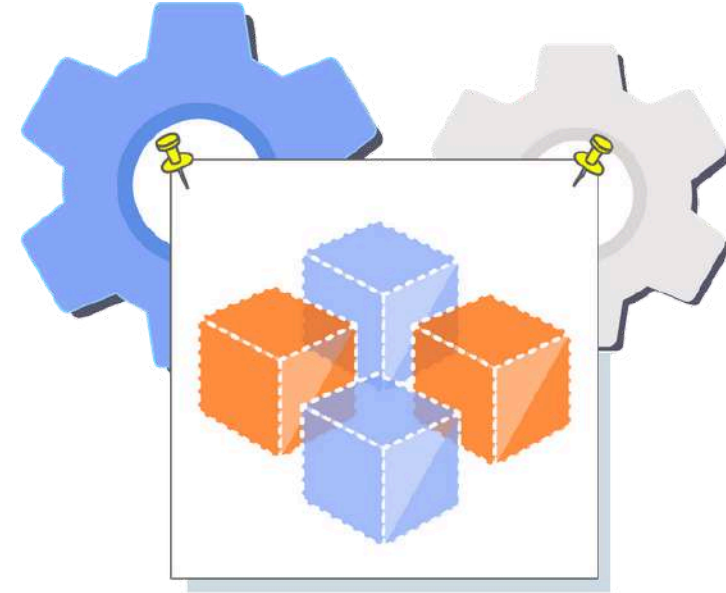
Managing a Managed Instance Group - Key Configuration

Rolling update - Gradual update to new instance template

- Specify new template
- When to update: Automatic (Proactive) or Selective (Opportunistic - when instance group is resized later)
- How to update
 - **Max surge:** Extra instances during update (e.g. 10% or 3)
 - **Max unavailable:** Instances allowed to be down during update

Rolling restart/replace: Restart or replace VMs without template change

- Configure Restart/Replace, Max unavailable, Max surge (for replace only)





Instance Group Scenarios [1/2]

Scenario	Solution
You are running a database and need to retain disk and identity	Use Stateful MIG to maintain instance names, disks, etc.
You want to gradually roll out a new version without updating all VMs immediately	Use rolling updates with Selective (Opportunistic - when instance group is resized later) update type
You want VM instances to automatically scale based on traffic or load	Configure autoscaling in a Managed Instance Group using CPU or other metrics
Your application takes time during startup and is incorrectly marked as unhealthy	Increase Initialization period (Cool-down period) to allow proper application startup before health checks



Instance Group Scenarios [2/2]

Scenario	Solution
You want to run exactly one VM instance at all times without scaling	Set minimum and maximum instances to 1
Your application needs to respond quickly to sudden traffic spikes	Reduce initialization period and lower autoscaling CPU target to trigger scaling faster
During deployment of a new version, you want to ensure that there is no reduction in capacity at any point in time. You want to make use of one new instance.	Use rolling update with maxSurge = 1 and maxUnavailable = 0
During deployment of a new version, you want to avoid using extra capacity while updating instances one at a time	Use rolling update with maxSurge = 0 and maxUnavailable = 1



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Getting Started with Cloud Architecture



What Is Availability?

Defines Uptime Percentage: Measures how often a service is available when users need it

- **Expressed as Percentages:** Example → 99.99% (four 9's availability)
- **Direct Impact on User Trust:** More downtime → lower customer satisfaction

Examples: Measuring Availability

- **99.95%:** 22 Minutes Downtime Per Month
- **99.99% (four 9s):** 4.5 Minutes Downtime Per Month (Common target for online apps)
- **99.999% (five 9s):** 26 seconds Downtime Per Month (Very hard to achieve)





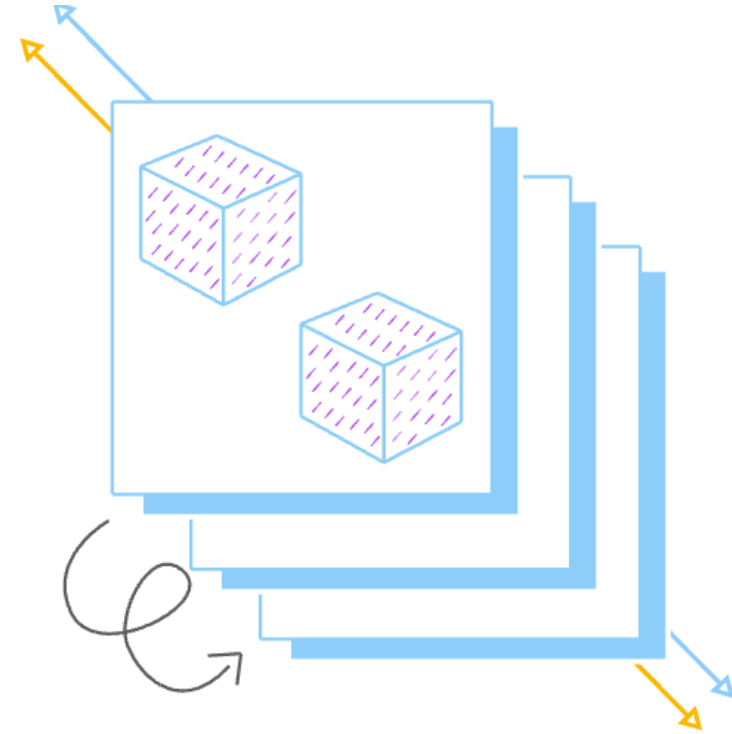
What Is Scalability?

Scenario: System handling 1000 transactions per second. Load expected to grow 10 times.

- **Question:** Can we handle **growth in users, traffic, or data size** without any drop in performance?
- **Question:** Does ability to serve more growth increase **proportionally** with resources?

Scalability: Ability to **adapt** to changes in demand (users, data)

- What are the options that can be considered?
 - Deploy to a bigger instance with bigger CPU and more memory
 - Increase the number of app instances and setup load balancer
 - And a lot more.





What Is Vertical Scaling?

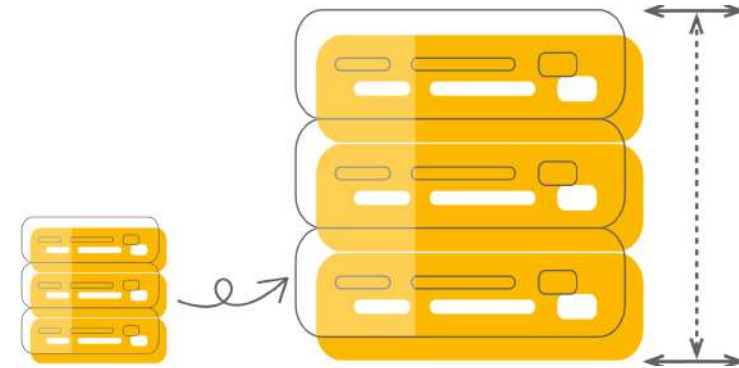
Increase Instance Size: Upgrade machine resources to handle more load

- **Add More Power:** More CPU, RAM, disk,...

Fast to Implement: No need for load balancers or distributed systems

Simple but Limited: There is always a maximum machine size

Can Be Costly: Larger instances are exponentially more expensive





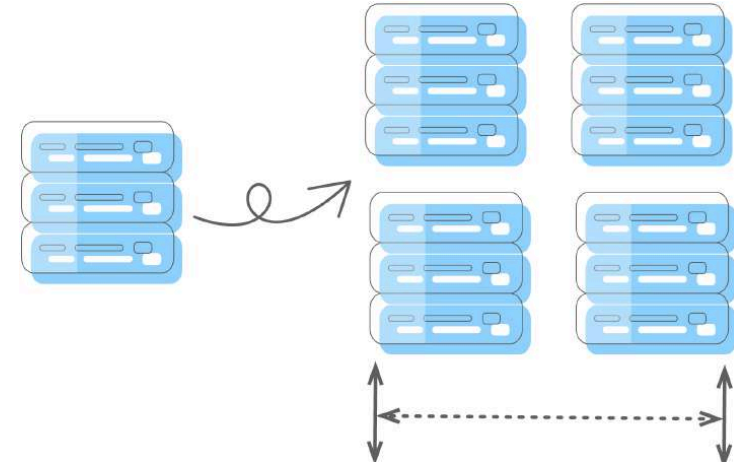
What Is Horizontal Scaling?

Add More Instances: Multiple copies of the application or database

- **Improves Availability:** If one instance fails, others continue serving
- **Preferred for Cloud Apps:** Especially microservices
- **Removes Vertical Limits:** Scale linearly instead of relying on bigger hardware

Requires Extra Components: Load balancers, service mesh, distributed data management

But More Cost-Effective: Use many small nodes rather than one large node





What Is Resiliency?

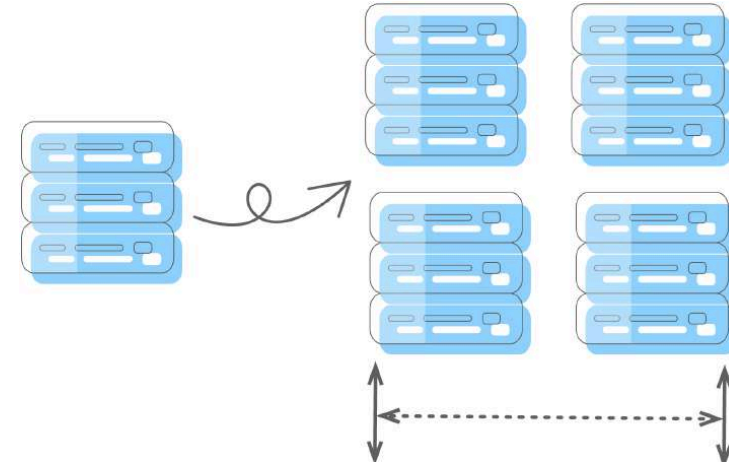
Resiliency: Ability of system to provide acceptable behavior when components fail

- **Goal:** Keep apps stable during outages
- **Focus Areas:** Surviving zone failures, VM crashes, ..

Outcome: Reduced downtime, stronger service continuity

Examples:

- Use VM Groups with health checks to auto-heal failed VMs
- Deploy across **multiple zones or regions** to avoid single-point failures



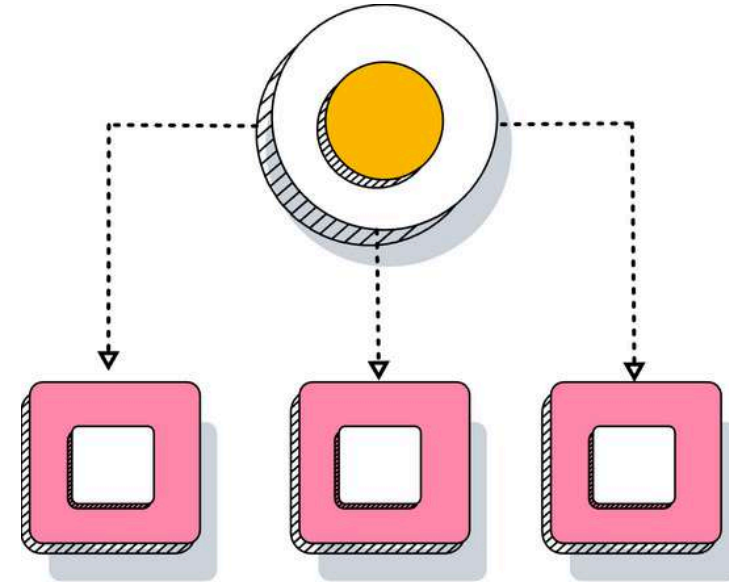


High Availability for Compute Engine VMs

Multiple Regional Instance Groups: Deploy microservices across multiple regions

- **Use Global HTTPS Load Balancer:** Distribute traffic worldwide with a single anycast IP
- **Configure Health Checks:** Enable auto-healing at instance group and load balancer level
- **Enable Live Migration:** Keep VMs running during host maintenance

Gain High Availability: Even if a region becomes unavailable, workloads continue to be available from other regions





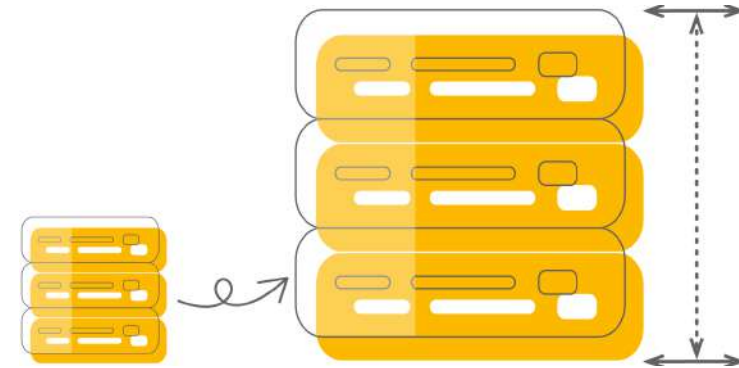
Vertical Scaling for Compute Engine VMs

Increase Machine Size: Move from smaller to larger machine types

- **Add More Resources:** More CPU, RAM, ..
- *e2-standard-2* to *e2-standard-4* or
- *e2-standard-16* to *e2-standard-32* or
- ...

Useful for Stateful Workloads: Traditional Relational Databases or apps that cannot easily scale horizontally

Quick to Apply: Stop and Resize a VM



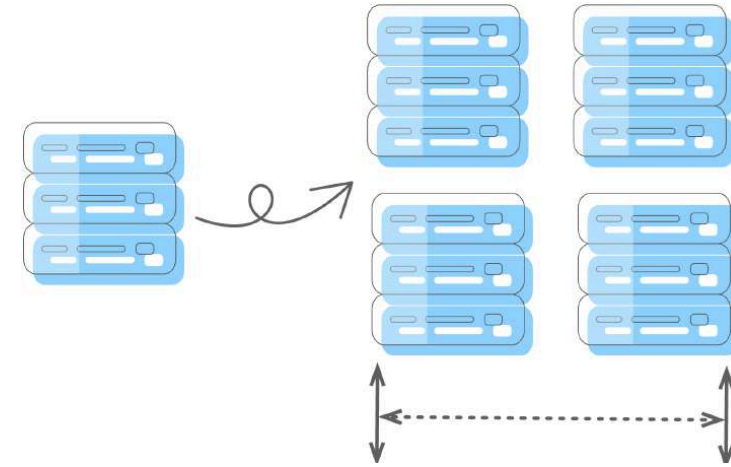


Horizontal Scaling for Compute Engine VMs

Use (Multiple) Managed Instance Groups:

Automatically create and scale VM replicas

- **Auto-scale Based on Metrics:** CPU, request count, or custom metrics
- **Distribute Across Zones or Regions:** In a single zone OR in multiple zones in single region OR in multiple zones across multiple regions
- **Use Load Balancers:** Route traffic evenly to all VM instances



Ideal for Stateless Services: Web apps, APIs, microservices, etc.



Compute Engine – A Review of Architectural Best Practices

Consideration	Best Practice
Performance	- Choose the right machine family for workload type - Use GPUs and TPUs as needed for graphics or ML workloads - Use hardened custom VM images to avoid slow startup and increase security
Security	- Use firewall rules to restrict traffic - Prefer internal IPs for internal communication - Use sole-tenant nodes as needed for compliance
Cost Efficiency	- Use Auto Scaling to optimize number/type of VMs - Use committed-use discounts for predictable long term workloads - Use Spot VMs for cost-sensitive, non-time-critical, fault-tolerant workloads
Resiliency	- Run VMs in managed instance groups (MIGs) behind global load balancers - Use Cloud Monitoring and Logging for visibility - Enable live migration and automatic restart - Configure health checks and maintain images across regions (for disaster recovery)



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Getting Started with

Gcloud - CLI



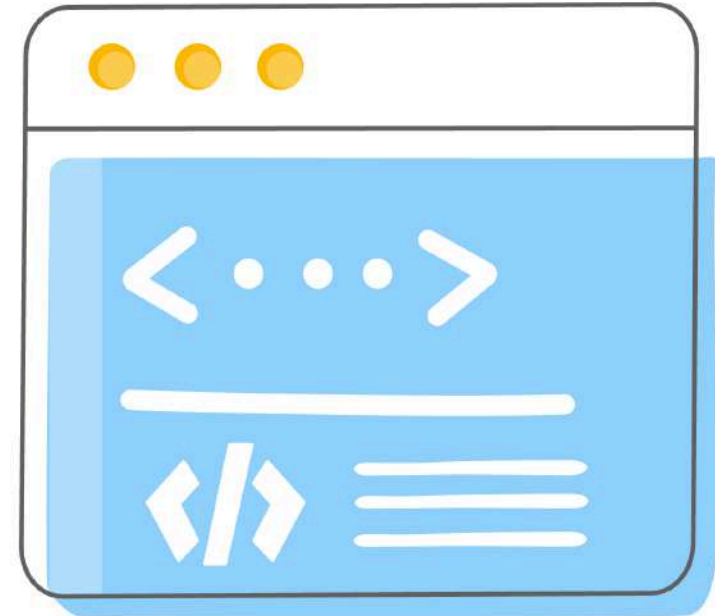
Gcloud

Gcloud: Primary CLI for Google Cloud

- **Wide Service Coverage:** Manage VMs, databases, instance groups, and many other resources
- **Full Resource Control:** Create, delete, update, and view cloud resources using commands

Specialized Tools: Some services have dedicated tools

- Cloud BigQuery - bq
- Cloud Bigtable - cbt
- Kubernetes - kubectl (in addition to Gcloud which is used to manage clusters)





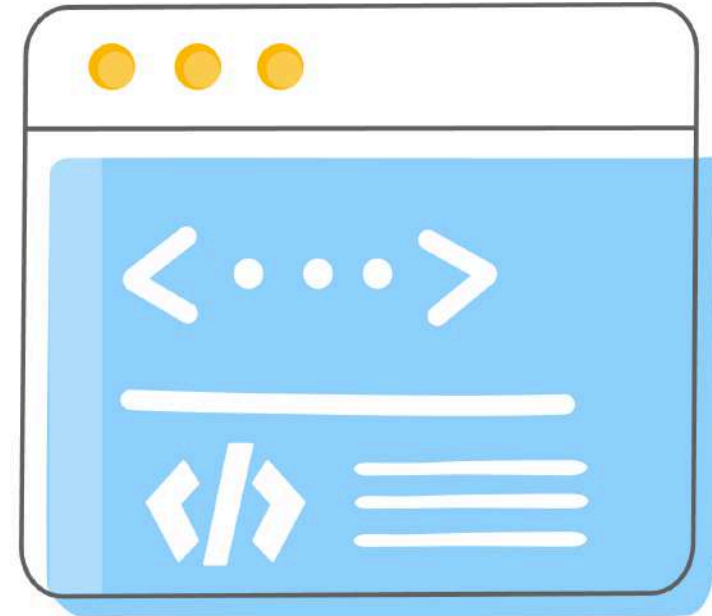
Gcloud - Getting Started

Installation

- **Gcloud:** Part of Google Cloud SDK
 - Cloud SDK requires Python
 - Instructions to install Cloud SDK (and Gcloud) =>
<https://docs.cloud.google.com/sdk/docs/install-sdk>
- (RECOMMEDED) Use Gcloud on Cloud Shell

Commands: To connect to Google Cloud

- **gcloud init** - initialize or reinitialize gcloud
 - Authorize gcloud to use your user account credentials
 - Setup configuration (Includes core.account, core.project)
- **gcloud config list** - lists all properties of the active configuration





gcloud config set

```
gcloud config set core/project VALUE
gcloud config set compute/region VALUE
gcloud config set compute/zone VALUE
gcloud config set core/verbosity VALUE(debug)
```

gcloud config set: Sets the specified property in your active configuration

- Syntax - `gcloud config set SECTION/PROPERTY VALUE`
 - `core, compute` - SECTIONS
 - `project, region, zone` - PROPERTIES
 - Specifying `core` is optional as it is the default SECTION!
 - `gcloud config set project VALUE`
 - `gcloud config set verbosity VALUE(debug)`
 - Get more details with `gcloud config set --help` (Look for **AVAILABLE PROPERTIES** in the content)

Opposite: `gcloud config unset`



Playing with gcloud config set

```
[compute]
gce_metadata_read_timeout_sec = 30
region = us-central1
zone = us-central1-b
[core]
account = testing@gmail.com
disable_usage_reporting = True
project = useful-device-303710
verbosity = info
[metrics]
environment = devshell
```

gcloud config set compute/region us-central1

gcloud config set compute/zone us-central1-a

gcloud config list

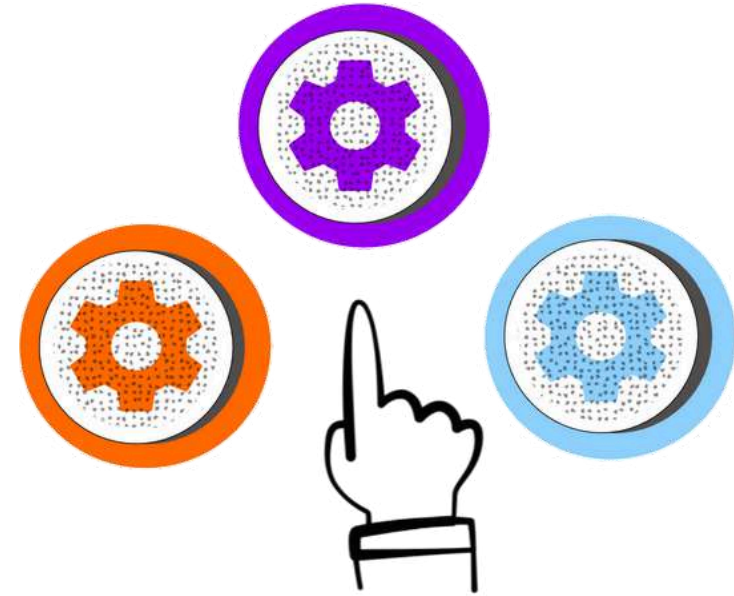


Gcloud - Managing Multiple Configurations

Scenario: You are working on multiple environments/projects. You want to switch between them easily.

- **gcloud config configurations create / delete / describe / activate / list**
 - Create: `gcloud config configurations create NAME(dev)`
 - Activate: `gcloud config configurations activate NAME(dev)`
 - List: `gcloud config configurations list`

Best Practice: Give the configuration a good name (dev, qa, .. OR project1-dev, project2-qa, ..)





gcloud command structure - Playing with Services

```
gcloud compute instances list
gcloud compute regions list
gcloud compute zones list
gcloud compute zones list --filter="region:us-central1"
gcloud compute zones list --filter="region:(us-central1 us-east1)"
gcloud compute machine-types list
```

gcloud GROUP SUBGROUP ACTION ...

- **GROUP** - config or compute or container or dataflow or functions or iam or ..
 - Which service group are you playing with?
- **SUBGROUP** - instances or images or instance-templates or machine-types or regions or zones
 - Which sub group of the service do you want to play with?
- **ACTION** - create or list or start or stop or describe or ...
 - What do you want to do?



Gcloud - A Few Scenarios

Scenario	Solution
What is typical structure of a gcloud command?	<code>gcloud GROUP SUBGROUP ACTION ...</code>
How can you set default region for compute engine without repeating it in every command?	<code>gcloud config set compute/region us-central1</code>
How can you set default zone for compute engine without repeating it in every command?	<code>gcloud config set compute/zone us-central1-a</code>
How to manage multiple configurations for different projects easily?	<code>gcloud config configurations create / delete / describe / activate / list</code>
How can you see the gcloud active project details?	<code>gcloud config list</code> - Lists all properties of the active configuration including the current project



Cloud Shell - Things to Remember

Storage: Runs on a temporary VM instance

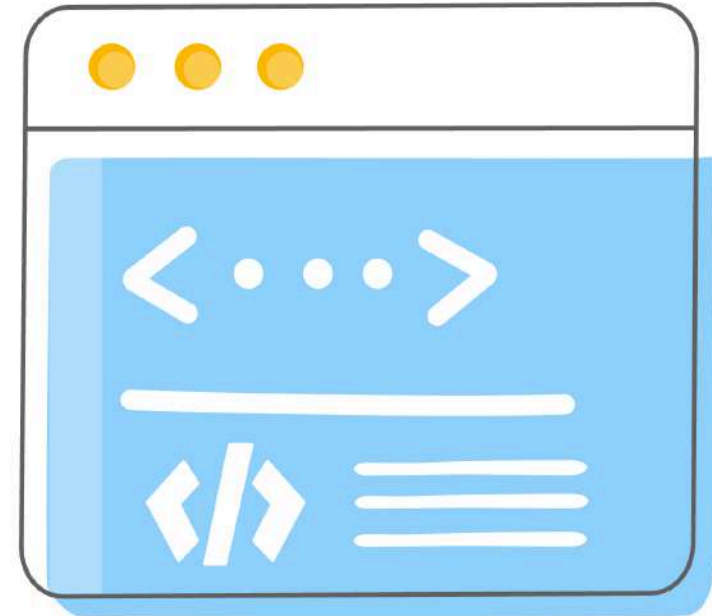
- **Created Automatically:** When you open Cloud Shell
- **Storage:** 5 GB persistent \$HOME directory storage
- **Tools:** Cloud SDK (gcloud), kubectl, Git, editors, ..

Session: Terminates after 20 inactive minutes

- **\$HOME:** Home directory (~) persists between sessions
 - **Scripts, user configuration files:** Like .bashrc and .vimrc etc
 - **120-Day Window:** Deleted after 120 days of inactivity
- **Data Loss:** Changes outside \$HOME are lost

Connectivity: Enables secure access to VMs

- **Private SSH:** SSH into VM instances using private IPs
(`gcloud compute ssh`)





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Getting Started with Load Balancing

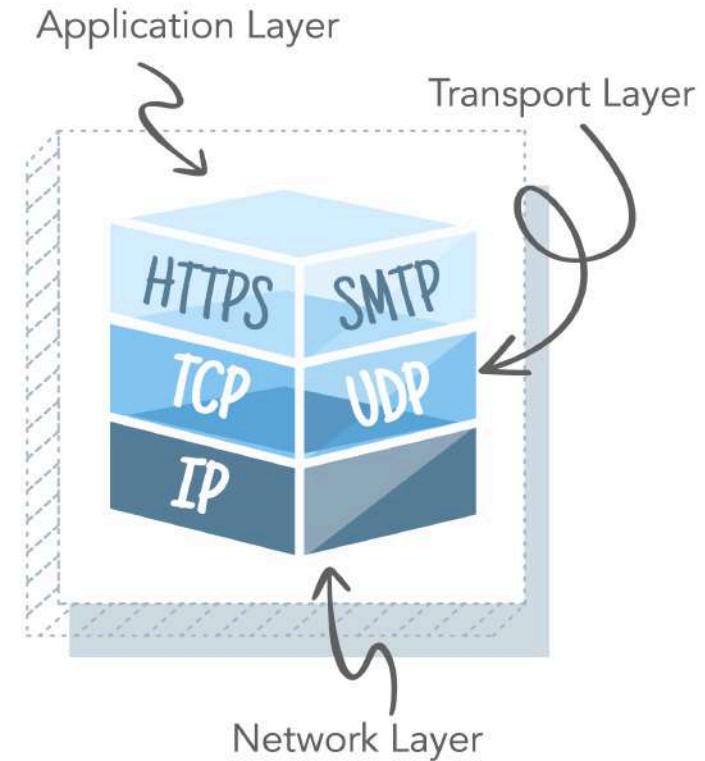


How Do Computers Communicate?

Scenario: How does data from your browser get transferred to a web server?

Multiple Layers: Each layer has a specific job

- **Network Layer (Layer 3)** – Transfers bits across the internet
- **Transport Layer (Layer 4)** – Ensures the bits arrive correctly
- **Application Layer (Layer 7)** – Handles what the app actually does (web sites, API calls, email, file transfers)





Different Layers use Different Protocols

Network Layer (Layer 3)

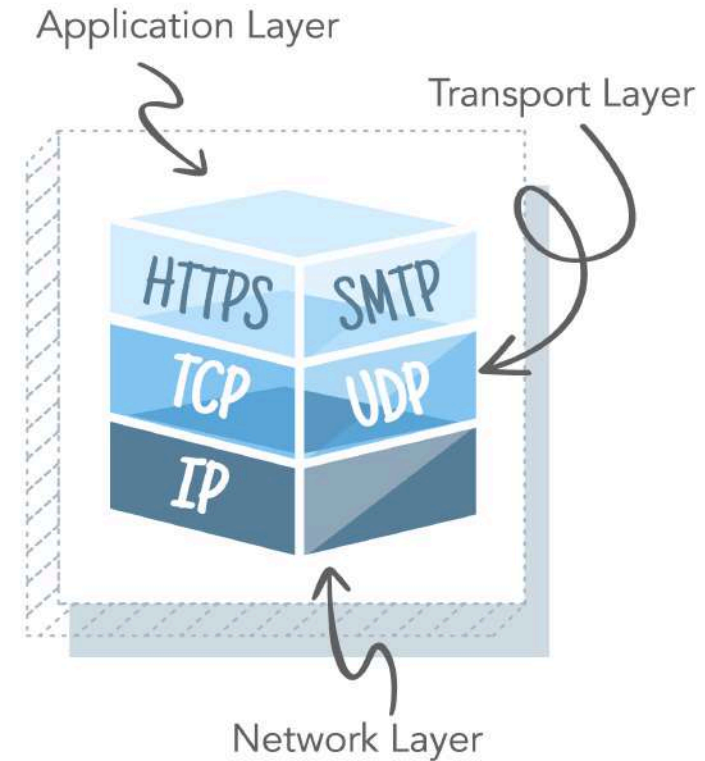
- **IP (Internet Protocol)** – Transfers packets (unreliable)

Transport Layer (Layer 4)

- **TCP (Transmission Control Protocol)** – Prioritizes reliability (retries in case of failures)
 - **TLS (Transport Layer Security)** – Secure TCP
- **UDP (User Datagram Protocol)** – Speed > Reliability

Application Layer (Layer 7)

- **HTTP (Hypertext Transfer Protocol)** – Web Apps
 - **HTTPS** – Secure HTTP (uses TLS)
- Other protocols: SMTP (Email), FTP (File transfer), etc.





Different Applications Have Different Needs [1/2]

Live gaming app: Needs ultra-fast response and tolerates a few lost packets

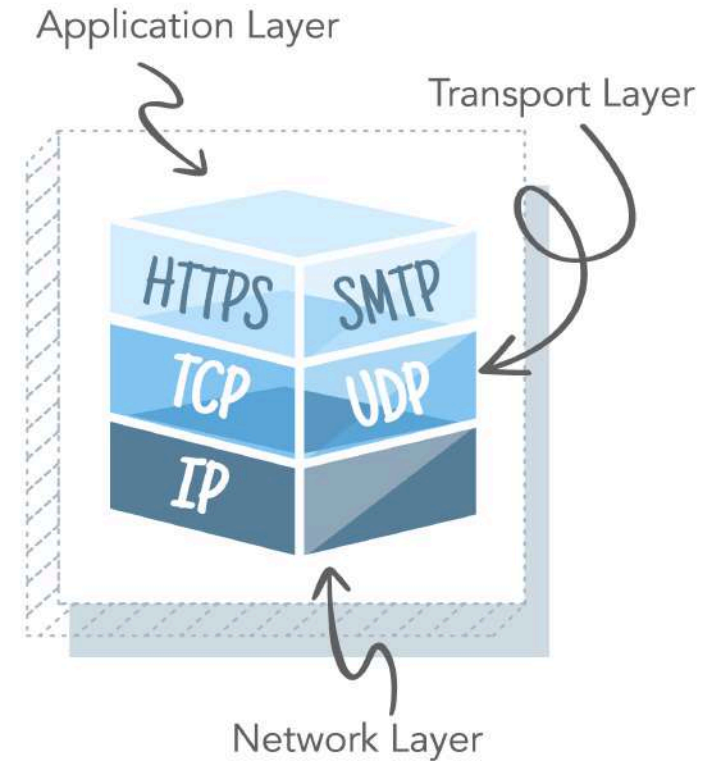
- **Choice:** Transport Layer (Layer 4) using UDP

Database connections: Needs ultra-fast reliable data transfer

- **Choice:** Transport Layer (Layer 4) using TCP

Web or email application: Needs standard communication protocols

- **Choice:** Application Layer (Layer 7) using HTTP, SMTP, FTP → Rich features and compatibility





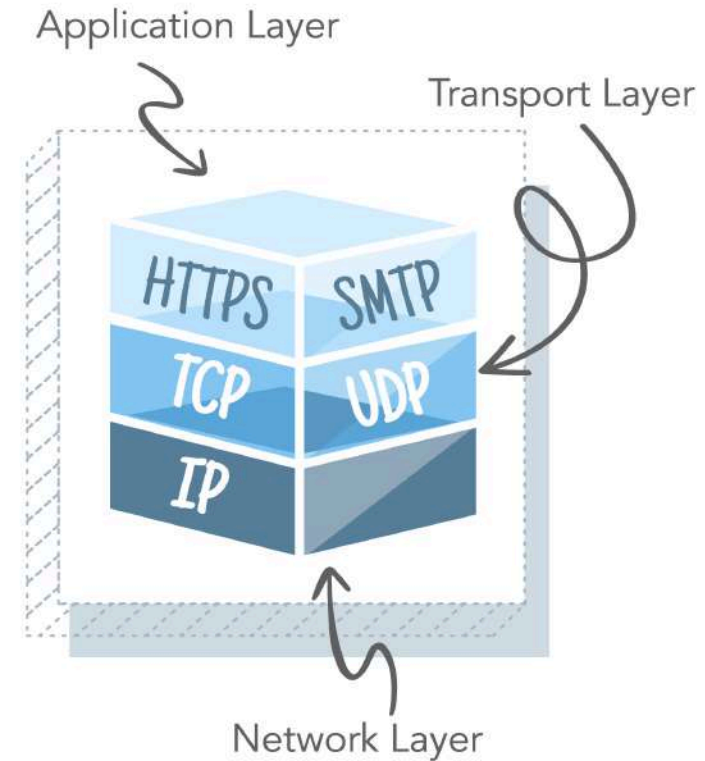
Different Applications Have Different Needs [2/2]

Speed critical: Communicate at Transport Layer (Layer 4)

- Minimal overhead, faster processing
- **Can tolerate small data loss:** Use UDP
- **Reliability is high priority:** Use TCP/TLS

Feature-rich communication (slightly slower): Use Application Layer (Layer 7)

- Web apps (HTTP/HTTPS), email (SMTP), file transfer (FTP)





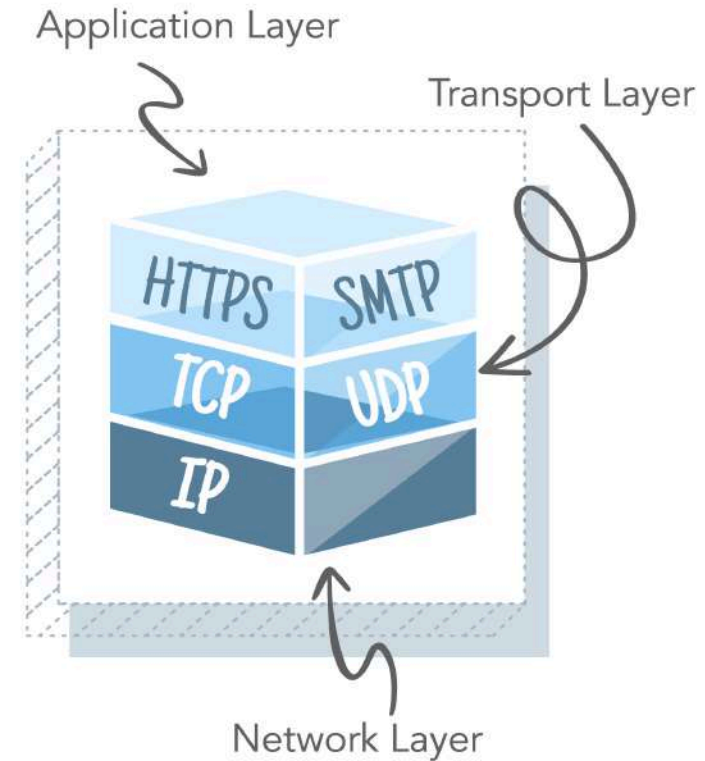
Layer 7 vs Layer 4 Load Balancing

Layer 4 Load Balancer: Works at Transport Layer

- **Routing based on:** IP address and port
- **Example:** Database connections
- **Advantage:** Fast and low overhead

Layer 7 Load Balancer: Works at Application Layer

- **Routing based on:** URL path, headers, host, etc.
- **Example:** Route `/api` vs `/login` to different services
- **Advantage:** Flexible and intelligent routing





Layer 4 Load Balancing (Transport Layer)

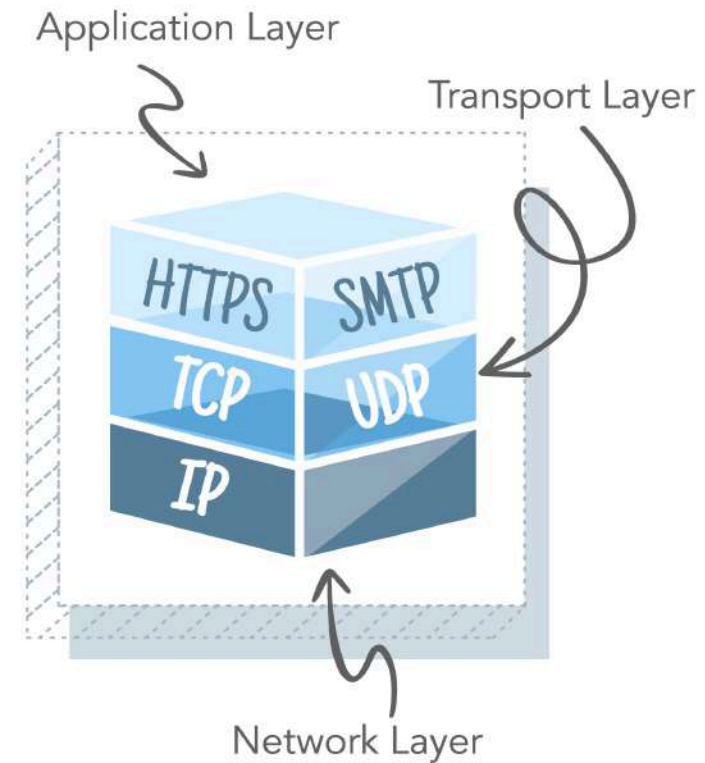
Operates at: Transport Layer

Routing decision based on: Source IP, Destination IP, Port

Does NOT understand application data

Fast and efficient: Minimal processing overhead

Use cases: High-performance, low-latency systems (databases, gaming, live streaming)





Layer 7 Load Balancing (Application Layer)

Operates at: Application Layer

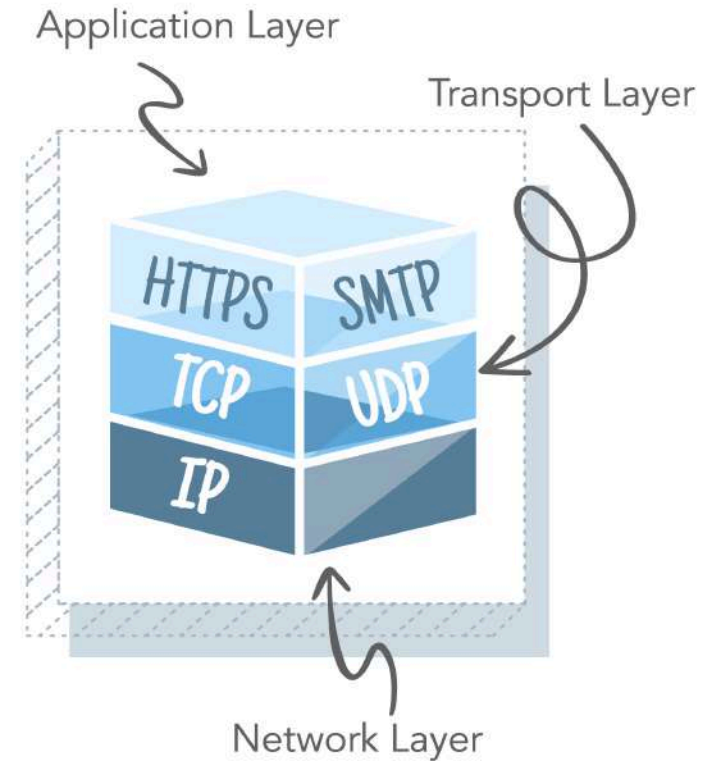
Routing decision based on:

- URL path (/api, /login)
- Hostname (api.example.com, login.example.com)
- Headers, cookies, etc.

Understands application content

More intelligent but slightly slower

Use cases: Web applications and APIs, Microservices, etc.





Key Differences - Layer 4 vs Layer 7 Load Balancing

Feature	Layer 4	Layer 7
Layer	Transport (TCP/UDP)	Application (HTTP/HTTPS)
Routing Criteria	IP + Port	URL, headers, content
Awareness	No data awareness	Has data awareness
Performance	Faster	Slightly slower
Flexibility	Limited	Highly flexible



Layers, Protocols and Load Balancing - A Few Scenarios

Scenario	Solution
Your application needs ultra-fast communication with minimal processing overhead. Recommend Layer.	Communicate at Transport Layer (Layer 4) instead of Layer 7
You are building a live gaming application where latency is critical and small packet loss is acceptable. Recommend Layer and Protocol.	Use UDP at Transport Layer (Layer 4) for fast, low-latency communication
You are building a web application that needs standard communication with browsers. Recommend Layer and Protocol.	Use Application Layer (Layer 7) protocols like HTTP/HTTPS
You want to route requests based on URL paths like <i>/api</i> or <i>/login</i>	Use Layer 7 Load Balancer for content-based routing
You want flexibility to route based on request content, headers, cookies, or hostnames	Use Layer 7 Load Balancer



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Getting Started with Load Balancing

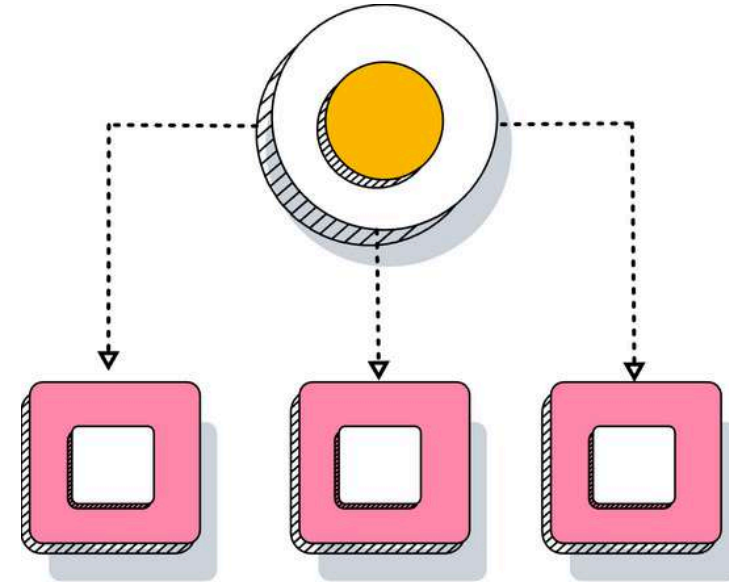


Cloud Load Balancing

Scenario: What if your application receives millions of user requests at the same time?

Cloud Load Balancing: Distribute traffic across multiple VMs

- Highly available (99.99% SLA)
- Automatically adjusts capacity with **auto scaling**
- **Global load balancing** with a single Anycast IP
- Supports **internal load balancing** for private traffic
- Routes traffic to healthy instances using **health checks**





Cloud Load Balancing - Terminology

Backend - Group of endpoints that receive traffic from LB

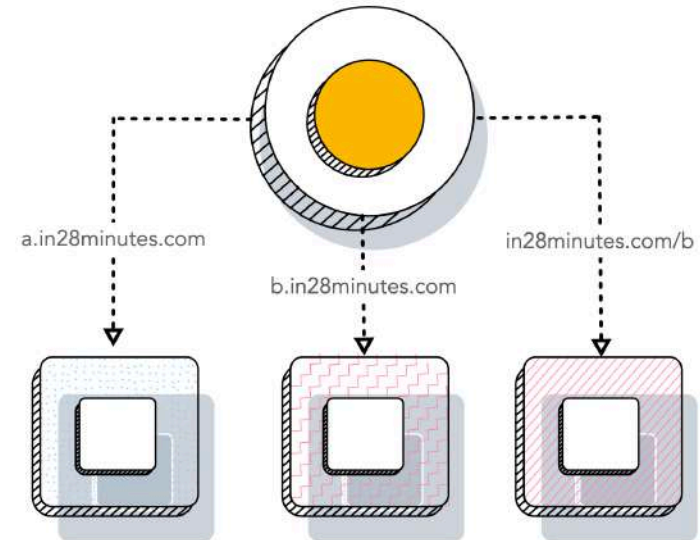
- **Examples:** MIG, Cloud Run instances, Kubernetes Pods, External backends

Frontend - Specify an IP addr, port & protocol

- **IP address:** Frontend IP for your clients requests
- **For SSL:** A certificate must also be assigned

For HTTP(S) LB : Define Routing rules using

- **path** - in28minutes.com/a vs in28minutes.com/b
- **Host** - a.in28minutes.com vs b.in28minutes.com
- **HTTP headers** and methods (POST, GET, etc), ..





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Getting Deeper with Load Balancing

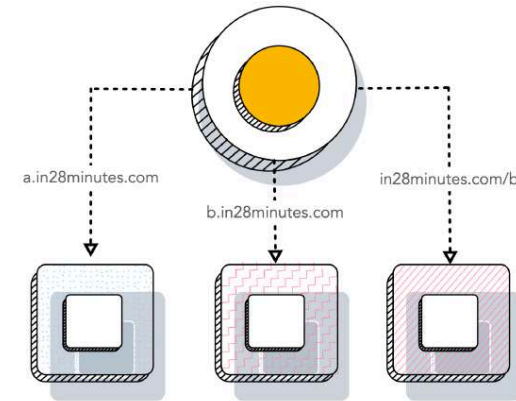


Proxy vs Passthrough load balancers

Proxy load balancers: Terminate connections at LB and open new connections from LB to backends

Passthrough load balancers: Do not terminate client connections

- Load-balanced packets received by backend as is (with source, destination, and, port info unchanged)
- Connections are terminated by the backend
- **Direct Server Return:** Responses from the backend go directly to the clients, not through the load balancer
- Recommended when you need to preserve client packet info

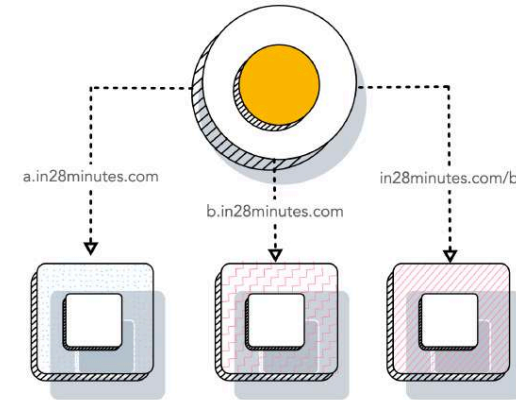




Cloud Load Balancing - Application vs Network LBs

Application LB: Proxy-based Layer 7 LB

- Provides an anycast IP address
- Distribute HTTP and HTTPS traffic to backends
- **External ALB:** Support clients from anywhere
- **Internal ALB:** Clients within/connected to VPC n/w

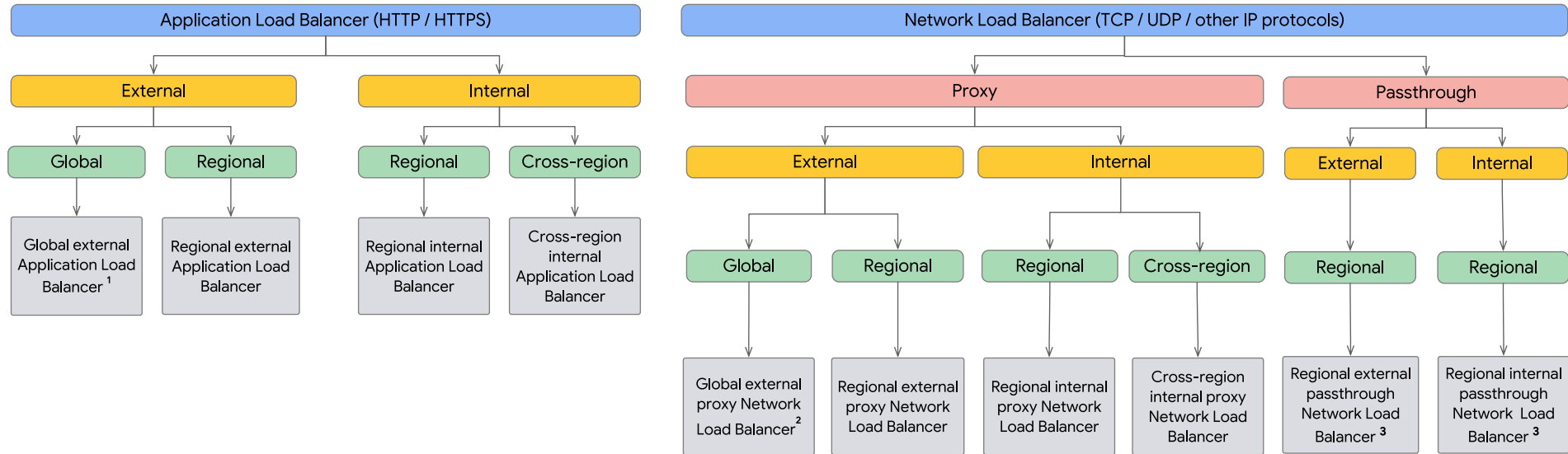


Network LB: Layer 4 LB

- Handle TCP, UDP, or other IP protocol traffic
- **Proxy NLB:** Reverse proxy LB - adv. traffic controls
- **Passthrough NLB:** Preserve source IP of client OR Direct server return for responses OR Support other IP protocols like ESP, GRE, ICMP, and ICMPv6



Cloud Load Balancing - Choosing a Load Balancer



<https://docs.cloud.google.com/load-balancing/docs/choosing-load-balancer>



Load Balancing - SSL/TLS Termination/Offloading

Client to Load Balancer: Over internet

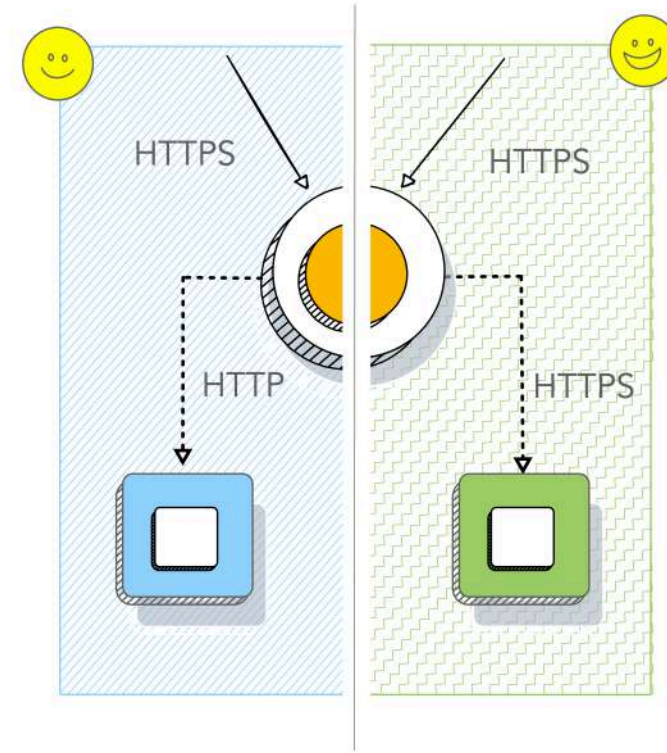
- HTTPS recommended

Load Balancer to VM instance: Through Google internal network

- HTTP or HTTPS (HTTPS is preferred)

SSL/TLS Termination/Offloading

- Client to Load Balancer: HTTPS/TLS
- Load Balancer to VM instance: HTTP/TCP





Cloud Load Balancing - Choosing a Load Balancer - Scenarios

Scenario	Solution
You are building REST API with path-based routing (<i>/api, /login</i>)	Use Application LB (Layer 7, Proxy)
You need TLS/SSL termination	Use Proxy Load Balancer (Application LB or Proxy Network LB)
You want to handle TCP/UDP workloads (databases, gaming)	Use Network Load Balancer (Layer 4)
You need ultra-low latency and want traffic sent directly to backend VMs	Use Passthrough Network Load Balancer
You want backend VMs to see original client IP and packets unchanged	Use Passthrough Network Load Balancer

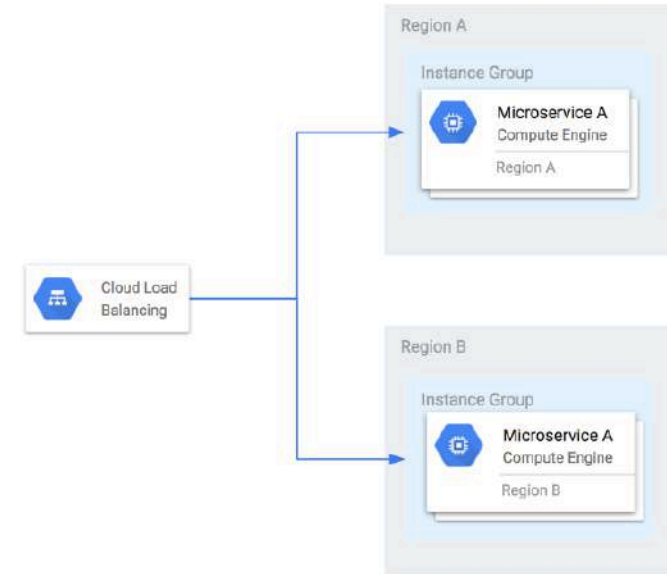


Load Balancing Across Multiple Regions

Global HTTP(S) LB: Can route to backends in multiple regions

Example: A Multi-Region Deployment

- **Multiple MIGs:** Setup Regional MIGs for your app in multiple regions
- **Global HTTP(S) LB:** Distribute load to MIGs in different regions
 - Single anycast IP address
 - Requests redirected to nearest region (Low latency)
 - Sends traffic to **healthy instances**
 - **If all backends within a region are unhealthy:** Traffic is distributed to healthy backends in other regions
 - (REMEMBER) Ensure that health check from load balancer can reach the instances in an instance group (Firewall rules)





Quick Review of HTTP(S) Load Balancing Concepts

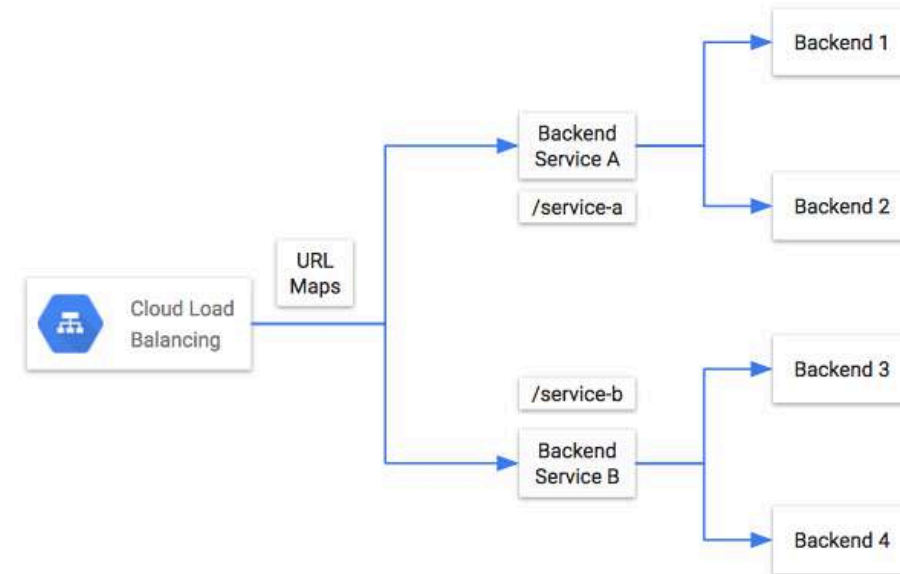
Frontend - Specify an IP addr, port & protocol

Backend - Group of endpoints that receive traffic from LB (Ex: MIG)

Backend Service - Group of backends (same or different regions)

URL Maps - Route requests to backend services

- URL `/service-a` maps to **Backend Service A**
- URL `/service-b` maps to **Backend Service B**





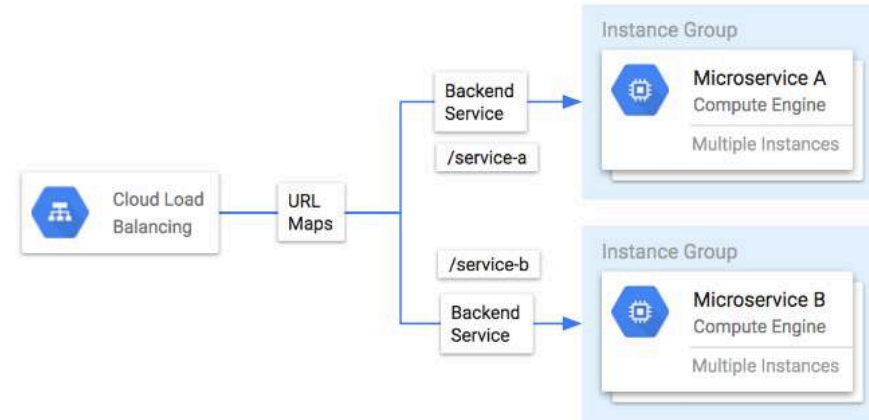
HTTP(S) Load Balancing Setup - 1 - Multiple Microservices

Backend Services: A Backend Service for each the Microservice

Backends: In this simple setup, we have one backend per microservice

URL Maps

- `/service-a` => Micro. A Backend Service
- `/service-b` => Micro. B Backend Service





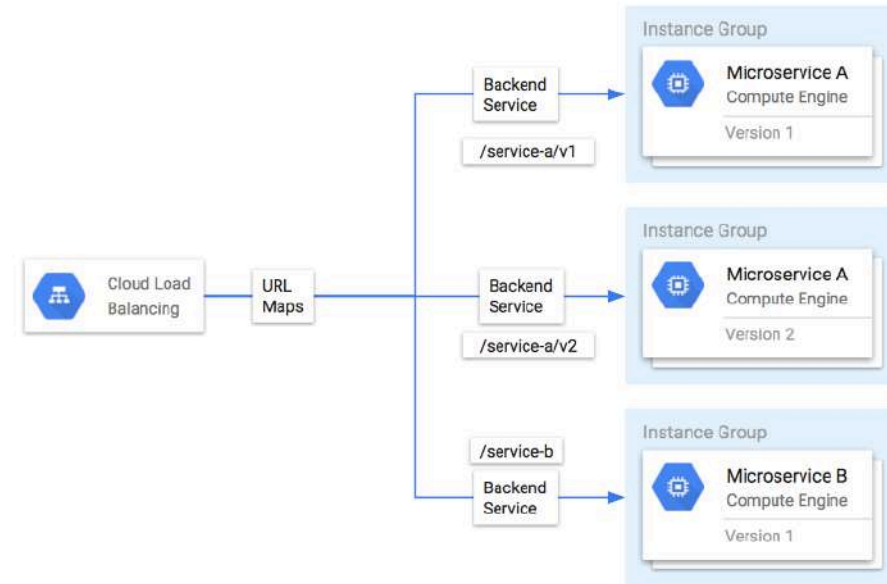
HTTP(S) Load Balancing Setup - 2 - Microservice Versions

Backend Services: A Backend Service for each Microservice version

Backends: In this simple setup, we have one backend per microservice version

URL Maps

- `/service-a/v1` => Micro. A V1 Backend Service
- `/service-a/v2` => Micro. A V2 Backend Service
- `/service-b` => Micro. B Backend Service





HTTP(S) Load Balancing Setup - 3 - Multi Regional Microservice

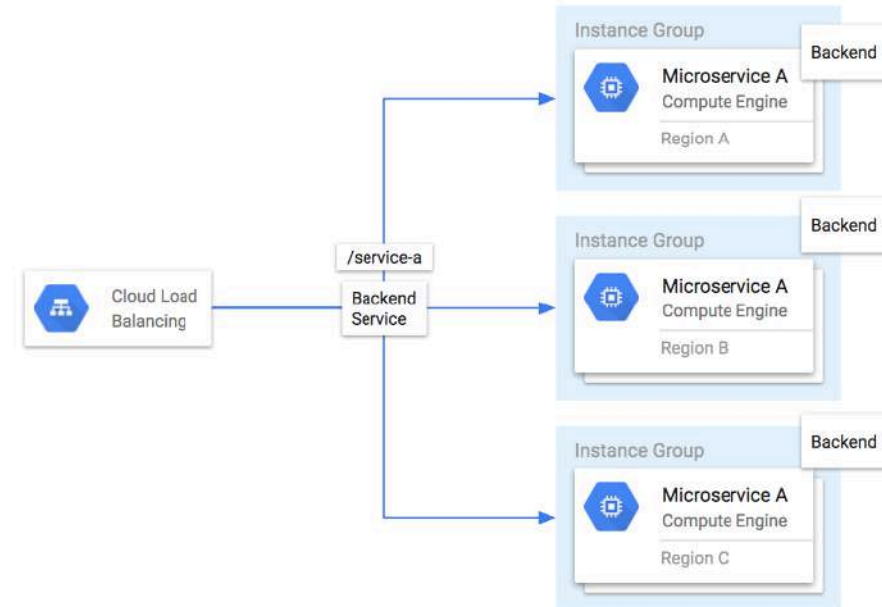
Backend Services: A Backend Service for each Microservice version

Backends: One backend for each regional microservice MIG

URL Maps: URL `/service-a` maps to the microservice **Backend Service**

Global routing: Route user to nearest instance (nearest regional MIG)

- Needs **Networking Premium Tier**: In STANDARD Tier, Forwarding rule and its external IP address are regional





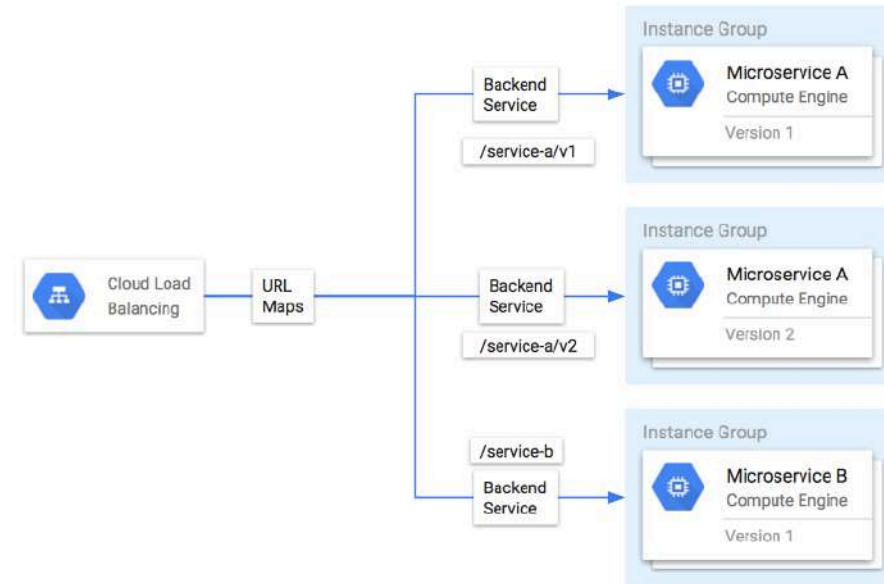
How Does Application Load Balancer Decide? [1/3]

Step 0: Maintain List of Healthy Backends/Endpoints

- Continuously performs health checks
- Ensures traffic is sent only to healthy backends/endpoints

Step 1: Listener (Frontend)

- Receives request on IP + Port (80/443)
- HTTP → HTTPS redirect (optional)

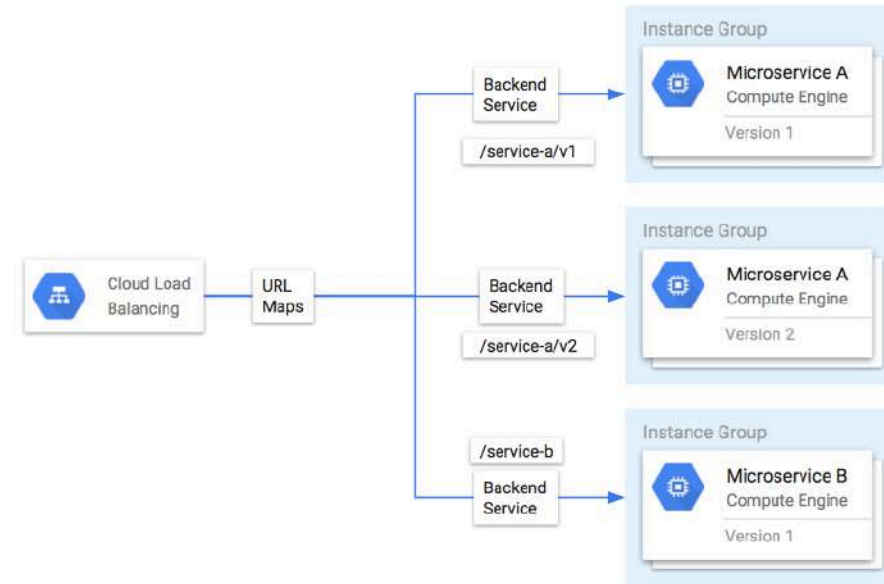




How Does Application Load Balancer Decide? [2/3]

Step 2: Choose Backend Service

- **URL Map:** Map to a **backend service**
 - Based on: **Host** (`api.example.com`) OR **Path** (`/api`, `/login`) OR **Headers** / **cookies** (advanced)
 - Versioning is also possible!
 - **Path-based:** `/service-a/v1` vs `/service-a/v2`
 - **Hostname-based:** `v1.api.example.com` vs `v2.api.example.com`
- **Weighted backends:** 70% → v1, 30% → v2



Step 3: Choose Backend Instance

- **Equal Distribution:** Bet. healthy instances
- **Session affinity:** Same user → routed to same instance (Based on cookies or IP)



How Does Application Load Balancer Decide? [3/3]

Configuration: Split traffic between two backend services using weights

- Backend Service 1 has a weight of 70 => It receives 70% of traffic
- Backend Service 2 has a weight of 30 => It receives 30% of traffic

```
defaultService: ${DEFAULT_SERVICE_URL}
name: matcher1
routeRules:
  - matchRules:
      - prefixMatch: /
    priority: ${PRIORITY}
    routeAction:
      weightedBackendServices:
        - backendService: ${SERVICE1_URL}
          weight: 70
        - backendService: ${SERVICE2_URL}
          weight: 30
```




What are Network Endpoint Groups?

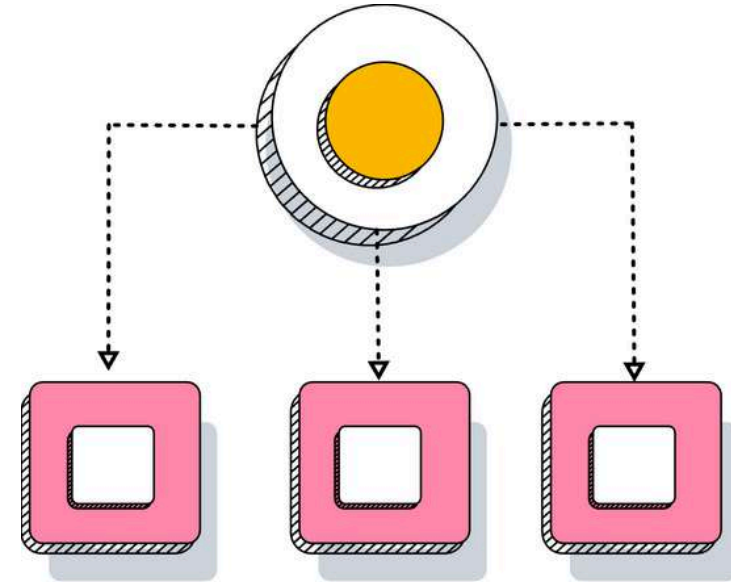
Instance Group Backend: Distribute traffic to VM instances (in an IG)

- Backend Service > Backend (MIG) > (VM1, VM2, VM3)

Question: How to route to Endpoints?

(10.0.1.5:8080, 10.0.1.6:8080, 10.0.1.7:8080)

- **Network Endpoint Groups:** Route traffic to specific endpoints (A NEG represents a collection of endpoints)
- Supports **pod-level routing** (GKE)
- Supports **serverless** (Cloud Run, App Engine) apps
- Supports **hybrid and external services**





Types of Network Endpoint Groups

NEG Type	Use Case
Zonal NEG	Route traffic to specific VM IP:Port or GKE Pods within a zone
Internet NEG	Route traffic to external services outside Google Cloud (public internet endpoints)
Serverless NEG	Route traffic to Cloud Run, App Engine, or serverless functions
Hybrid NEG	Route traffic to on-prem or multi-cloud services via VPN/Interconnect
Private Service Connect NEG	Privately access Google APIs or services across VPCs



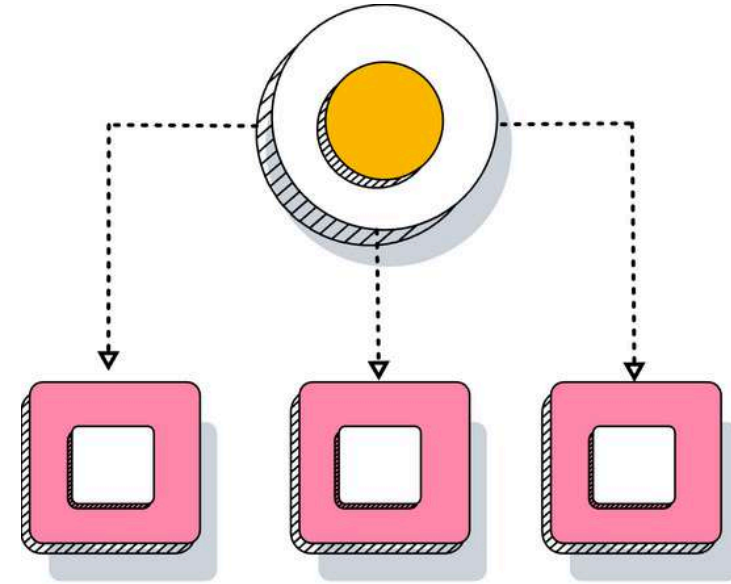
Setup Application LB For Serverless Apps [1/2]

Step 1: Deploy Backend Application

- Deploy Cloud Run / App Engine / Cloud Functions

Step 2: Configure a Listener

- Reserve a static global external IP address
- Create Google-managed SSL certificate
- Configure frontend IP + Port (HTTPS)





Setup Application LB For Serverless Apps [2/2]

Step 3: Setup Backend

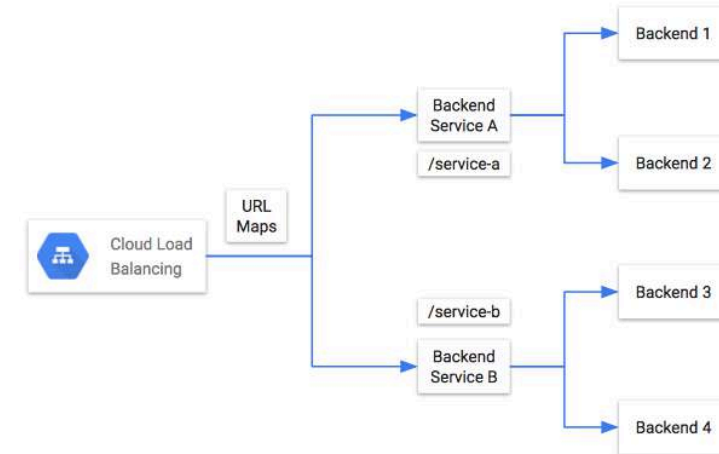
- Create a **Serverless Network Endpoint Group**
- Connect it to your **serverless service**

Step 4: Setup Backend Service

- Attach **Serverless NEG** to backend service

Step 5: Routing Configuration (Traffic Management)

- Create **URL Map** with routes
 - **Host-based routing** OR **Path-based routing** OR ..





Cloud Load Balancing - Key Integrations

Scenario	Solution
Cloud CDN	Enable Cloud CDN with Application Load Balancers for optimizing static content delivery for your end users
Google Cloud Armor	Cloud Armor policies can be applied to Application Load Balancer backends to provide DDoS protection, WAF capabilities, and rate limiting.
Cloud Logging	Log all the load balancing requests to analyze traffic and debug problems. Export logs to Cloud Storage, BigQuery, or Pub/Sub for analysis.
Identity-Aware Proxy (IAP)	Secure access to web applications without requiring VPNs. Authenticate users and enforce IAM-based access control.
Private Service Connect	Expose services privately across VPCs or consume Google APIs without traversing the public internet



Cloud Load Balancing Scenarios [1/2]

Scenario	Solution
You want only healthy instances to receive traffic	Configure a health check
Which layer does Network Load Balancer operate at?	Layer 4 (Transport Layer)
When do you use Passthrough Network Load Balancer?	When you need ultra-low latency and want traffic sent directly to backend VMs without proxying
You want high availability for your VM instances in MIGs	Create Multiple MIGs for your VM instances in multiple regions. Load balance using a Global external Application Load Balancer.
You want to route requests to multiple microservices using the same load balancer	Create a MIG and a Backend Service for each microservice. Create URL Maps with rules to route based on the path (/microservice-a, /microservice-b etc).



Cloud Load Balancing Scenarios [2/2]

Scenario	Solution
How can you implement versioning using a Application Load Balancer	Use Layer 7 routing with paths like <code>/service-a/v1</code> → V1 backend service, <code>/service-a/v2</code> → V2 backend service
How can you perform A/B testing using a Load Balancer	Use traffic splitting (weighted routing) in HTTP(S) Load Balancer
How can you enable SSL encryption for HTTP(S) Load Balancer	Configure a Google-managed SSL certificate for HTTPS



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Getting Started with

Managed Compute Services

Getting Started with Managed Services



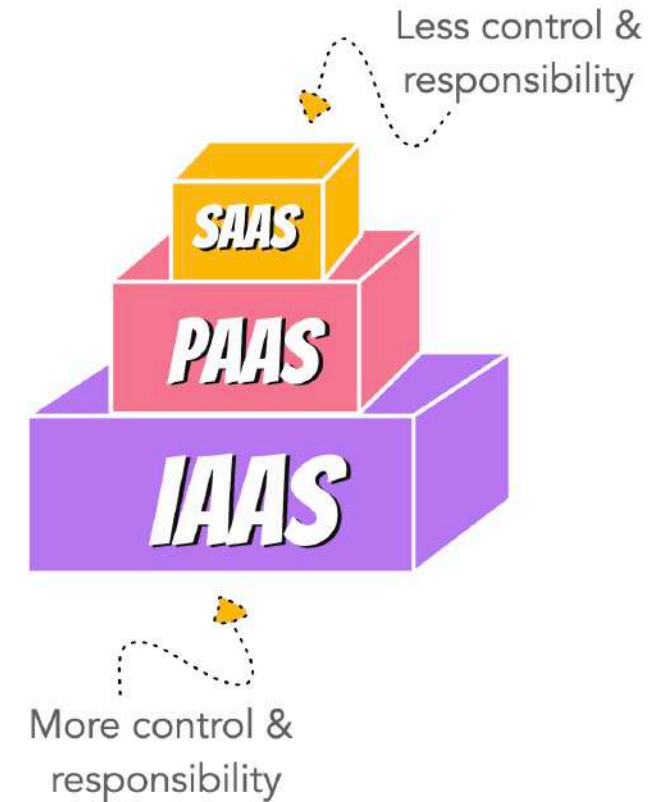
Question: Do you want to run apps in the cloud like you do in a data center?

- **Alternative:** Or are there better cloud-native approaches?

You should **understand some terminology:**

- **IaaS** (Infrastructure as a Service)
- **PaaS** (Platform as a Service)
- **SaaS** (Software as a Service)

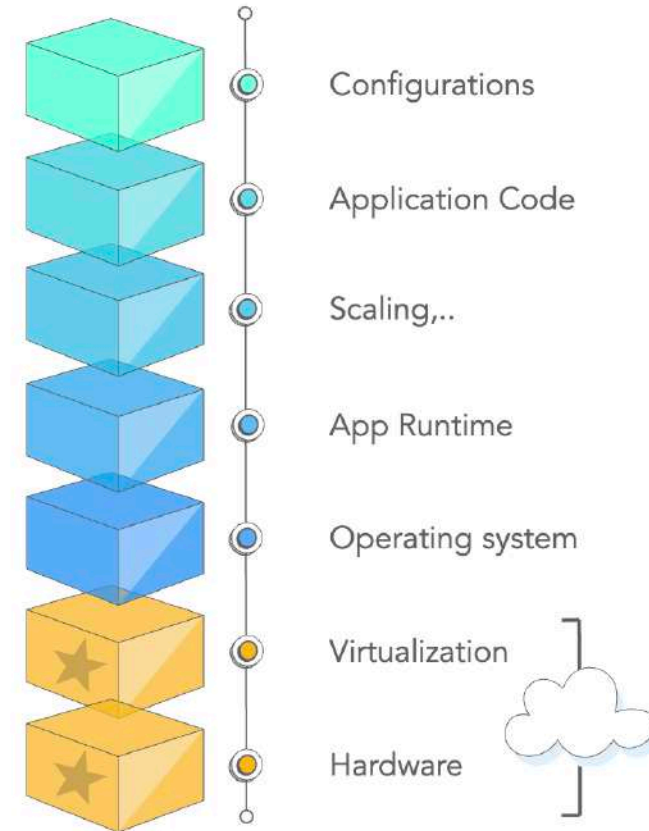
Ready for a journey to understand these?





Example 1A: Setting up an App - IaaS Approach

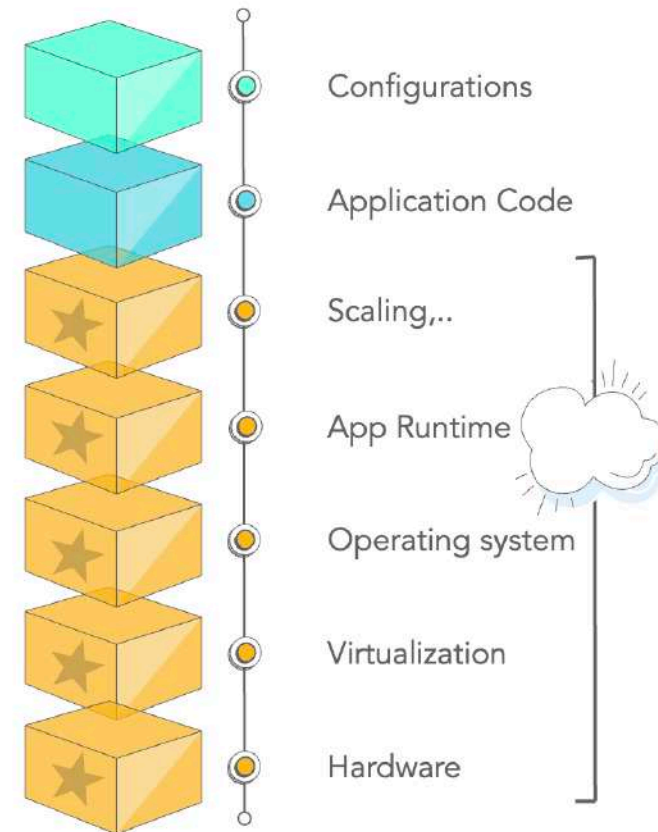
- 1: Provision VMs – Create VMs on cloud
- 2: Install Software – Install OS, runtimes etc
- 3: Deploy Code – Upload code and configure your application manually
- 4: Configure Ops – Set up load balancing and auto scaling yourself
- 5: Maintain – You are responsible for OS patching, runtime upgrades and server health





Example 1B: Setting up an App - PaaS Approach

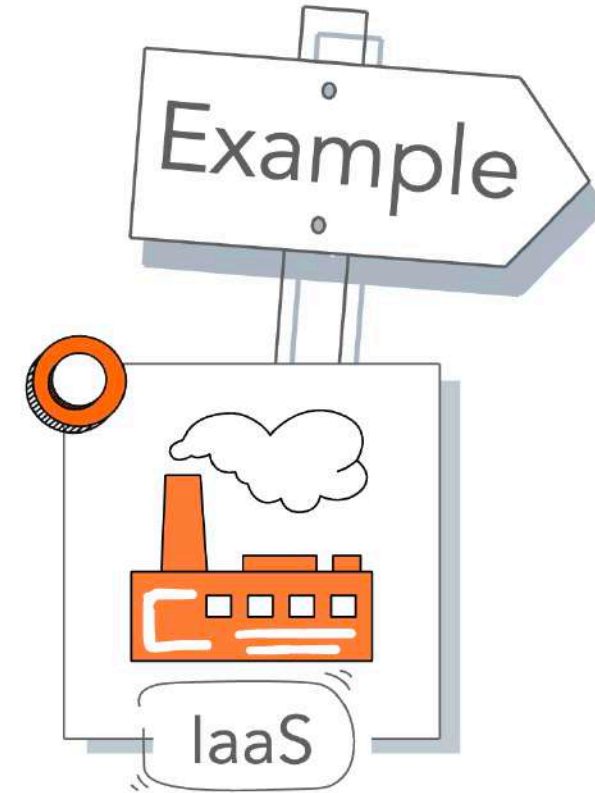
- 1: Pick a Platform** – AWS Elastic Beanstalk, Azure App Service, or Google App Engine
- 2: Upload Code** – Upload your code artifact
- 3: Automatic Setup** – Platform provisions VMs, installs OS, and starts app
- 4: Built-in Ops** – Scaling and Load Balancing are easily configurable
- 5: Zero Maintenance** – No Manual OS patching or server management required





Example 2A: Setting up a SQL Database - IaaS Approach

- 1: Provision VMs** – Create VMs to host your database
- 2: Install Database** – Set up MySQL, PostgreSQL, or any DB you need
- 3: Configure Storage** – Attach disks
- 4: Set Up Ops** – Manually configure backup jobs and replication
- 5: Maintenance** – Manage scaling, patches, and monitor performance

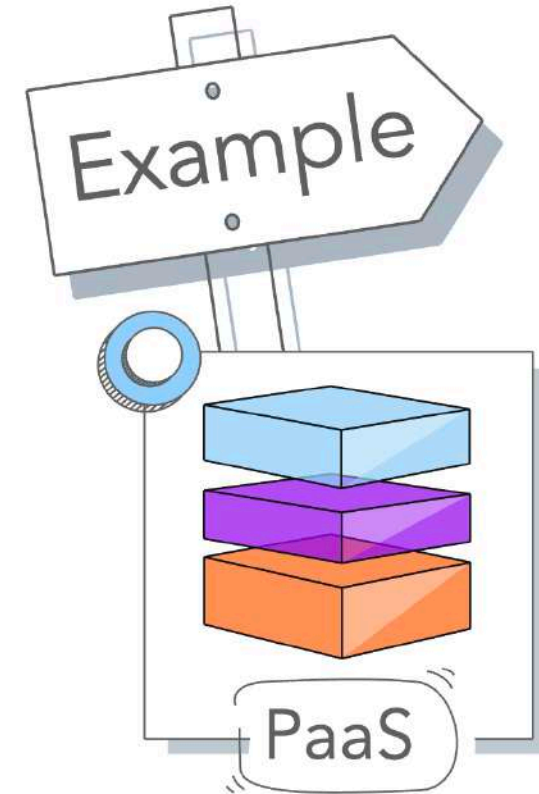




Example 2B: Setting up a SQL Database - PaaS Approach

- 1: **Pick a Service** – AWS RDS, Azure SQL Database, or Google Cloud SQL
- 2: **Choose Type** – MySQL, PostgreSQL, SQL Server, etc
- 3: **Configure** – Decide CPU, memory, and storage size
- 4: **Enable Ops** – Turn on built-in backups & high availability
 - **Ready** – Database is ready to use

Focus: Data and queries, NOT servers or maintenance





What is IaaS?

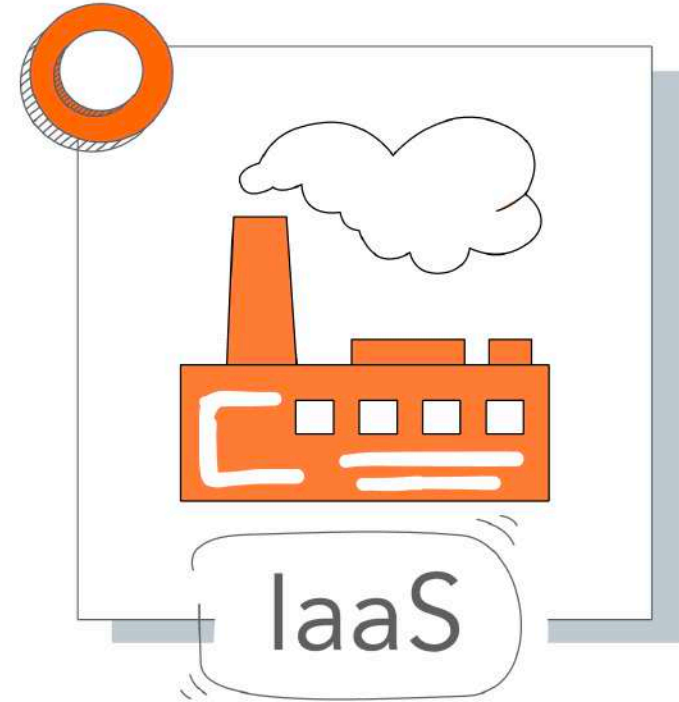
IaaS: Infrastructure as a Service

- **Concept:** Use only infrastructure from the cloud provider

Customer Responsibility: High

- **Components:** OS, runtime, and system software
- **Operations:** Availability, durability, and scaling

Summary: You manage almost everything except the hardware





What is PaaS?

PaaS (Platform as a Service): Use a complete platform provided by the cloud

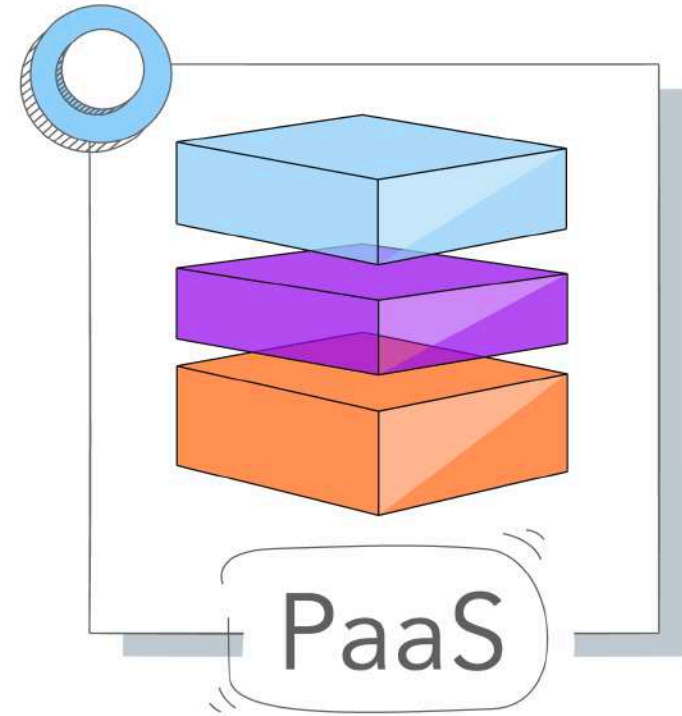
Cloud Provider Responsibility: High

- **Components:** OS, runtime, availability, durability, scaling

Customer Responsibility: Low

- **Focus:** Code, data, and configuration

Summary: Cloud provider manages the routine work





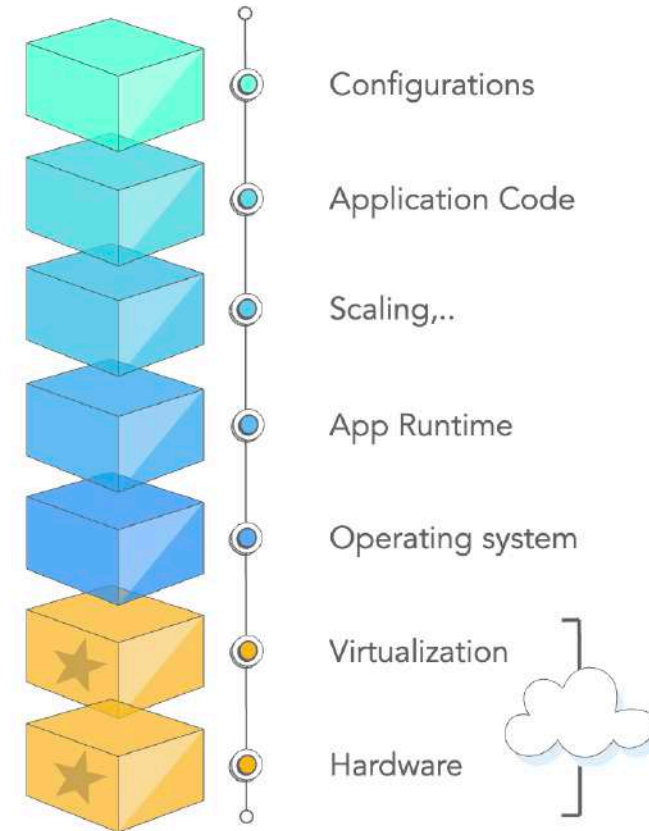
Who is Responsible for What? - IaaS

Cloud platform: The Basics

- Virtualization, Hardware and Networking

Customers of cloud: Everything Else

- **Application:** Code and Runtime
- **Operations:** Load balancing, Auto scaling, Availability
- **Database:** Software, upgrades, configuration (tables, views,..), and data
- **OS:** Upgrades and patches





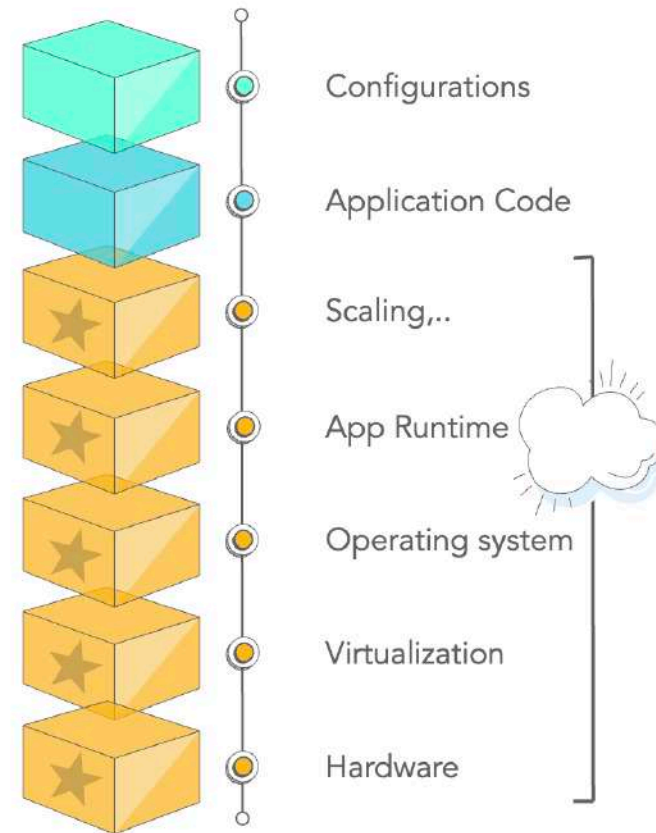
Who is Responsible for What? - PaaS

Cloud Platform: Heavy Lifting

- **Infrastructure:** Virtualization, Hardware, Networking, *OS: Upgrades and patches
- **Platform:** Runtime, Database software, Scaling, Availability, Durability, Load Balancing

Customer: Business Logic

- **Application:** Code & configuration
- **Database:** Schema, Tables, and Data
- **Restricted Access:** Usually no access to OS or VM instances





PaaS Services Can Do a Lot More

Scope: PaaS is NOT limited to running apps and databases

Managed Services: Includes common building blocks

- **Storage:** Amazon S3, Azure Blob Storage, Google Cloud Storage
- **Analytics:** BigQuery, Azure Synapse, AWS Redshift

Goal: Let developers focus on logic and innovation

- **Benefit:** Skip the setup and maintenance





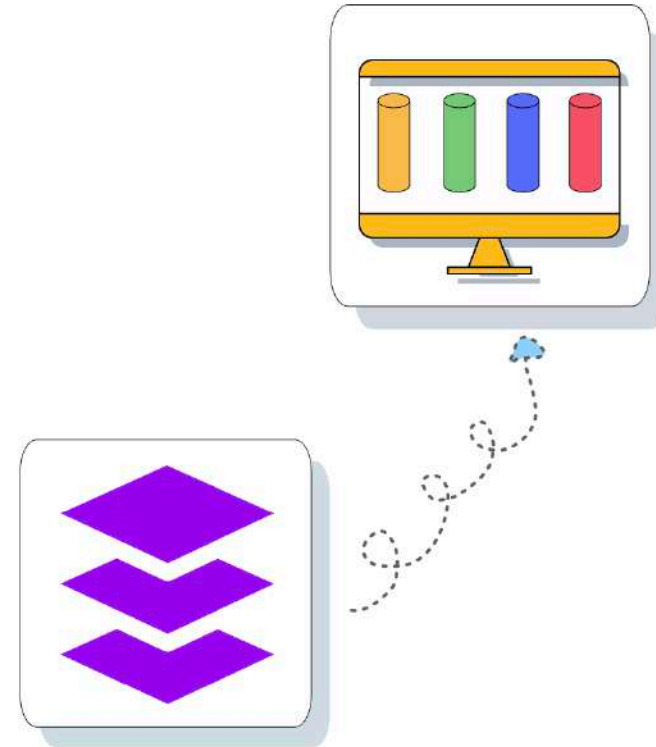
From Platforms to Complete Applications

IaaS: You manage most things yourself (infra, scaling, app, data)

PaaS: The platform helps you focus on your application and data

SaaS: You use complete software - no setup needed!

Next Step: Let's explore SaaS (Software as a Service)





What is SaaS (Software as a Service)?

Definition: Centrally hosted software

- **Hosting:** Mostly on the cloud

Billing: Offered on a subscription basis (Pay as you go)

Examples:

- **Email:** Gmail, Outlook 365
- **Office tools:** Microsoft Office 365, Google Docs
- **Calendar:** Google Calendar





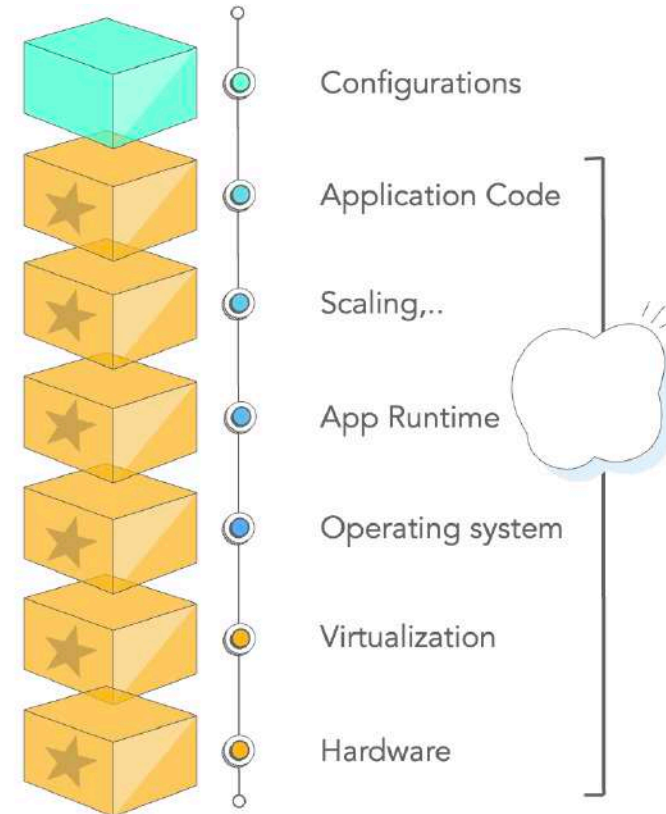
Who is Responsible for What? SAAS

Service Provider: Almost Everything

- **Infrastructure:** OS, Runtime, Networking
- **Application:** Code updates and maintenance
- **Operations:** Auto scaling, Availability, Load balancing

Customer: Usage

- **Content:** Documents, emails, sheets
- **Configuration:** Users, permissions etc





Quick Recap - IaaS vs PaaS vs SaaS

IaaS: Platform provides infra

- **You:** Manage all other layers

PaaS: Platform handles infra, setup, and scaling

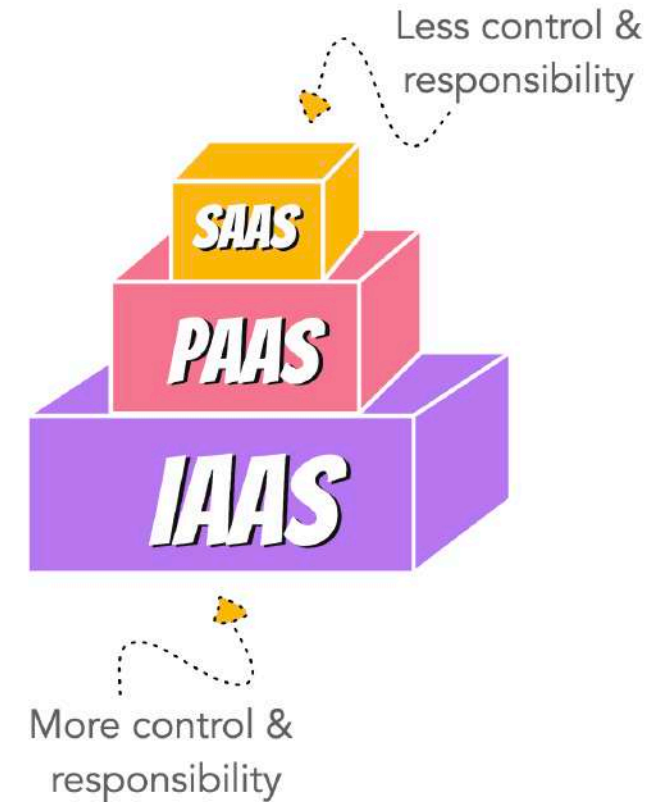
- **You:** Handle code, configuration, and data

SaaS: Platform provides everything

- **You:** Just use the app

Trend: Moving from IaaS → PaaS → SaaS

- **Control:** Decreases
- **Ease of Use:** Increases





Service Categories - Scenarios

Scenario	Solution
IaaS or PaaS or SaaS: Deploy Custom Application in Virtual Machines	IaaS
IaaS or PaaS or SaaS: Deploy a Database in Virtual Machines	IaaS
IaaS or PaaS or SaaS: Using Gmail	SaaS
IaaS or PaaS or SaaS: Using a Managed Service to setup a database	PaaS
True or False: Customer is responsible for OS updates when using PaaS	False
True or False: Customer is completely responsible for Availability when using PaaS	False
True or False: In PaaS, customer has access to VM instances	False
True or False: In PaaS, customer can customize OS and install custom software	False
True or False: PaaS services only offer Compute services	False
True or False: In PaaS, customer can configure hardware needs (memory, cpu etc)	True



Choosing Between IaaS, PaaS, and SaaS - Scenarios

Scenario	Solution
Legacy Migration: You must migrate a 10-year-old app that requires a specific, older version of the OS with customization	IaaS
Rapid Development: A startup wants to deploy a new NodeJS web app immediately. They have no Ops team and don't want to manage servers	PaaS
Productivity: You need email, calendar, and document collaboration tools for 500 employees starting tomorrow	SaaS
Custom Database: You need to run a specialized database engine that needs customization of Operating System	IaaS
CRM Needs: Your sales team needs a tool to track leads and customers. You want zero maintenance overhead	SaaS



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Google Cloud

Managed Compute Services



Cloud Managed Services for Compute - Models

Model	Explanation
IaaS (Infrastructure as a Service)	Use only infrastructure from the cloud provider Customer manages almost everything (OS, runtime, availability, durability, scaling) except the infrastructure
PaaS (Platform as a Service)	Use a complete platform provided by the cloud Delegate a lot of responsibility (OS, runtime, availability, durability, scaling) to the cloud provider
SaaS (Software as a Service)	Centrally hosted software offered on a subscription basis Examples: Gmail, Outlook 365, Google Docs, Google Calendar Customer has minimal responsibility



Cloud Managed Services for Compute - Key Concepts

Concept	Explanation
Serverless	• When you build an app, you think about: "Where to deploy? What server? How to scale and keep it available?" What if you did not have to worry about any of these? • Concept: You focus on code - the cloud platform handles everything else • Zero Infrastructure Worries: Automatic Scaling + Pay Only for Usage (Ideally Zero requests → Zero cost)
Containers	• Package entire app and dependencies as one unit: code + runtime + dependencies • Run the same package, the same way everywhere (Laptop, test, and cloud) • Language or application type does NOT matter
Container Orchestration	• Microservices: Need a lot of features and automation • Container Orchestration platforms like Kubernetes: Enable running 100s of microservices (each with multiple instances) with minimum management headache • Key Features: Auto scaling, Load Balancing, Release Management, Automatically recover from failures, etc.



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Getting Started with Serverless



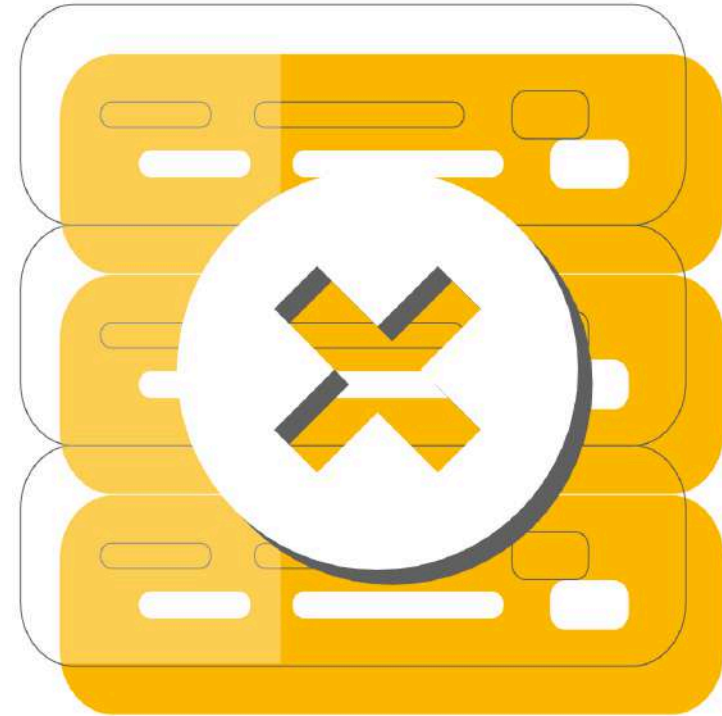
What if You Didn't Have to Manage Servers?

Scenario: When you build an app, you think about:

- Where to deploy? What kind of server?
- How to scale and keep it available?

Serverless: What if you did not have to worry about any of that?

- **Concept:** You focus on code - the cloud handles everything else
- **Remember:** Serverless $\neq$ No Servers
- **Servers still exist:** You just don't manage them





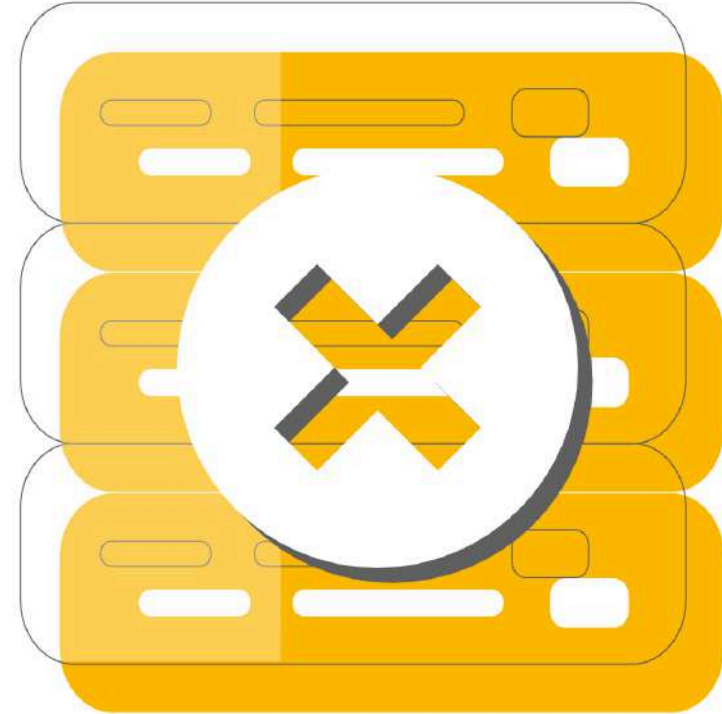
Serverless Explained – The Uber Analogy

Think of Uber:

- You don't own or maintain a car
- You just request a ride when you need it
- Uber handles cars, drivers, and maintenance
- Cars, drivers and maintenance exist, BUT you don't worry about them!

Serverless works the same way

- You focus **ONLY** on your app logic
- The cloud handles infrastructure, scaling, and uptime
- You **pay only for what you use**





What Does Serverless Mean for You?

Zero Infrastructure Worries

- You have **no visibility** into servers, VMs, or scaling

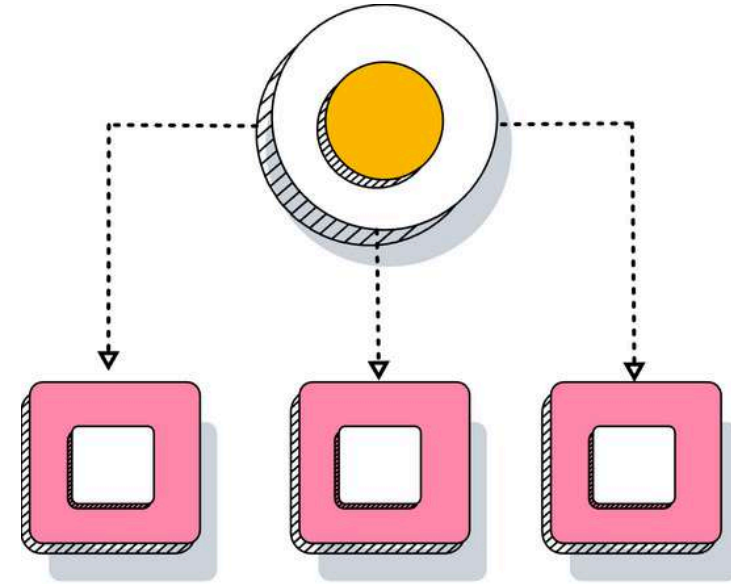
Automatic Scaling & High Availability

- The cloud handles it for you

Pay Only for Usage

- Zero requests → Zero cost
- You pay for requests, **not for servers**

Focus on Code – the rest is managed automatically





Misconception: Serverless $\neq$ Just Compute

Serverless is not limited to compute services

- It includes databases, APIs, messaging, and more

Examples:

- **Databases** – Amazon DynamoDB, Google Cloud Firestore, Azure Cosmos DB (Pay per operation)
- **Messaging** – Google Cloud Pub/Sub, AWS EventBridge, Azure Service Bus (Pay per message)
- And a lot of other types of managed services





Serverless - My Perspective

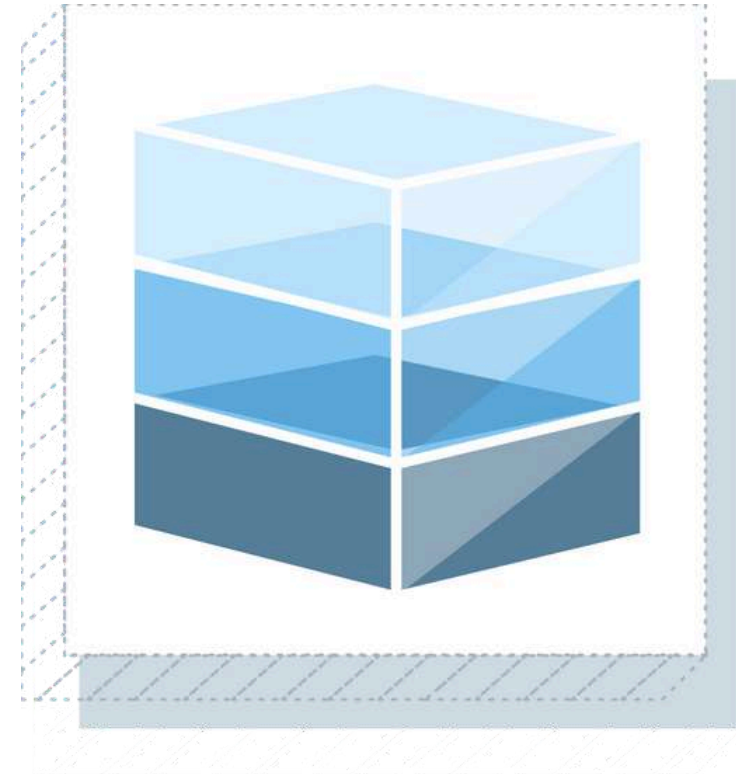
Level 1 (Managed): Handles scaling and availability BUT you pay for instances

- Can scale down to zero instances
- **Examples:** Google App Engine, AWS Fargate

Level 2 (True Serverless): Pay per invocation, not per instance

- **Examples:** Google Cloud Run Functions, AWS Lambda, Azure Functions

When I say “Serverless,” I mean Level 2: Zero infra management + Pay per request





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Getting Started with Containers



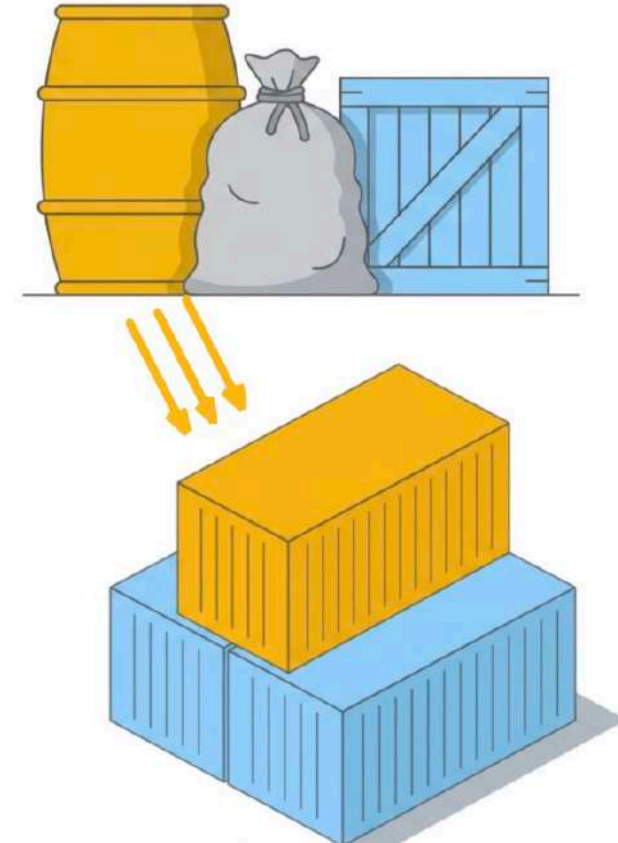
Evolution of Shipping Containers

Before: Goods shipped as loose items in different shapes and sizes

- **No Standard:** Each shipment needed custom handling
- **Unsafe:** Goods damaged or lost during transfers

Breakthrough: A single **standard box** called a shipping container

- **Universal:** Same box works on ships, trucks etc.
- **Fast:** Easy to load, stack, and move
- **Cheap:** Lower handling cost





Containers Bring Same Revolution for Software

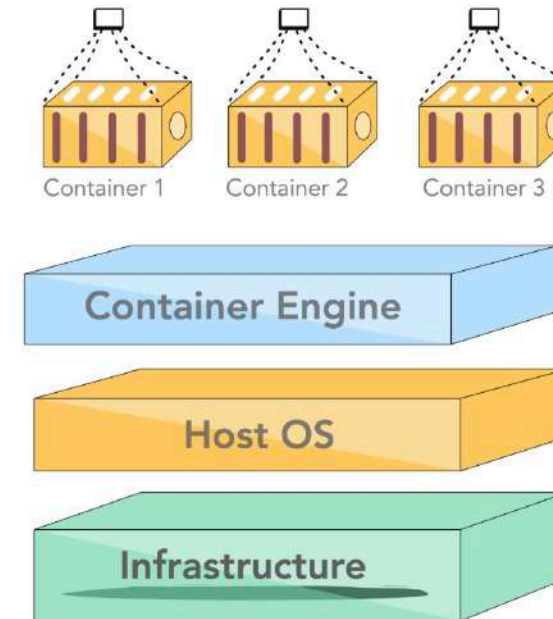
Before: Every application had a different deployment process

Solution: Package entire application and dependencies as one unit

- Code, runtime, and dependencies in one unit

Result: Run the same package, the same way everywhere

- Laptop, test, and production (cloud) environments
- Language or application type does NOT matter



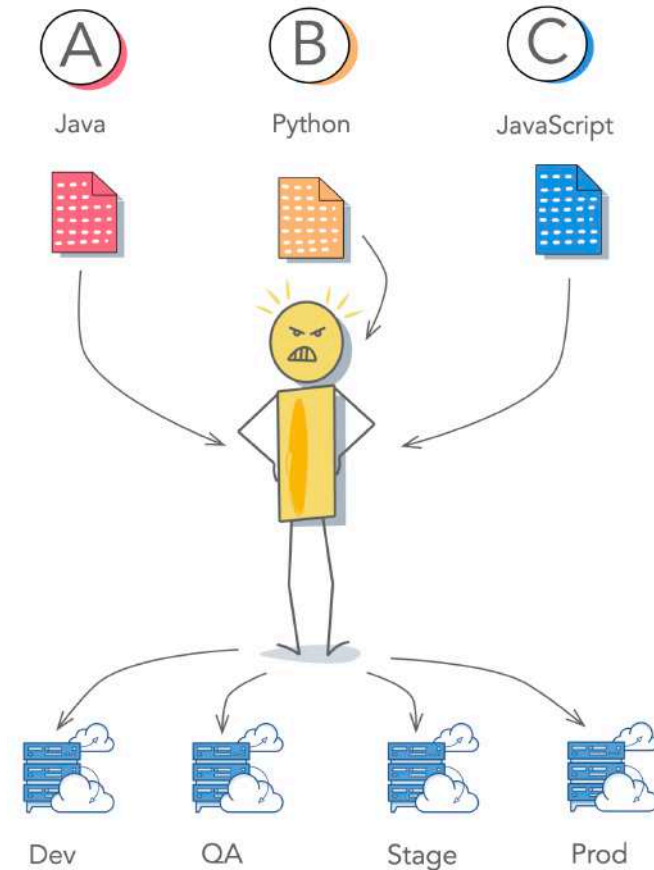


Traditional Deployment Before Containers

Documentation Driven: Deployment steps are described in a document

Manual Steps:

- **Hardware:** Setup Server
- **OS:** Install Linux/Windows/Mac/..
- **Runtime:** Install the right version of Java/Python/Node.js
- **Application:** Copy application code or binary files
- **Dependencies:** Setup dependencies





Challenges with Traditional Deployment

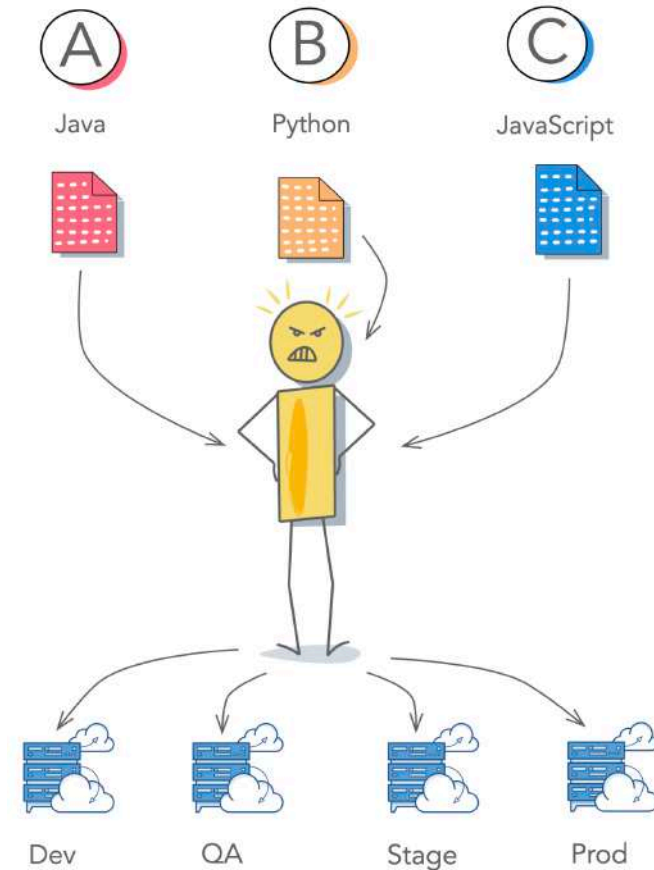
Manual Setup: Install OS > Setup software > Setup app > dependencies

Slow: Takes hours per environment

Error-Prone: Missed steps, mismatched versions

Different Docs: For different applications!

Hard to Debug: Works on my machine problems





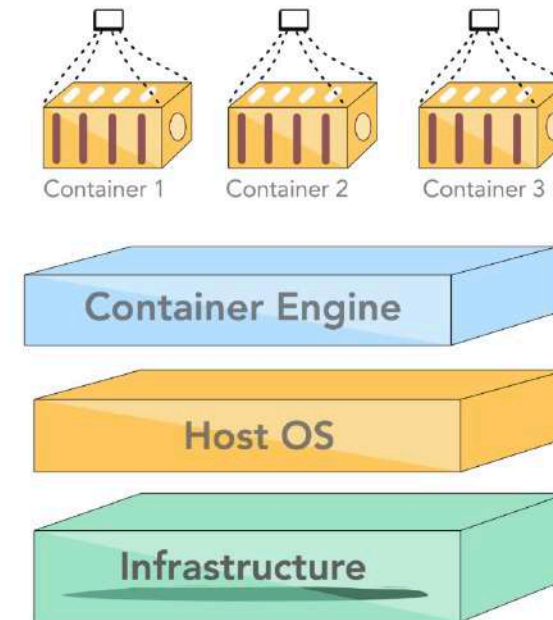
Enter Containers – A Better Way

Concept: Deliver a Complete Package

All-in-One:

- **OS:** Included
- **Runtime:** Included
- **Code:** Included
- **Dependencies:** Included

Simple: If the container runs on your laptop, it runs on the server





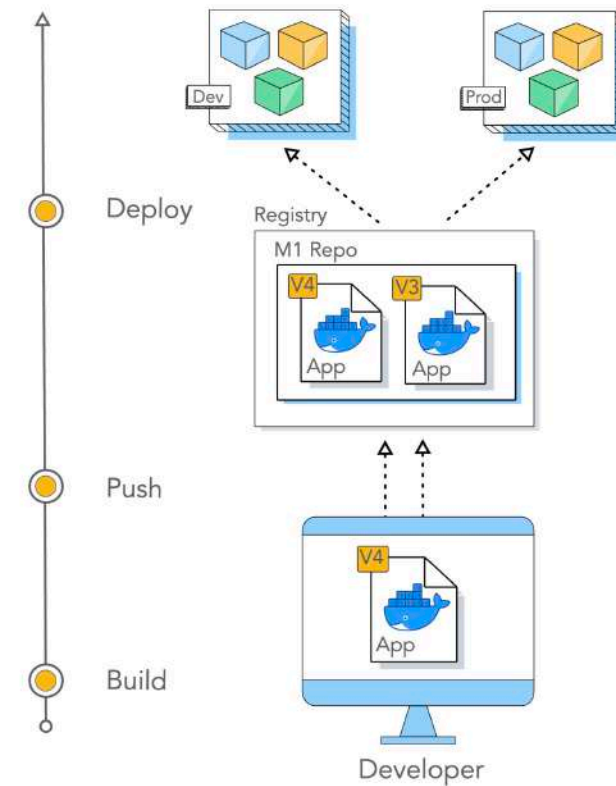
How do Containers work?

Idea: Package your application and its dependencies into a **container image**

Steps:

- **Build** – Create a container image with everything needed to run the app
- **Push** – Upload the image to a container registry
- **Deploy** – Download (Pull) the image and run it anywhere (local, data center, cloud)

Goal: Run the same container everywhere - no surprises!





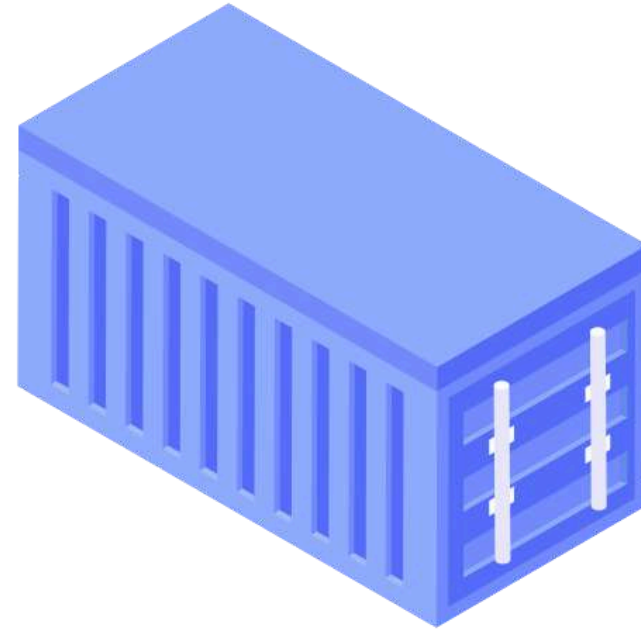
Component 1: Container Image

What is it?: A single file (essentially) containing:

- OS layer
- Your Runtime
- Your code
- Dependencies (Libraries)

Key Trait: Immutable (Read-only)

- Once built, it never changes
- Create new versions as needed - v1.0.0, v1.0.1, v1.1.0, ..(tags)





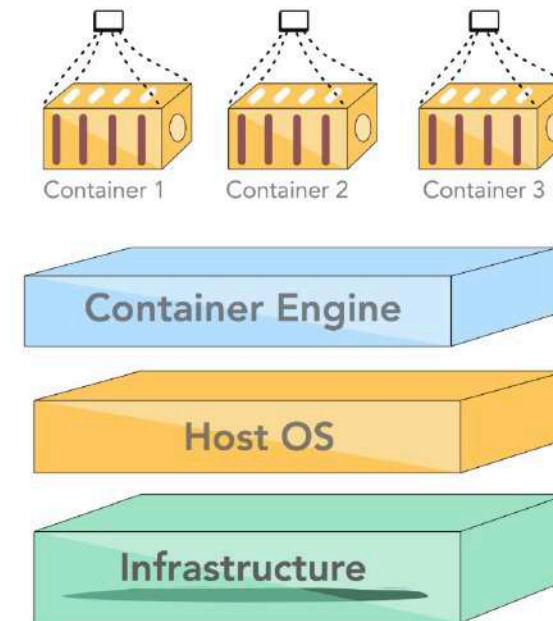
Component 2: Container Engine

Role: The software installed on your laptop or server that runs containers

Examples:

- **Docker:** The most famous one
- **Podman:** A popular alternative
- **Containerd:** The industry standard runtime

Job: "Take this image and run it as a container"





Component 3: Container Registry

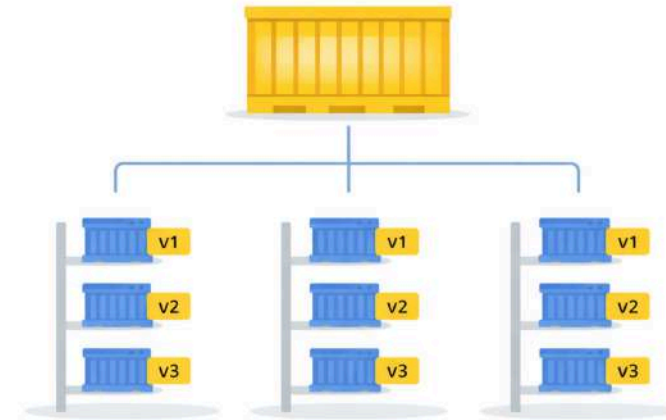
Role: A secure place to store your images

Public Registries:

- **Docker Hub:** The default public library

Private Cloud Registries:

- **AWS:** Elastic Container Registry
- **Azure:** Azure Container Registry
- **Google:** Google Artifact Registry





Containers – The Big Picture Recap

Step 1: Create a Container Image (Dev)

- Package code, runtime, and dependencies

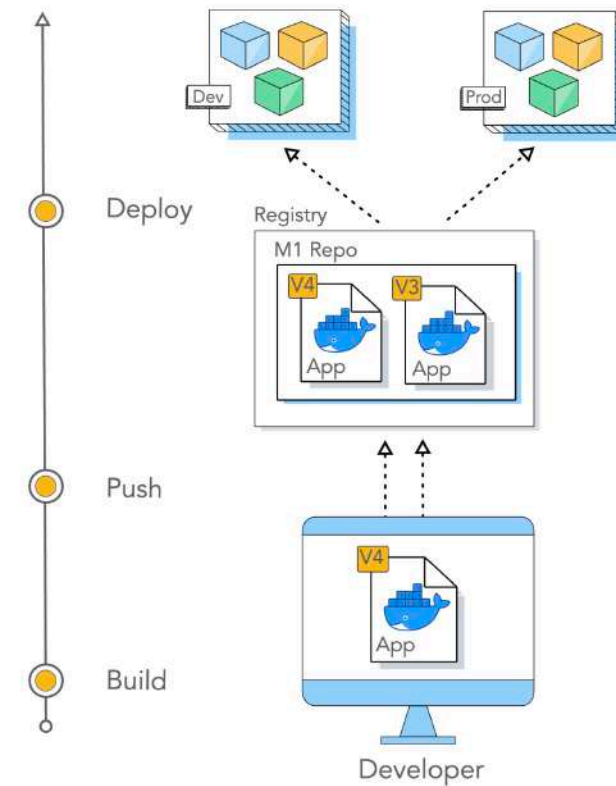
Step 2: Push it to Container Registry (Dev)

- Share the image across teams and environments

Step 3: Deploy to an Environment (Ops)

- Pull the image and run it as a container (using Container Engine)

Outcome: Same image, Same behavior,
Everywhere you run it





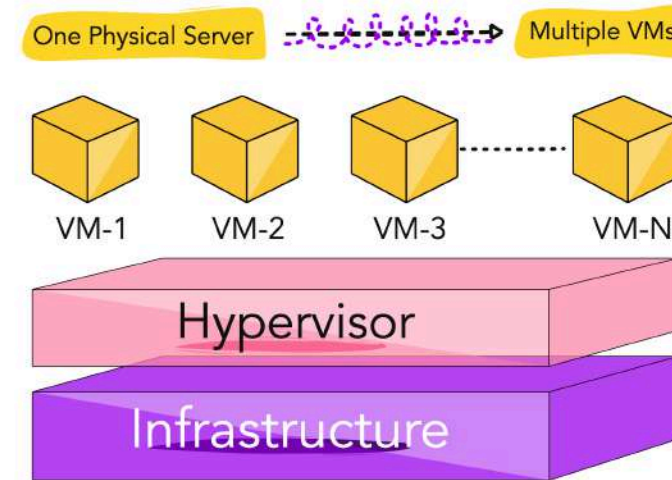
Virtual Machines - How do they work?

Key Layers: 3 key layers

- **Base Layer:** The physical server
- **Hypervisor:** Software that creates and manages VMs
- **VMs:** Each VM runs its own OS (Windows, Linux, ..)
 - Each VM contains application runtime, libraries, and the app

Key Benefits: Multiple VMs on one server →
Better resource utilization

Key Limitation: Each VM includes a full OS →
Heavy and slow to start





Containers - How do they work?

Base Layer: The physical server

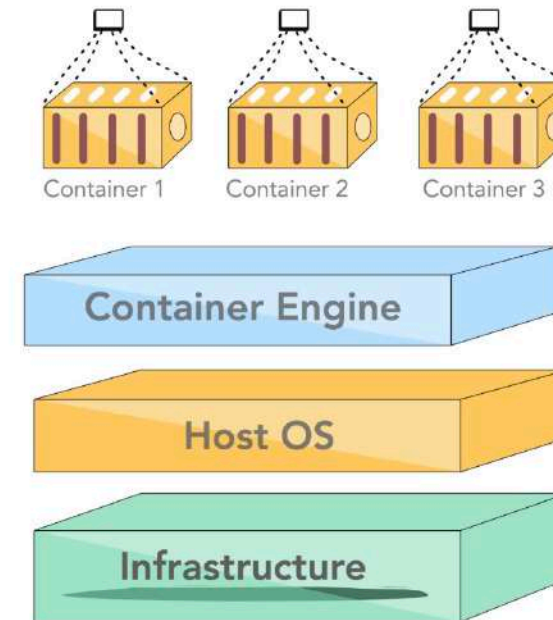
Host OS: Physical server runs a base OS

Container Engine / Runtime: Runs on top of the host OS

- Manages images, containers, etc.

Containers: Running instances created from container images

- Share the host OS kernel BUT isolated from other containers





Virtual Machines vs Containers

Feature	VMs	Containers
Architecture	Complete virtualized computers with their own OS, running on a hypervisor (Hardware Virtualization)	Lightweight, sharing the host OS core, running using Container Engine (OS layer Virtualization)
Performance	Resource-intensive (Each VM has full OS)	Lightweight (Host OS is shared)
Boot Time	Slower (Full OS startup)	Faster (Shares Host OS)
Resource Usage	Consumes more CPU, memory, storage	Efficient
Image Size	Large (includes OS)	Smaller (app + deps only)
Security	Stronger, separate OS for each VM	Shared OS kernel BUT Container engine ensures isolated processes for each container



Containers - A Summary

Standard Unit: Like shipping containers

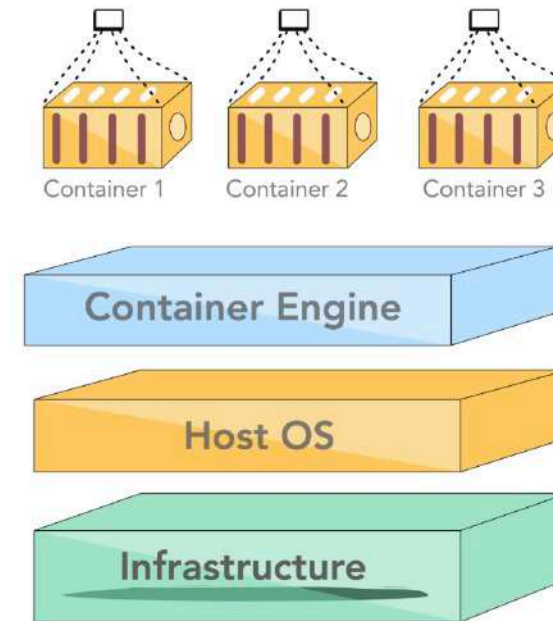
- **Before:** Different documents for Java, Python, NodeJs, ML, Analytics, etc.. (Manual)
- **After:** Everything is inside the container image
- **Ops Teams:** Only need to know how to run containers
 - **No Surprises:** Same steps everywhere for every app

Portable: Build once, run anywhere

- **Environments:** Local, on-prem data center, cloud

Fast Startup: Seconds or less

High Density: Many containers per server





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Getting Started with Microservices



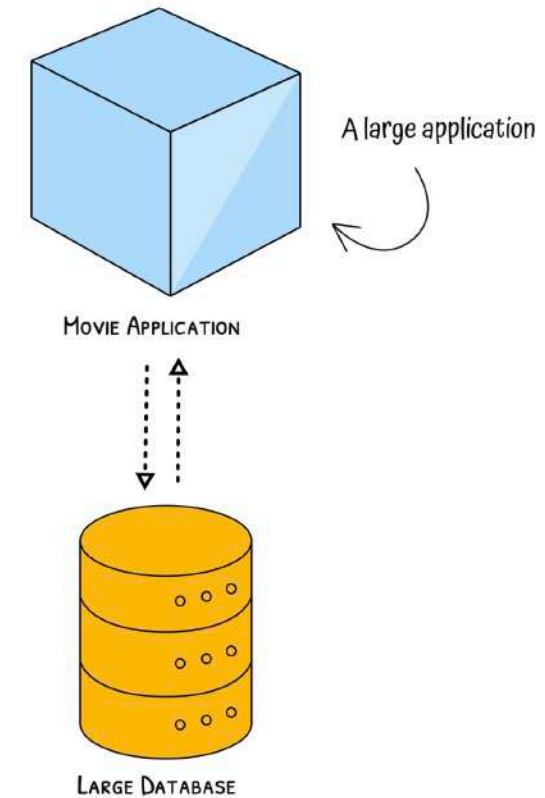
Understanding the Monolith Application

Monolith: A single large application

- **Database:** Typically connects to one large database
- **Codebase:** Can have millions of lines of code

Development: Entire team works on the same codebase

Challenge: Difficult to maintain and release





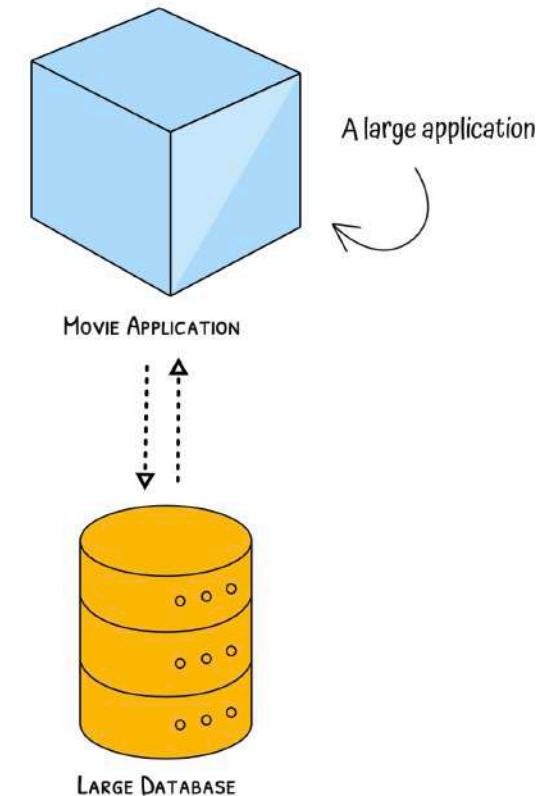
Challenges with Monolith Architecture

Deployment Complexity: Must deploy entire app for small changes

Release Cycles: Very long (months) due to regression testing

Scalability: Must scale entire app (expensive), not just components

Technology Lock-in: Hard to adopt new languages or frameworks



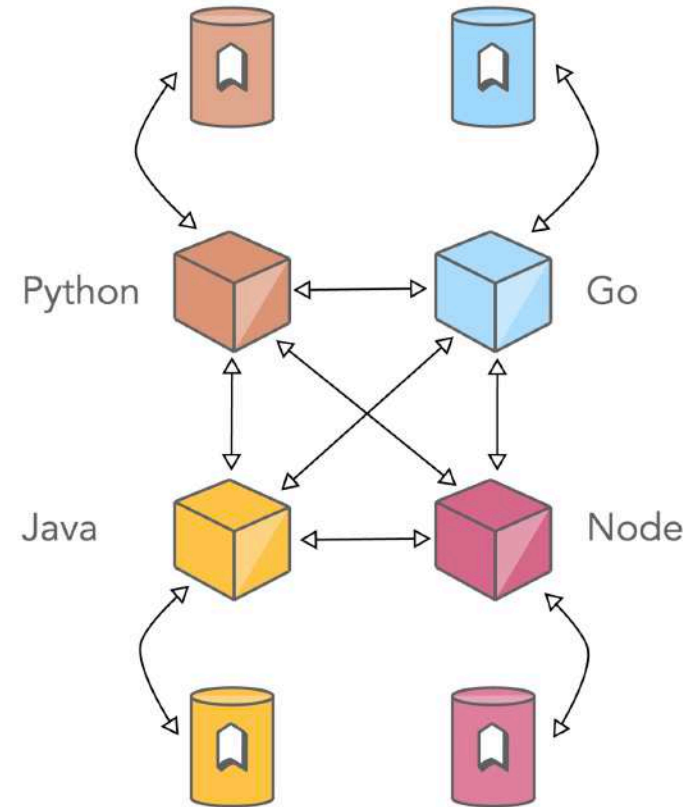


What Exactly is a Microservice?

Sam Newman: Small, autonomous services working together

James Lewis and Martin Fowler: An architectural style for developing apps as a suite of small, independently deployable, and scalable services

- Each running in its own process and organized around business capabilities
- These services communicate via lightweight mechanisms and feature decentralized data management and governance





Three Pillars of Microservices

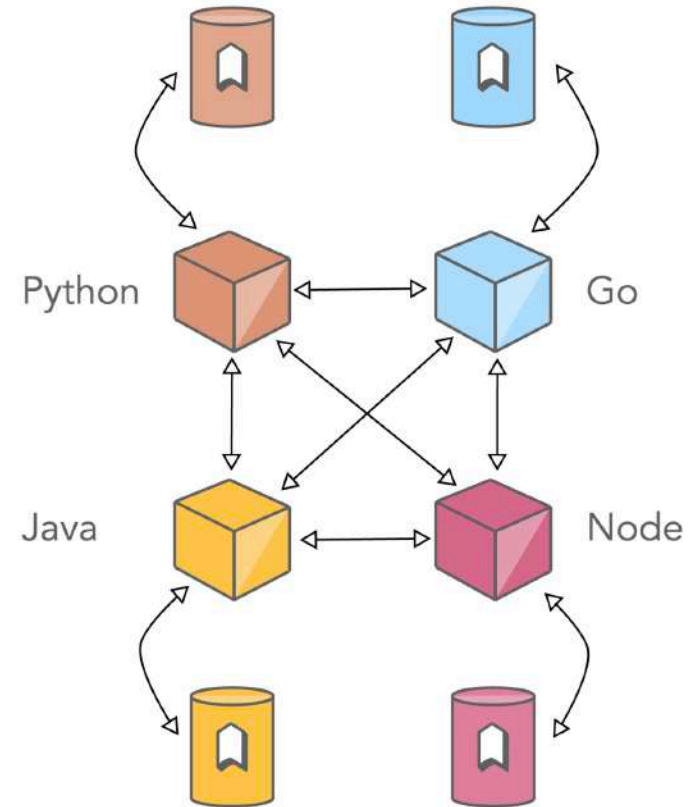
API-Based Communication: REST-first, lightweight service-to-service communication

Small Units: Well-chosen, independently deployable units

- **Boundaries:** Clearly defined functionality

Dynamic Scaling: Scale instances up/down independently

- **Benefit:** Adjust based on demand for specific services



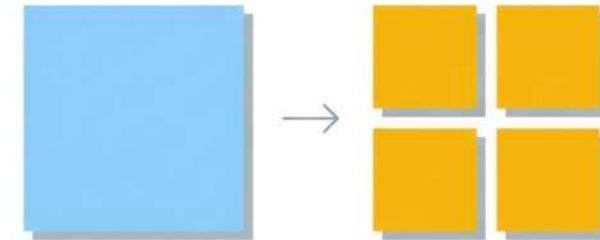


Monolith vs Microservices - An Example

Monolith: One massive app handling everything

Microservices: Decomposed into specific services

- **Movie Service:** Manages details and showtimes
- **Booking Service:** Handles tickets, seats, payments
- **Customer Service:** Handles profiles and authentication
- **Review Service:** Manages ratings and comments





Why Choose Microservices?

Flexibility: Adopt new technologies and processes easily

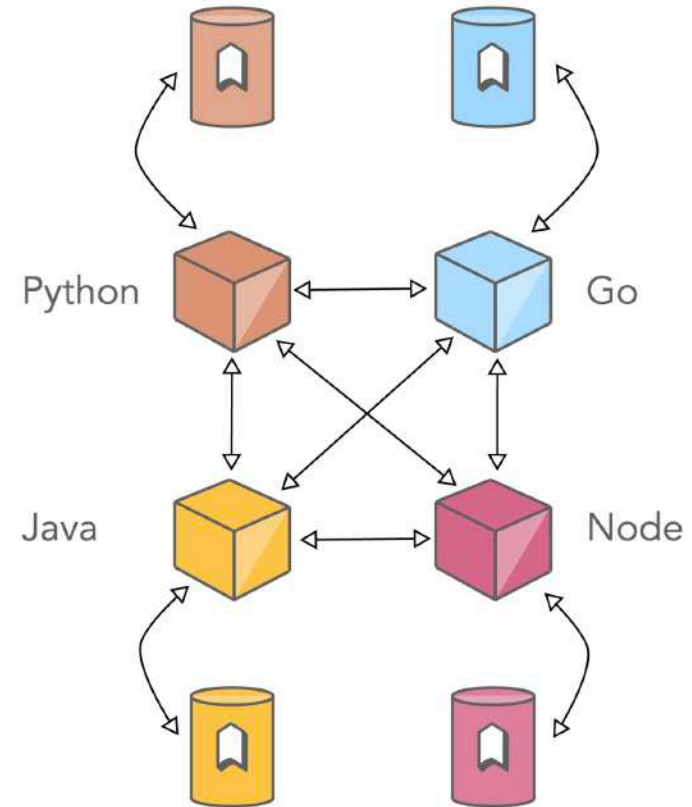
- **Innovation:** Experiment with different languages per service

Dynamic Scaling: Scale only what is needed

- **Cost:** More efficient than scaling a monolith

Faster Releases: Small independent units allow quick deployment

- **Agility:** Respond faster to market demands





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Getting Started With

Microservices Challenges



Challenges With Microservices Architectures

Standardization: Need standardized operations

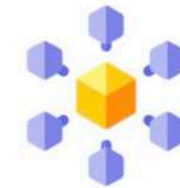
Auto Scaling: Services must scale automatically

Load Balancing: Distribute traffic between instances

Service Discovery: How do services find each other?

Release Management: Flexible releases

Resiliency: Handle failures automatically



MICROSERVICE A



MICROSERVICE B



MICROSERVICE C

Standardized Operations for Microservices

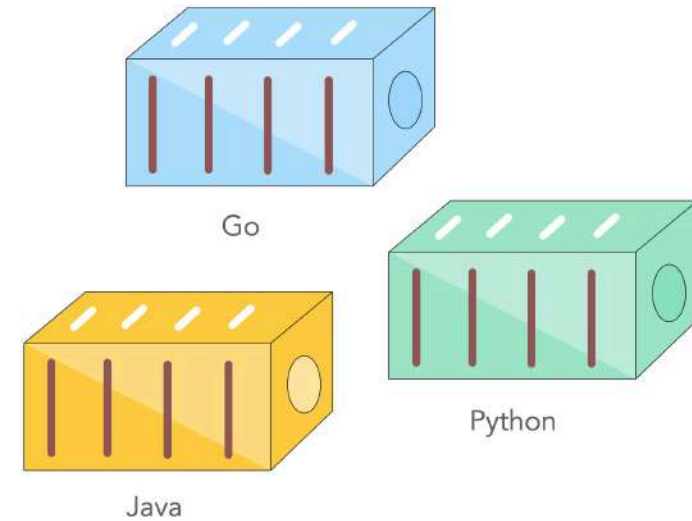


Challenges:

- **Diverse Tech Stack:** Java, Python, Node.js, etc.

Need: Standardized Operations

- Release Management
- Observability
 - Logs
 - Metrics
 - Traces
- Etc.





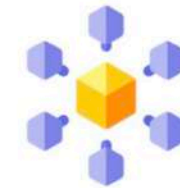
Automatic Scaling for Each Microservice

Multiple Microservices: A number of microservices with varying load

Varying no of instances: Each microservice will have different number of instances

Auto Scaling: Need to scale each microservice automatically based on load

Efficiency: Scale up and down automatically, saving costs



MICROSERVICE A



MICROSERVICE B



MICROSERVICE C



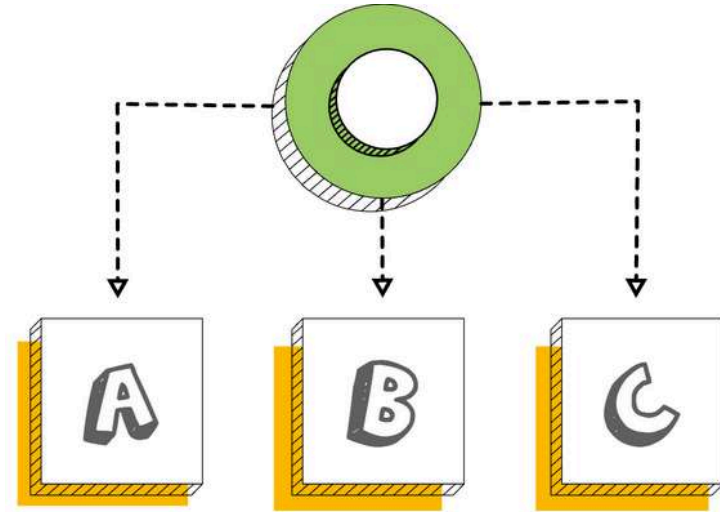
Load Balancing between Microservice Instances

Multiple Microservices: A number of microservices with varying load

Varying no of instances: Each microservice will have different number of instances

Load Balancing: Distribute traffic across instances

Performance: Ensure no single instance is overwhelmed





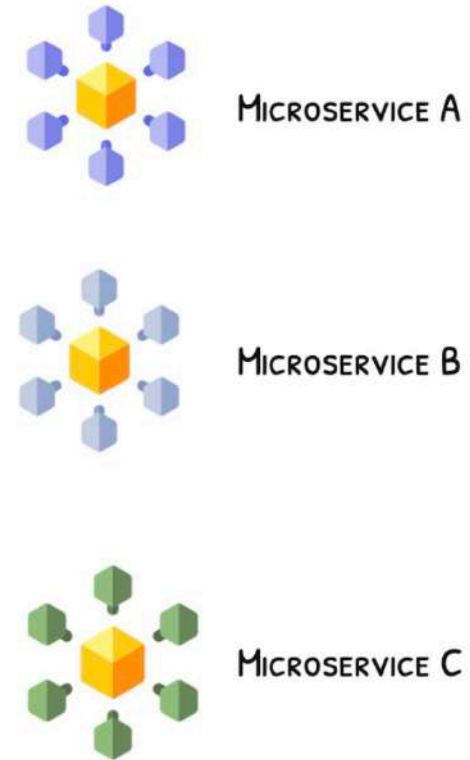
Help Microservices Locate Each Other - Service Discovery

Communication: Microservices need to talk to each other

Challenge: Microservice instances are created dynamically - as and when they are needed

- **Dynamic IPs:** IP addresses might keep changing
- **Best Practice:** Avoid hardcoding URLs

Need: Microservices should be able to locate each other whenever needed





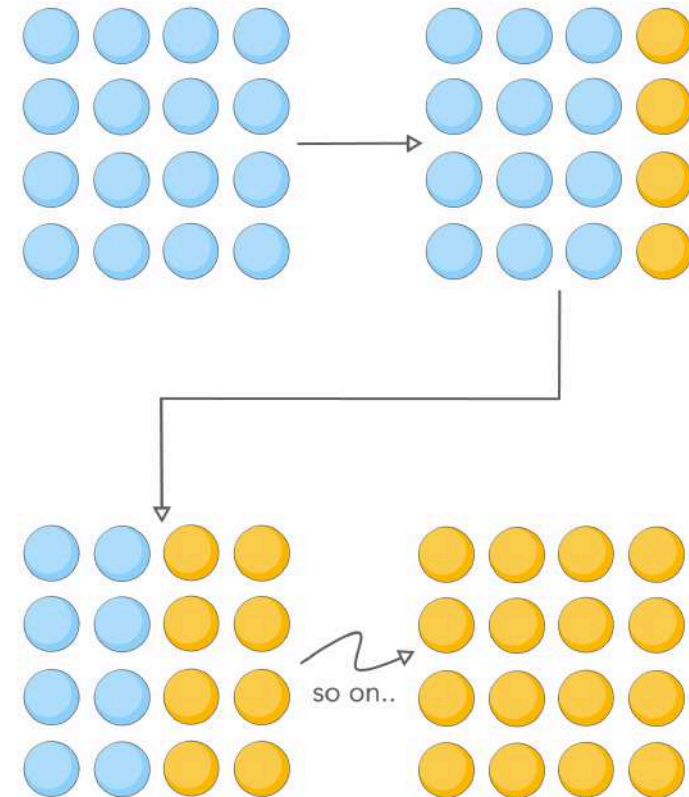
Manage Releases of Microservices

Frequent Releases: New versions of microservices are released very often

Need: Zero downtime releases with rollback possibilities

Example Strategy:

- **Rolling Updates:** Update instances in batches



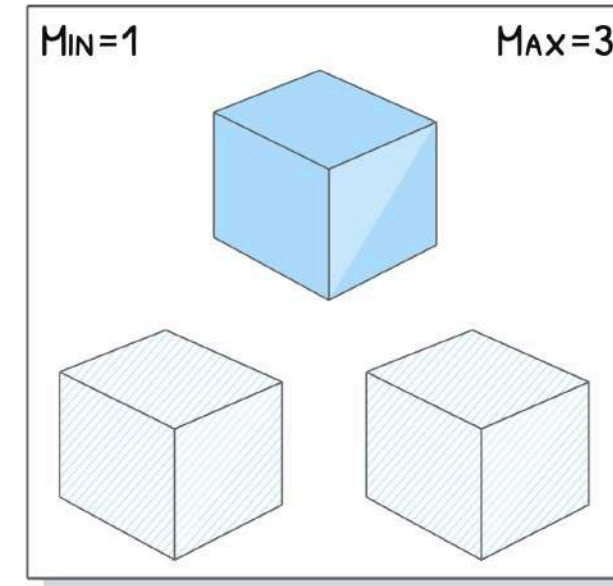


Resiliency - Identify and Act on Failures

Reality: Systems, nodes and applications fail often

Need: Recover automatically

- **Isolation of Failures:** Prevent one failure from crashing the whole system
- **Self-Healing:** Detect and recover from failures automatically





Challenges With Microservices Architectures - Summary

Standardization: Need standardized operations

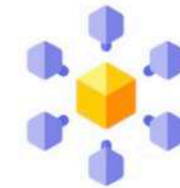
Auto Scaling: Each service must scale automatically

Load Balancing: Distribute traffic between instances

Service Discovery: How do services find each other?

Release Management: Flexible releases

Resiliency: Handle failures automatically



MICROSERVICE A



MICROSERVICE B



MICROSERVICE C



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Exploring Solutions

Containers and Orchestration



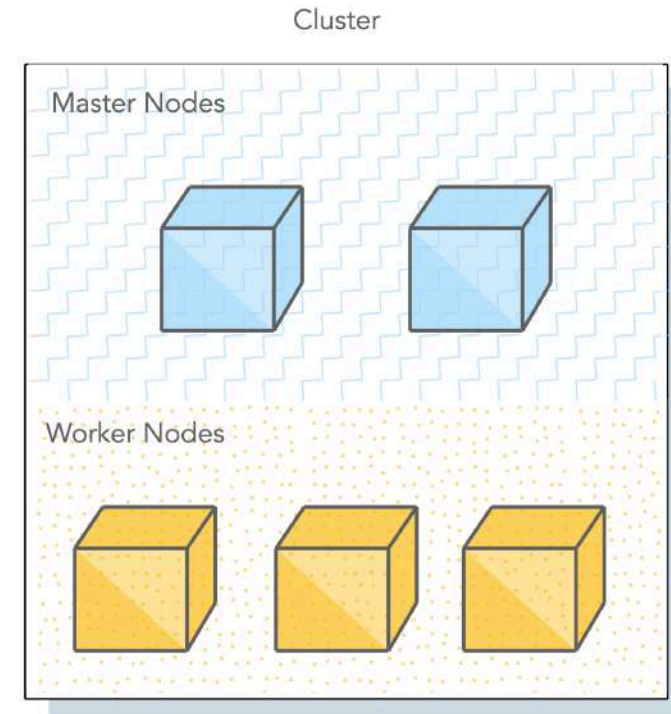
Challenges With Microservices Architectures - Solutions

Containers: Enable

- **Standardization:** Need standardized operations

Container Orchestration: Enable

- **Auto Scaling:** Scale services automatically
- **Load Balancing:** Distribute traffic between instances
- **Service Discovery:** Help services located other services
- **Release Management:** Flexible release strategies
- **Resiliency:** Handle failures automatically





Standardization with Containers

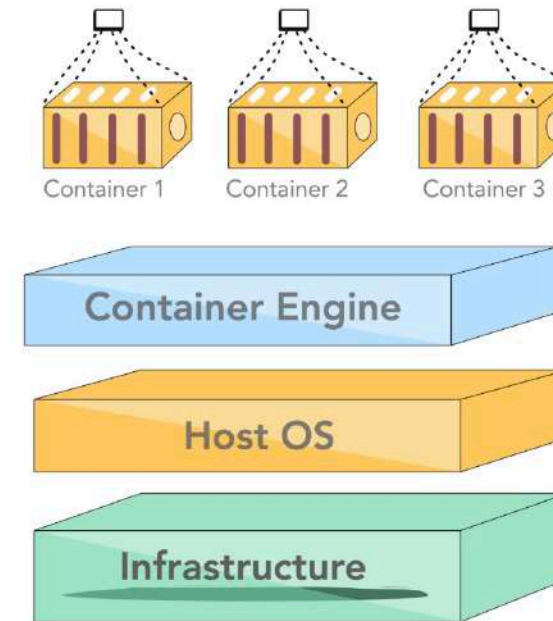
Diverse Tech Stack: Java, Python, Node.js, etc.

Need: Standardize Operations

- **Solution:** Containers
- **How?:** Build a container image for each microservice

Benefits:

- **Deployment:** Deploy all microservices the same way
- **Observability:** Standardize how services expose logs, metrics and traces



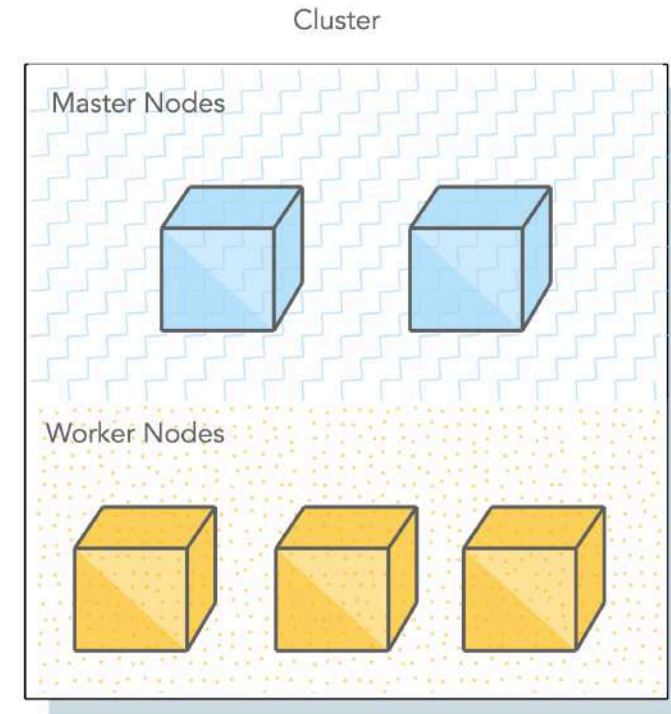


What is Kubernetes?

Kubernetes: An open-source container orchestration platform

How It Works (Simplified):

- You create a Cluster
- Declare your needs: **Microservice A** (3 instances), **Microservice B** (4 instances) and ...
 - Kubernetes deploys and manages these microservices
 - Kubernetes continuously monitors and reconciles the desired state





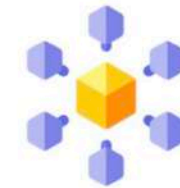
Automatic Scaling for Each Microservice

Multiple Microservices: A number of microservices with varying load

Auto Scaling: Need to scale each microservice automatically based on load

Solution: Kubernetes Auto Scaling Solutions

- **Declarative:** Tell Kubernetes that you need auto scaling for Microservice A



MICROSERVICE A



MICROSERVICE B



MICROSERVICE C



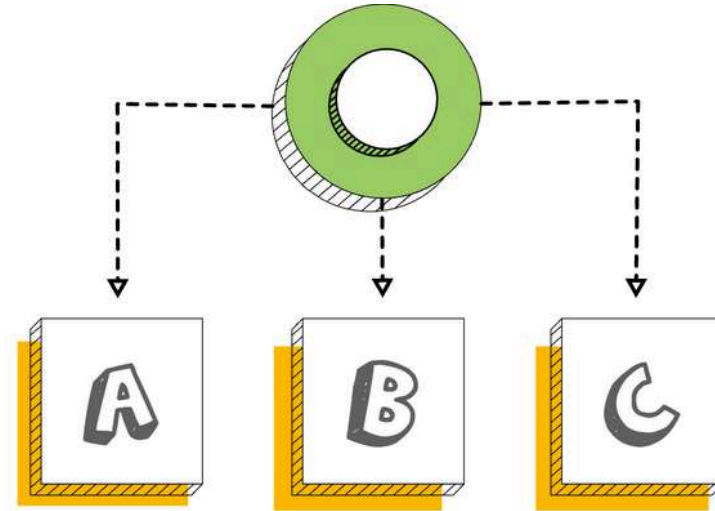
Load Balancing between Microservice Instances

Varying no of instances: Each microservice will have different number of instances

Need: Distribute traffic across instances

Solution: Kubernetes Service

- **How?:** Use Kubernetes Service Name to distribute traffic automatically among the microservices instances





Help Microservices Locate Each Other - Service Discovery

Communication: Microservices need to talk to each other

Challenge: Dynamic IP addresses of containers

Solution: Kubernetes service discovery

- **Mechanism:** Use Kubernetes Service Name to discover other microservices
- **Benefit:** No need to hard-code IP addresses





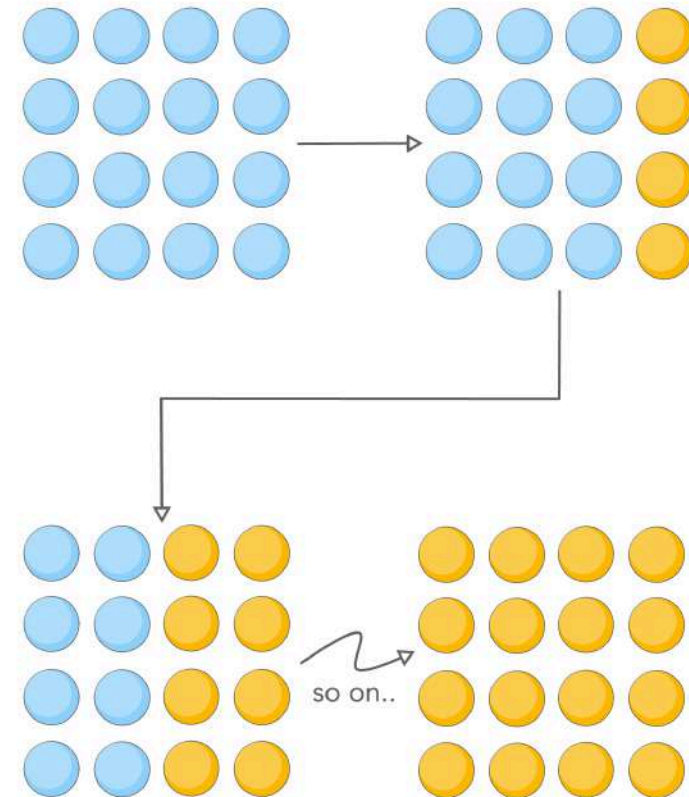
Manage Releases of Microservices

Challenge: Microservices need frequent releases with zero downtime

Solution: Kubernetes Deployments

Example Strategy:

- **Rolling Updates:** Update instances in batches





Resiliency - Identify and Act on Failures

Reality: Systems, nodes and applications fail often

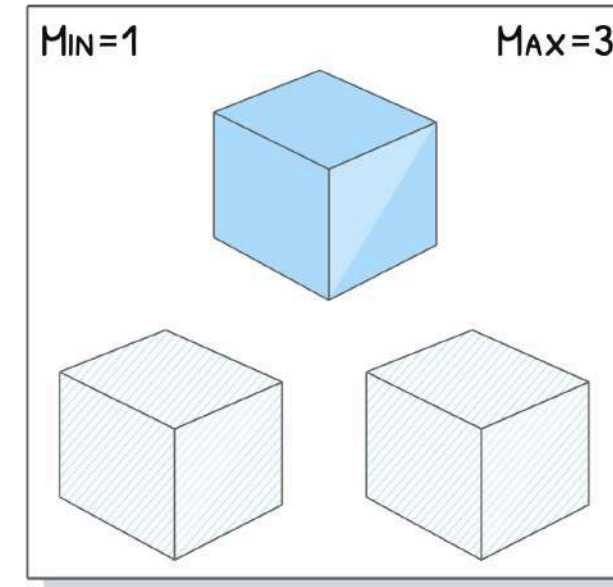
- **Need:** Recover automatically

Isolation: Prevent one failure from crashing the whole system

- **Solution:** Kubernetes ensures that a failure of microservice instance does not impact other microservices

Self-Healing: Restart or replace failed containers

- **Solution:** Kubernetes health checks



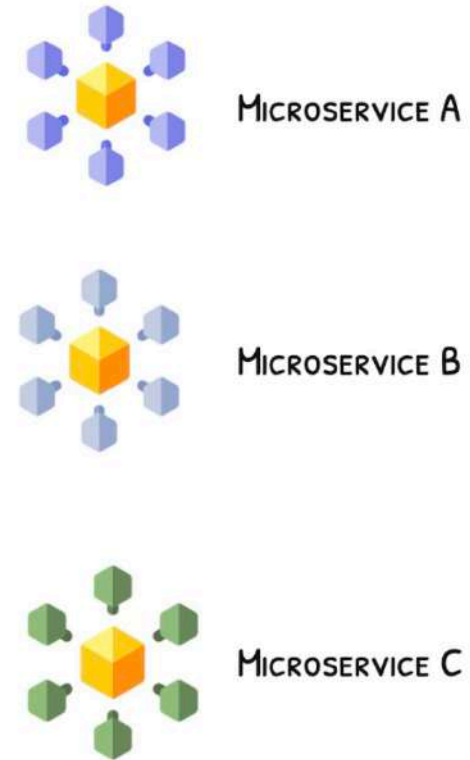
Challenges With Microservices Architectures - Solutions



Standardization: Containers (Docker)

Container Orchestration: With Kubernetes

- **Auto Scaling:** Kubernetes Auto Scaling Solutions
- **Load Balancing:** Kubernetes Services
- **Service Discovery:** Kubernetes Service Discovery
- **Release Management:** Flexible Deployment Strategies
- **Resiliency:** Kubernetes Self-Healing



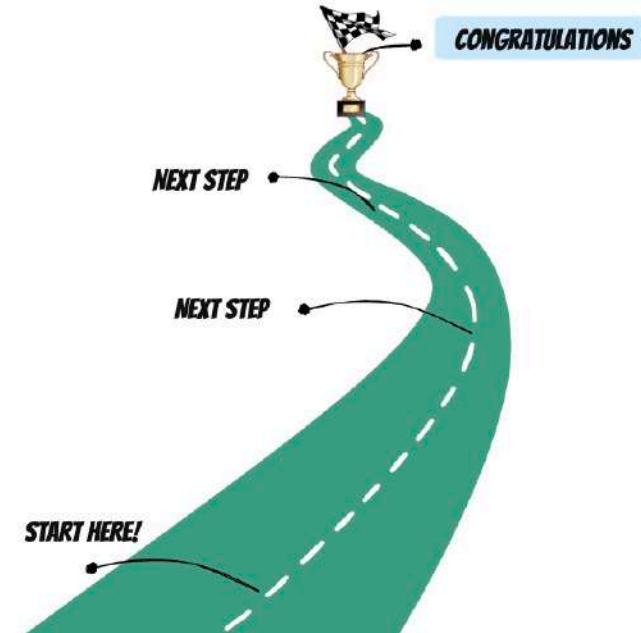


Summary - Journey to Microservices

The Journey:

- **Problem:** Challenges with monolith applications
- **Solution:** How microservices address the challenges
- **New Challenges:** Complexity introduced by microservices
- **Solution:** Containers and container orchestration

Key Outcome: Understand microservices and the need for containers and orchestration





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Google Cloud

Compute Managed Services



Google Cloud Managed Services for Compute

Service	Details	Category
Google Compute Engine	Create VMs and VM groups	IaaS
Google App Engine	Build and deploy highly scalable applications on a fully managed platform	PaaS (Containers, Serverless)
Google Cloud Run	Build and deploy highly scalable containerized applications. Does NOT need a cluster!	PaaS (Containers, Serverless)
Google Cloud Run Functions	Build event driven applications using simple, single-purpose functions	PaaS (Functions, Serverless)
Google Kubernetes Engine	Orchestrate containerized microservices on Kubernetes Needs advanced cluster configuration and monitoring	PaaS (Container Orchestration)



Scenarios - Google Cloud Compute Services

Scenario	Google Cloud
How do you create Virtual Machines?	Google Compute Engine
How do you create a group of similar VMs?	Managed Instance Groups
How do distribute load among VMs?	Cloud Load Balancing
What is the easiest way to run a single container based application in Google Cloud?	Google Cloud Run
How do you orchestrate containers in Google Cloud?	Google Kubernetes Engine (GKE)
How do you build serverless event driven functions in Google Cloud?	Cloud Run Functions



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Getting Started with Google App Engine



Getting Started with App Engine

Scenario: Deploy an app quickly (even without containerizing) to Google Cloud

- **Solution:** App Engine

Features:

- **Standard:** Deploy Go, Java, .NET, Node.js, PHP, Python, Ruby, .. apps using pre-configured runtimes
- **Flexible:** Deploy Apps built in any language
- **Deploy to Scaling:** Handles everything
- **Advanced Features:** Automatic LB, Auto scaling, App versioning, Traffic splitting, App health monitoring





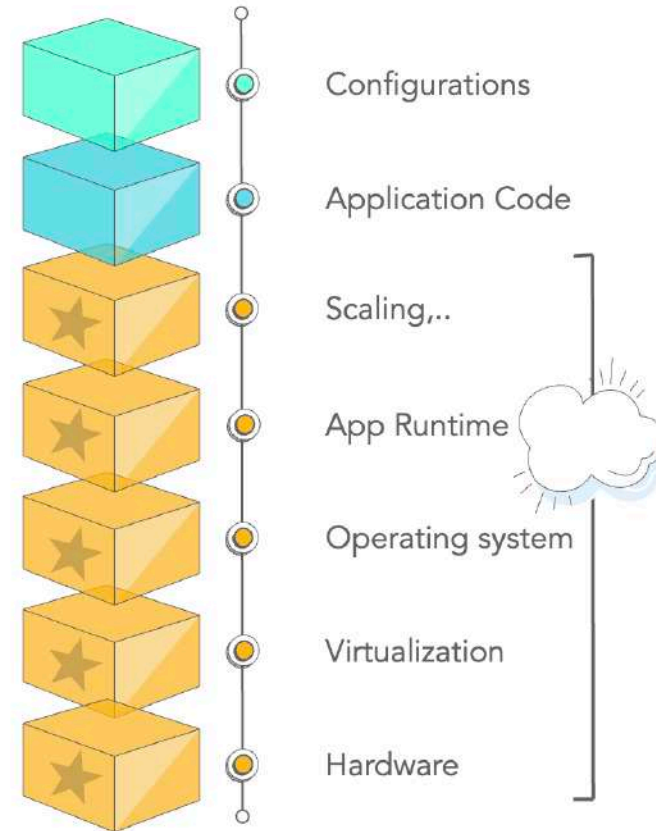
Compute Engine vs App Engine

Compute Engine: IaaS

- MORE Flexibility
- BUT MORE Responsibility
 - Choosing Image, Installing Software, Choosing Hardware
 - Fine grained Access/Permissions (Certificates/Firewalls)
 - Availability etc

App Engine: PaaS

- Serverless
- LESSER Responsibility
- LOWER Flexibility

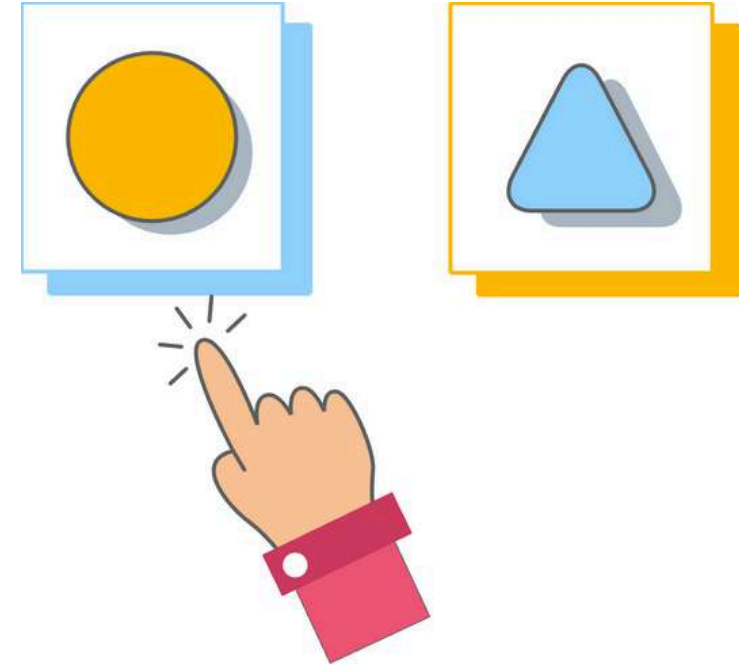




App Engine environments - Standard

Standard: Applications run in language specific sandboxes

- Supported Language Examples: Java, Python, PHP, Node.js, Ruby, Go
- Instance startup time: Seconds
- Complete isolation from OS/Disk/etc.
- Recommended if your want to:
 - Run your apps at very low cost (Pay only for what you need and when you need it. Example: scale to 0 instances when there is no traffic) OR
 - Your app experiences sudden and extreme spikes of traffic and you need immediate scaling





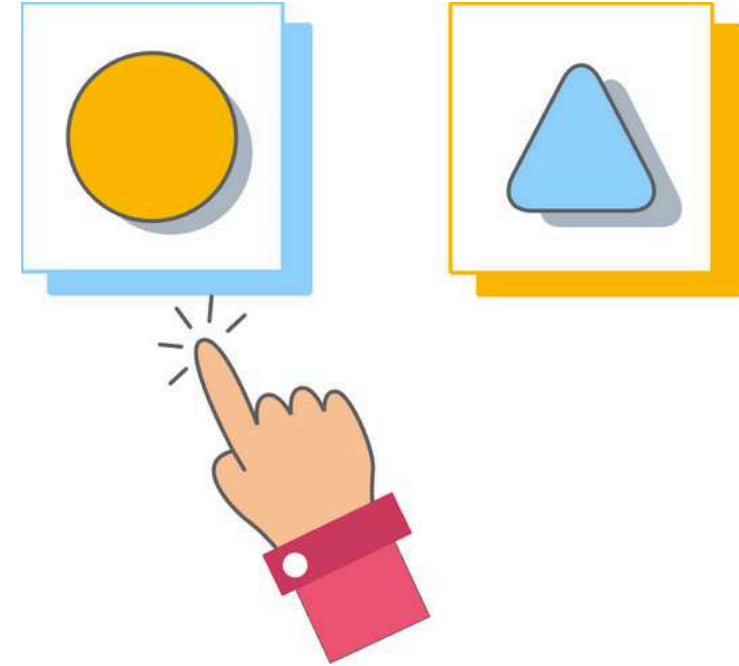
App Engine environments - Flexible

Flexible: Application instances run within containers

- **GCE VMs:** Makes use of GCE VMs
- **No Runtime Restrictions:** Support ANY runtime
- **Built-in support:** For Python, Java, Node.js, Go, Ruby, PHP, .NET
- **Get Access to:** SSH, Background processes and local disks

Recommended For Containerized Apps:

- Receiving consistent traffic (gradual scaling)
- Needs access to Background processes





App Engine environments - Standard vs Flexible

Feature	Standard	Flexible
Makes use of GCE VMs	No	Yes
Key Pricing Factors	Instance hours	vCPU, Memory & Persistent Disks
Scaling to zero	Yes	No. Minimum one instance
Instance startup	Seconds	Minutes
Rapid Scaling	Yes	No
SSH for debugging	No	Yes



App Engine - Application Component Hierarchy

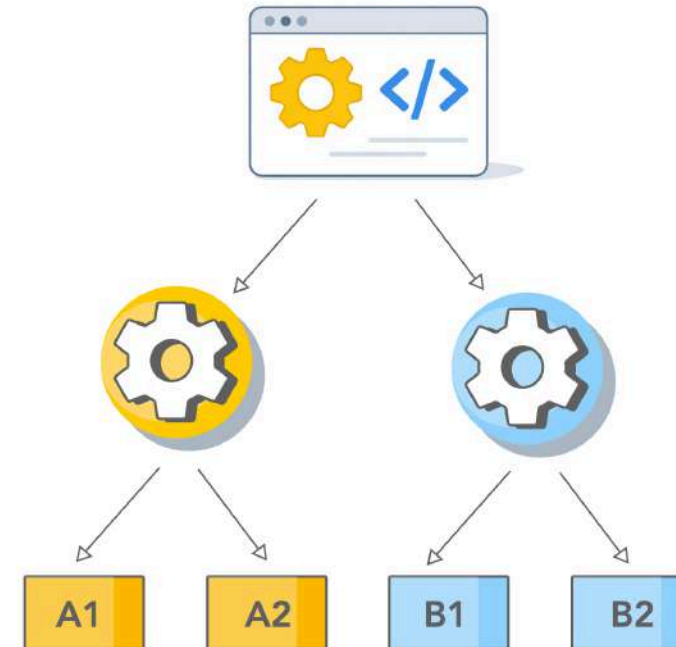
Application: One App per Project

Service(s): Multiple Microservices or App components

- You can have multiple services in a single application
- Each **Service** can have different settings
- Earlier called Modules

Version(s): Each version associated with code and configuration

- Each **Version** can run in one or more instances
- Multiple versions can co-exist
- Options to rollback and split traffic





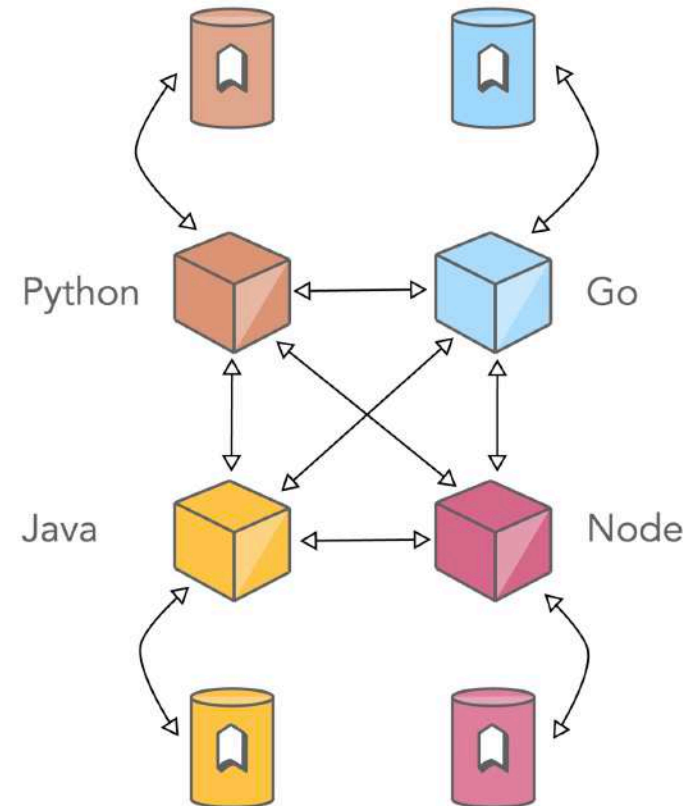
App Engine - Remember

Regional: Services deployed in one region across zones

- You **CANNOT** change an Application's region

Good option: For simple microservices (multiple services)

- Use **Standard** when you are using supported languages
- Use **Flexible** if you are building containerized apps
 - **ATLEAST one instance** is always running when using **Flexible**
 - **Go for Standard** if you want to scale down to **zero** instances when there is NO load





Getting Started with App Engine - Scenarios

Scenario	Solution
Which of these minimizes operations (App Engine or Compute Engine?)	App Engine
Maximum Number of Applications in a Project	One
You want rapid scaling. Should you go for App Engine Standard or Flexible?	Standard
Which of these supports scaling to zero - App Engine Standard or Flexible?	Standard
Explain the structure - You want to create multiple versions of a Microservice using App Engine	Create a service for each microservice. Each service can have multiple versions.



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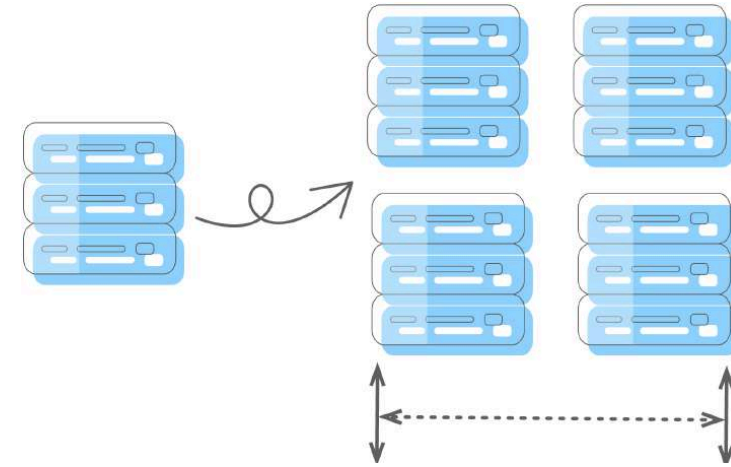
Getting Deeper with Google App Engine



App Engine - Options For Scaling Instances [1/2]

1: Automatic: Auto scale instances

- **Scaling:** Based on
 - **Target CPU Utilization:** Configure a CPU usage threshold
 - **Max Concurrent Requests:** Configure max concurrent requests an instance can receive
 - **Target Throughput Utilization:** Configure a threshold
- **Best Used For:** Continuously running workloads like simple websites, REST APIs, etc.
 - Resident Instances: Run continuously [Configure **Min Instances**]
 - Dynamic Instances: Added based on load [Configure **Max Instances**]
 - Use all dynamic instances if you are cost sensitive [Min=0]
 - If you are not very cost sensitive, keep a set of resident instances running always [Example: Min=5]

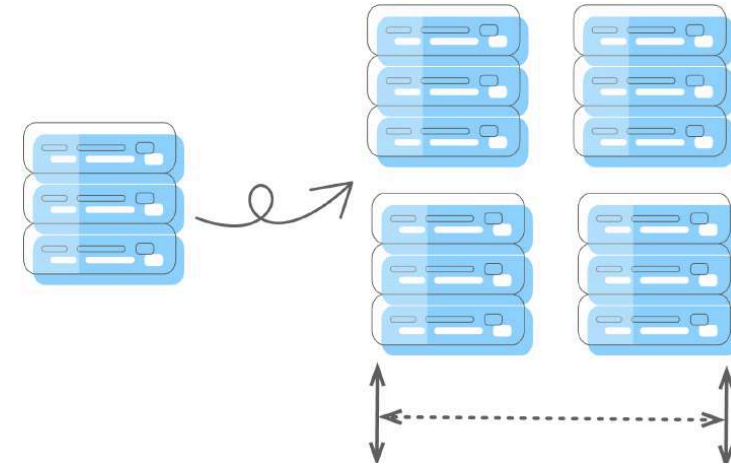




App Engine - Options For Scaling Instances [2/2]

2: Basic: Instances created on receiving requests

- Recommended for Adhoc Workloads
- Instances are shutdown if ZERO requests
 - Tries to keep costs low [High latency is possible]
- NOT supported by App Engine Flexible Environment
- **Best Used For:** Long-running background tasks, batch jobs, etc.



3: Manual: Configure specific number of instances to run

- Adjust number of instances manually over time



app.yaml Reference

runtime: pythonXY # python39 OR python49 OR .. The name of the runtime environment that is used by your app

instance_class: F1

service: service-name

#env: flex

inbound_services:

- warmup # Enables warmup requests.

App Engine sends a blank request to launch a resident instance before live user traffic hits it, minimizing cold starts

env_variables:

ENV_VARIABLE: "value"

automatic_scaling:

target_cpu_utilization: 0.65

min_instances: 5

max_instances: 100

max_concurrent_requests: 50

#basic_scaling:

#max_instances: 11

#idle_timeout: 10m #amount of time an instance will remain running after it processes its last request before its shut down

#manual_scaling:

#instances: 5



AppEngine - Request Routing

Combination of three approaches

- Routing with **URLs**:
 - `https://PROJECT_ID.REGION_ID.r.appspot.com` (default service called)
 - `https://SERVICE-dot-PROJECT_ID.REGION_ID.r.appspot.com` (specific service)
 - `https://VERSION-dot-SERVICE-dot-PROJECT_ID.REGION_ID.r.appspot.com` (specific version of service)
 - Replace -dot- with . if using custom domain
- Routing with a **dispatch file**:
 - Configure `dispatch.yaml` with routes
 - `gcloud app deploy dispatch.yaml`
- Routing with **Cloud Load Balancing**:
 - Configure routes on Load Balancing instance

dispatch:

1. Route a subdomain to a mobile service

- url: "://example.com*"

service: mobile-api

2. Route specific paths to customer service

- url: "*/customers/*"

service: customer-service

3. Route to billing service

- url: "*/checkout/*"

service: billing-service

4. Route static assets

- url: "*/static/*"

service: static-file-handler

5. Catch-all: Route all other remaining traffic to the main service

- url: "*/*"

service: default



AppEngine - Deploying new versions without downtime [1/2]

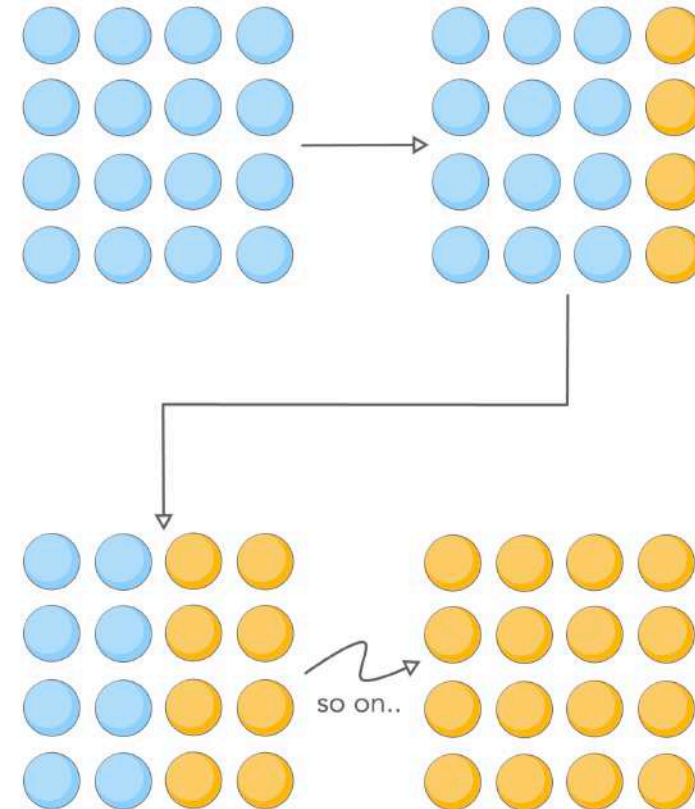
Scenario: How do I go from V1 to V2 without downtime?

Option 1: I'm very confident - Deploy & shift all traffic at once:

- Deploy and shift all traffic at once from v1 to v2:
gcloud app deploy

Option 2: I want to manage the migration from v1 to v2

- **STEP 1:** Deploy v2 without shifting traffic (`--no-promote`)
 - *gcloud app deploy --no-promote*

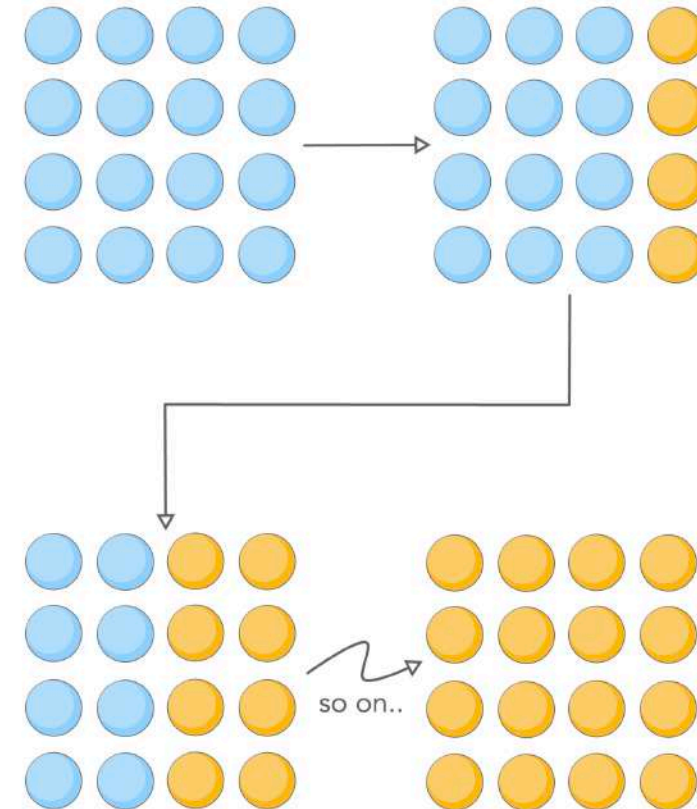




AppEngine - Deploying new versions without downtime [2/2]

Option 2 (Continued): I want to manage the migration from v1 to v2

- **STEP 2: Shift traffic to V2**
 - **Option 1 (All at once Migration):** Migrate all at once to v2
 - `gcloud app services set-traffic s1 --splits V2=1`
 - **Option 2 (Gradual Migration):** Gradually shift traffic to v2. Add `-migrate` option.
 - Gradual migration is not supported by App Engine Flexible Environment
 - **Option 3 (Splitting):** Control the pace of migration
 - `gcloud app services set-traffic s1 --splits=v2=.5,v1=.5`
 - Useful to perform A/B testing
 - Ensure that new instances are warmed up before they receive traffic (app.yaml - `inbound_services > warmup`)





App Engine - Scenarios [1/2]

Scenario	Solution
I want to create two Google App Engine Apps in the same project	Not possible. You can only have one App Engine App per project. However you can have multiple services and multiple versions for each service.
I want to create two Google App Engine Services inside the same App	Yup. You can create multiple services under the same app. Each service can have multiple versions as well.
I want to move my Google App Engine App to a different region	App Engine App is region specific. You CANNOT move it to different region. Create a new project and create new app engine app in the new region.
I don't want to allow more that 10 instances for an App Engine Service	Set max_instances in app.yaml. Example: <code>automatic_scaling > max_instances: 10</code>



App Engine - Scenarios [2/2]

Scenario	Solution
I want a minimum of 5 instances always running for an App Engine Service	Set min_instances in app.yaml. Example: <code>automatic_scaling > min_instances: 5</code>
Deploy new version to App Engine Service without shifting traffic	<code>gcloud app deploy --no-promote</code>
How can I split traffic between two versions of a Microservice	<code>gcloud app services set-traffic s1 --splits v1=0.9,v2=0.1</code>
Perform Canary deployments	Deploy v2 without shifting traffic (<code>gcloud app deploy -no-promote</code>) Shift some traffic to V2 (<code>gcloud app services set-traffic s1 --splits v1=0.9,v2=0.1</code>)



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Getting Started with Cloud Run



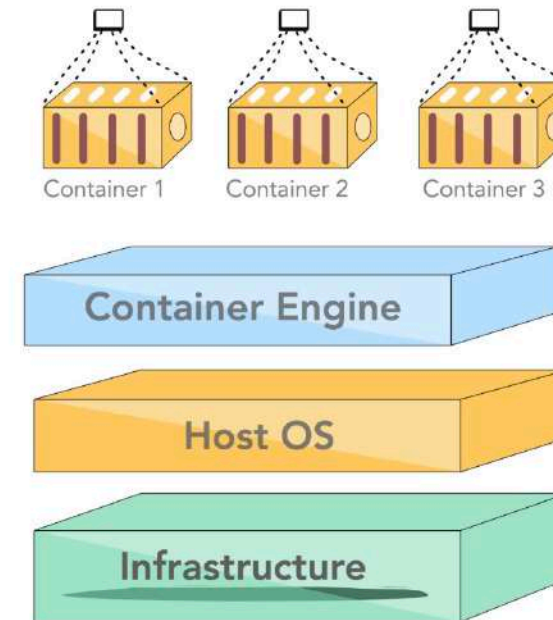
Getting Started with Cloud Run

Cloud Run: Deploy Containers in Seconds

- Built on top of an open standard - **Knative**
- **Fully managed** serverless platform
 - ZERO infrastructure management
 - Pay-per-use (Key cost factors: Used CPU, Memory and Request count)

Fully integrated **end-to-end developer experience**:

- **No limitations**: in languages, binaries and dependencies
- **Simple Integrations**: Cloud Build, Cloud Monitoring, Cloud Logging etc.

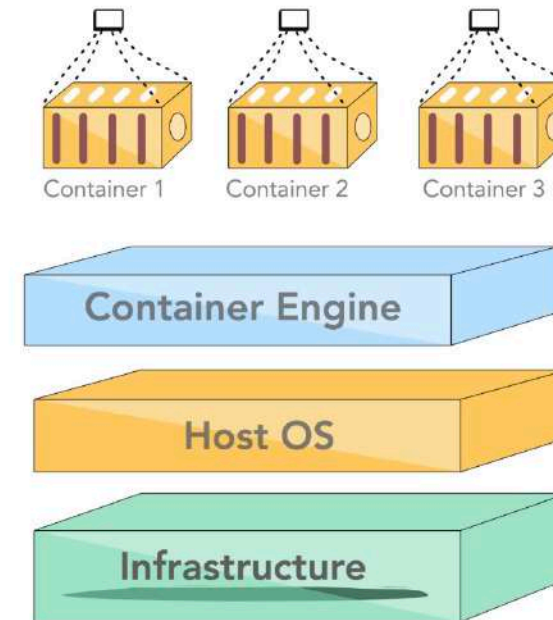




App Engine vs Cloud Run

Newer and More Flexible: Cloud Run is the newer, more flexible alternative

- **Built on Experience:** Cloud Run builds upon over a decade of operational learnings from App Engine
- **Deployment Options:** Both support deploying from source code or container images
- **Scaling and Cost:** Both scale to zero (except App Engine Flexible). However, Cloud Run offers faster startup times and superior cost efficiency for spiky traffic.
- **Example Features in Cloud Run But NOT in App Engine:** Multi Region Support, Committed Use Discounts (CUDs)





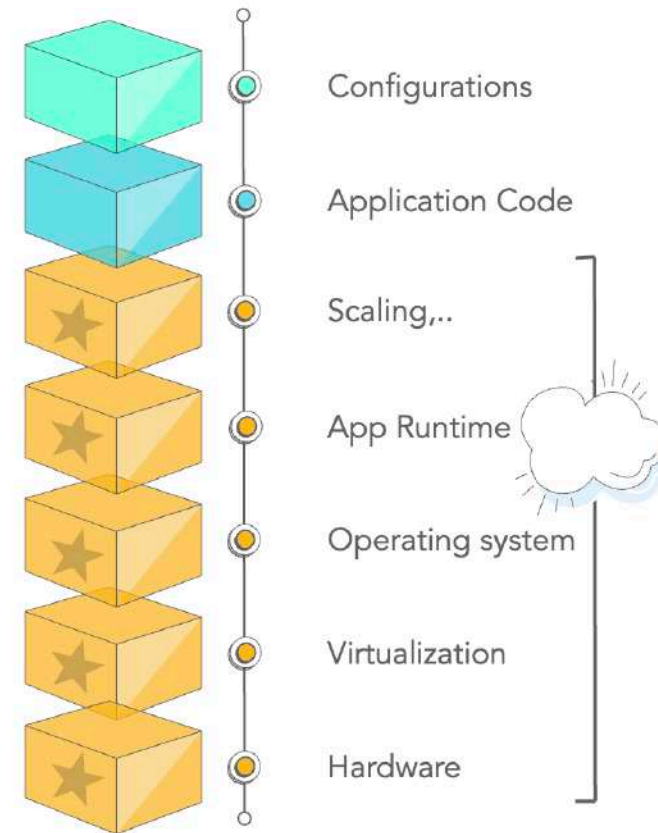
Compute Engine vs Cloud Run

Compute Engine: IaaS

- MORE Flexibility
- BUT MORE Responsibility
 - Choosing Image, Installing Software, Choosing Hardware
 - Fine grained Access/Permissions (Certificates/Firewalls)
 - Availability etc

Cloud Run: PaaS (Serverless Container Platform)

- Serverless
- LOWER Flexibility
- LESSER Responsibility





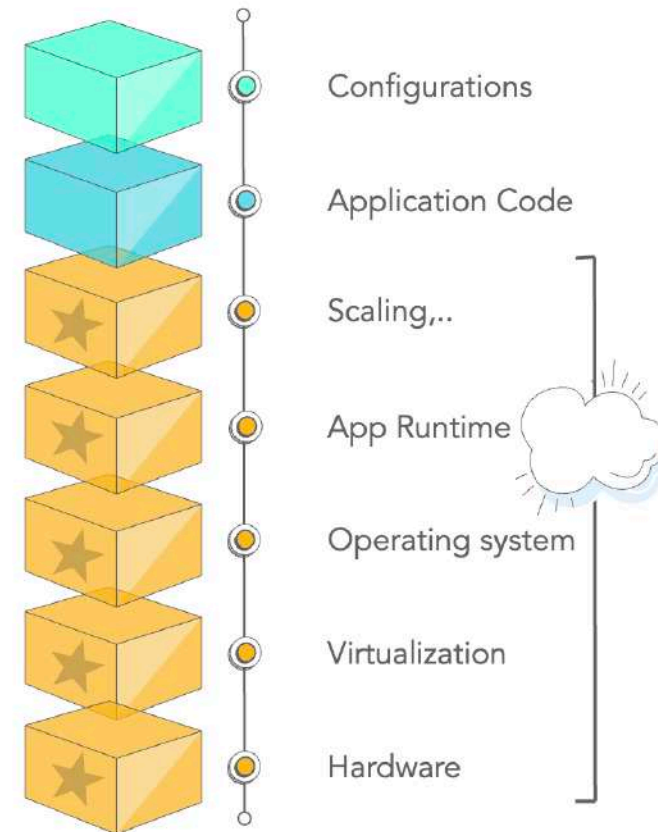
Advantages of Cloud Run

Language neutral: Applications can be built using any programming language or framework

Minimal ops: No infra, OS, or runtime management

Automatic scaling: Handle variable traffic efficiently (Scale to zero)

Cost efficient: Pay only for actual usage





Getting Started with Cloud Run - Scenarios

Scenario	Solution
Which of these is the newer, more flexible serverless option - App Engine or Cloud Run?	Cloud Run
Hosting containerized workloads with highly intermittent or spiky traffic at absolute minimum cost	Cloud Run
Migrating existing containerized application to Google Cloud with minimal effort	Cloud Run
Deploying a Knative-compatible application with zero infrastructure management	Cloud Run
Deploying a simple Python application with low or unpredictable traffic cost-effectively without managing runtimes or web servers	Cloud Run



Typical Steps in Using Cloud Run

Containerize: Package a container image

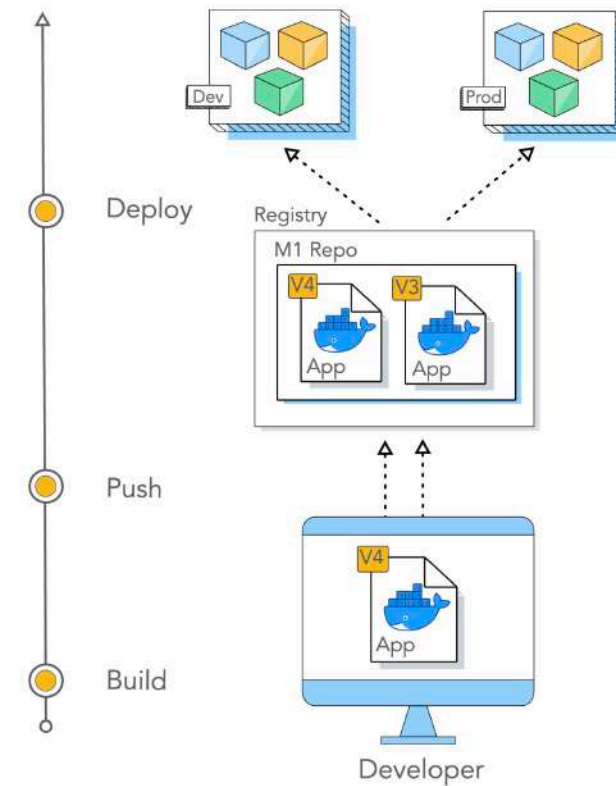
- OR let Cloud Run build a image from source code using a Dockerfile or buildpacks

Push: Push the container image to Artifact Registry

Deploy: Run your container on Cloud Run

Automatic scaling: Scales based on incoming requests

Release management: Deploy versions and split traffic





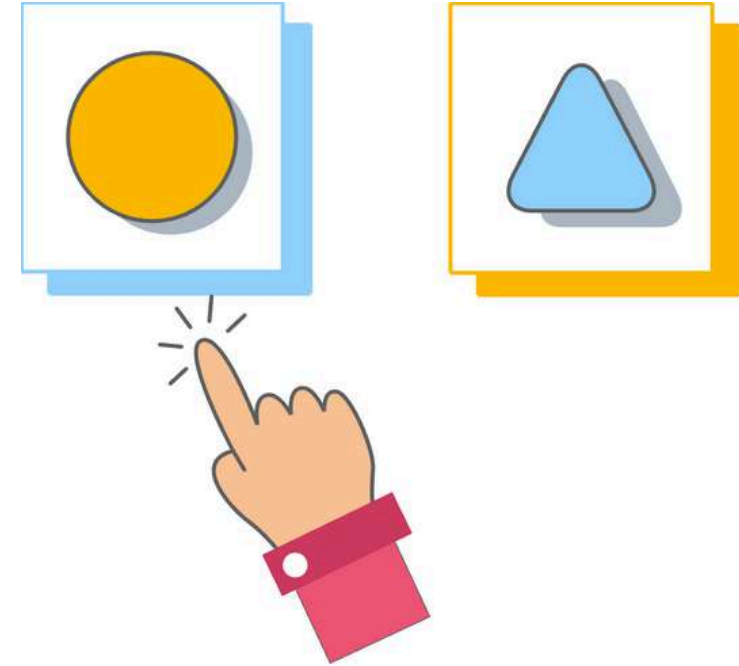
Cloud Run - Multiple Options

Web apps & APIs - Cloud Run Services

Handle Event Triggers (From Pub/Sub, Storage) - Cloud Run Functions

Run a batch job (Run-to-completion) - Cloud Run Jobs

Perform continuous background work - Cloud Run Worker Pools





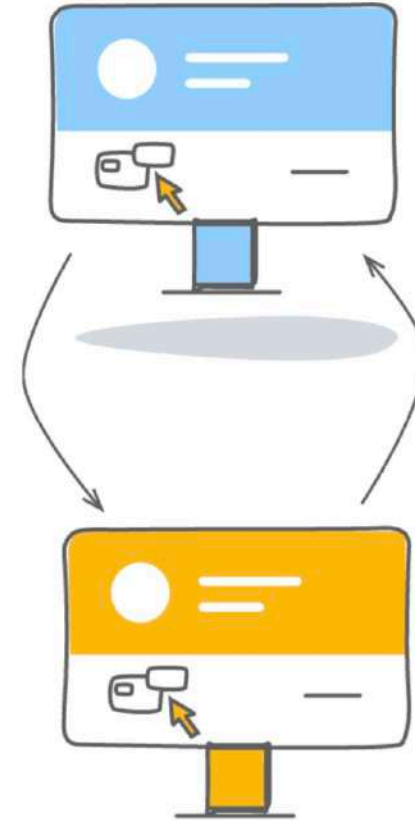
Cloud Run Services

Use for web apps & APIs: Handle HTTP/2, WebSockets, gRPC, etc.

Auto scaling: Automatically scales instances based on load (Scale to zero when there are no requests)

Traffic management: Seamlessly deploy new revisions, roll back instantly, or split traffic

Target Workloads: Ideal for e-commerce backends, REST APIs, and simple microservices





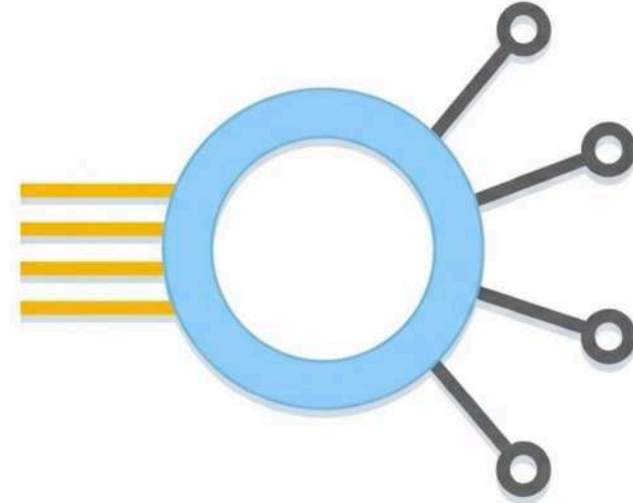
Cloud Run Functions

Lightweight, Single-Purpose Logic: Deploy small, focused functions

Event-Driven Architecture: Automatically triggers in response to cloud events

Source-to-Deploy Workflow: Deploy code directly; Google handles the underlying work

Example Workloads: Generating image thumbnails or sending an email when a file is uploaded to Cloud Storage



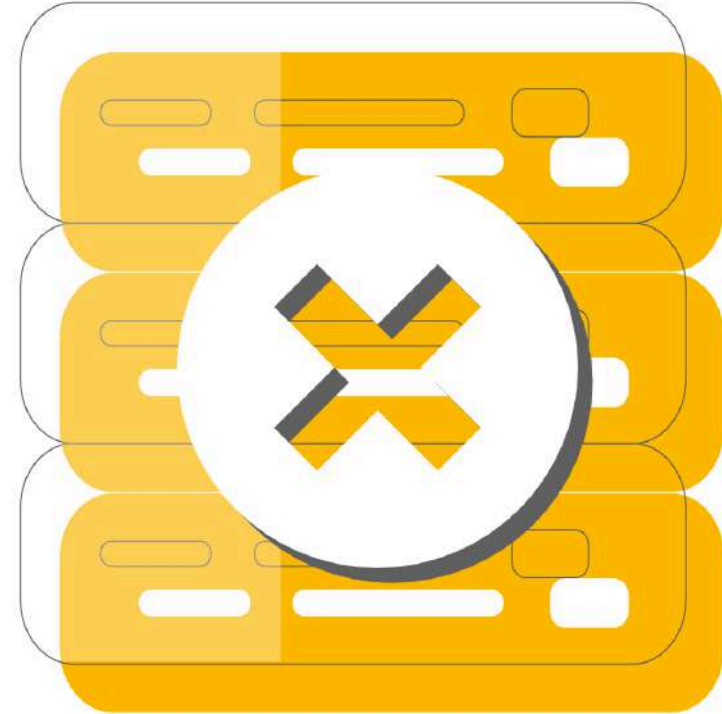


Cloud Run Jobs

Run-to-Completion Tasks: Built specifically for workloads that execute a finite set of steps and then cleanly shut down

Time-Based Scheduling: Seamlessly integrates with Cloud Scheduler to run routine tasks automatically on a schedule

Example Workloads: Nightly database backups, end-of-month financial reporting, or database migrations





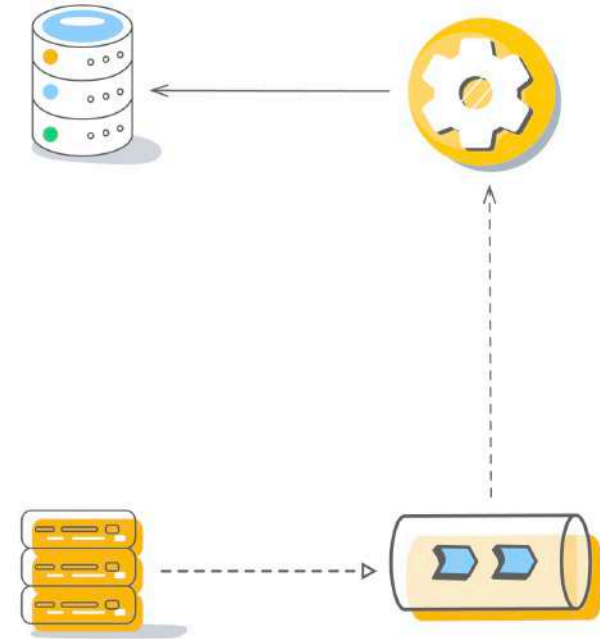
Cloud Run Worker Pools

Continuous Background Workers: Built for non-request, pull-based workloads with no public HTTP endpoints

- **Streamlined Queue Consumption:** Ideal for architectures that pull tasks directly from message brokers (e.g., Pub/Sub Pull, Apache Kafka).

Optimized Costs: Eliminates the overhead of public endpoints and load balancers

Example Workloads: Process messages continuously from Pub/Sub, Kafka, or other message queues





Cloud Run - Services vs Jobs vs Worker Pools vs Functions

Scenario	Recommendation
Build a web application, REST API, or handle streaming connections (WebSockets/gRPC)	Cloud Run Service
Run continuous background workers consuming messages from Pub/Sub or Kafka	Cloud Run Worker Pools
Execute a run-to-completion task (batch data processing, database migration)	Cloud Run Job
Run a scheduled task like a daily cleanup script or report generation	Cloud Run Job <i>(via Cloud Scheduler)</i>
Deploy simple, event-driven lightweight logic without managing containers	Cloud Run Function



Cloud Run - Component Hierarchy

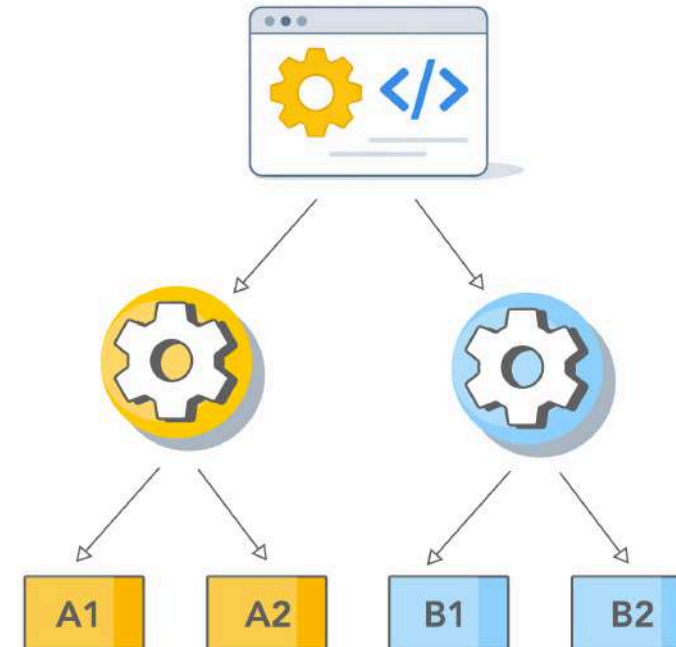
Project: The Google Cloud boundary wrapping your resources

Service: The top-level Cloud Run resource representing a microservice or application

Revision: Represent a Service Version

- Multiple revisions can co-exist simultaneously
- Allows for flexible release management and granular traffic splitting

Instance: The underlying container runtime environment created to handle requests for a specific Revision





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Getting Deeper with

Google Cloud Run

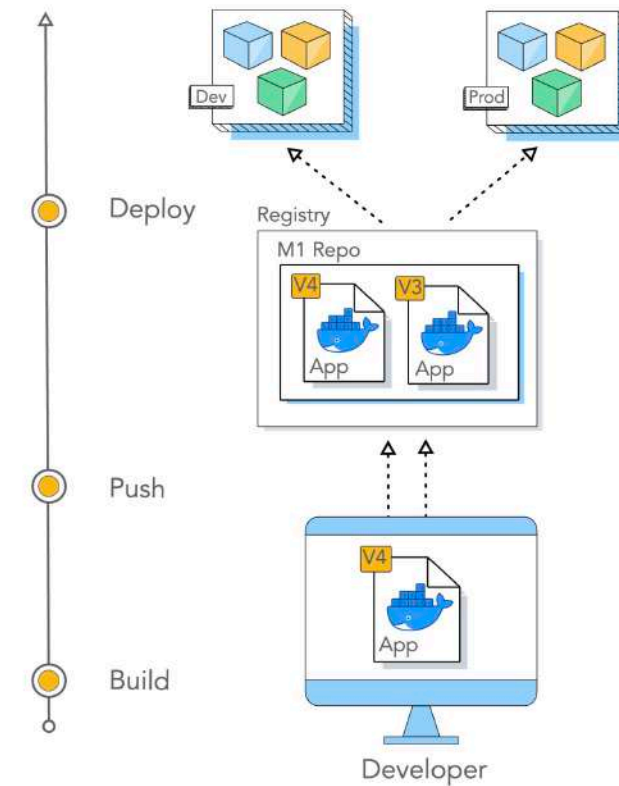


Cloud Run - Deployment Options

Option 1: Deploy a Pre-Created Container Image

Option 2: Deploy source → Cloud Run builds Container Image and Deploys it

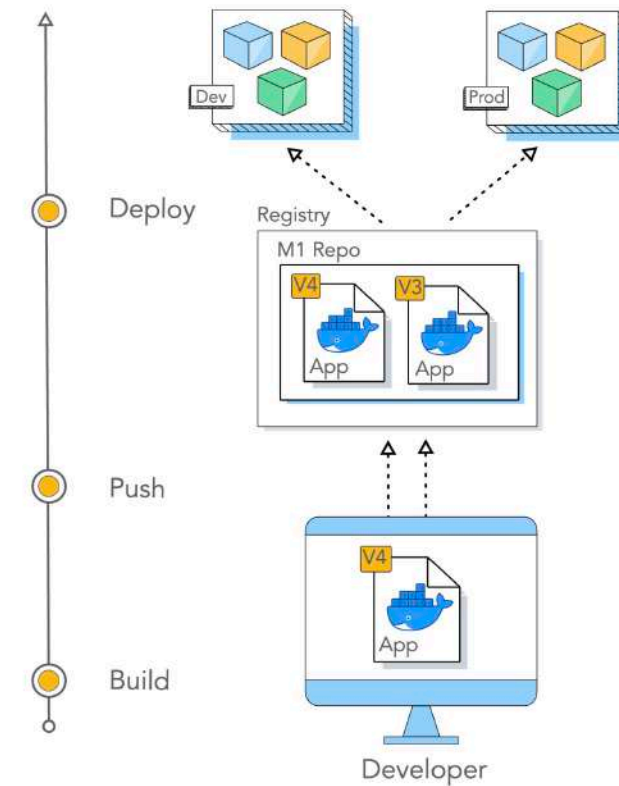
Option 3: Deploy source → Cloud Run deploys app (using a base container image) without a build step



Cloud Run Deployment Option 1 - Pre-Created Container Image



- 1: Build your app using your favorite language/framework
- 2: Create a container image for your app
- 3: Push it to Artifact registry
- 4: Deploy container to Cloud Run
 - Example: `gcloud run deploy SERVICE --image IMAGE_URL` or use cloud console



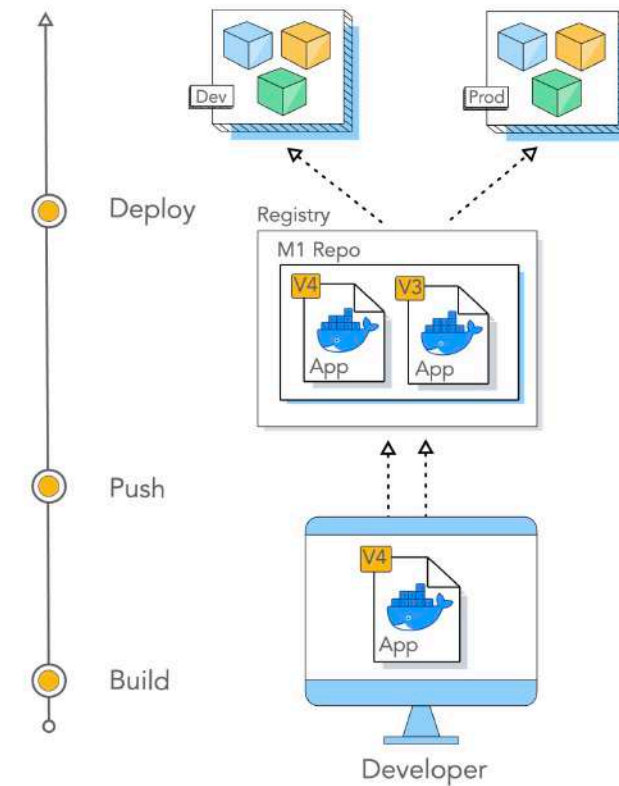


Cloud Run Deployment Option 2 - Deploy Source with Build

1: Build your app using your favorite language/framework

2: Deploy using `--source` option: `gcloud run deploy SERVICE --source` .

- If you are using a supported language (Python, Node.js, Java, etc.), buildpacks are used to build your container image
 - OR You can provide a `Dockerfile`. Container image is built using your `Dockerfile`.
- Uses Cloud Build in the background (No need to install container software on your machine)

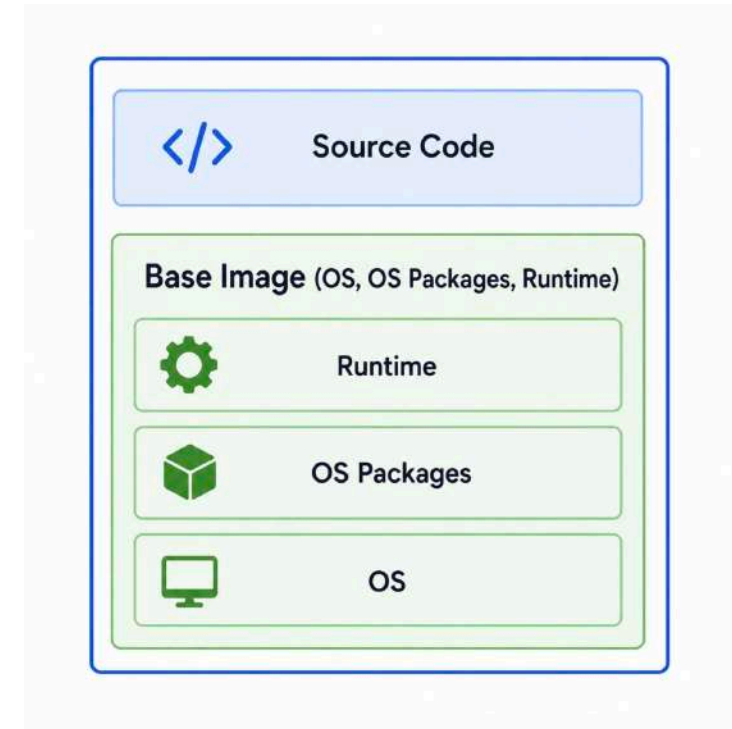




Cloud Run Deployment Option 3 - Deploy Source without Build

Option 3: Deploy source → Cloud Run deploys app without build step

- Run deploy with `--source` and `--no-build`: `gcloud run deploy SERVICE --source . --no-build --base-image=python:3.11 --command=python --args=app.py`
- Deploy your source code (including pre-packaged artifacts) directly
- Faster deployment as container build step is skipped
- **In the Background:**
 - Source folder (`--source .`) uploaded to Cloud Storage
 - Source code folder directly mounted on top of base image





Cloud Run Deployment Options - Scenarios

Scenario	Recommended Option
You already have container images built by your CI/CD pipeline	Deploy Pre-Created Container Image (<code>--image</code>)
You have a Dockerfile and want Cloud Run to use it during deployment	Deploy Source with Build (<code>--source .</code>)
You are building a Python application and want Cloud Run to build and deploy the container for you - without need a Dockerfile	Deploy Source with Build (<code>--source .</code>)
You have a JAR, Python script, or pre-packaged artifact ready to run and you do not want to build a new image for each revision	Deploy Source without Build (<code>--source . --no-build</code>)
You need full control over container image creation	Deploy Pre-Created Container Image (<code>--image</code>)

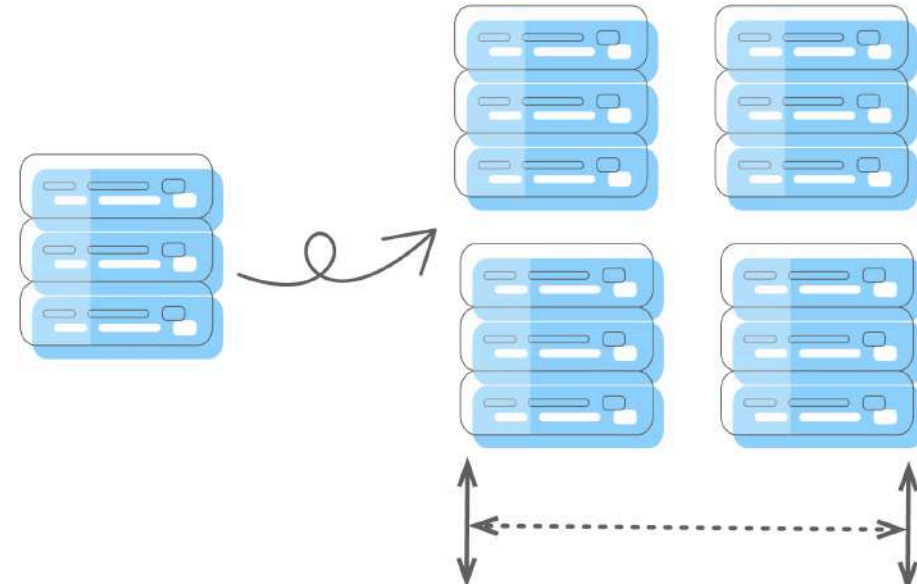


Exploring Cloud Run Scaling [1/2]

Automatic Scaling: Cloud Run adds instances as traffic increases

- **Scale In:** Removes instances when traffic decreases
- **Scale To Zero:** Revisions with no traffic are scaled down to zero instances
- **Minimum Instances:** Keep instances warm and reduce cold starts
- **Maximum Instances:** Prevent excessive scaling

Manual Scaling: Useful when you need a fixed number of instances





Exploring Cloud Run Scaling [2/2]

Description	Command
Configure minimum and maximum instances	<code>gcloud run deploy SERVICE --image IMAGE_URL --min MIN_VALUE --max MAX_VALUE</code>
Configure manual scaling with a fixed number of instances	<code>gcloud run deploy SERVICE --scaling=INSTANCE_COUNT --image IMAGE_URL</code>
Disable a service	<code>gcloud run services update SERVICE --scaling=0</code>



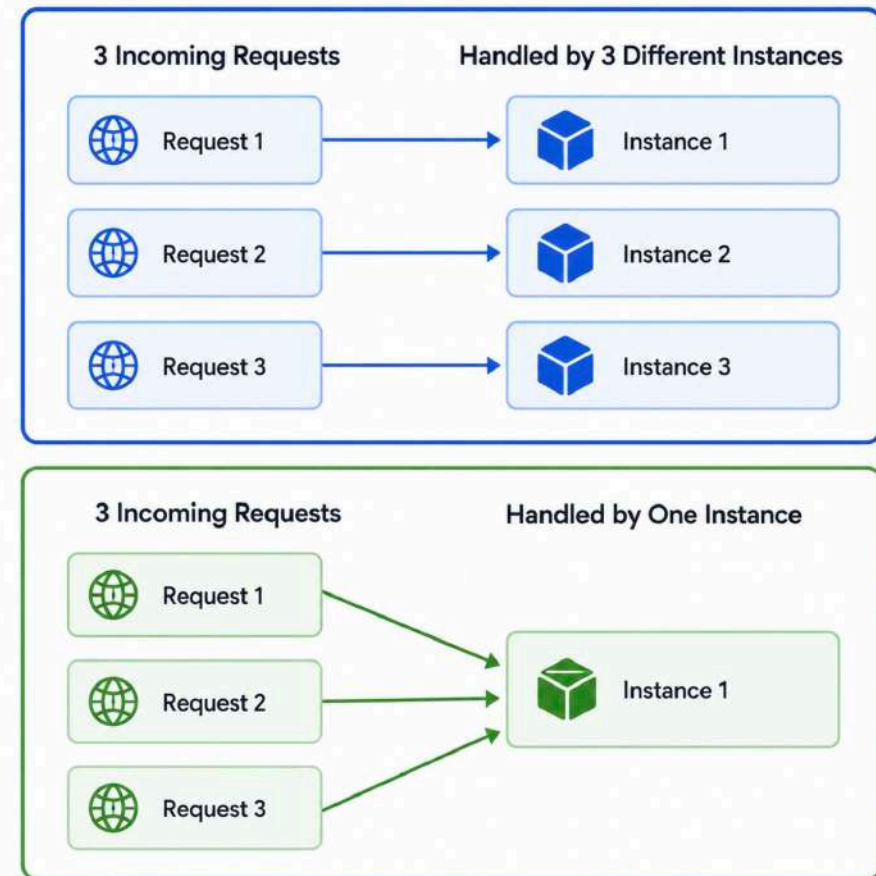
Why Does Concurrency Matter?

Scenario: 800 requests arrive at the same time

- **Concurrency = 1:** Cloud Run needs about 800 instances
- **Concurrency = 80:** Cloud Run needs about 10 instances

Higher Concurrency: Fewer instances needed

Benefit: Lower costs and faster scaling





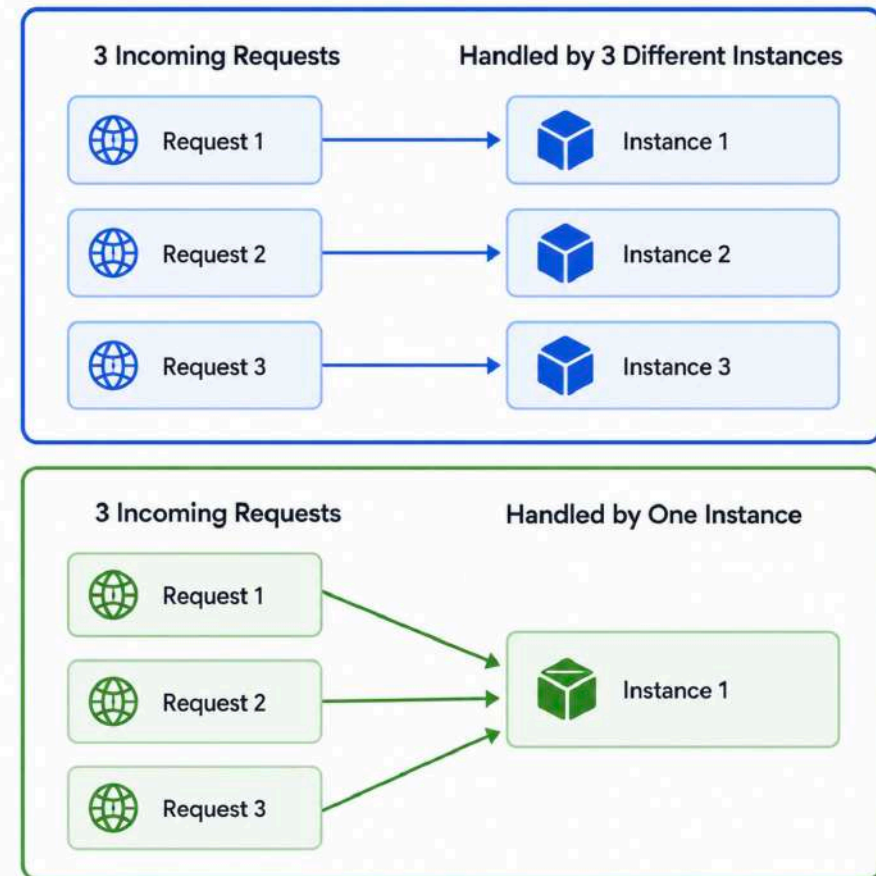
Configuring Concurrency For a Cloud Run Service

Cloud Run allows you to configure **maximum concurrent requests per instance**

- If your code cannot process parallel requests, set concurrency to **1**
- Default concurrency is **80 requests per instance**

Configuring concurrency:

- `gcloud run deploy SERVICE --concurrency CONCURRENCY`
- `gcloud run services update SERVICE --concurrency CONCURRENCY`





Exploring Pricing Options for Cloud Run

Request-based billing(default): Charge per request + Per-second rate for vCPU & memory consumed during request processing

- **Ideal For:** Event-driven or highly variable traffic

Instance-based billing: Charged for the entire lifetime of an instance (no per-request fee)

- **Recommended For:** Stable traffic and continuously available applications
- **Required When:** Attaching specialized hardware (like GPUs), maintaining active minimum instances, or running asynchronous background threads





Exploring Cloud Run - Scenarios

Scenario	Solution
For services with steady traffic - instance-based or request-based billing	Consider using instance-based billing
For services with sporadic, or spiky traffic - instance-based or request-based billing	Consider using request-based billing
Avoid cold starts and reduce latency	Set a minimum number of instances
Prevent unbounded autoscaling	Configure maximum instances
Run a fixed number of instances always (without autoscaling)	Manual scaling - configure a specific number of instances



Serving Traffic From Multiple Regions with Cloud Run

Global Users: Users may access your service from any part of the world

Multi Region Deployment: Deploy in multiple regions for low latency (*gcloud run deploy SERVICE_NAME --image=IMAGE_URL --regions=REGIONS*)

- **High Availability:** If one region fails, traffic automatically shifts to another
- **Single IP Access:** Use a Global HTTPS Load Balancer with a single anycast IP
- **Routing Logic:** Serverless NEGs map Cloud Run revisions to backend targets





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Continuing Your Journey With Containers



Creating Images with Dockerfile (Python)

Dockerfile: A text file with image build instructions

- **FROM:** Lightweight Python base image
- **WORKDIR:** Set working directory inside the container
- **COPY:** Copy application files into the image
- **RUN:** Install Python dependencies
- **EXPOSE:** Document the application port
- **CMD:** Command to start the application

Build It: `docker build -t my-python-app:1.0.0 .`

```
FROM python:{version}-alpine
WORKDIR /app
COPY . /app
RUN pip install -r requirements.txt
EXPOSE 5000
CMD python ./launch.py
```



Creating Images with Dockerfile (Node.js)

Dockerfile: A text file with image build instructions

- **FROM:** Lightweight Node.js base image
- **WORKDIR:** Set working directory inside the container
- **COPY:** Copy application files into the image
- **RUN:** Install Node.js dependencies
- **EXPOSE:** Document the application port
- **CMD:** Command to start the application

Build It: `docker build -t my-node-app:1.0.0 .`

```
FROM node:{version}-alpine
WORKDIR /app
COPY . /app
RUN npm ci
EXPOSE 5000
CMD node index.js
```



Understand How Container Images Are Built As Layers

Base Image Layer: Minimal OS + runtime

- Example: `FROM node:{version}-alpine`

Instruction Layers: Each Dockerfile instruction adds a layer

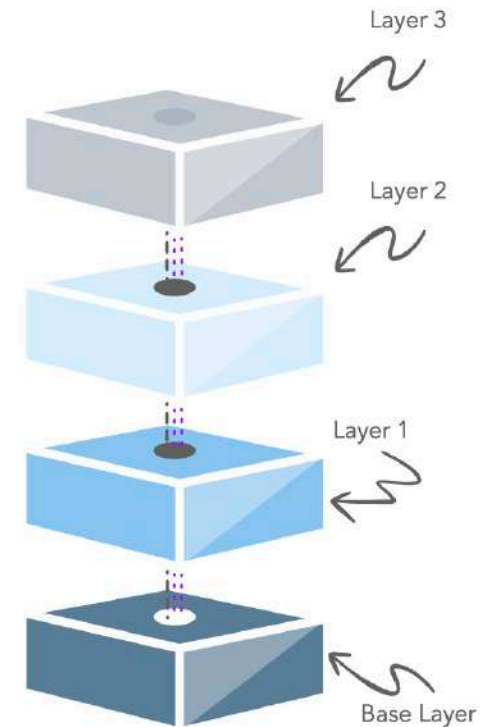
- Example: `WORKDIR /app`, `EXPOSE 5000` etc.

App Layer: Source code copied as a layer

- Example: `COPY . /app`

Dependency Layer: Installed libraries cached as a reusable layer

- Example: `RUN npm ci`





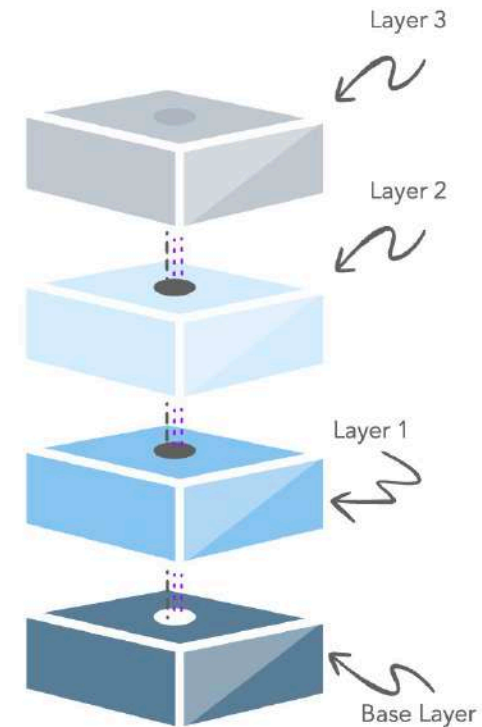
Understand How To Reuse Container Layers

Layer Caching: Unchanged layers from previous builds are reused

- **Faster Builds:** Only layers on top of the first modified layer are rebuilt
- **Efficient Storage:** Shared layers reduce disk usage

Best Practices:

- **Order Matters:** Place rarely changing instructions first
- **Dependency First:** Copy dependency descriptors and install dependencies before copying source code





Understand How To Reuse Container Layers - Node.js

Best Practice: Always copy dependency files before application code

- **Dependency Layer First:** package.json and lock file copied before source code

Stable Layers: Base image and dependencies reused across builds (if dependencies do not change)

Faster Rebuild: npm install runs only when dependencies change

```
FROM node:{version}-alpine
WORKDIR /app
# COPY . /app
# RUN npm ci

# 1: Copy dependency files first (better caching)
COPY package.json package-lock.json ./
RUN npm ci

# 2: Copy source code last
COPY . .

EXPOSE 5000
CMD ["node", "index.js"]
```



Best Practices with Docker

Best Practice 1: Keep It Light

- **Alpine Image:** Smaller base image
- **.dockerignore:** Do not copy temp files
- **Layer Caching:** Optimize Dockerfile

Best Practice 2: Semantic Versioning

- MAJOR.MINOR.PATCH (e.g., 1.2.5)
 - MAJOR (1.x.x): Breaking changes
 - MINOR (x.2.x): New features (Compatible)
 - PATCH (x.x.5): Bug fixes only

Best Practice 3: Pin Base Image Versions

- **Avoid latest:** Prevent unexpected changes

```
FROM node:{version}-alpine
WORKDIR /app

# 1: Copy dependency files first (better
#      caching)
COPY package.json package-lock.json ./
RUN npm ci

# 2: Copy source code last
COPY . .

EXPOSE 5000
CMD ["node", "index.js"]
```



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Getting Started with Kubernetes



Challenges With Microservices Architectures

Large No of Services: Need for automation

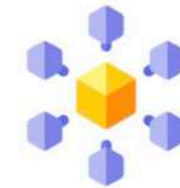
Auto Scaling: Must scale automatically

Load Balancing: Distribute traffic between instances

Release Management: Flexible releases

Resiliency: Automatically recover from failures

And a lot's more: Service Discovery, Centralized Configuration Management, etc.



MICROSERVICE A



MICROSERVICE B



MICROSERVICE C



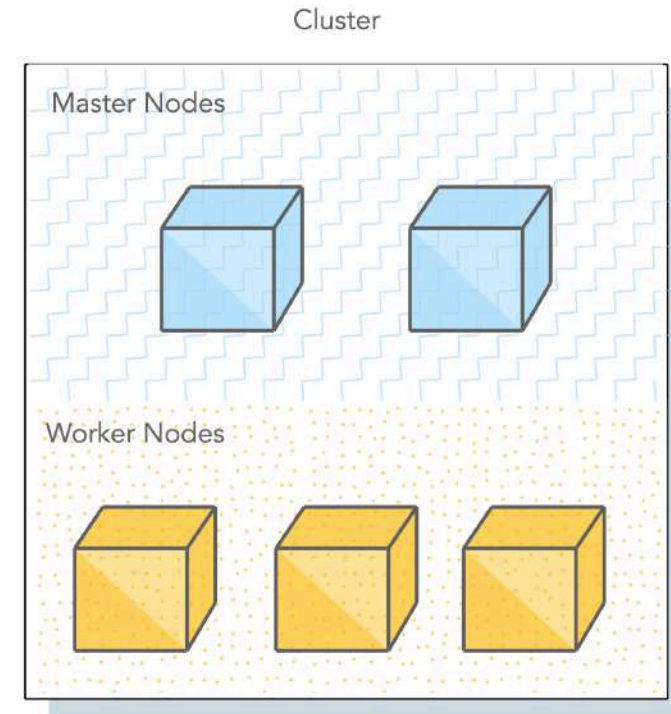
What is Kubernetes?

Kubernetes: An open-source container orchestration platform

- **Goal:** Automate deployment, scaling, and management of containerized apps
- **Origin:** Created at Google, now managed by CNCF

How It Works (Simplified):

- Create a Cluster
- Declare your needs: I need 2 instances of **Microservice A**, 5 instances of **Microservice B**, etc.
- Kubernetes will do its magic:
 - Deploys the containers
 - Monitors their health





Understanding Kubernetes Architecture

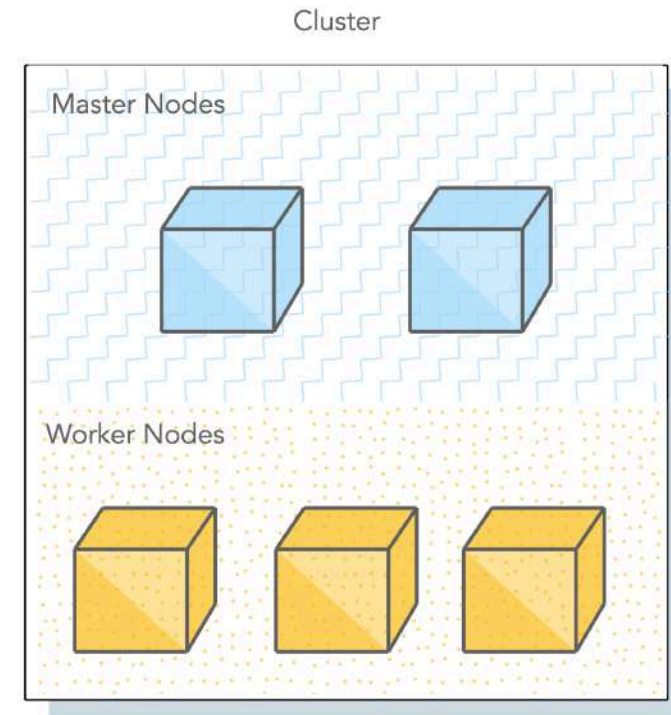
Control Plane (Master Nodes): Manages the cluster

Worker Nodes: Runs your applications

Declare your needs: I need 2 instances of **Microservice A** (and ..)

- Control plane schedules workloads
- Monitors and ensures high availability

Recommendation: Multiple Master Nodes For High Availability





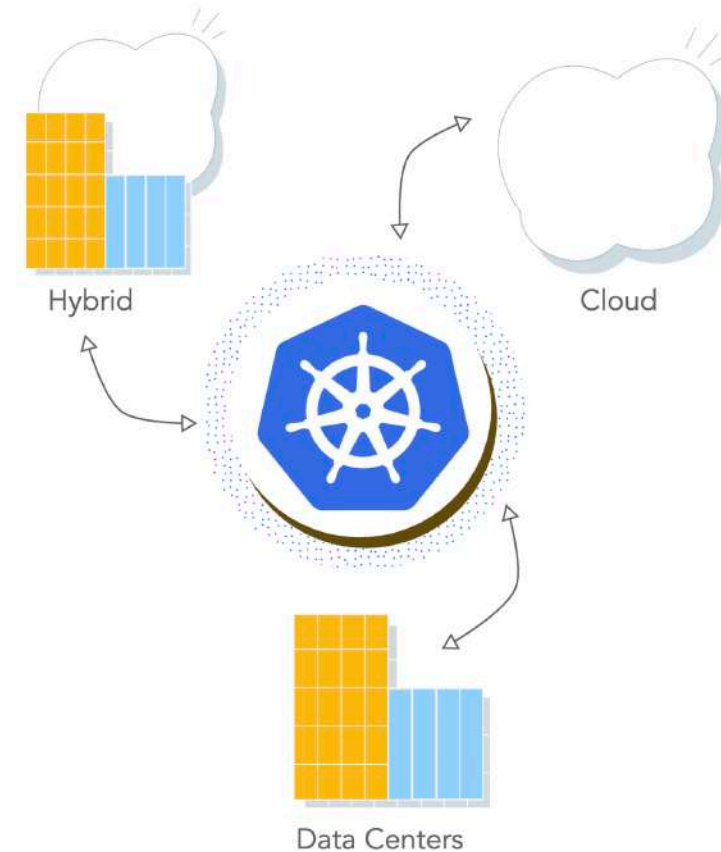
Kubernetes is Cloud and Platform Agnostic

Run Anywhere: On-premises, Public Cloud, Hybrid, or Edge

- **On-premises:** Use OpenShift, Rancher, etc.
- **Cloud:** AWS (EKS), Azure (AKS), Google Cloud (GKE)
- **Hybrid Cloud:** Combine on-prem and cloud clusters

Provides: Portability and scalability

- **Avoid Vendor Lock-in:** Works across providers
- **Language Agnostic:** Works with any language or framework
- **Run any workload:** Microservices, batch jobs, ML, analytics, etc.





Why Kubernetes Is Called Declarative

Desired State: You tell what you want

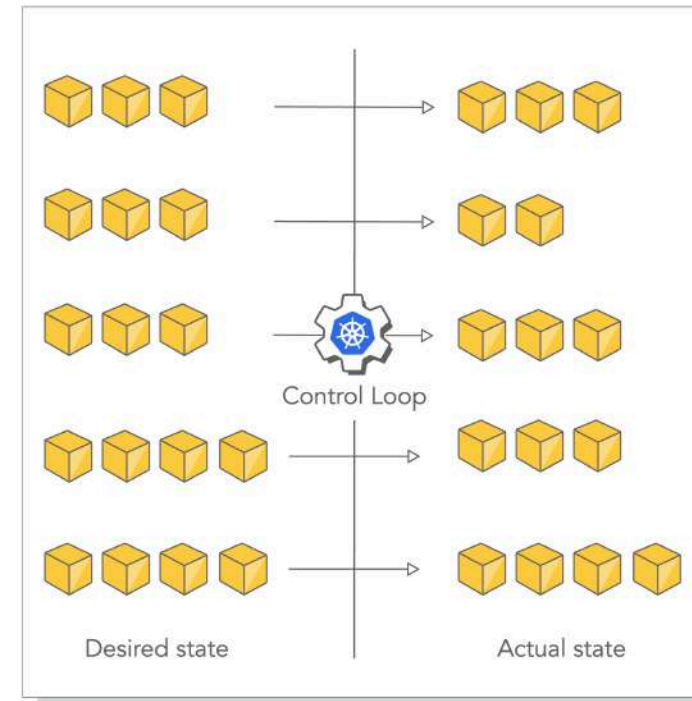
- Example: “Run 3 instances of Microservice A”

Actual State: Current state of the cluster

- Example: “2 instances of Microservice A are running in the cluster”

Control Loop: Reconciliation Loop

- **Continuous Comparison:** Is actual state $\neq$ desired state?
- **If there is a difference:** Automatically takes action to fix it





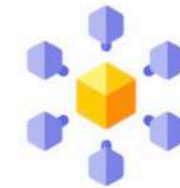
Summary - Getting Started with Kubernetes

Large Number of Services: Strong need for automation in microservices architectures

Architecture: Master Nodes (Control Plane) + Worker Nodes

Declarative: Define desired state, Kubernetes makes it happen (and maintains the desired state)

Flexible: Runs anything, anywhere with zero vendor lock-in



MICROSERVICE A



MICROSERVICE B



MICROSERVICE C



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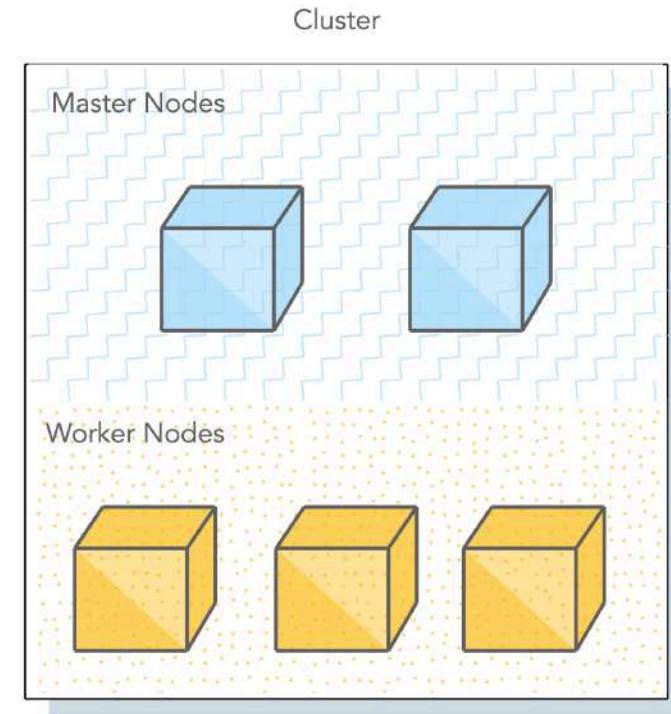
Getting Started with Managed Kubernetes



Managing Kubernetes is NOT Easy

Kubernetes Clusters: Not easy to setup and manage

- **Infrastructure Setup in Data Centers:** Servers + Networking + Security
- **Kubernetes Installations:** Master Nodes + Worker Nodes
- **Deploying Your Applications:** Deploy and manage applications
- **Kubernetes Configuration:** Scaling of Nodes, networking, storage
- **Observability & Troubleshooting:** Logs, metrics, alerts and cluster health





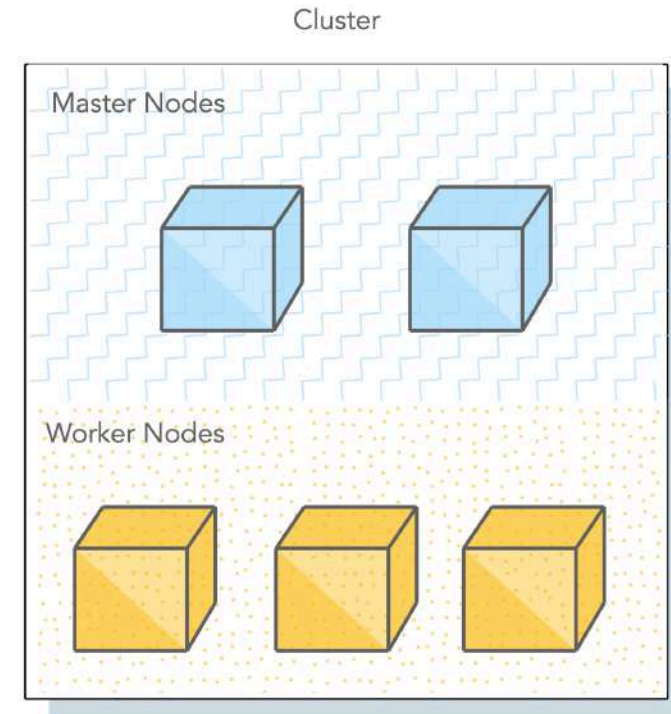
What is Self Managed Kubernetes (On-premises)?

Self-Managed: You manage everything

- **Infrastructure:** Setup physical servers in data centers
- **Kubernetes Setup:** Install control plane and worker nodes
- **Kubernetes Configuration:** Scaling of Nodes, etc.
- **Workloads:** Deploy and manage applications
- **Upgrades:** Maintain K8S versions and patch clusters
- **Observability & Troubleshooting:** Logs, metrics, alerts and cluster health

Pro: Full control

Con: High operational overhead





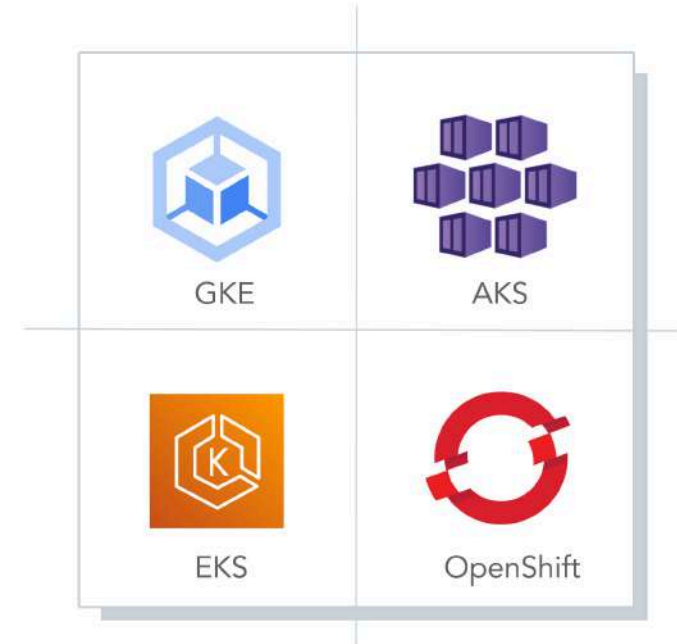
What is Cloud Managed Kubernetes?

Cloud Managed: Cloud provider manages infrastructure and Kubernetes control plane

- **Cloud Provider manages:** Infrastructure, Kubernetes Installation, Control plane, availability, observability and upgrades
- **You manage:** Kubernetes configuration (Node type, scaling, etc.) and workloads

Benefit: Reduced operational effort

- **Avoid undifferentiated heavy lifting:** Focus on your applications
- **Built-in features:** Built-in observability and security integrations





Managed Kubernetes – How It Solves Real Problems

Challenge	How Managed Kubernetes Helps
Setting up infrastructure in data centers	Cloud provider provisions and manages compute, networking and storage
Installing Kubernetes	Control plane, storage and networking are preconfigured
Node maintenance	Nodes are auto-patched, upgraded and auto-scaled
Observability setup	Integrated dashboards and alerts for logs, metrics and traces
High ops overhead	Teams focus on applications instead of infrastructure work
Cost management	Autoscaling ensures you pay only for resources you actually need



Managed Kubernetes Services in the Cloud

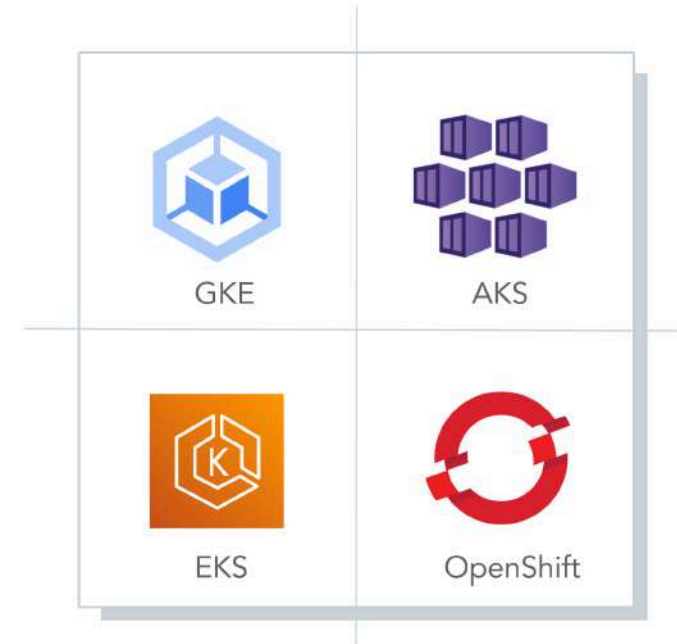
GKE: Google Kubernetes Engine - Managed Kubernetes from Google Cloud

EKS: Amazon Elastic Kubernetes Service - Managed Kubernetes from AWS

AKS: Azure Kubernetes Service - Managed Kubernetes from Microsoft Azure

OpenShift: Enterprise Kubernetes platform from Red Hat

- **Available as Managed Service:** Offered on multiple clouds





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Getting Started with

Google Kubernetes Engine



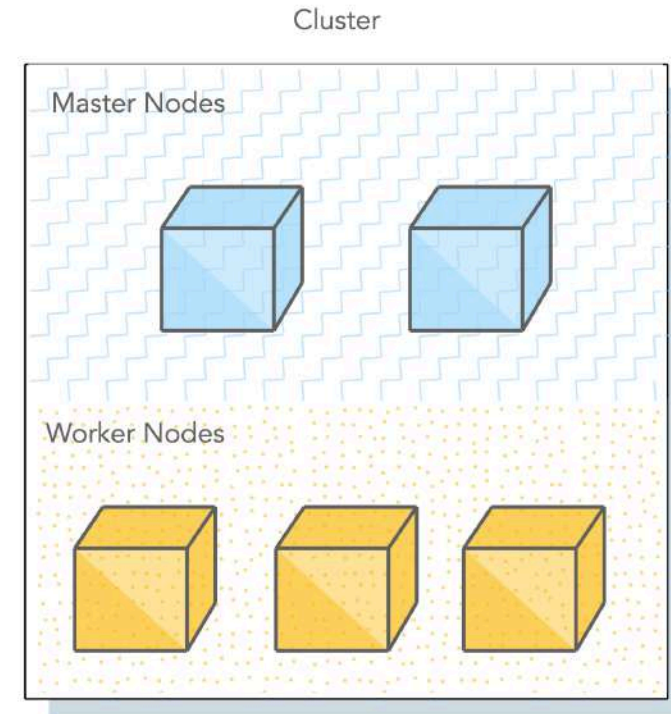
What Is Google Kubernetes Engine (GKE)?

Full Managed Kubernetes: Fully managed Kubernetes service from Google Cloud

- **Purpose:** Run, scale, and manage containerized applications with minimal effort
- **Design:** Based on Google's operational experience with large-scale container workloads

Modes:

- **Standard Mode:** Customer decides the number and type of worker nodes
- **Autopilot Mode:** Google manages everything in the cluster





GKE Standard vs GKE Autopilot – A Quick Comparison

Feature/Task	GKE Standard	GKE Autopilot
Setting up infra. - servers etc.	Google Cloud's Responsibility	Google Cloud's Responsibility
Installing Kubernetes	Google Cloud's Responsibility	Google Cloud's Responsibility
Setting up Control Plane & Worker Nodes	Google Cloud's Responsibility	Google Cloud's Responsibility
Configuration of Worker Nodes (Type, Scaling)	Customer's Responsibility	Google Cloud's Responsibility
Workloads	Customer's Responsibility	Customer's Responsibility
Key Cost Factor	Nodes (You pay for entire node)	CPU/Memory requests
Operational effort	Higher	Very low
Best for	Custom infra & special workloads	Most other workloads



Using GKE Standard Mode – An Overview

1: Create Cluster

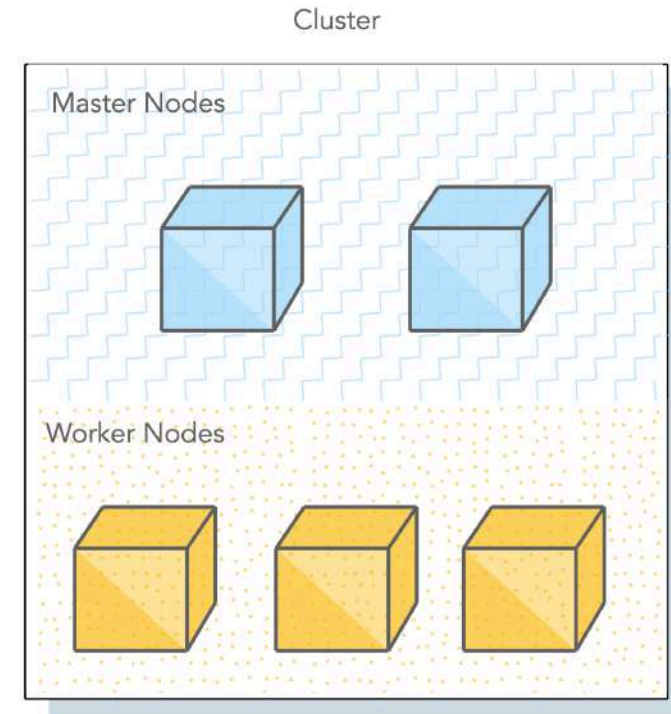
- A: Choose Region and/or Zone
- B: Configure nodes (no of nodes, machine type, size)

2: Deploy Applications

3: Manage Applications

- A: Manage Nodes Manually

Key Cost Factor: Nodes (You pay for entire node)





Using GKE AutoPilot Mode – An Overview

1: Create Cluster (No need to configure nodes)

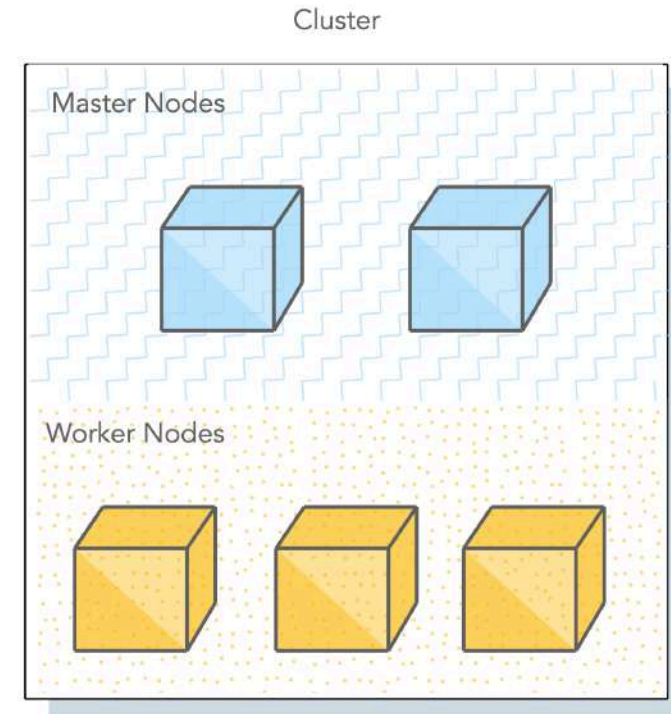
- A: Choose Region
- B: Take Defaults

2: Deploy Applications

3: Manage Applications

- No need to manage nodes

Key Cost Factor: CPU/Memory requests





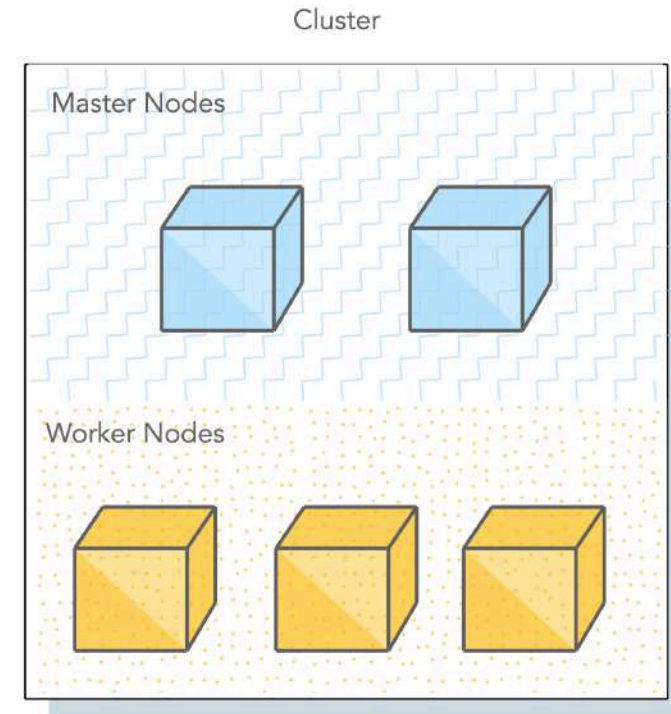
GKE Autopilot Mode – What It Is

Operating model: A Kubernetes mode where Google manages everything in the cluster for you

Nodes: Automatically created, sized, secured, and repaired

Billing: Primarily based on requested CPU and memory

GOAL: Reduce Kubernetes operational effort to ZERO!





GKE Autopilot Mode – Why Teams Use It

Less management: No node configuration, patching, or right-sizing

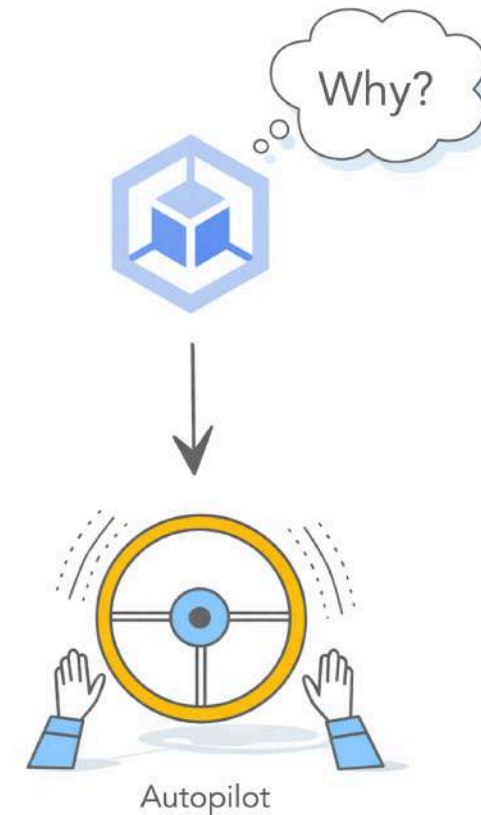
- Upgrades are automatic and safe
- Security best practices are enforced by default

Higher efficiency: Automatic resource scaling

- No idle VM cost because billing is based on requests

Better reliability: Resilience built into the platform

- Automatic monitoring and repairs





GKE Standard – Why Teams Still Use It

Custom Infrastructure Control: Full control over node configuration

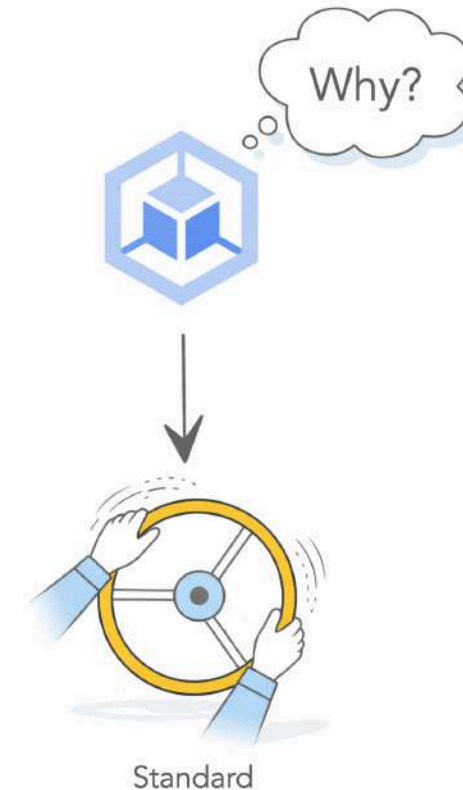
- Choose machine types, disk types, etc.
- Attach GPUs or TPUs for ML and HPC workloads

Node-Level Access: For direct control

- SSH access to nodes, Install custom agents

Run Hybrid Clusters: On-prem integration

Spot VMs: Run fault-tolerant, non-immediate workloads on Spot VMs





Kubernetes - A Journey - Getting Started

Step	What You Do	Example Command
1	Create a Cluster	<code>gcloud container clusters create my-cluster</code>
2	Log in to Cloud Shell	Open Cloud Shell in the console
3	Connect to the Cluster	<code>gcloud container clusters get-credentials my-cluster --zone us-central1-a --project solid-course-258105</code>
4	Deploy your microservice	<code>kubectl create deployment hello-world-rest-api --image=in28min/hello-world-rest-api:0.0.1.RELEASE</code>
	Expose as a service	<code>kubectl expose deployment hello-world-rest-api --type=LoadBalancer --port=8080</code>
5	Increase instances	<code>kubectl scale deployment hello-world-rest-api --replicas=2</code>
6	Enable auto-scaling for the microservice (HPA)	<code>kubectl autoscale deployment hello-world-rest-api --max=10 --cpu=70%</code>



Kubernetes - A Journey - The End!

Step	What You Do	Example Command
7	Increase number of nodes (Standard Mode Only)	<code>gcloud container clusters resize my-cluster --node-pool my-node-pool --num-nodes=5</code>
8	Enable auto-scaling for the Kubernetes cluster (Standard Mode Only)	<code>gcloud container clusters update cluster-name --enable-autoscaling --min-nodes=1 --max-nodes=10</code>
9	Delete the microservice	<code>kubectl delete service SERVICE_NAME</code> <code>kubectl delete deployment DEPLOYMENT_NAME</code>
10	Delete the cluster	<code>gcloud container clusters delete CLUSTER_NAME</code>



Key Kubernetes Logical Concepts - Simplified

Deployment: Manages your app (Defines container image and number of instances)

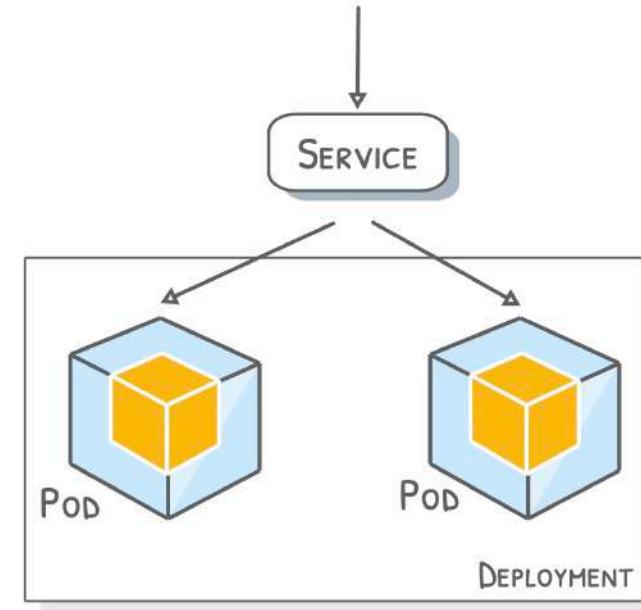
- `kubectl create deployment hello-world-rest-api --image=CONTAINER_IMAGE --replicas=2`

Pod: Created when you create a deployment

- 2 instances => 2 pods

Service: Expose pods in a deployment to outside world

- `kubectl expose deployment hello-world-rest-api --type=LoadBalancer --port=8080`





Key Kubernetes Concepts – Quick Review [1/2]

K8S Concept	Simple Explanation
Control Plane	The “brain” of the cluster (Master Nodes)
Worker Nodes	Machines that run your applications
Deployment	Manages Your App - <code>kubectl create deployment hello-world-rest-api --image=CONTAINER_IMAGE</code>
Pod	Created when you create a deployment. Smallest deployable unit in Kubernetes.
Service	Expose pods in a deployment to outside world. <code>kubectl expose deployment hello-world-rest-api --type=LoadBalancer --port=8080</code>



Key Kubernetes Concepts – Quick Review [2/2]

K8S Concept	Simple Explanation
GKE Autopilot Mode	A Kubernetes mode where Google manages everything in the cluster for you. Nodes are automatically created, sized, secured, and repaired. You just need to worry about the workloads.
Horizontal Pod Autoscaling (HPA)	Automatically increases or decreases the number of pods based on CPU utilization, memory usage, or custom metrics
Cluster Autoscaling	Automatically adds or removes worker nodes based on load



Google Kubernetes Engine - Introduction - Scenarios

Scenario	Solution
You want to keep your costs low	Consider AutoPilot mode If using Standard mode - Consider Spot VMs & Committed-use discounts. Choose right machine type to fit your workload type.
You want an efficient, completely auto scaling GKE solution	Configure Horizontal Pod Autoscaler for deployments and Cluster Autoscaler for node pools (or go for GKE AutoPilot Mode)
The smallest unit in Kubernetes, running your microservice	Pod
A stable way to expose pods in a deployment to outside world	Service
A Kubernetes mode where Google manages everything in the cluster	GKE Autopilot Mode



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Google Cloud Certification

Most Important Service



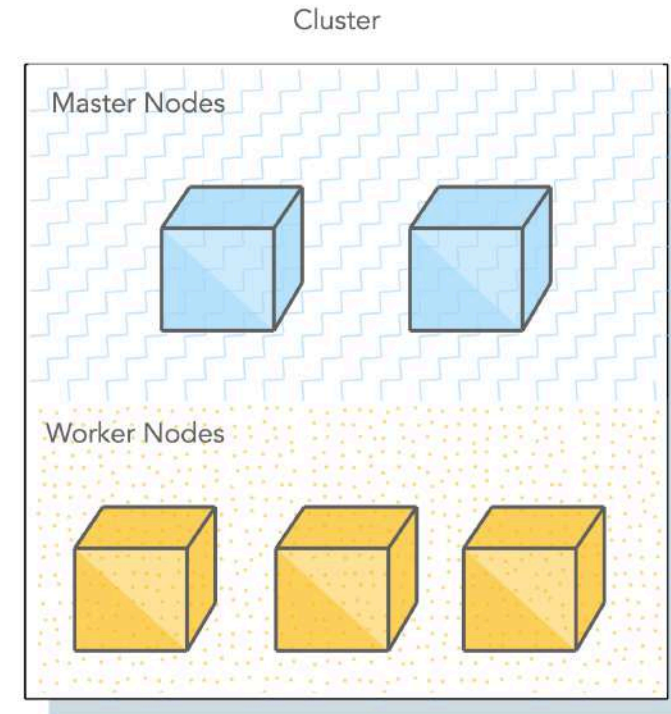
Most Important Service For This Certification!

Origin: Kubernetes was introduced by Google

Managed Kubernetes Leader: Google
Kubernetes Engine (GKE)

Certification Focus: Google Cloud
certifications place strong emphasis on GKE

What This Means For You: Understanding
Kubernetes concepts is critical for success in
this certification





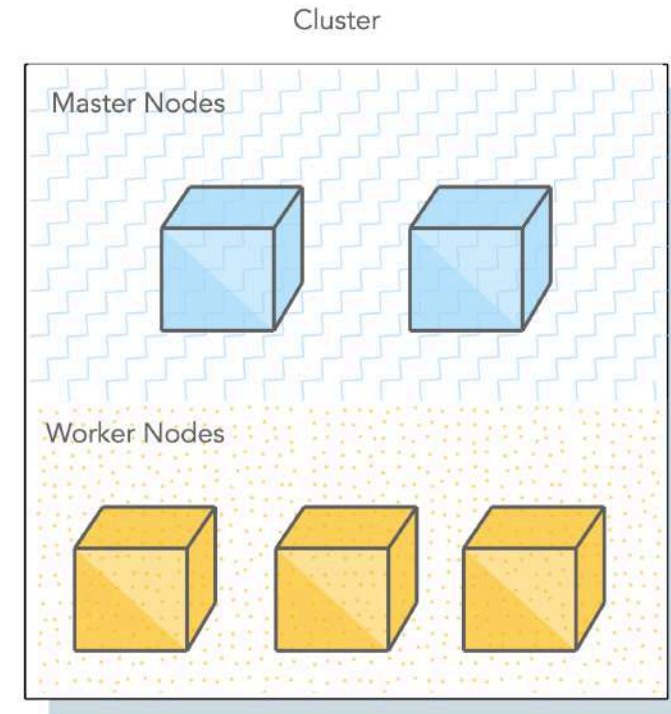
This Section Took a Lot of Time to Design

Challenge: Kubernetes is one of the toughest topics to teach

Goal: Break down a complex ecosystem into simple, easy-to-learn sections

How We Structured It: Security, Networking, Hybrid Kubernetes, Storage, YAML and more

Expectation: You understand Kubernetes concepts step by step without getting overwhelmed





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Digging Deeper Into Kubernetes Concepts



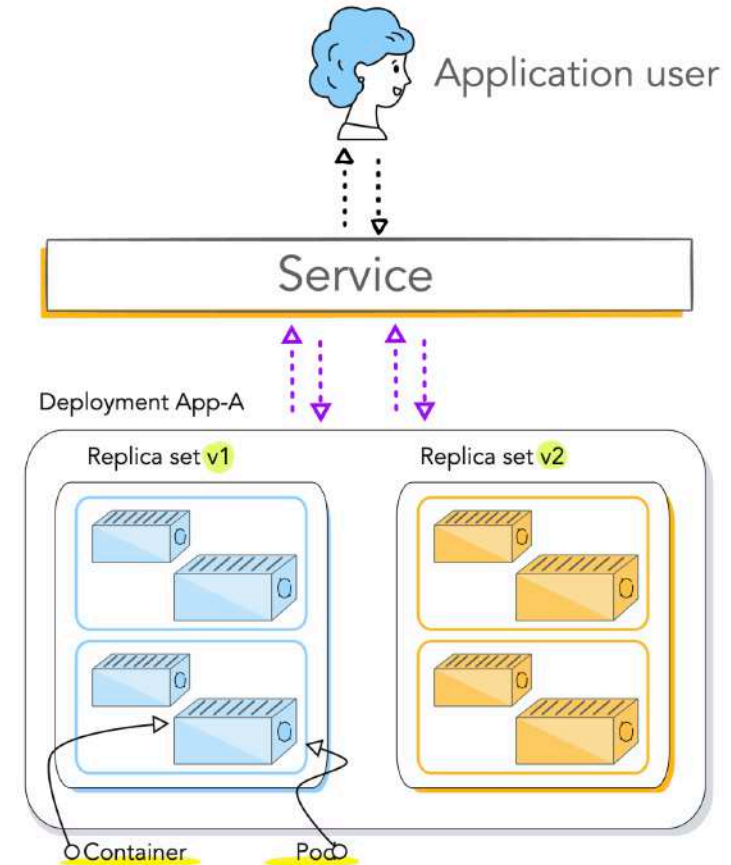
Microservices Example

Typical Microservices Architecture: Multiple Microservices

- Multiple versions
- Multiple instances

Example: Let's consider a simple example

- **Microservice:** Hello World Rest Api
- **Two Versions:** 0.0.1.RELEASE and 0.0.2.RELEASE
- **Multiple instances:** For each version





Deployment vs ReplicaSet [1/2]

ReplicaSet: Ensures the **right number of Pods** are always running

- Recreates Pods if they fail
- Represents **one version** of a microservice
 - 2 versions => 2 ReplicaSet's
- Updating the deployment image creates a **new ReplicaSet**

Deployment: Manages ReplicaSets

- One Deployment represents all versions of a microservice
- Ensures smooth, zero-downtime releases

```
kubectl create deployment hello-world-rest-api  
--image=in28min/hello-world-rest-  
api:0.0.1.RELEASE
```

Step 2: Scale it to 16 replicas

```
kubectl scale deployment hello-world-rest-api --  
replicas=16
```

*# Step 3: (Optional) Update to a new version
(e.g., newer tag)*

```
kubectl set image deployment hello-world-rest-  
api hello-world-rest-api=in28min/hello-world-  
rest-api:0.0.2.RELEASE
```



Deployment vs ReplicaSet [2/2]

Updating from v1 → v2

- Deployment creates **v2 ReplicaSet**
- Gradually scales down **v1 ReplicaSet** & scales up **v2**
- Typical progression:
 - v1:16 → v2:4
 - v1:12 → v2:8
 - v1:8 → v2:12
 - v1:4 → v2:16
 - v1:0 → v2:16
- Result: **Zero-downtime rolling update**
- Current Scenario: 1 Deployment managing 2 ReplicaSet's

```
kubectl create deployment hello-world-rest-api  
--image=in28min/hello-world-rest-api:0.0.1.RELEASE
```

Step 2: Scale it to 16 replicas

```
kubectl scale deployment hello-world-rest-api --  
replicas=16
```

*# Step 3: (Optional) Update to a new version
(e.g., newer tag)*

```
kubectl set image deployment hello-world-rest-api  
hello-world-rest-api=in28min/hello-world-rest-api:0.0.2.RELEASE
```



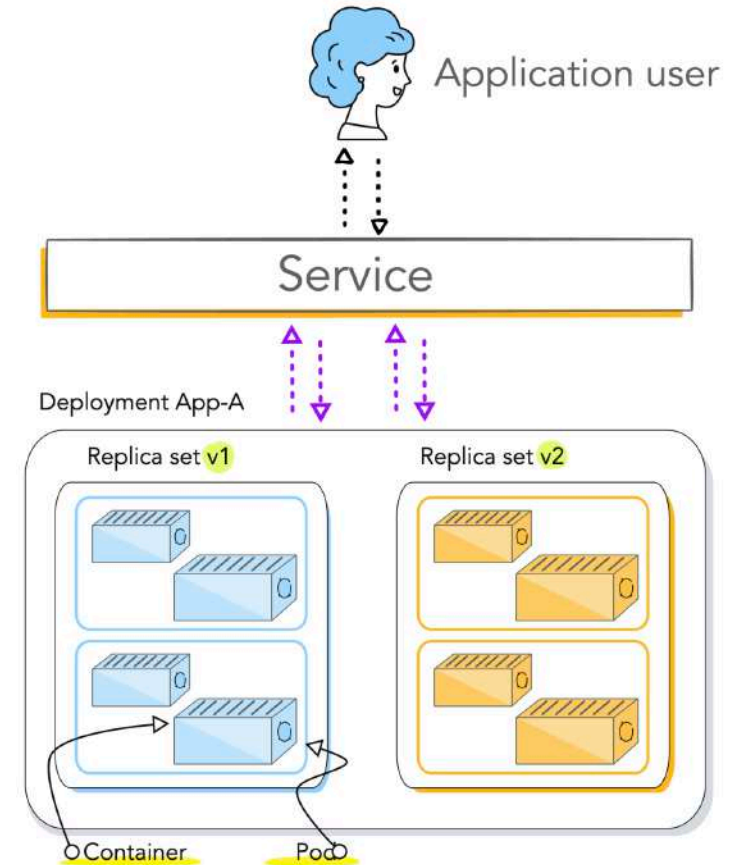
K8S Fundamentals - Pod - A Review

Pod – Smallest deployable unit

- **Runs one or more containers** (typically one microservice instance)
- **Ephemeral** – Kubernetes creates/replaces Pods automatically
- **Shared Resources** – Containers in a Pod share network, IP, ports, volumes
- **Statuses** – Running, Pending, Succeeded, Failed, Unknown

Example

- v1 Pods → `hello-world-rest-api:0.0.1.RELEASE`
- v2 Pods → `hello-world-rest-api:0.0.2.RELEASE`





When Does a Pod Have Multiple Containers?

Sidecar Pattern – Most common

- Example: Logging container next to main app

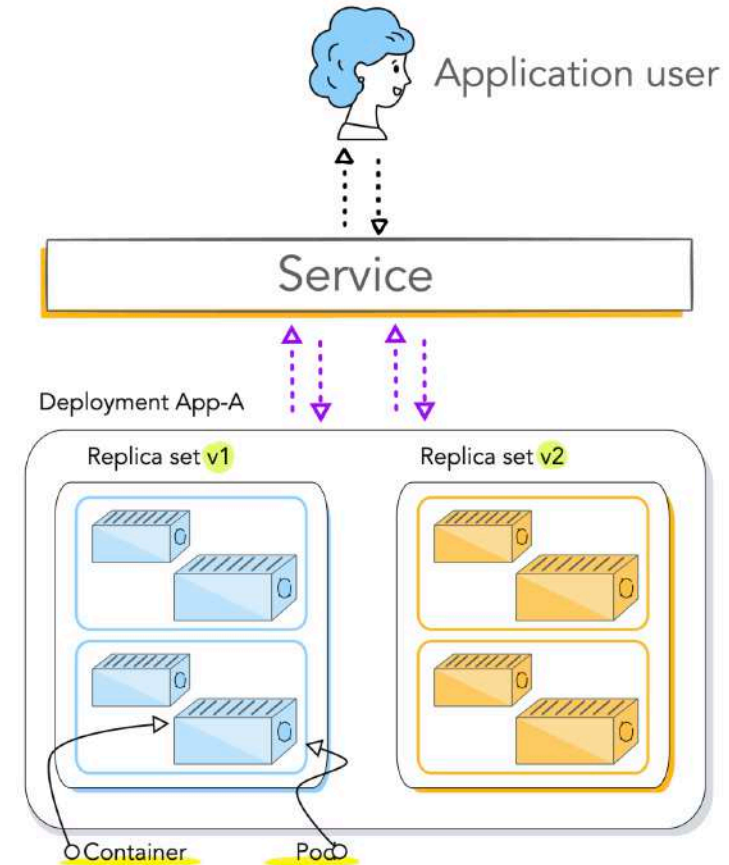
Helper Containers – Background or setup tasks

- Example: Data sync or file processor

Tightly Coupled Tasks – Must share lifecycle & storage

- Example: Main app + helper sharing a volume

Key Idea – One Pod = One logical unit, even if multiple containers exist





Kubernetes Logical View - For a Microservice [1/2]

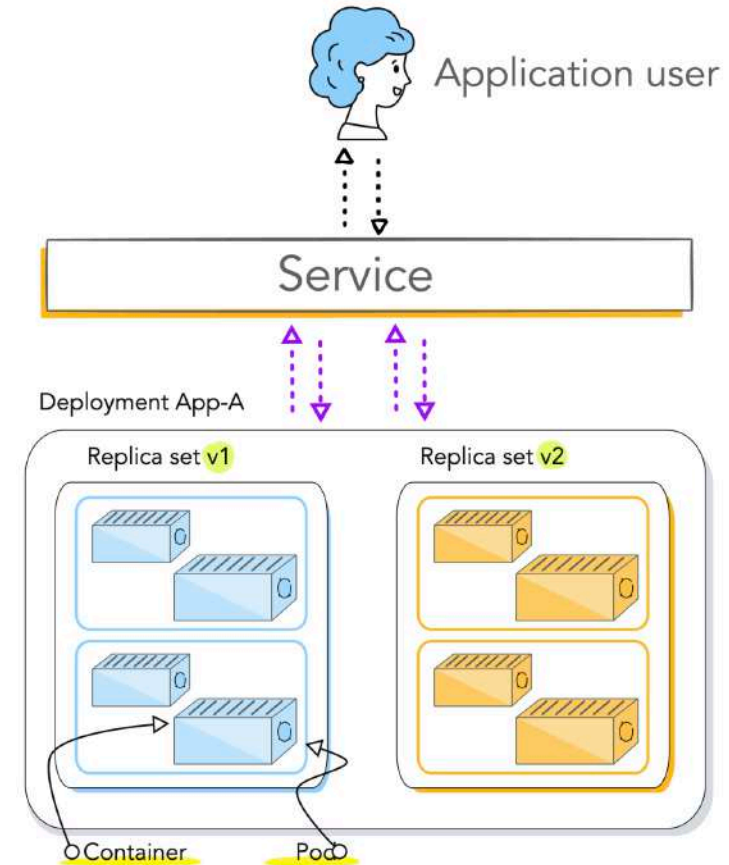
Deployment: High-level configuration that

- Creates and manages ReplicaSets
- Manages the rollout and rollback process

ReplicaSet: Ensures a fixed number of Pods are running for a specific version.

- v1 → 0.0.1.RELEASE
- v2 → 0.0.2.RELEASE

Pods: Actual units of deployment that run the application containers

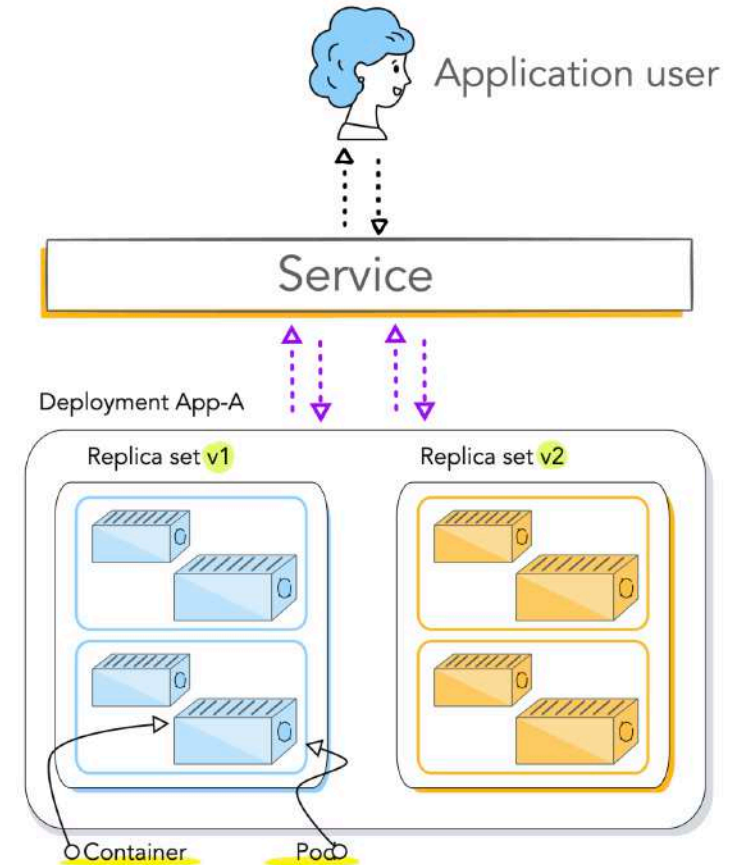




Kubernetes Logical View - For a Microservice [2/2]

Service: Acts as a **stable IP and DNS name**

- **Routes traffic:** To multiple microservice versions
- **Receives Requests:** From Users and Other Microservices
- **Load Balancing:** Between Multiple Pods





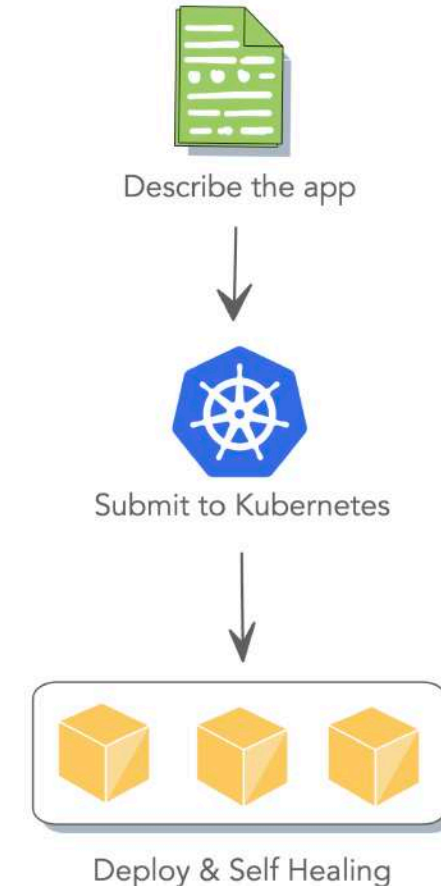
How Does Kubernetes Work?

Step 1 – Describe the app: Define what app, how many copies, what resources it needs

Step 2 – Submit to Kubernetes: Control plane receives the request

- **Scheduling:** Scheduler chooses the best worker nodes
- **Run the app:** Worker nodes run the app
- **Report status:** Nodes continuously share status

Step 3 – Self Healing: Kubernetes replaces anything that fails





Key Components in the Kubernetes Cluster

Cluster : Group of Compute Engine instances:

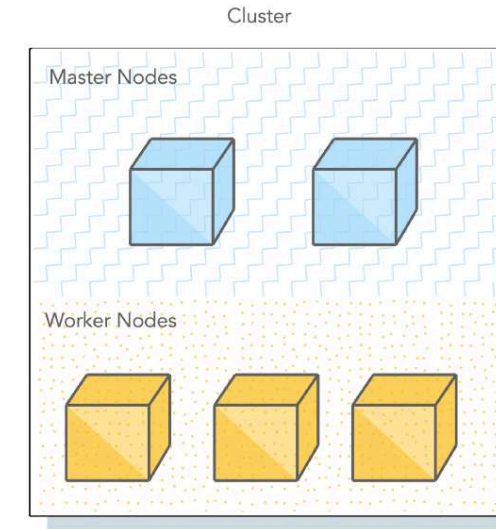
- **Master Node(s)** - Manages the cluster
- **Worker Node(s)** - Run your workloads (pods)

Master Node (Control plane) key components:

- **API Server** - Handles communication (from nodes and outside)
- **Scheduler** - Decides placement of pods
- **Controller Manager** - Manages core controllers
- **etcd** - Distributed database storing the cluster state

Worker Node key components:

- **Kubelet** - Manages communication with master node(s)
- **Container Engine** - Runs your pods



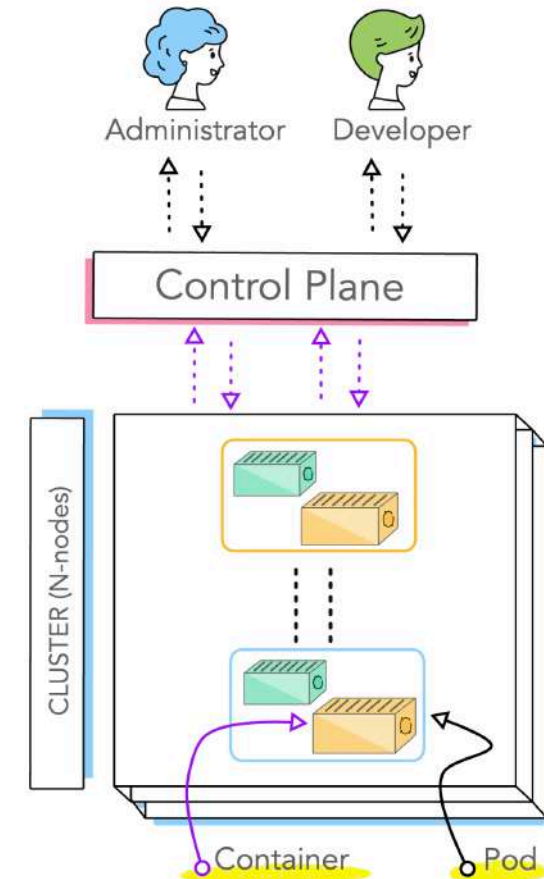
K8S Physical Architecture - Control Plane - Controller Manager



Controller Manager: Runs core controllers (e.g., Node Controller, Replication Controller) to maintain desired state

Example Controllers: Include

- **Node Controller:** Monitors the health of nodes
- **Replication Controller:** Ensures the specified number of pod replicas are running at all times
- **Deployment Controller:** Handles release management
- **Cloud Controller Manager:** Interfaces with cloud provider APIs to handle cloud-specific resources like persistent disks and external load balancer





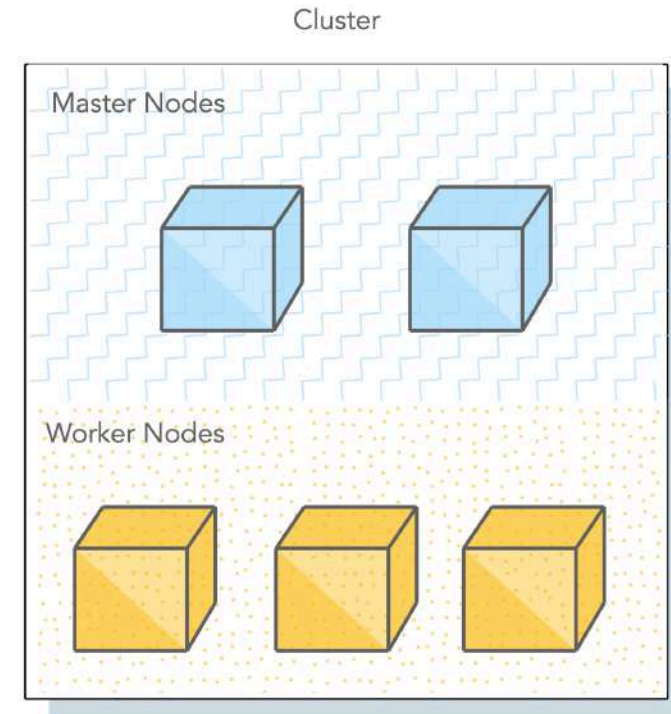
Understanding Kubernetes Node Pools

Hierarchy: Cluster > Node Pool(s) > Nodes

Why Node Pools?: Not all workloads are the same - Some need more CPU, Some need more memory, Some need GPUs or TPUs

- **Node Pool:** A group of nodes with identical configuration (Same machine type, Same disk type, ..)
 - **Example: Node Pool 1** → General workloads (standard CPU machines)
 - **Example: Node Pool 2** → ML workloads (GPU-enabled machines)

Key Benefit: One Cluster (easy to manage) and Multiple Node Pools (Right node/vm for the right workload)





Kubernetes Node Pools - Scenarios

Scenario	Solution
Reduce infrastructure cost for non-time-critical workloads while keeping critical services stable	Create two node pools : Use a node pool with Spot VMs for non-time-critical workloads, Use one with standard VMs for critical services
Application has both CPU-intensive and normal workloads	Create a node pool with compute-optimized machines for CPU-heavy workloads Use general-purpose nodes for other workloads
Some workloads require large amounts of memory	Create a node pool with memory-optimized machines Use general-purpose nodes for normal workloads
Run GPU-based AI/ML or video workloads along with normal workloads	Create a node pool with GPU-enabled nodes for these workloads Run normal workloads on standard node pools



GKE Cluster Types

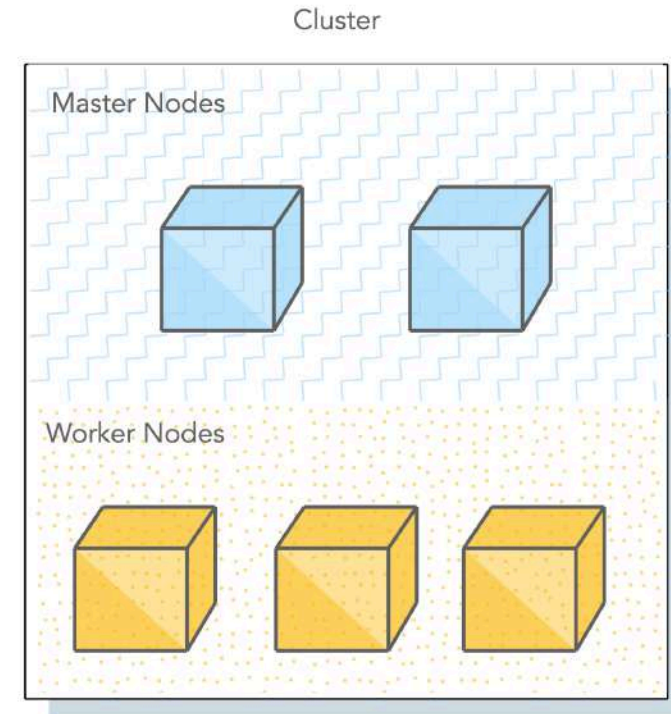
Type	Description
Zonal Cluster	Single control plane in a single zone. Worker nodes can be in one zone (single-zone clusters) or multiple zones in a region(multi-zonal clusters)
Regional Cluster	Replicates the control plane across multiple zones in a region => High Availability. Worker nodes are distributed across multiple zones in a region (multi-zonal clusters). GKE Autopilot clusters are always regional. GKE Standard clusters can be zonal or regional.
Private Cluster	Nodes use only internal IPs (no external IPs). Control plane and node communication occurs through private networking.
Confidential GKE Node Pool	Nodes run on Confidential VMs using memory encryption (designed for highly sensitive workloads)
Alpha Cluster	Includes alpha K8S APIs for testing pre-GA features. NOT for production.



Creating a Deployment - Background Activity

Command Execution: User runs `kubectl create deployment hello-world-rest-api`

- **API Server:** `kubectl` sends request to API Server
- **Validation and Storage:** API Server authenticates, validates, and stores the Deployment in `etcd`
- **Deployment Controller Triggered:** Notices the new Deployment and begins processing it
- **ReplicaSet:** Deployment Controller creates ReplicaSet
- **Pod Creation Initiated:** ReplicaSet requests creation of the required Pod(s) - default is 1.
- **Scheduler:** Assigns Pod to a healthy, available Node





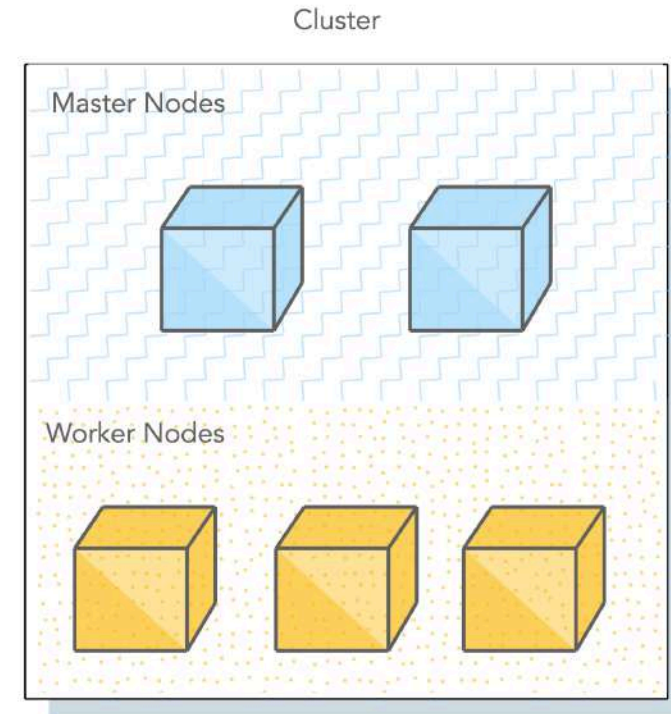
Creating a Deployment - Background Activity - 2

Command Execution: User runs `kubectl create deployment hello-world-rest-api`

- **Kubelet and Container Start:** Kubelet on that Node pulls the image and starts the container
 - **Pod State: Running:** Pod transitions to **Running** once the container is fully started
 - **Deployment Ready:** Deployment is marked ready after the desired number of Pods are successfully running

Controllers: Watch Pods and Nodes

- **Automatic Restart:** Unhealthy Pod => Replaced
- **Rescheduling:** Node Failed => Pods moved
- **Maintaining Desired State:** Controllers ensures no of running Pods always matches what you declared





Google Kubernetes Engine - Digging Deeper - Scenarios

Scenario	Component
Handles communication in a Kubernetes cluster (from nodes and outside)	API Server
Decides placement of pods on worker nodes	Scheduler
Distributed database storing the cluster state	etcd
Ensures the right number of Pods are always running	ReplicaSet
Manages Release Strategies through ReplicaSets	Deployment
Acts as a stable IP and DNS name	Service



Troubleshooting Kubernetes Deployment Errors

Error	Meaning	How to Troubleshoot
ImagePullBackOff / ErrImagePull	Kubernetes cannot pull the container image	• Verify image name and tag • Ensure image exists in registry • Check if node service account has artifact registry access
CrashLoopBackOff	Container starts → crashes → restarts repeatedly	• Check logs: <code>kubectl logs POD_NAME</code> • Run commands inside Pod: <code>kubectl exec -it POD_NAME -- ls</code>
PodUnschedulable	Kubernetes cannot place the Pod on any node	• Check cluster resource availability • Validate Pod resource requests/limits
Use Describe for Full Info	Shows detailed current state + events	• <code>kubectl describe pod POD_NAME</code>



Troubleshooting Kubernetes - Scenarios

Scenario	Solution
You are making a new deployment. You get ImagePullBackOff error	Is image name correct? Is image tag correct? Does the node pool service account have access to the container image?
You are making a new deployment. You get PodUnschedulable error	Are there sufficient resources on the cluster?
My pod stays pending	Most probably Pod cannot be scheduled onto a node(insufficient resources)
My pod stays waiting	Most probably failure to pull the image



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Google Kubernetes Engine

More Important Features



GKE - A Few More Best Practices

Title	Description
Shielded GKE Nodes	Automatically protect your nodes against advanced threats. Uses Secure Boot to ensure that only "clean," verified software is allowed to start up on the machine. Guards against "rootkits" - deeply hidden viruses that try to infect a computer before its security software even turns on. Default for GKE Autopilot and Standard.
Kubernetes Releases and Cluster Upgrades	Managed upgrade experience with release channels (rapid, regular, stable). Stable channel is preferred for production workloads. Node auto-upgrade ensures nodes are regularly updated with stable Kubernetes versions.
Confidential GKE Nodes	Enforce encryption of data in-use in your nodes and workloads
GKE Sandbox	Sandbox for Untrusted Code - Add a second layer of defense for untrusted containerized workloads. Creates a secure boundary so that if the application breaks, it only breaks its own "sandbox" (the cage) and cannot affect the rest of your server or other applications.



GKE - A Few Advanced Features

Title	Description
Config Connector	Open source Kubernetes add-on that lets you manage Google Cloud resources through Kubernetes. Configure Google Cloud services using Kubernetes tooling and APIs.
Config Sync	Simplify GitOps Implementation (Automate the synchronization of your configuration and policies across any number of clusters. Use a single source of truth, such as a Git repository.)
Policy Controller	Policy Controller acts as an automated gatekeeper in Kubernetes, using "Constraints" (rules) to validate resource requests against security standards during the admission phase. It intercepts object creation requests and immediately rejects those that violate pre-set compliance rules. Examples: Prevent unverified images, Restricting Image Repositories, Disallowing Public Load Balancers, etc.
Binary Authorization	Security feature that ensures only trusted container images are deployed to GKE. Images must be signed or verified by trusted authorities before deployment. Helps prevent running unapproved or vulnerable container images.



GKE - A Few More Scenarios

Scenario	Solution
You want to execute un-trusted third-party code in a Kubernetes cluster	Enable GKE Sandbox and run these workloads in sandbox-ed nodes to isolate them from the rest of the cluster
You need encryption of sensitive workloads even while they are being processed in memory	Use Confidential GKE Nodes to enable encryption of data in-use
You want to manage Google Cloud resources like Pub/Sub or Cloud SQL using Kubernetes YAML	Use Config Connector to manage Google Cloud resources through Kubernetes manifests
Your organization wants all clusters to automatically follow the configuration stored in a Git repository	Use Config Sync to implement GitOps and keep cluster configuration synchronized with Git
Your production cluster must receive stable Kubernetes upgrades automatically	Use Stable Release Channel with node auto-upgrade enabled



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Getting Started with

Declarative Kubernetes



What are Labels?

Labels: Key value pairs attached with Kubernetes resources (Pods, Deployments, ReplicaSet, Service, ...)

Assign multiple labels: You can assign multiple labels to a single resource

Examples:

- app: a
- environment: prod
- tier: backend

```
labels:  
  app: a  
  environment: prod  
  tier: backend  
  version: v1  
  team: payments  
  billinggroup: finance
```



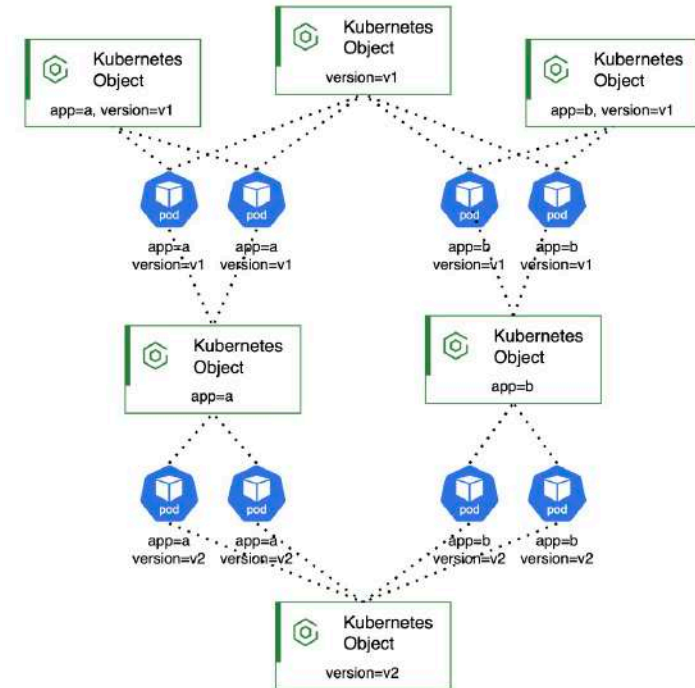
Why are Labels Needed? [1/2]

Logical Grouping of Resources: Group and organize resources (like Pods, Services, Deployments)

- Group all resources of a application with `app=a`

App Versioning: Use labels (`version=v1`) to track app versions

- Attach labels `app=hello-world-rest-api version=v1` with V1 and `app=hello-world-rest-api version=v2` with V2.



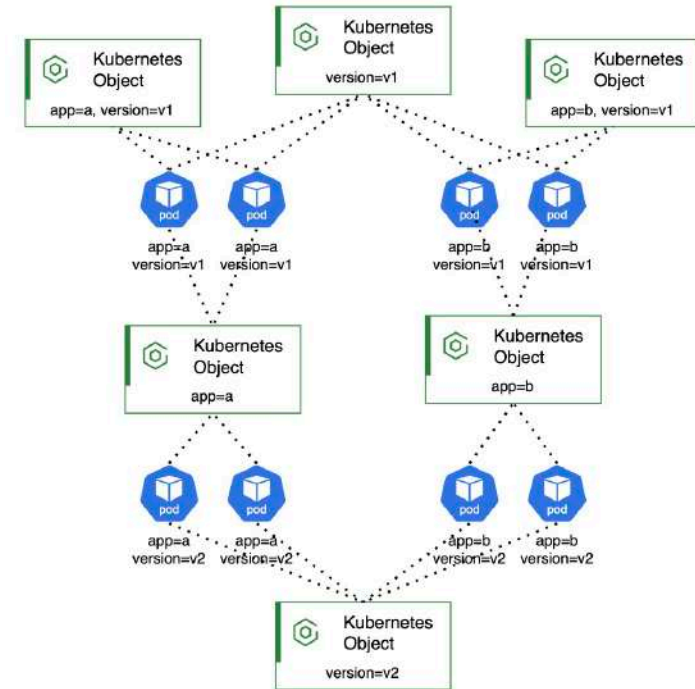


Why are Labels Needed? [2/2]

Organize Workloads: Labels can categorize workloads by team, app, or function
`team=frontend, app=webapp`

Route Traffic: Services use **label selectors** to route traffic to the correct set of Pods

- A Service with **selector: app=a** forwards traffic to both v1 and v2 pods
- A Service with **selector: app=a & version=v1** forwards traffic only to v1 pods





Labels Provide Flexibility for Operations

Example 1: Delete all staging resources using `kubectl delete pod -l environment=staging`

Example 2: Tools like Prometheus, Grafana can use labels to **group, filter, and visualize** metrics or logs - Show CPU usage grouped by `app`, `team`, or `version`

Example 3: Rollouts and Updates - Route 90% traffic to Pods with `version=v1`, and 10% to `version=v2`

```
labels:  
  app: a  
  environment: prod  
  tier: backend  
  version: v1  
  team: payments  
  billinggroup: finance
```



Understanding Kubernetes YAML - Basics

Each configuration YAML contains:

- **apiVersion:** Specifies the version of the Kubernetes API you're using (e.g., `v1` for Service, `apps/v1` for Deployment)
- **kind:** What type of configuration is it? (Deployment, Service, ConfigMap, Secret etc)
- **metadata:** What metadata do you want to associate with a resource?
 - **name:** name of the resource
 - **namespace:** namespace of the resource
 - **labels:** key/value pairs attached with the resource
- **spec:** Configuration of the resource

```
apiVersion: v1
kind: Service
metadata:
  name: hello-world-rest-api
  namespace: default
  labels:
    app: hello-world-rest-api
spec:
  selector:
    app: hello-world-rest-api
  ports:
    - port: 8000
      targetPort: 8000
  type: LoadBalancer
```



An Example Deployment YAML

kind: Object type (Deployment)

metadata.name: Name of the Deployment

spec: Deployment Specification

- **spec.replicas:** How many Pods to run?
- **spec.selector:** Labels used to find Pods
- **template:** Pod blueprint
 - **template.metadata.labels:** Pod Labels
 - **template.spec.containers:** Containers inside the Pod
 - **container.image:** Docker image to run
 - **containerPort:** Port exposed inside the container

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: hello-world-rest-api
spec:
  replicas: 2
  selector:
    matchLabels:
      app: hello-world-rest-api
  template:
    metadata:
      labels:
        app: hello-world-rest-api
        version: v1
    spec: # POD SPEC
      containers:
        - name: hello-world-rest-api
          image: IMAGE_PATH
          ports:
            - containerPort: 8000
```



Kubernetes - Deployment Strategy

Rolling Updates: Update Pods gradually

- **maxSurge**: Max no of Pods that can be created over the desired number of Pods
- **maxUnavailable**: Max no of Pods that can be unavailable during the update process
- **Ex 1**: 1 new instance at a time, No reduction in capacity: maxSurge - 1, maxUnavailable - 0
- **Ex 2**: Minimum cost, 1 at a time (No increase in no of instances): maxSurge - 0, maxUnavailable - 1

Recreate: Replace all Pods at once (Not recommended)

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: hello-world-rest-api
spec:
  replicas: 2
  minReadySeconds: 45
  selector:
    matchLabels:
      app: hello-world-rest-api
      version: v1
  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxSurge: 25%
      maxUnavailable: 25%
  template:
    # TEMPLATE GOES HERE!
```



Understanding Deployment Probes [1/2]

Liveness Probe: Detect application failures and restart containers

- Kubernetes automatically restarts unhealthy containers
- Keeps only healthy instances serving traffic

Startup Probe: Allow time for startup

- Prevents liveness checks from restarting slow-starting containers

Readiness Probe: Send traffic only when the instance is ready

- Pod receives traffic only after readiness succeeds

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: hello-world-rest-api
spec:
  template:
    spec:
      containers:
        - name: hello-world-rest-api
          image: IMAGE
          startupProbe:
            httpGet:
              port: 8000
              path: /startup
          readinessProbe:
            httpGet:
              port: 8000
              path: /readiness
          livenessProbe:
            httpGet:
              port: 8000
              path: /liveness
```




Understanding Deployment Probes [2/2]

Probe	Correct Scenario to Use
Startup Probe	When an application takes time to start (loading configs, warming caches, connecting to databases) and you want to avoid premature restarts
Readiness Probe	When a Pod should receive traffic only after the dependencies are fully ready (for example, after dependencies like DB or external services are available) When an application temporarily cannot handle requests due to high load, dependency outages, or maintenance tasks
Liveness Probe	When you want Kubernetes to detect a stuck or unhealthy application and automatically restart the container



More Deployment YAML Examples

Environment Variables: Provide config values to the app at runtime

- Example: `DATABASE_HOST=my-db`

Resource Requests: Minimum resources guaranteed for the container

- `memory: 128Mi`: Minimum 128Mi memory
- `cpu: 250m`: Minimum 0.25 CPU core

Resource Limits: Max resources

- `memory: 512Mi`: Maximum 512Mi memory
- `cpu: 500m`: Maximum 0.5 CPU core

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: hello-world-rest-api
spec:
  template:
    spec:
      containers:
        - name: hello-world-rest-api
          image: IMAGE
          env:
            - name: DATABASE_HOST
              value: my-db
          resources:
            requests:
              memory: "128Mi"
              cpu: "250m"
            limits:
              memory: "512Mi"
              cpu: "500m"
```



Exploring Graceful shutdown - Kubernetes

Recommended: Clean up your connections and shutdown properly

Step By Step: When Kubernetes decides to terminate a pod

- **Runs preStop Hooks:** Execute cleanup logic (closing DB connections, ..)
- **Sends SIGTERM signal:** To the pod
 - Kubernetes waits for a grace period (default: 30 seconds)
 - **If POD is still running:** SIGKILL signal is sent to forcibly remove the pod
- **Recommended:** Define preStop Hook and/or handle SIGTERM signal

```
process.on('SIGTERM', () => {  
  console.info('Time to clean up and  
shutdown.');
```

```
});  
  
spec:  
  containers:  
    - name: hello-world-rest-api  
      image: IMAGE  
      lifecycle:  
        preStop:  
          exec:  
            command:  
              ["/bin/sh",  
               "-c",  
               "YOUR_COMMAND"]
```



How Is a Secret Different from a ConfigMap? [1/2]

ConfigMaps Store Configuration: App settings such as URLs, ports,

- **ConfigMaps Are Plain Text:** Easier to read but not meant for confidential data

Secrets Store Sensitive Data: Passwords, API keys, tokens, certificates

- **Secrets Use Base64 Encoding:** Values are encoded

Typical Pattern: ConfigMap → hostnames, ports | Secret → passwords, credentials

```
apiVersion: v1
kind: Secret
metadata:
  name: my-app-secrets
type: Opaque
data:
  my-app-db-password:
    cGFzc3dvcmQxMjM=

apiVersion: v1
kind: ConfigMap
metadata:
  name: db-config
data:
  db-host: my-database.svc.cluster.local
  db-port: "5432"
  db-name: my-app-db
  db-user: appuser
```



How Is a Secret Different from a ConfigMap? [2/2]

Injected as Environment Variables:

Applications can read them as normal environment variables

- **ConfigMap Example:** DB_HOST, DB_PORT, DB_NAME, DB_USER
- **Secret Example:** DB_PASSWORD stored securely

Same Usage Pattern: Refer to them using `valueFrom`

Key Idea: Separate sensitive data from normal configuration

```
apiVersion: apps/v1
kind: Deployment
spec:
  template:
    spec:
      containers:
        - name: hello-world-rest-api
          image: in28min/hello-world-rest-api:latest
          env:
            - name: DB_HOST
              valueFrom:
                configMapKeyRef:
                  name: db-config
                  key: db-host
              # Others: DB_PORT,DB_NAME,..
            - name: DB_PASSWORD
              valueFrom:
                secretKeyRef:
                  name: my-app-secrets
                  key: my-app-db-password
```



An Example Service YAML

apiVersion: Core API group

kind: Kubernetes Service Object

metadata.name: Name of the Service

spec: Service Specification

- **spec.selector:** Labels to identify backend Pods
- **spec.ports.port:** Port exposed by the Service (Frontend)
- **spec.ports.targetPort:** Pod/container port to forward traffic to (Backend)
- **spec.type: LoadBalancer:** Creates an external Load Balancer for public access

```
apiVersion: v1
kind: Service
metadata:
  name: hello-world-rest-api
spec:
  selector:
    app: hello-world-rest-api
    version: v1
  ports:
    - port: 8000
      targetPort: 8000
  type: LoadBalancer
```



What are Namespaces?

Namespaces: Kubernetes resources can be assigned a namespace

Default Namespace: If no namespace is specified, Kubernetes assigns objects to the **default** namespace

Logical Separation: Provide a way to group cluster resources logically making management easier

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: demo
  namespace: my-namespace
# Other details
```

```
# View all namespaces
# kubectl get namespaces
```

```
# Run commands in a specific namespace
# kubectl get pods -n my-namespace
```



Namespaces - Use Cases

Isolation of resources: In same cluster

- **Different teams:** Or projects
- **Access Control:** Restricting user access
 - Dev team - full dev access (limited prod access)

Resource Management: Resource limits

- **CPU & memory quotas:** For a namespace

Shared Resources: Common resources can be shared across multiple namespaces

- **Ex:** Monitoring, logging or ingress controllers

```
apiVersion: v1
kind: ResourceQuota
metadata:
  name: compute-limits
  namespace: dev-team
labels:
  policy: enforced
spec:
  hard:
    requests.cpu: "2"
    requests.memory: 4Gi
    limits.cpu: "4"
    limits.memory: 8Gi
```




Using Kubernetes Namespaces

Command	Purpose
<i>kubectl get namespaces</i>	View all namespaces in the cluster
<i>kubectl create namespace my-namespace</i>	Create a new isolated namespace
<i>kubectl run nginx --image=nginx --namespace=my-namespace</i>	Run a Pod inside a specific namespace
<i>kubectl get pods --namespace=my-namespace</i>	List Pods within a chosen namespace
<i>kubectl config set-context --current --namespace=my-namespace</i>	Set a default namespace for future kubectl commands
<i>kubectl delete namespace my-namespace</i>	Remove an entire namespace and all its resources
<i>kubectl get all -n my-namespace</i>	View all resources inside a namespace



Exploring Kubernetes YAML - An Example

Deployment V1

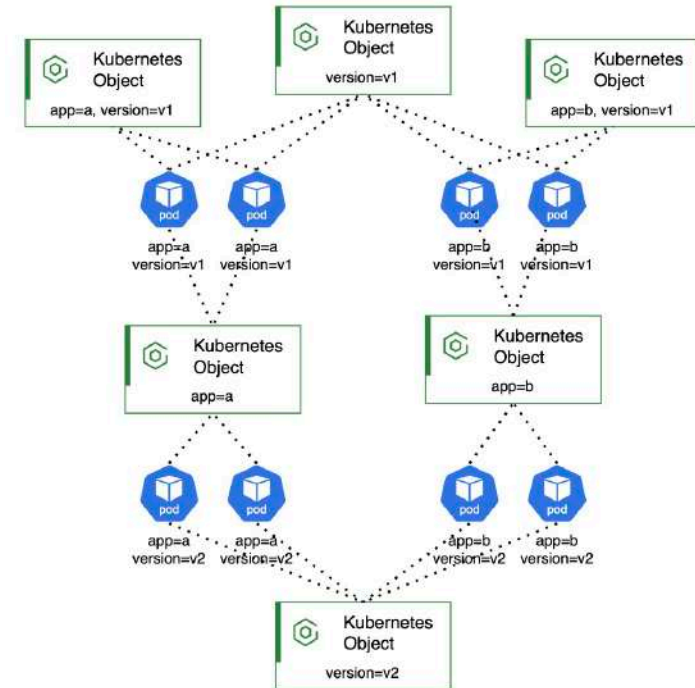
- POD Labels: `app:hello-world-rest-api` and `version:v1`

Deployment V2

- POD Labels: `app:hello-world-rest-api` and `version:v2`

Service

- `selector: app: hello-world-rest-api` (Load balance between v1 and v2 pods)
- `selector: app: hello-world-rest-api` and `version:v2` (Load balance between v2 pods only)





Why Use StatefulSet Instead of Deployment for Kafka?

Stable Identity Required: Kafka brokers need fixed hostnames

Persistent Storage Per Broker: Each broker needs its own durable volume

Deployment Does Not Guarantee Stability: Pod names change (e.g., kafka-abc123)

StatefulSet Guarantees Stability: Predictable names like kafka-0, kafka-1

```
apiVersion: apps/v1
kind: StatefulSet
metadata:
  name: kafka
```



Why Use DaemonSet?

Run Pod On Every Node: Ensures exactly one Pod runs on each node

- **Automatic Scaling With Nodes:** When a node joins the cluster, a Pod is created automatically

Ideal For Node-Level Services: Common for logging, monitoring, and networking agents

- **Examples:** Fluentd for logs, Prometheus Node Exporter for metrics, CNI networking plugins

```
apiVersion: apps/v1
kind: DaemonSet
metadata:
  name: log-agent
spec:
  selector:
    matchLabels:
      app: log-agent
  template:
    metadata:
      labels:
        app: log-agent
    spec:
      containers:
        - name: log-agent
          image: fluentd
```



What is Pod Disruption Budget?

Scenario: Node maintenance reduces the number of available Pods

- **Solution:** Use a Pod Disruption Budget to enforce a minimum availability for Pods

Options: Configure one of these

- **minAvailable:** Minimum number of Pods that must remain running
 - **Example:** `minAvailable: 80%` ensures at least 80% of Pods stay running during maintenance
- **maxUnavailable:** Maximum number of Pods allowed to be unavailable at a time
 - **Example:** Deployment has 10 Pods, `maxUnavailable: 1` ensures only one Pod can be disrupted at a time

```
apiVersion: policy/v1
kind: PodDisruptionBudget
metadata:
  name: hello-world-rest-api-pdb
spec:
  minAvailable: 10 # OR 80%
  selector:
    matchLabels:
      app: hello-world-rest-api
```



Kubernetes - Scenarios [1/2]

Scenario	Solution
How can you divide one physical Kubernetes cluster into multiple virtual clusters with customized access to different teams?	Use namespaces
You are making use of Kubernetes Deployment Strategy - Rolling Updates. When releasing new version, you want to update 1 instance at a time without reduction in capacity	maxSurge set to 1, maxUnavailable set to 0
What does Kubernetes do to ensure that your application has completely started up and is ready to accept traffic?	Configure startup probe
Your application sometimes becomes unresponsive but the container is still running	Configure a liveness probe
During cluster maintenance you want at least 3 Pods always running	Use PodDisruptionBudget with <i>minAvailable: 3</i>



Kubernetes - Scenarios [2/2]

Scenario	Solution
How to manage resource sharing for multiple teams in the same Kubernetes cluster?	Create namespaces per team with resource quotas and specify resource requests and limits per names
How to secure a multi-tenant cluster so each customer accesses only their own blog?	Create a separate Kubernetes namespace per tenant and enforce isolation using Network Policies
Stateful Workloads needing stable identity - Deployment or StatefulSet	StatefulSet
A node joins the cluster and you want an observability Pod automatically deployed there	DaemonSet
You want to perform controlled rolling upgrades without downtime	Use Deployment with RollingUpdate strategy (default)



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Getting Started with

Kubernetes Networking



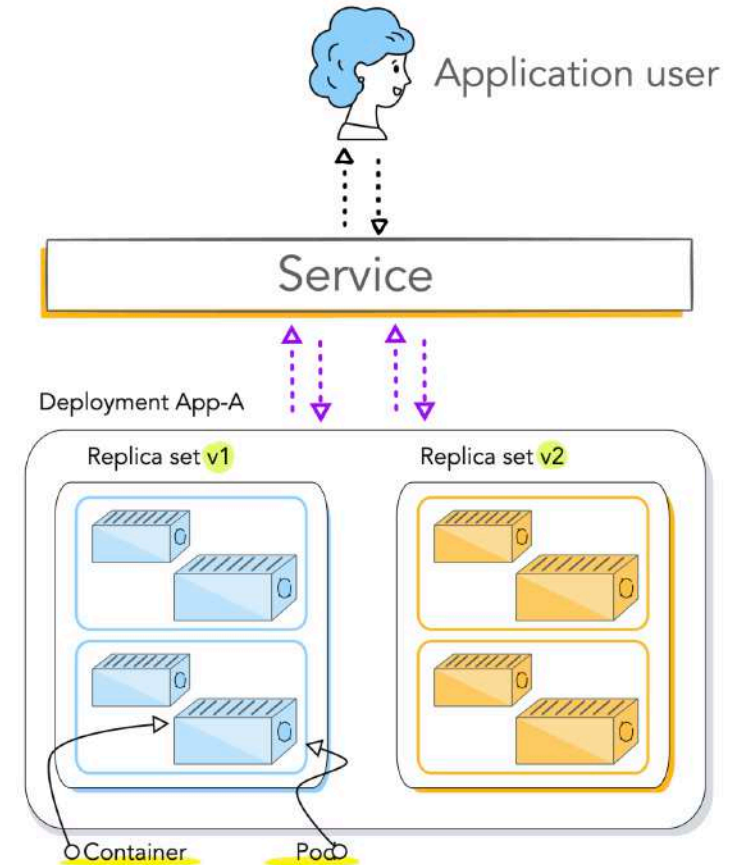
Why do we need a Kubernetes Service?

Do You Know?: Pods are ephemeral

- **Key Question:** How to provide a stable address for a set of pods?

Kubernetes Service: Provides a stable IP & DNS name for a set of Pods (abstract ephemeral nature) `kubectl expose deployment name --type=TYPE`

- **Load Balancing:** Distributes network traffic evenly across multiple Pods to ensure reliability and performance





What are Different Service Types?

ClusterIP: Internal Access Only (DEFAULT)

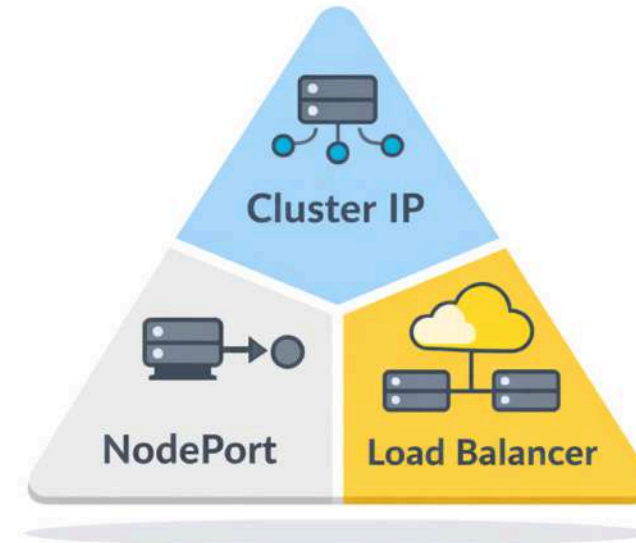
- **NO external access:** Only within the cluster
- **Microservices:** Ideal for internal communication

NodePort: Exposes the service on a static port (30000–32767) on every node

- **External Access:** `http://<NodeIP>:<NodePort>`
- **Dev/Test Friendly:** Great for quick testing

LoadBalancer: Load Balancing with Cloud LB

- **Enable Public Access:** With Public IP





Scenarios - Kubernetes Service Types

Scenario	Recommended Service Type
Your backend API should only be accessible by other services inside the cluster. No external users should reach it directly.	ClusterIP
You are testing a new microservice in a development environment and want quick external access using a node's IP and port.	NodePort
You are deploying a production web application that must be publicly accessible with high availability and load balancing.	LoadBalancer
A database service should only be reachable by application pods inside the cluster.	ClusterIP



What is a Fully Qualified Domain Name?

Do you know?: `www.in28minutes.com`
represents a hierarchy

- `com` → Top Level Domain (TLD)
- `in28minutes` → Second Level Domain (SLD)
- `www` → Subdomain

FQDN: Specifies the precise location of a resource in the DNS hierarchy

- **FQDN Structure:** `<subdomain>.<domain>.<tld>.`
- **Ends with a Dot (.):** Denotes the root of the DNS tree
- **Usually Omitted:** Commonly omitted in day-to-day usage





Why Does FQDN Matter in Kubernetes?

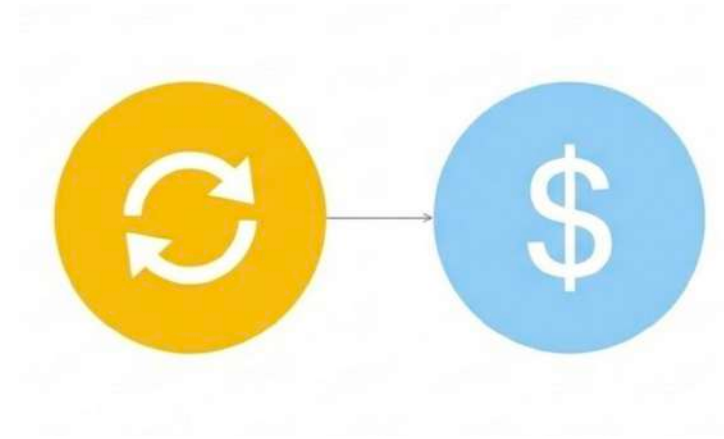
Question: Why Does FQDN Matter in Kubernetes?

- **FQDN:** Used to address services in Kubernetes

Format: `<service-name>.
<namespace>.svc.cluster.local.`

Example: `currency-exchange.default.svc.cluster.local.`

Cluster-Wide Access: Used to access a service across the cluster





How does a Service help with Service Discovery?

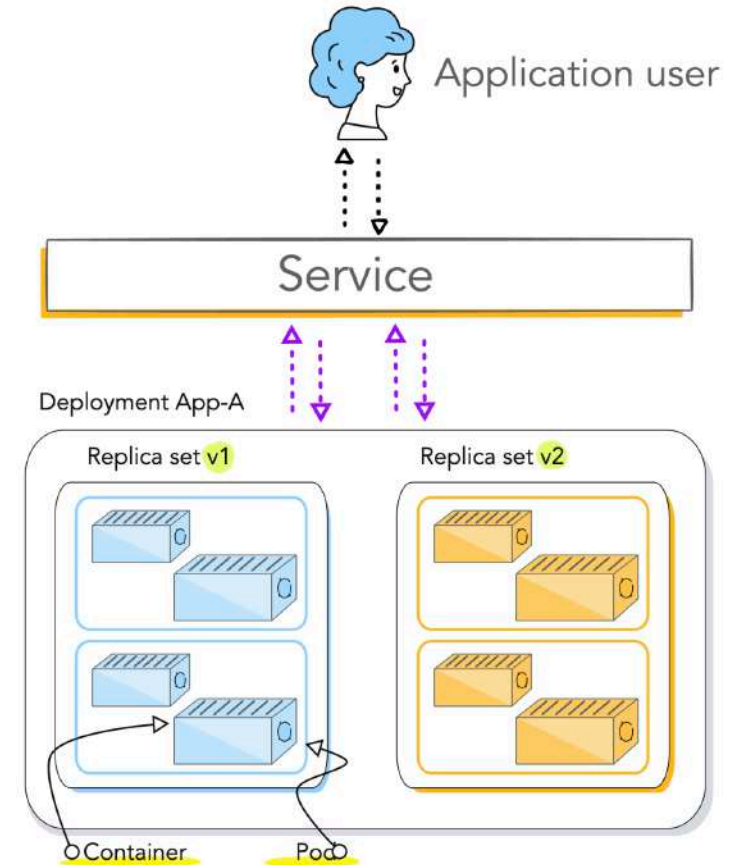
Pods are ephemeral: Can restart, move to other nodes, receive new IPs

Service: Stable way to reach Pods

- **Assigned a ClusterIP:** Virtual IP that does not change
- **Traffic Distribution:** Requests to this IP load-balanced across healthy Pods selected by the Service

Automatic DNS Setup: For each Service

- **FQDN:** `<service>.<namespace>.svc.cluster.local`.
- **Name-Based Access:** Access other services using DNS names - no need to hard-code IP addresses

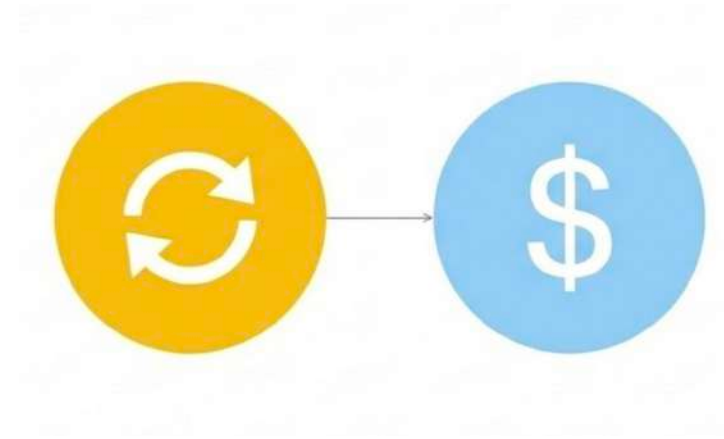




Example - Service Discovery

Example: How does `currency-conversion` talk to `currency-exchange`?

- Pod (`currency-conversion`) issues DNS lookup (e.g., `currency-exchange.default.svc.cluster.local`.)
- Service name is resolved to ClusterIP address of the service



Service Name Format: Can be simplified

- `currency-exchange.default.svc.cluster.local`. (fully qualified domain name)
- `currency-exchange` (within the same namespace)
- `currency-exchange.default` (across namespaces)



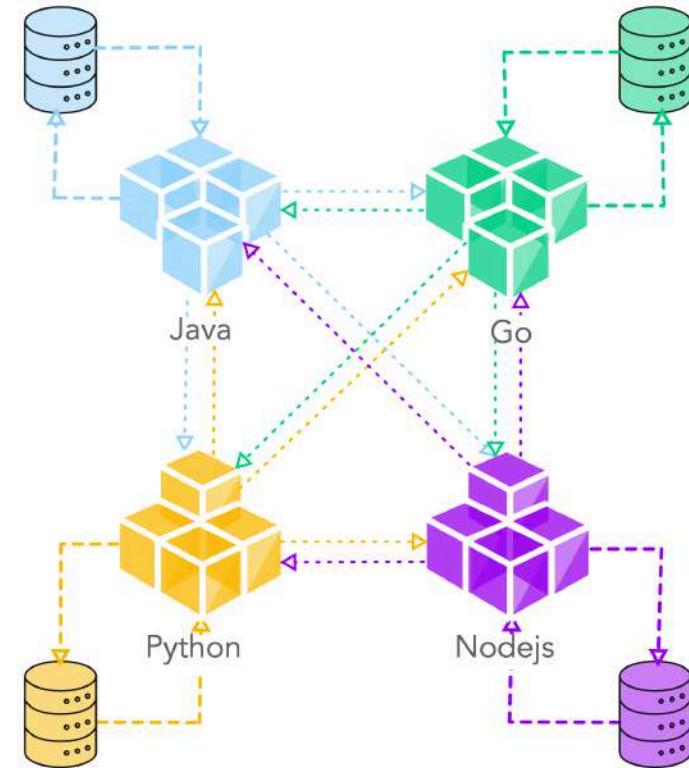
Why do we need an Ingress?

Service: If each microservice is exposed using **LoadBalancer** Service type

- Separate load balancer is provisioned per microservice (especially in cloud environments)

Ingress: Multiple Microservices => One LB

- Routes incoming HTTP(S) traffic to backend services
- **Path-Based Routing:** Forwards requests based on URL paths (e.g., `/currency-exchange`, `/currency-conversion`)
- **Host-Based Routing:** Directs traffic based on host names (e.g., `exchange.example.com`, `conversion.example.com`)





Using an Ingress - An Example

Path of Request: Ingress > Service > Pod(s)

- Ingress Configuration: Path Based
 - /currency-exchange > Service currency-exchange
 - /currency-conversion > Service currency-conversion

Needs:

- Service currency-exchange - Type ClusterIP or NodePort
- Service currency-conversion - Type ClusterIP or NodePort

```
# NOT COMPLETE YAML
kind: Ingress
spec:
  rules:
  - http:
      paths:
      - path: /currency-exchange
        pathType: Prefix
        backend:
          service:
            name: currency-exchange
            port:
              number: 80
      - path: /currency-conversion
        pathType: Prefix
        backend:
          service:
            name: currency-conversion
            port:
              number: 80
```



Why do we need an Ingress Controller?

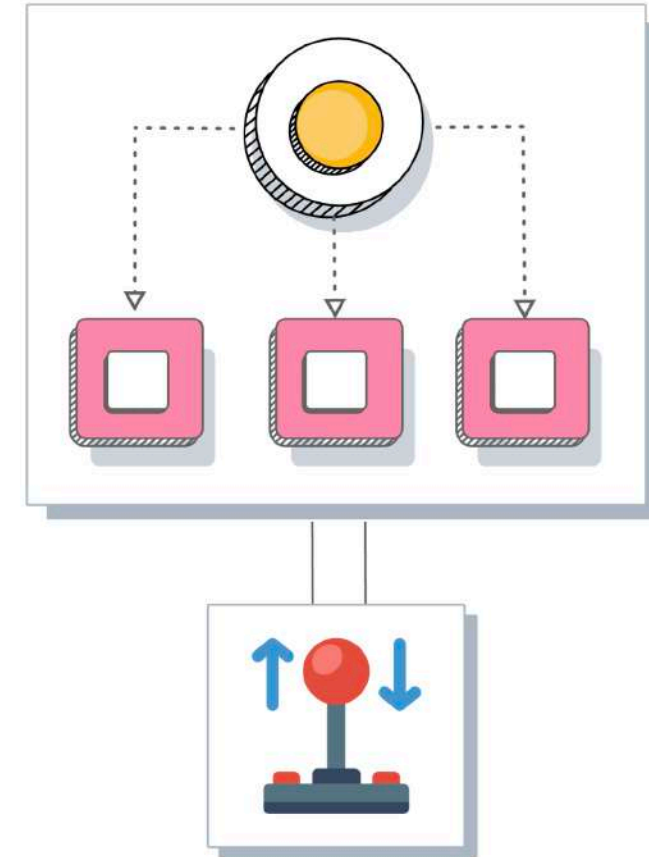
Ingress Is Only a Configuration

- **No Execution Logic:** A declarative object - defines routing rules but doesn't perform any routing.
- **Stored in etcd:** But has no effect

Ingress Controller Applies the Rules

- **Monitors Resources:** For changes to Ingress objects
- **Translates Rules:** Converts Ingress definitions into actual configurations

Examples: NGINX Ingress Controller (most widely used), GKE built-in Ingress controller, AWS ALB Ingress Controller (for Amazon EKS)





Ingress vs LoadBalancer Service

Feature	Ingress	LoadBalancer Service
Purpose	Acts as an router for multiple Services	Exposes a single Service externally
Scope	Can route to many Services based on host/path	Tied to one Service only
External IP	Shared across all routes via Ingress controller	Each Service gets a separate external IP
Best For	Web applications, microservices behind shared entry point	Exposing a standalone API, database, or service directly



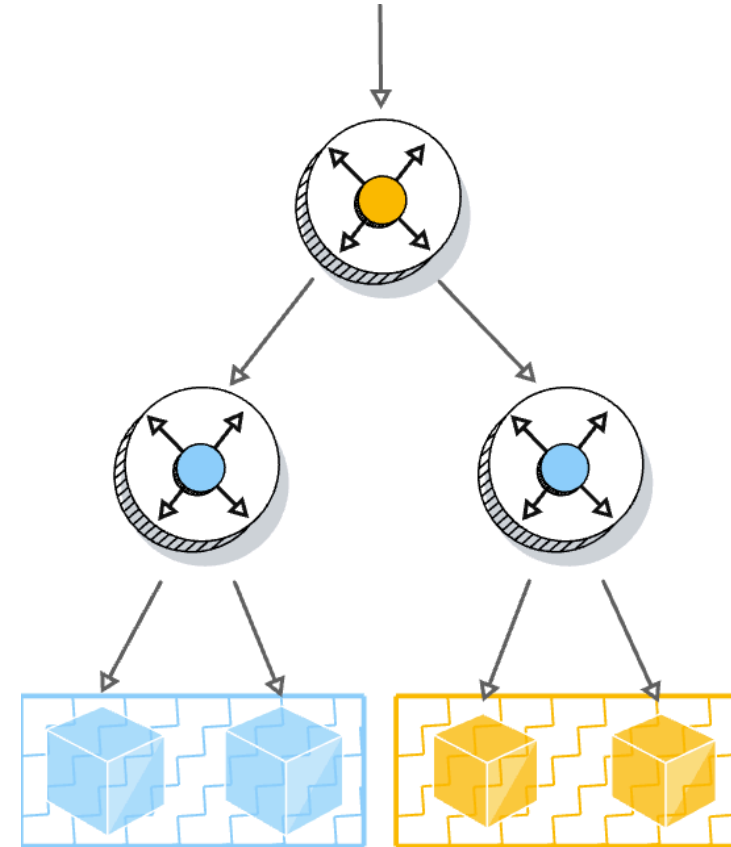
What is the need for Gateway API?

Gateway API: New way to manage traffic into Kubernetes

- More flexible and powerful than the Ingress API
- **Supports More Than HTTP(S)**: Works with HTTP, HTTPS, TCP, UDP, gRPC, and TLS
- **Better Debugging Info**: Gives clear status updates, making it easier to monitor and troubleshoot

Clear Team Roles: Clear separation of roles

- Platform teams manage Gateways (**GatewayClass** - Can be shared across the Cluster - Routes from different namespaces can share the same Gateway)
- App teams define routes





Gateway Multi-tenant Traffic Routing example [1/4]

```
# OVERVIEW
# FLOW: Internet
# -> Gateway (port 80, in namespace 'infra')
# -> HTTPRoute (/app in namespace 'team-a',
#           /metrics in namespace 'team-b')
# -> Service (in team-a and team-b)
#
# GatewayClass (`nginx.org/gateway-controller`) >
# Gateway (`allowedRoutes: from all namespaces`)
# HTTPRoute (`/currency-exchange` to `currency-exchange` service)

# WHAT: Defines a reusable gateway configuration
# BACKGROUND: Gateway controller detects it
# The controller takes ownership of any Gateway objects
# that reference this GatewayClass.
# No infra is provisioned YET!
apiVersion: gateway.networking.k8s.io/v1
kind: GatewayClass
metadata:
  name: shared-gateway-class
spec:
  controllerName: nginx.org/gateway-controller
```



Gateway Multi-tenant Traffic Routing example [2/4]

```
# WHAT: Shared external entry point defined in 'infra'
# BACKGROUND: Gateway Controller configures the
# Gateway (NGINX, Envoy, Cloud LoadBalancer, ..)
# to listen on the configured port
apiVersion: gateway.networking.k8s.io/v1
kind: Gateway
metadata:
  name: shared-gateway
  namespace: infra
spec:
  gatewayClassName: shared-gateway-class
  listeners:
    - name: http
      protocol: HTTP
      port: 80
      allowedRoutes:
        namespaces:
          from: All
  # Allows HTTPRoute objects from any namespace to bind
```



Gateway Multi-tenant Traffic Routing example [3/4]

```
# WHAT: Routing rules created in 'team-a' namespace
# BACKGROUND: This HTTPRoute links the shared Gateway
# Gateway (NGINX, Envoy, Cloud LoadBalancer, ..) is updated
# Match any HTTP request with prefix /currency-exchange
# Forward requests to `currency-exchange`
apiVersion: gateway.networking.k8s.io/v1
kind: HTTPRoute
metadata:
  name: route-to-app
  namespace: team-a
spec:
  parentRefs:
    - name: shared-gateway
      namespace: infra
  rules:
    - matches:
        - path:
            type: PathPrefix
            value: /currency-exchange
      backendRefs:
        - name: currency-exchange # PRE EXISTING SERVICE
          port: 80
```



Gateway Multi-tenant Traffic Routing example [4/4]

```
# WHAT: Another route in 'team-b' namespace
# BACKGROUND: This HTTPRoute also attaches to the
# shared Gateway and matches /metrics. Shows how
# multiple teams can use one Gateway
# in different namespaces
apiVersion: gateway.networking.k8s.io/v1
kind: HTTPRoute
metadata:
  name: route-to-metrics
  namespace: team-b
spec:
  parentRefs:
    - name: shared-gateway
      namespace: infra
  rules:
    - matches:
        - path:
            type: PathPrefix
            value: /metrics
  backendRefs:
    - name: metrics-service # PRE EXISTING SERVICE
      port: 80
```




How is Gateway different from Ingress?

Feature	Ingress	Gateway API
Protocol Support	HTTP, HTTPS ONLY	HTTP, HTTPS, TCP, UDP, gRPC, and TLS
Path Matching	✓ Yes	✓ Yes
Role Separation	✗ No	✓ Yes (Infra teams manage GatewayClass & Gateway App teams define HTTPRoute and other routes)
Multi Namespace Support	⊘ Limited	✓ Namespaced Routes
Status Reporting	⚠ Minimal (Silent failures)	✓ Detailed



Kubernetes Networking - Scenarios [1/2]

Scenario	Solution
How to prevent API consumers from being impacted by API changes in a Kubernetes microservices app?	Use an Ingress or Gateway API with path based or host based routing so different API versions (for example <code>/v1</code> , <code>/v2</code>) can route to different services
You want enable ONLY internal communication between your microservice deployments in a Kubernetes Cluster	Create a Service of type ClusterIP so the service is reachable only inside the cluster
Your frontend application must be publicly accessible from the internet while backend services remain internal	Expose the frontend using LoadBalancer Service or Ingress or Gateway API, while backend services use ClusterIP Service
You want to test a service quickly from outside the cluster without configuring a load balancer	Use a NodePort Service and access it using <code>http://<NodeIP>:<NodePort></code>



Kubernetes Networking - Scenarios [2/2]

Scenario	Solution
Your application consists of many microservices such as <i>orders</i> , <i>payments</i> , and <i>inventory</i> and each should be reachable through different paths	Use Ingress or Gateway API Path Based Routing (for example <i>/orders</i> , <i>/payments</i> , <i>/inventory</i>)
Different teams deploy services in separate namespaces, but all must share a single entry point	Use Gateway API with HTTPRoute objects in each namespace attached to the same Gateway
A company platform team wants to control infrastructure while application teams only define routing rules	Use Gateway API role separation where platform team manages GatewayClass and Gateway, and app teams create HTTPRoute



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Getting Started with

Multiple Regional Clusters



Need for Multiple Regional GKE Clusters

You've deployed a critical application on **Google Kubernetes Engine (GKE)**

Can a regional outage affect your application's availability?

- Yes, if your workload is limited to a single region

What if you need **global reach** for your application?

- Use **Multiple Regional Clusters in GKE**





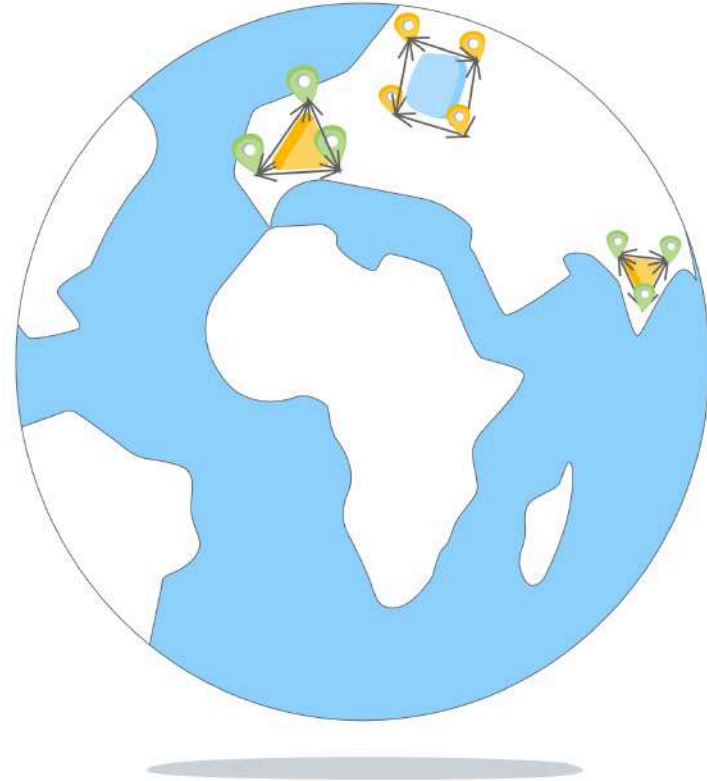
What are Multiple Regional GKE Clusters?

Multiple Regional Clusters: Deploy separate GKE clusters in **two or more Google Cloud regions**

- You are protected from regional failures
- You can route traffic to the nearest healthy region

Useful for: Global availability, Compliance with data residency, Failover across regions

Global Load Balancing: Can be used to load balance between multiple clusters across different regions





Load Balancing For Multiple Regional GKE Clusters

Multi-cluster Ingress: Deploy a single, global ALB that routes traffic to Service objects running in multiple GKE clusters across different regions

Multi-cluster Gateways: Manage traffic for services deployed across multiple GKE clusters





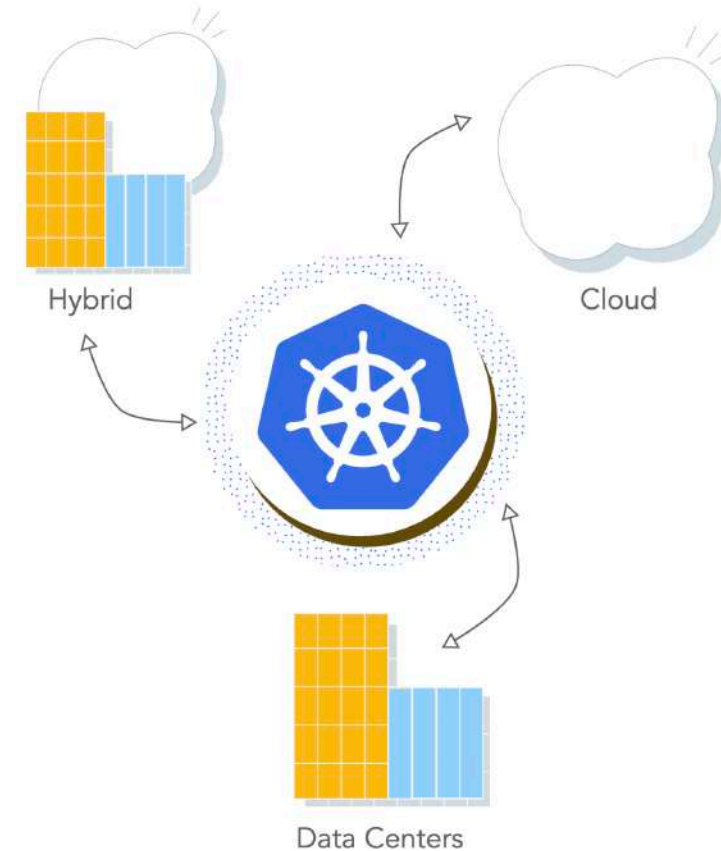
Multi-Cluster Ingress – GKE Multi-Cluster Load Balancing

Multi-Cluster Ingress (MCI) : Extends **Ingress** to **deploy shared ALB across clusters**

Key Capabilities:

- **Single Global IP (VIP)** — consistent entry point across clusters and regions
- **Cross-region traffic failover** — based on health checks
- **Proximity-based routing** — via **Anycast** to reduce client latency

Fully managed and integrates with **Cloud Load Balancing**





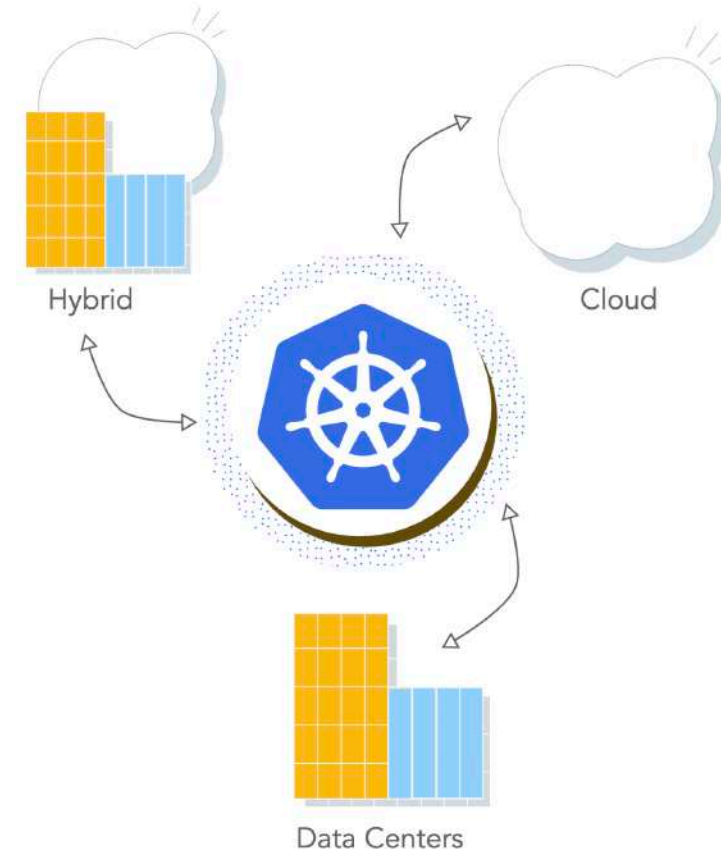
Multi-Cluster Gateway – Modern GKE Load Balancing

Multi-Cluster Gateway (MCG): Built on the Kubernetes Gateway API

Designed for **flexibility**, **separation of concerns**, and **future extensibility**

Key Benefits:

- **Standardized API** – Vendor-neutral, extensible traffic routing
- **Global L7 Load Balancing** – Works with Google's global load balancer





Multi-Cluster Ingress vs Multi-Cluster Gateway in GKE

Feature	Multi-Cluster Ingress (MCI)	Multi-Cluster Gateway (MCG)
API Base	Kubernetes <i>Ingress</i> API	Kubernetes Gateway API (next-gen standard)
Global IP	✅ Single global IP	✅ Single global IP
Configuration Model	Simpler, <i>Ingress</i> -only YAML	Modular: <i>GatewayClass</i> , <i>Gateway</i> , <i>HTTPRoute</i> , etc.



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Getting Started with Scaling Kubernetes



What are the Scaling Options in Kubernetes?

Scaling Layer	What It Scales	Example Resource	Applicable to AutoPilot
Pod-level (Horizontal)	Number of Pod replicas	HorizontalPodAutoScaler (HPA)	Yes
Pod-level (Vertical)	CPU/Memory of each Pod	VerticalPodAutoScaler (VPA)	Yes
Cluster-level	Number of Nodes in cluster	Cluster AutoScaler	Managed automatically by Autopilot



What is Horizontal Pod AutoScaler (HPA)?

Purpose: Automatically scales the number of Pod replicas in a Deployment based on observed metrics

- CPU, memory, or custom

Requirement: Metrics Server add-on should be installed & running to expose Pod or Node CPU/memory usage

```
apiVersion: autoscaling/v2
kind: HorizontalPodAutoscaler
metadata:
  name: hello-world-rest-api-hpa

spec:
  scaleTargetRef:
    apiVersion: apps/v1
    kind: Deployment
    name: hello-world-rest-api

  minReplicas: 2
  maxReplicas: 10

  metrics:
  - type: Resource
    resource:
      name: cpu
      target:
        type: Utilization
        averageUtilization: 70
```



What is Vertical Pod AutoScaler?

Purpose: Automatically recommends or updates CPU/memory resource requests & limits for Pods, adjusting “vertically” instead of adding replicas

Requirement 1: A component exposing the resource.metrics.k8s.io API (commonly the Metrics Server) must be running so VPA can fetch usage data

Requirement 2: Needs VPA Controllers need to be installed

```
apiVersion: autoscaling.k8s.io/v1
kind: VerticalPodAutoscaler
metadata:
  name: hello-world-rest-api-vpa

spec:
  targetRef:
    apiVersion: apps/v1
    kind: Deployment
    name: hello-world-rest-api

  resourcePolicy:
    containerPolicies:
      - containerName: "*"
        minAllowed:
          cpu: "100m" # (0.1 vCPU)
          memory: "128Mi"
        maxAllowed:
          cpu: "1"
          memory: "1Gi"
```



How is HPA different from VPA?

Aspect	HPA	VPA
Purpose	Scales the number of pod replicas based on load	Adjusts CPU and memory resources for pods dynamically
Scaling Mechanism	Adds/removes pod replicas to meet demand	Increases/decreases resources (CPU/memory) of individual pods
Integration	Built-in to Kubernetes by default	Not built-in; installed separately as an add-on
Impact on Pods	Creates/destroys pods to handle load	Typically applies resource limits on new pods
What metrics does it use?	Current usage metrics - CPU, memory, or custom metrics	Historical resource usage (CPU/memory)

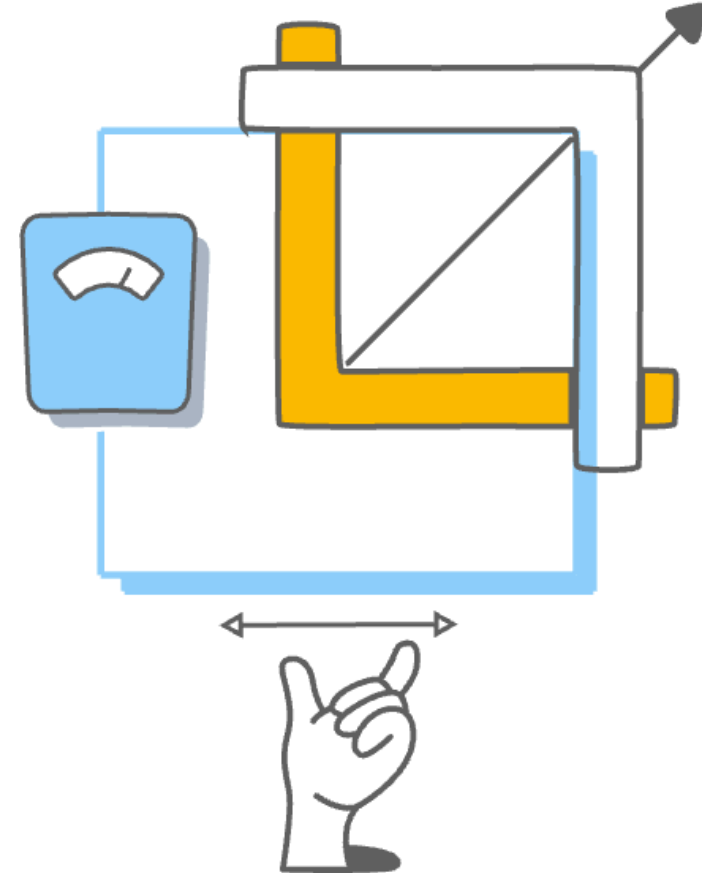


What is Cluster AutoScaler?

Cluster AutoScaler: Automatically adjusts the number of nodes in your cluster based on the pending workload

- If Pods can't be scheduled due to insufficient resources, it **adds nodes**
- If nodes are underutilized and their Pods can be moved elsewhere, it **removes nodes**

Integrates with Cloud API: Works directly with your **cloud provider's API** (e.g., AWS, GCP, Azure) to scale **infrastructure** (not just Pods)





How Does Cluster AutoScaler Works?

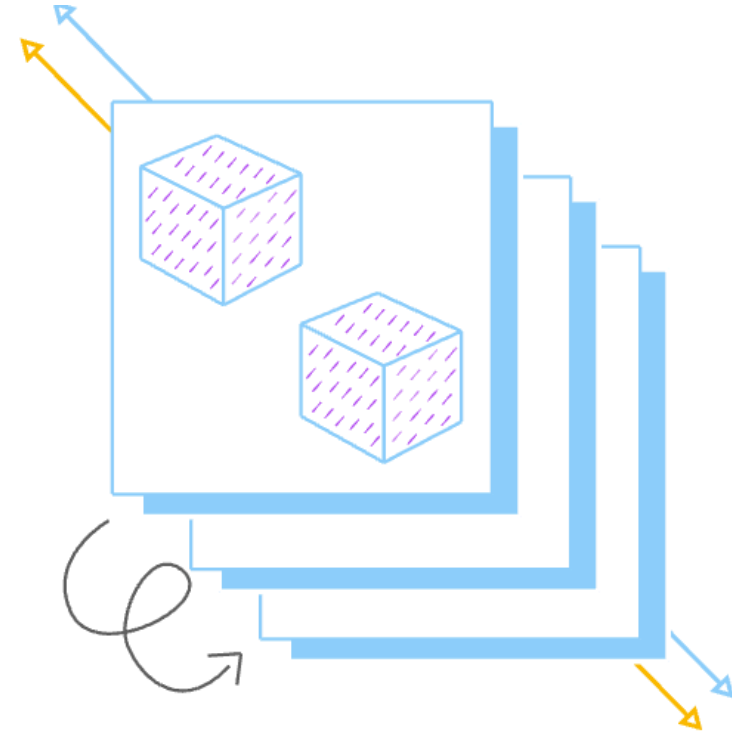
Scale Up: A new Pod is unschedulable (e.g., not enough memory or CPU)

- AutoScaler requests a new node

Scale Down: A node is underutilized

- Pods should be moved to other nodes
- Over a period of time, the node is slowly drained and removed

Supported Cloud Providers: Google Cloud (GKE), Amazon Web Services (EKS), Microsoft Azure (AKS), ...





How does HPA compare to VPA and Cluster AutoScaler?

Feature	HPA	VPA	Cluster AutoScaler
What is scaled?	Pod replicas	Pod CPU/memory requests	Nodes in the cluster
Metrics Used	Current utilization - CPU, memory, custom	Historical usage	Current metrics - Node Utilization, Pending Pods (scheduling status), etc.
Setup Complexity	Low	Medium (Needs VPA components)	Medium (Needs support from Cloud Provider)



Kubernetes Scaling - Scenarios

Scenario	Solution
Your cluster has many nodes that remain mostly idle for long periods	Use Cluster Autoscaler to automatically remove underutilized nodes
An e-commerce platform receives sudden traffic spikes during a sale event	Combine HPA (scale pods) with Cluster Autoscaler (scale nodes)
A batch processing application requires more memory, but does not need multiple replicas	Use VPA to recommend or update resource requests
New pods remain in Pending state because the cluster does not have enough CPU or memory	Use Cluster Autoscaler to automatically add new nodes



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Getting Started with Kubernetes Storage



Why Is Persistence Important?

Ephemeral Pod Storage: Data disappears when Pods restart, move, or terminate

Persistence: Store data beyond the Pod lifecycle

- **Protect critical data:** Databases, logs, and user files stay safe
- **Support stateful apps:** Databases and message brokers require stable storage
- **Enable backups:** Allow recovery after failures
- **Provide read-only data:** Share static files safely across containers





What is the need for Volumes?

Provides Storage: Volumes provide storage for Pods

Shared filesystem: Multiple containers in a Pod can read and write shared data from a Volume

Enable persistence: Data in a Volume can survive container restarts





Useases for Kubernetes Volumes [1/2]

hostPath: Mount a file or directory from the node filesystem

nfs: Shared storage accessible by Pods across multiple nodes

persistentVolumeClaim: Request persistent storage managed by Kubernetes

configMap: Provide configuration values to containers as files

```
## Usual Deployment Spec
spec:
  containers:
    - name: app-container
      image: busybox
      command: ["sleep", "3600"]

    volumeMounts:
      - name: host-vol
        mountPath: /mnt/host
        readOnly: true # for safety

  volumes:
    - name: host-vol
      hostPath:
        path: /var/log
        type: Directory
```



Useases for Kubernetes Volumes [2/2]

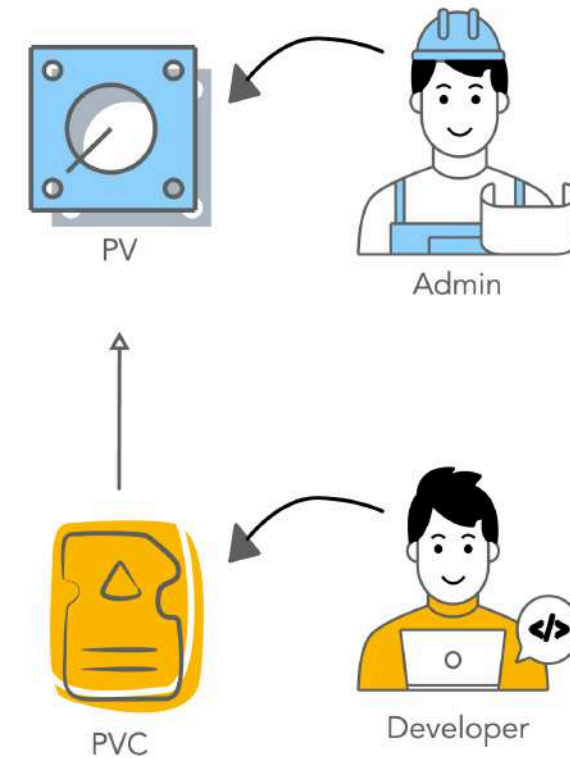
```
## Usual Deployment Spec
spec:
  containers:
    - name: app-container # Followed by Usual Container Definition
      volumeMounts:
        - name: config-vol
          mountPath: /etc/config # CONFIG MAP
        - name: nfs-vol
          mountPath: /mnt/nfs # NFS
        - name: pvc-vol
          mountPath: /mnt/data # PERSISTENT VOLUME CLAIM

  volumes:
    - name: config-vol
      configMap:
        name: app-config # ConfigMap must exist in the same namespace
    - name: nfs-vol # NFS volume
      nfs:
        server: 10.0.0.12
        path: /shared-data # NFS server must allow client access
    - name: pvc-vol
      persistentVolumeClaim:
        claimName: data-pvc # The PVC must exist before Pod creation
```


Static vs Dynamic Storage Provisioning - Static [1/2]



- 1: Create Physical Storage:** Admin manually provisions a disk (e.g., GCE Persistent Disk)
- 2: Create PV:** Admin creates a PersistentVolume pointing to the disk
- 3: Request Storage:** Developer creates a PersistentVolumeClaim asking for storage
- 4: Binding:** Kubernetes finds the matching PV and "binds" it to the PVC
- 5: Mount:** The Pod uses the PVC to mount the storage into its container



Static vs Dynamic Storage Provisioning - Dynamic [2/2]



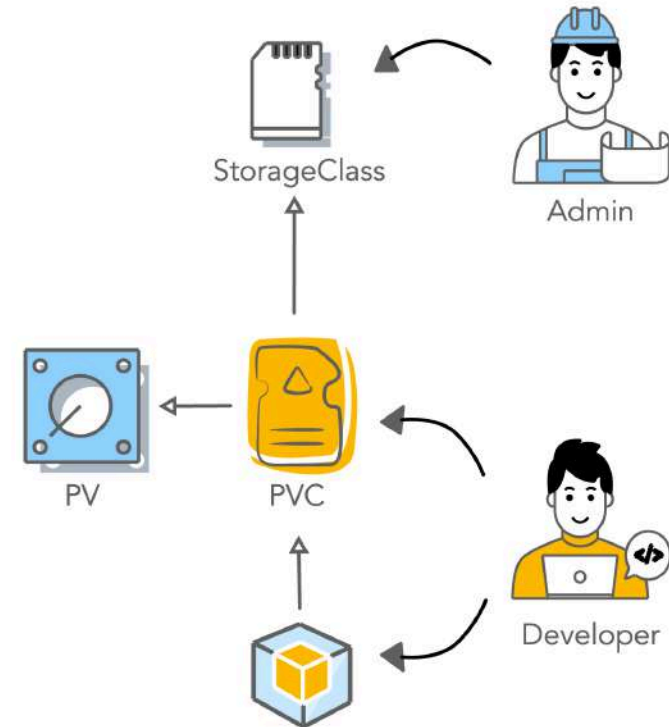
1: Create StorageClass: Admin defines a StorageClass specifying how to create storage

2: Request Storage: Developer creates a PVC that references the storageClassName

3: Auto-Provision: Kubernetes uses the StorageClass to provision a new disk

4: Auto-PV: K8S creates a PersistentVolume object for the new disk and binds to PVC

5: Mount: The Pod starts and uses the newly created storage without admin intervention





What Is Static Volume Provisioning?

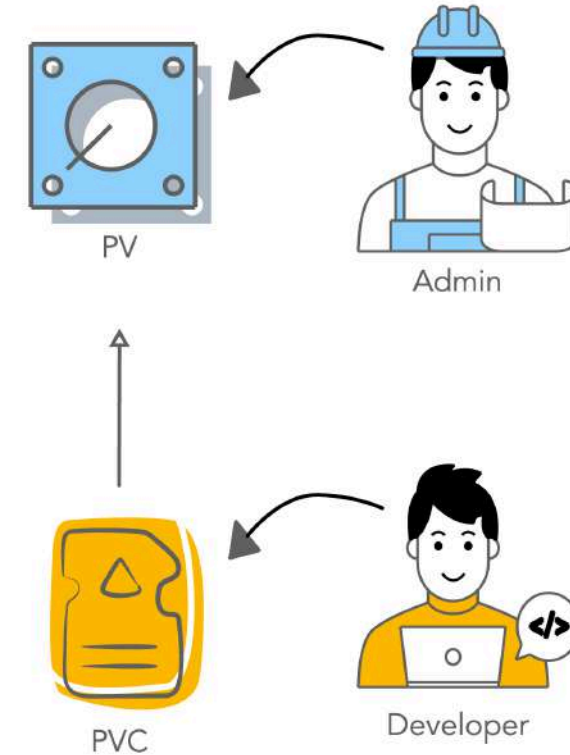
Admin-Created PVs: Predefined storage with fixed size & access rules

- **No automation:** PVs must be created manually

PVCs Request Storage: PVC binds only to suitable PVs

Steps

- STEP 1: Create Disk and PersistentVolume (Admin)
- STEP 2: Create PersistentVolumeClaim (Dev)
- STEP 3: Map Pod To PersistentVolumeClaim (Dev)





STEP 1: Create Disk and PersistentVolume

apiVersion/kind: PersistentVolume

metadata.name: Name of the PV

spec - PV Specification

- **spec.capacity.storage:** Size of the volume (1Gi)
- **spec.accessModes:** How Pods can use the volume (ReadWriteOnce)
- **spec.csi:** Defines the connection to an external storage resource

```
apiVersion: v1
kind: PersistentVolume
metadata:
  name: static-pv
spec:
  capacity:
    storage: 1Gi
  accessModes:
    - ReadWriteOnce # One node RW
  storageClassName: manual
  persistentVolumeReclaimPolicy: Retain
  csi:
    driver: pd.csi.storage.gke.io
    volumeHandle:
projects/PROJECT_ID/zones/us-central1-
a/disks/static-disk
    fsType: ext4
```



STEP 2: Create PersistentVolumeClaim

kind: PVC: User request for persistent storage

accessModes: Must match PV's access mode

resources.requests.storage: Minimum required size

storageClassName: Must match the PV's StorageClass to bind

Kubernetes finds the matching PV → binds PVC to PV

```
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: static-pvc
spec:
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 1Gi
  storageClassName: manual
```



STEP 3: Map Pod To PersistentVolumeClaim

Note: Example is a Pod Spec

volumeMounts.mountPath: Directory in the container where the PVC is mounted

volumes.name: Logical volume name that links Pod → PVC

persistentVolumeClaim.claimName: PVC name (`static-pvc`)

Pod automatically gets access to the PV bound to that PVC using path `/usr/share`

```
apiVersion: v1
kind: Pod
metadata:
  name: pvc-pod
spec:
  containers:
    - name: app
      image: busybox
      command: ["sleep", "3600"]
      volumeMounts:
        - name: storage-vol
          mountPath: /usr/share
  volumes:
    - name: storage-vol
      persistentVolumeClaim:
        claimName: static-pvc
```



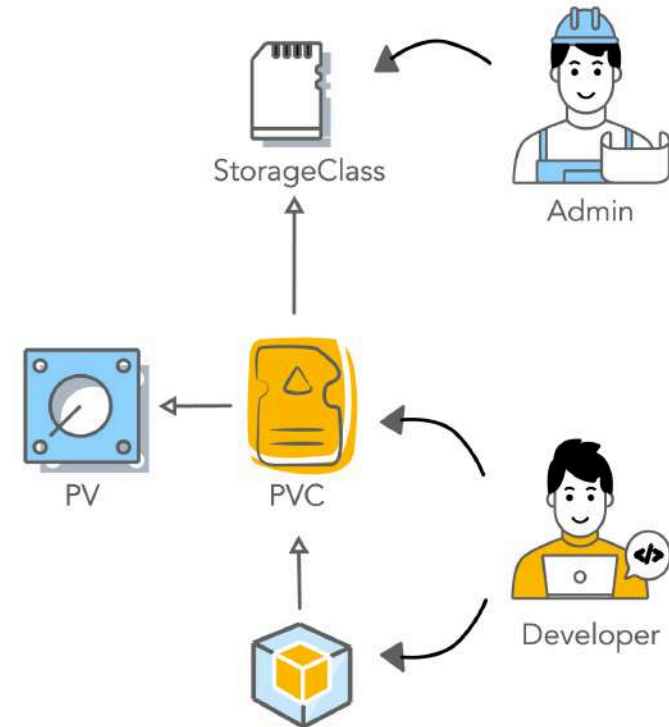
What Is Dynamic Volume Provisioning?

Automatic Volume Creation: PVs are created when a PVC is created

- **Uses StorageClass:** Defines provisioner type, disk type, filesystem
- **No Manual Work Needed:** Removes admin burden for most workloads

Steps

- STEP 1: Create StorageClass (Admin)
- STEP 2: Create PersistentVolumeClaim (Dev)
- STEP 3: Map Pod to PersistentVolumeClaim (Dev)





STEP 1: Create StorageClass (Dynamic Provisioning Rules)

StorageClass: Defines *how* Kubernetes should create storage dynamically

- Specifies the provisioner (AWS EBS, GCP PD, Azure Disk, etc.)

Configuration

- **provisioner:** CSI driver that creates actual disk
- **parameters:** Defines storage behavior (type, filesystem)
- **reclaimPolicy: Delete:** Delete the disk when PVC is deleted
- **volumeBindingMode: WaitForFirstConsumer:** Creates disk only when Pod is scheduled

```
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
  name: fast-storage
provisioner: pd.csi.storage.gke.io
parameters:
  type: pd-ssd      # Persistent Disk
  fsType: ext4      # Filesystem
reclaimPolicy: Delete
volumeBindingMode: WaitForFirstConsumer
`
```




STEP 2: Create PersistentVolumeClaim

PVC requests storage: 5Gi volume needed

Matches StorageClass: **fast-storage** - matches name of StorageClass

- *Kubernetes automatically creates a PV that satisfies this PVC when needed*

```
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: dynamic-pvc
spec:
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 5Gi
  storageClassName: fast-storage
```



STEP 3: Map Pod to PersistentVolumeClaim

volumeMounts.mountPath: Where the container sees the storage (/data)

persistentVolumeClaim.claimName: PVC name (dynamic-pvc)

Kubernetes attaches the dynamically created PV → PVC → Pod automatically

```
apiVersion: v1
kind: Pod
metadata:
  name: dynamic-pod
spec:
  containers:
    - name: app
      image: busybox
      command: ["sleep", "3600"]
      volumeMounts:
        - name: dynamic-vol
          mountPath: /data # Storage visible inside
  container
  volumes:
    - name: dynamic-vol
      persistentVolumeClaim:
        claimName: dynamic-pvc
```



Key Concepts in Kubernetes Storage

PersistentVolume (PV): Cluster-level storage resource created by **admin**

- Defines size, storage type, access mode

PersistentVolumeClaim (PVC): User request for storage

- Kubernetes binds PVC → matching PV

Pod Mapping: Pod uses PVC to mount storage

- Pod never talks directly to PV
- PVC becomes the “bridge” between Pod and underlying storage





Scenarios - Kubernetes Storage

Scenario	Solution
A database Pod must keep data even after rescheduling	Use persistent storage backed by a PV/PVC instead of container filesystem
Application logs must survive Pod restarts	Write logs to persistent storage instead of only inside the container
Storage disk to be attached with a Pod is provisioned manually by the platform team	Use Static Provisioning with pre-created disk and PV
Each new workload should get a disk automatically	Use Dynamic Provisioning with a StorageClass
Teams want faster self-service of storage for Pods without creating tickets	Admin creates StorageClass and teams create PVCs



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Getting Started with

Kubernetes Observability



What is Observability?

Scenario: Your application is running slowly or returning errors

- You know something is wrong but do not know **where** or **why**

Observability: Ability to understand what is happening inside a system

- Based on **external signals**
 - **Logs** – Records of events
 - **Metrics** – Numerical measurements over time
 - **Traces** – End-to-end request flow across services





Why Do We Need Observability in GKE?

Workloads are deployed on (GKE)

Teams need **visibility into cluster health and application behavior**

- Monitor **resource utilization** – CPU, memory, network
- Detect **errors and failures quickly**
- Diagnose problems across **Pods, containers, and nodes**

Need **monitoring, logging, tracing, dashboards, and alerting**

- Without building custom observability solutions





GKE – Built-in Monitoring and Logging

Monitoring & Logging: Enabled by default

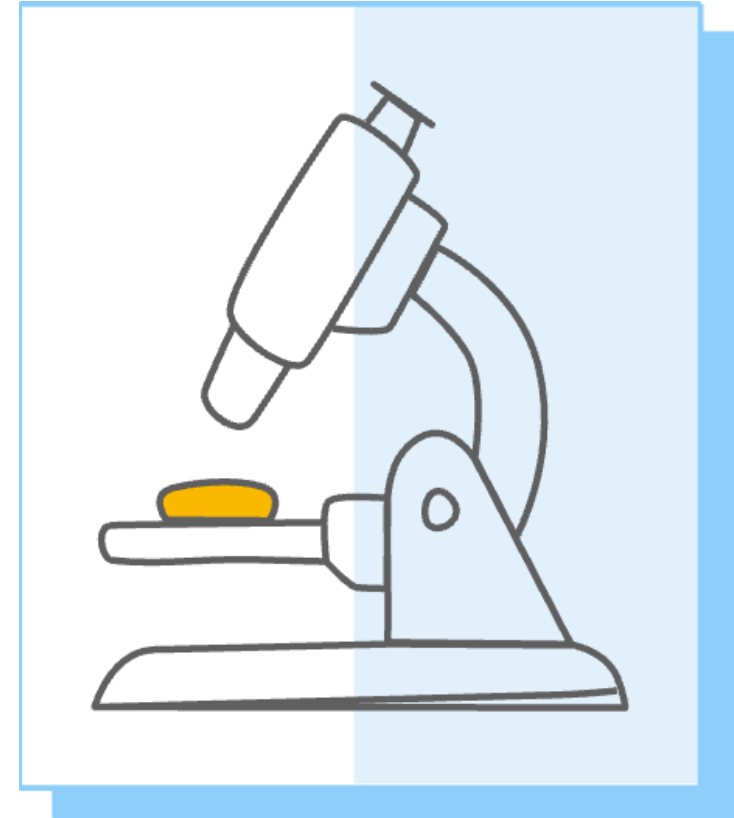
- Collects: System/App logs, Pod and node metrics

Pre-configured Dashboards: Cluster utilization, Pod health, App performance

No manual agents required: GKE automatically configures everything needed

Customize As You Need: Create

- Custom Dashboards: Metrics, Logs, SLO tracking, etc.
- Your Own Alerts: Based on metrics





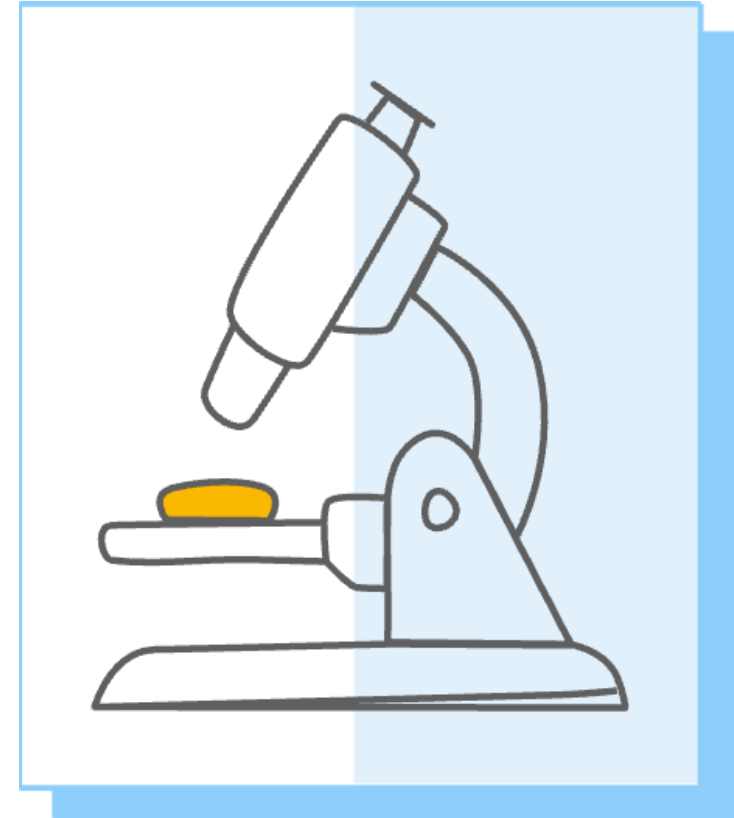
Centralized Insights for Kubernetes Workloads

Centralized Visibility: Monitor multiple microservices in one place

Track important signals

- CPU and memory usage
- pod restarts
- latency spikes
- error rates

Faster troubleshooting: Correlate logs, metrics, and events





Observability for GKE – Built-In Integrations [1/2]

Observability for GKE provides built-in monitoring and logging

Integrated with Google Cloud Observability:

- **Cloud Monitoring** – Metrics, dashboards, alerting
- **Cloud Logging** – Centralized log collection
- **Managed Service for Prometheus** – Prometheus-compatible metrics collection

Enables **cluster and application visibility** without extra setup





Observability for GKE – Built-In Integrations [2/2]

Default Configuration

- System/app logs and audit logs → **Cloud Logging**
- System metrics → **Cloud Monitoring**
- Prometheus metrics → **Managed Service for Prometheus**

Provides unified observability for clusters and workloads





Additional Observability Metrics in GKE

Additional metrics packages: Can be enabled

- **Control Plane Metrics:** Monitor Kubernetes components: API Server, Scheduler, Controller Manager
- **Kube State Metrics:** Monitor Kubernetes objects - Deployments, Pods, Nodes

Third-Party Application Metrics: Use Prometheus exporters

- Example workloads: MongoDB, Redis





Why OpenTelemetry?

Challenge: Different tools and formats for logs, metrics, and traces

Solution: One standard for collecting telemetry (logs, metrics, and traces)

OpenTelemetry:

- Open-source, cloud-neutral project under CNCF
- Provides standard APIs and SDKs to collect logs, metrics, and traces
- Works across cloud platforms and supports most programming languages



Microservices Telemetry Using Managed OpenTelemetry for GKE

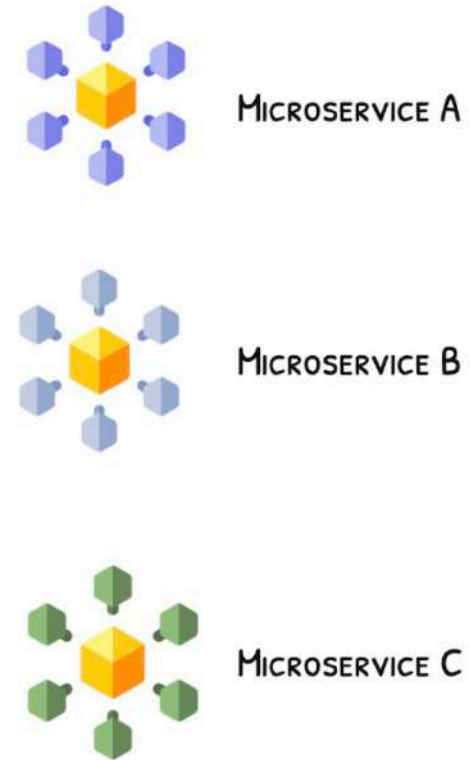


Instrumentation (Microservices): Use OpenTelemetry SDKs to generate telemetry

- Telemetry types: logs, metrics, traces
- **OTLP (OpenTelemetry Protocol):** Telemetry is sent using OTLP (Standard format)

Managed OpenTelemetry for GKE:

- Collecting OLTP data on GKE
- **Exports** to multiple backends
 - **Metrics:** Cloud Monitoring
 - **Logging:** Cloud Logging
 - **Tracing:** Cloud Trace





Export and Integrate Observability Data

Observability data can be exported to other Google Cloud services

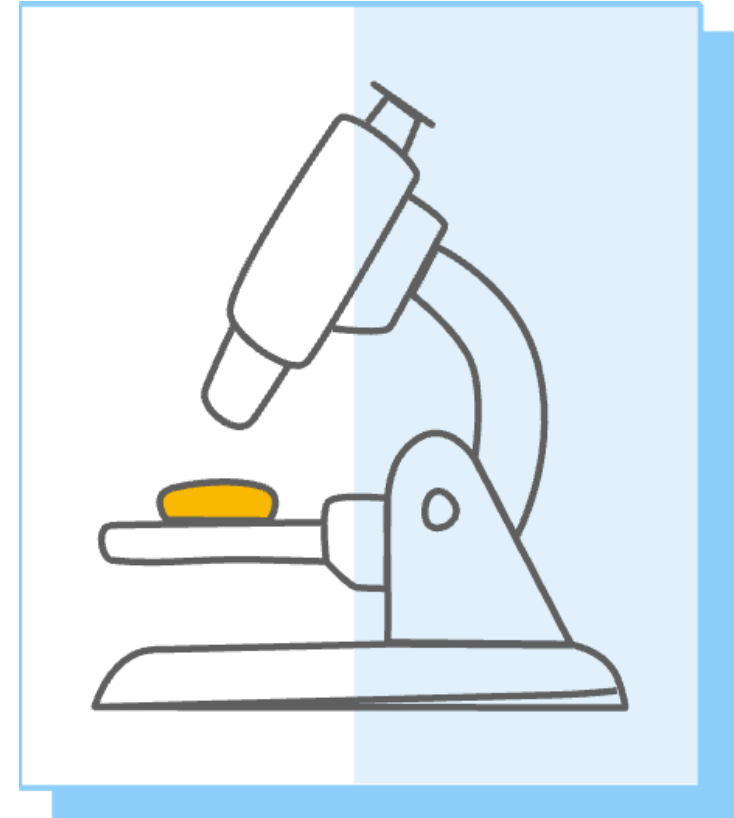
BigQuery: Long-term log storage

- Advanced analytics and reporting

Pub/Sub: Real-time streaming pipelines

- Trigger automation and alerts

Enables **integration with external monitoring workflows**





Observability for GKE – Common Scenarios

Scenario	Solution
You want to solve a production issue using logs & metrics	Use Cloud Logging and Cloud Monitoring dashboards to correlate pod logs with system metrics and detect root causes
You want to set up alerts on pod failures or CPU spikes	Define alerting policies in Cloud Monitoring using GKE metrics
You want a central view of workload health across the cluster	Use pre-configured observability dashboards or build custom ones to monitor deployments etc.
You want to track if your services meet reliability targets	Create SLOs to monitor your service performance goals using collected GKE metrics
Trigger real-time alerts based on logs based on complex automation	Export logs to Pub/Sub and set up alerts based on automation



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Getting Started with Service Mesh



Microservices Challenges

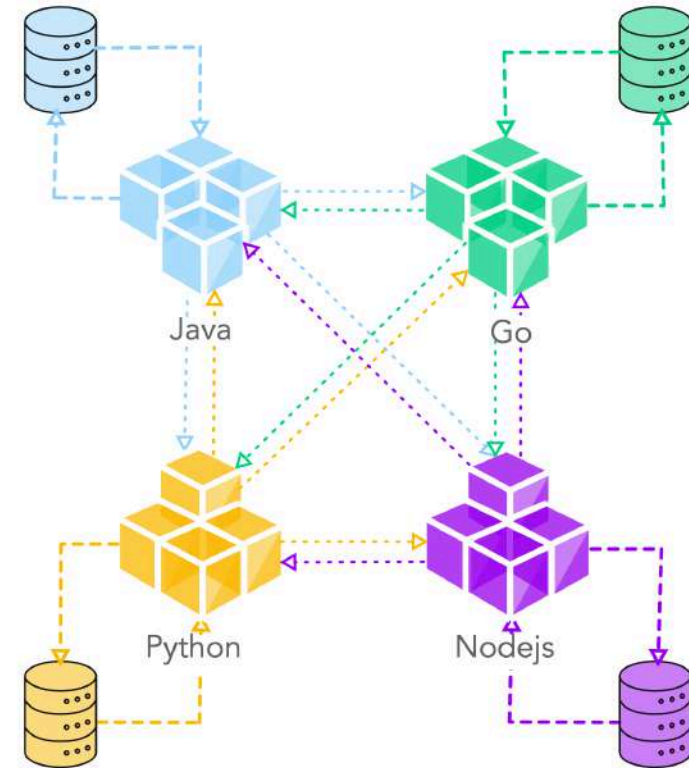
Microservices: Multiple independent services working together

- **Common Needs:** Security, observability, reliability, and release management

Challenge:

- Each service must implement these features individually
- Leads to complexity, & duplication

Question: How do we simplify this?





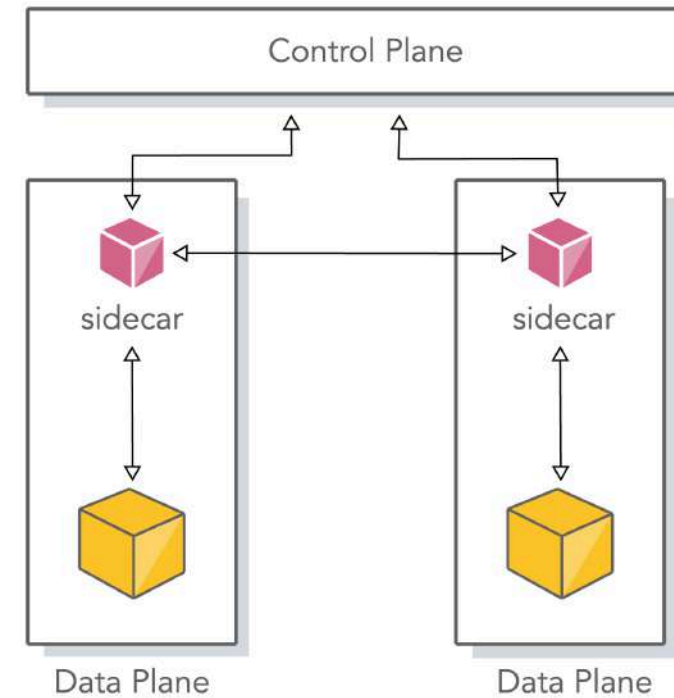
Enter Service Mesh

Service Mesh: Layer that handles common microservice features automatically

Goal: Offload security, observability, and reliability concerns from application code

How It Works:

- **Control Plane** – Configures and manages proxies
- **Data Plane** – Proxy (sidecar) beside each container
 - Handles all common features for microservices





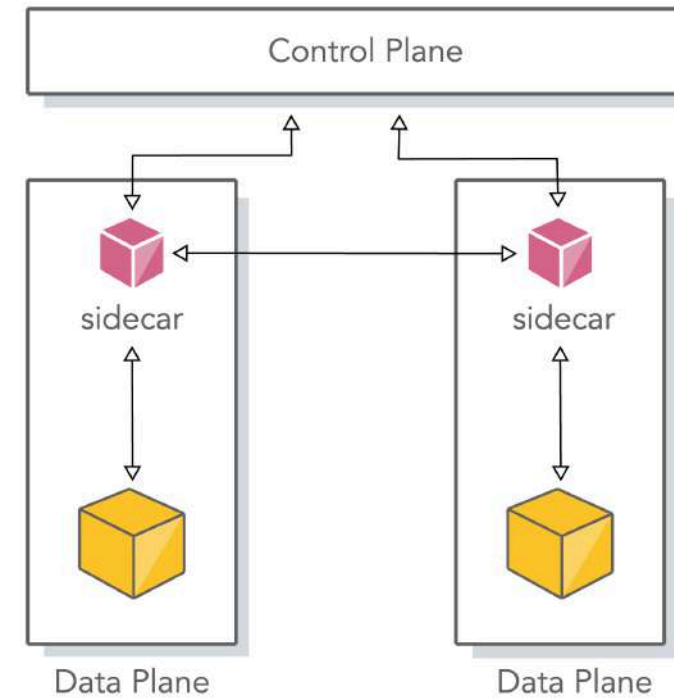
Getting Started with Istio

Istio: Popular open-source Service Mesh

- Integrates seamlessly with **Kubernetes**

Key Features:

- Security** – Authentication, authorization, mTLS, encryption
- Observability** – Metrics, logging, tracing
- Reliability** – Rate limiting, circuit breakers, retries
- Release Management** – Canary releases, A/B testing, traffic shadowing



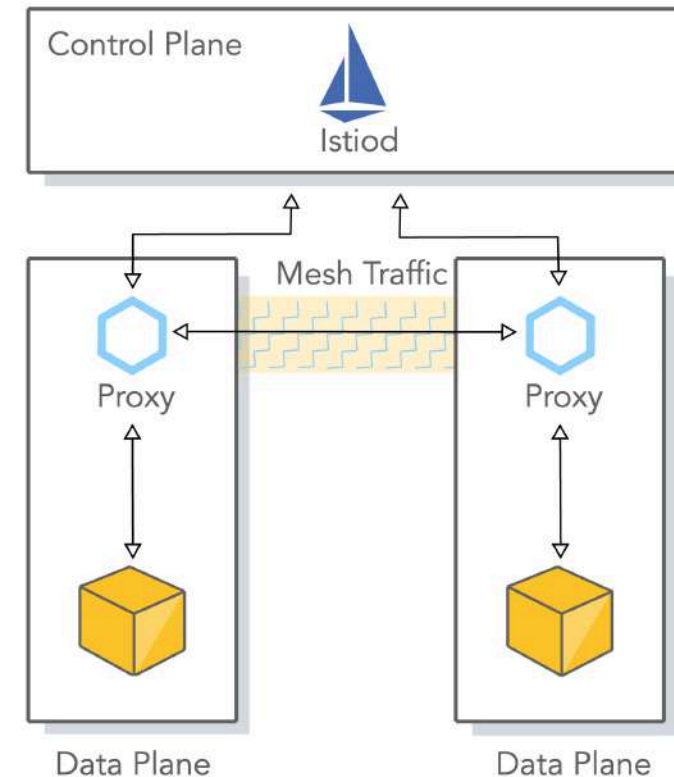


Istio in Action

Scenario: A user request flows through multiple microservices

Architecture:

- **Istiod (Control Plane):** Configure all common features
 - Sends **config** to Envoy proxies
- **Envoy Proxies (Data Plane):** Deployed as **sidecars** beside each service
 - Handle observability, traffic routing, load balancing, etc.
- **Result:** Consistent security and observability across all services
 - Developers focus on **business logic**, not infrastructure





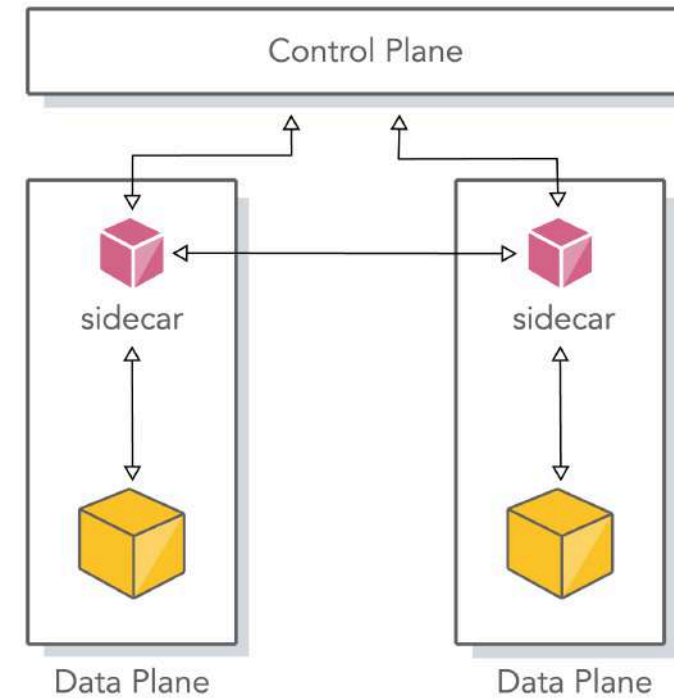
Cloud Service Mesh - Service Mesh in Google Cloud

Cloud Service Mesh: Google's fully managed service mesh

- **Offers Unified APIs:** Google Cloud (GKE, GCE), Off-cloud environments (using Istio APIs)

Features:

- **Traffic management:** Service discovery, load balancing, canary and blue green deployments, circuit breakers
- **Observability:** Metrics, Logs and Traces (including SLO tracking)
- **Security:** mTLS authentication, encryption in transit





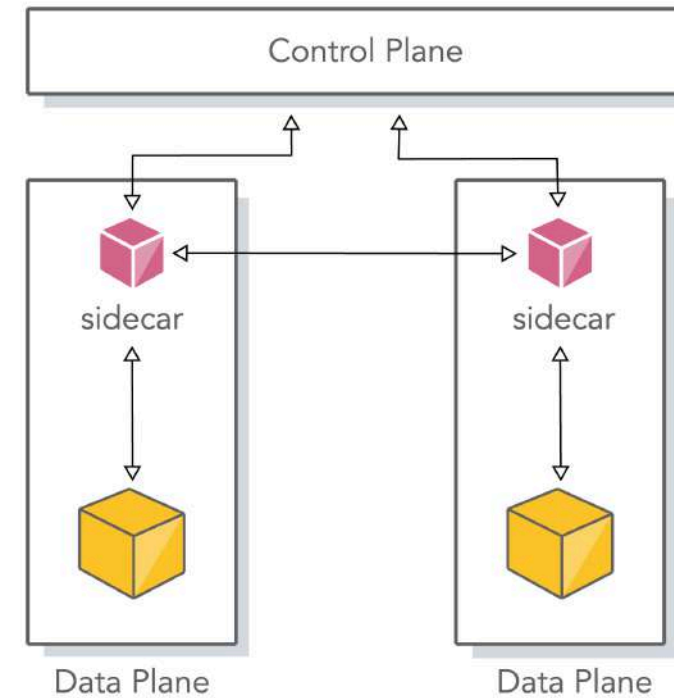
Cloud Service Mesh - Deployment options

On Google Cloud:

- **GKE:** Managed Cloud Service Mesh - managed control and data plane for GKE (recommended)
- **GCE:** Managed Cloud Service Mesh - managed control and data plane for Compute Engine with VMs (recommended)
- **Cloud Run:** Managed Cloud Service Mesh - managed control and data plane for Cloud Run (recommended)
- **GKE:** In-cluster control plane for GKE with Istio APIs (Discouraged)

Off Google Cloud:

- In-cluster control plane for Kubernetes with Istio APIs





Cloud Service Mesh - Use cases

Title	Description
Monitor SLOs	Define SLOs (Service Level Objectives). Set up alerting policies to monitor SLOs and receive alerts.
Fault injection capabilities	Allow you to simulate failures, such as crashes or delays, in specific microservices to test how the system behaves under failure conditions (test resilience)
Identify bottlenecks and debug problems	Cloud Service Mesh provides telemetry (metrics, logs and traces) between services, helping identify bottlenecks
How to perform A/B testing with zero downtime in GKE?	Service Mesh supports fine-grained routing and gradual rollouts, which are ideal for A/B testing
Encrypt all in-cluster app traffic with mutual authentication	Enable mTLS on Service Mesh - Enables transparent encryption of all pod-to-pod traffic without modifying app code



A Quick Review - Putting things into context

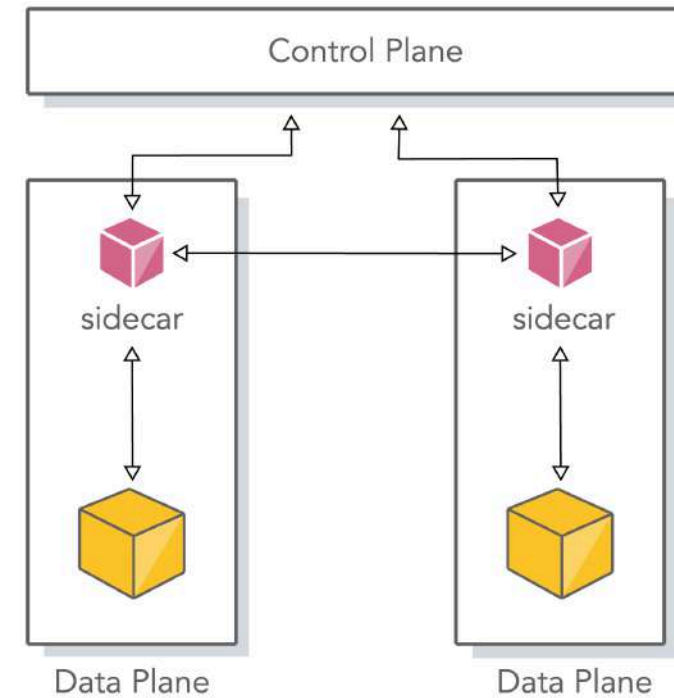
Docker: Create Container Images

Kubernetes: Container Orchestration

Google Kubernetes Engine: Kubernetes Managed Service in Google Cloud

Cloud Service Mesh: Google Cloud Service Mesh

- **Use cases:** Cross cutting concerns and testing patterns such as canary, shadow, and A/B testing





Scenarios - Service Mesh

Scenario	Solution
Every microservice team implements authentication differently	Use a Cloud Service Mesh to centralize authentication and security policies
Microservice developers should focus only on business logic	Use a Cloud Service Mesh to handle observability, load balancing, etc.
Traffic between services must be encrypted automatically (with mutual authentication)	Enable mTLS so Cloud Service Mesh proxies handle encryption
Control traffic distribution between service versions	Configure routing rules in Cloud Service Mesh
You need visibility into service-to-service communication without manual setup	Use Cloud Service Mesh's preconfigured dashboards
You want to monitor request rates, error rates, and latencies for services	Cloud Service Mesh automatically collects telemetry and feeds into Cloud Observability with no extra config



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Getting Started with Hybrid Kubernetes



The Challenge of Managing Multiple Clusters

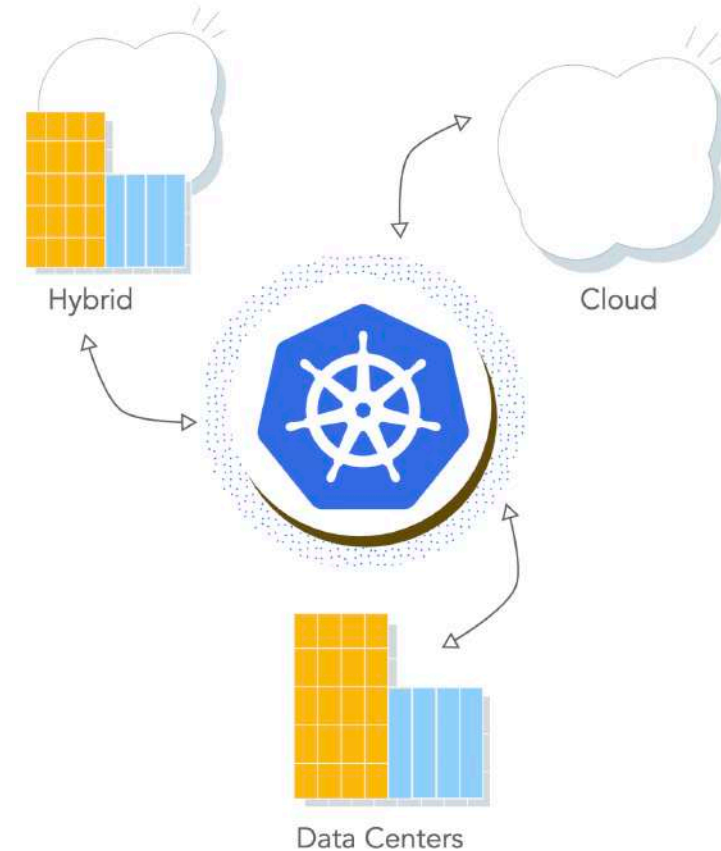
Scenario: Your applications run across multiple Kubernetes clusters

- Some clusters run on **GKE**
- Some clusters run **outside Google Cloud**

The Challenge:

- Manage clusters consistently
- Apply changes across environments
- Maintain visibility and governance

Question: Can we manage them together in a simpler way?





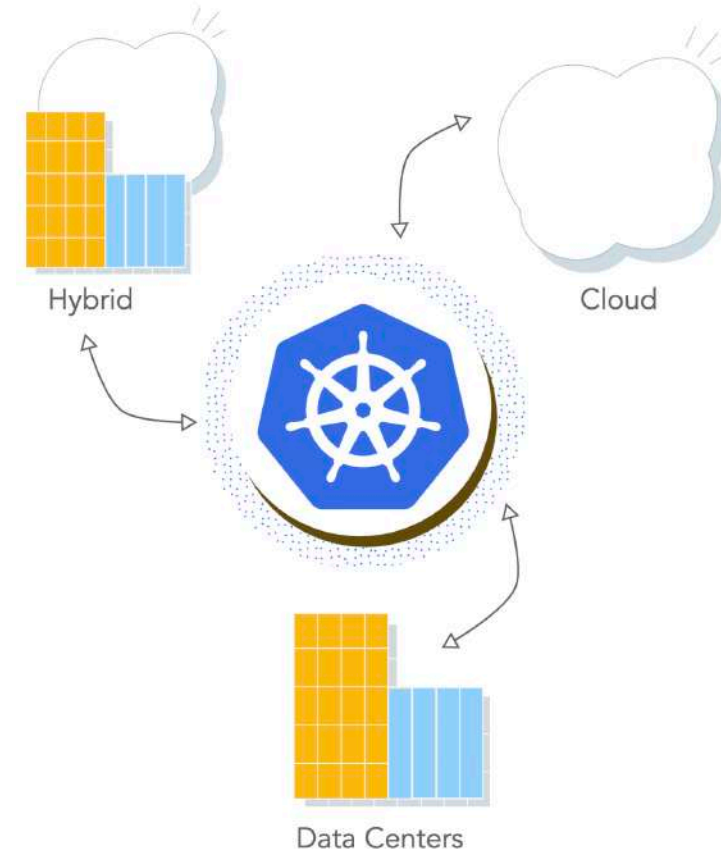
The Solution: Fleets

Fleet: A way to simplify managing multiple Kubernetes clusters together

Clusters Can Be:

- **GKE clusters** on Google Cloud
- **Kubernetes clusters** outside Google Cloud
 - Other cloud platform (AWS, Azure, ..) OR
 - On-premises

Goal: Treat related clusters as part of one logical group





Why Do We Need Fleets?

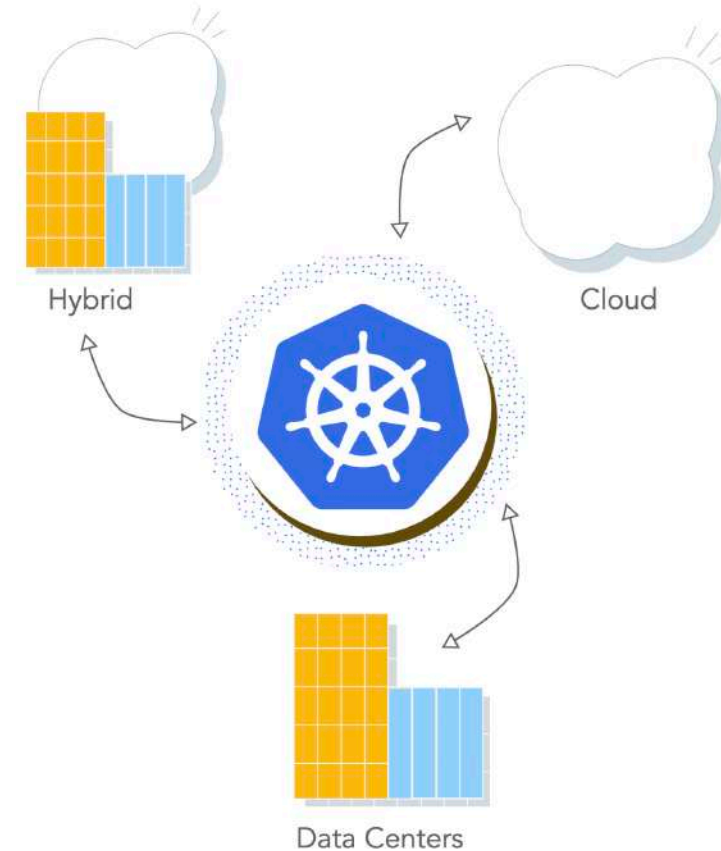
Scenario: You want to make a production-wide change across clusters

- Without fleets, you update clusters one by one
- Changes may span **multiple projects**

Problems Without Fleets:

- Repeated manual effort
- Inconsistent configuration
- Harder governance and visibility

Fleet Advantage: Manage related clusters together



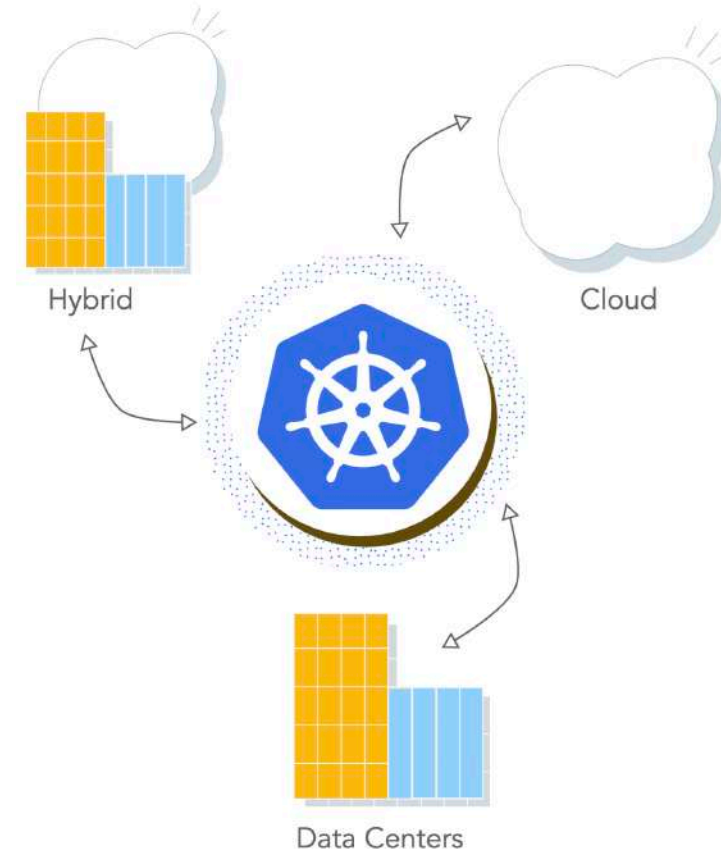


Constraints on Fleets

One Project, One Fleet: A Google Cloud project can have only **one fleet** associated with it

- **Fleet Host Project:** The project associated with a fleet is called the **fleet host project**
- **One Fleet, Many Projects:** A fleet can include clusters from **multiple projects**
 - You can include clusters from other cloud platforms (AWS, Azure, ..) AND On-premises

One Cluster, One Fleet: A cluster can belong to only **one fleet** at a time





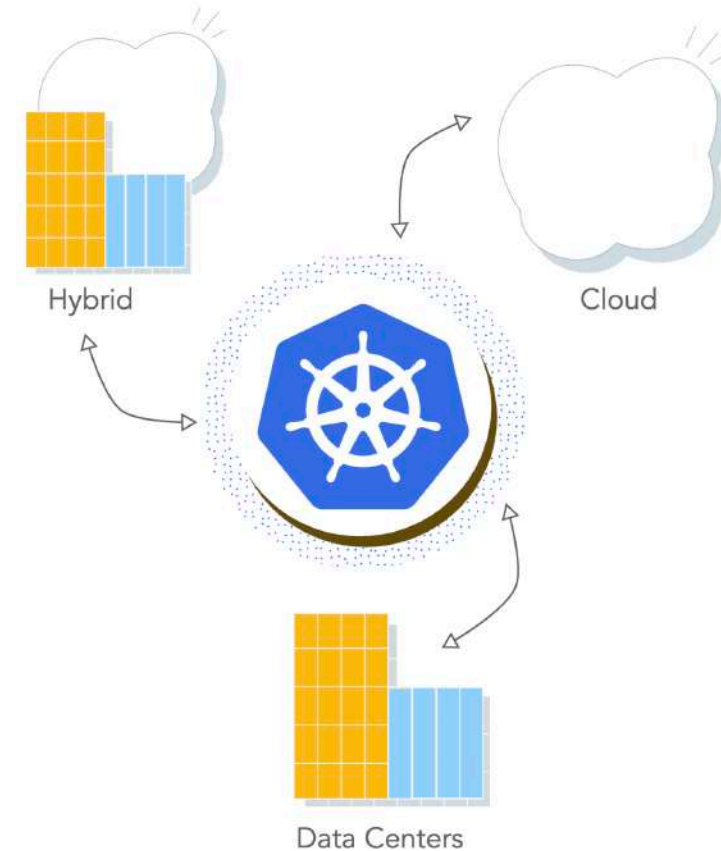
What is a Connect Agent?

Connect Agent: Software installed on clusters outside Google Cloud

- **Purpose:** Establishes control plane connectivity between the cluster and Google Cloud
- **Used For:** Non-GKE clusters added to a fleet

Advantages: Connect Agent enables

- **Unified Access:** Interact with clusters through the **Google Cloud console**
- **Simpler Operations:** Use a common interface across different clusters
- **Big Benefit:** Consistent experience across hybrid and multi-cluster environments

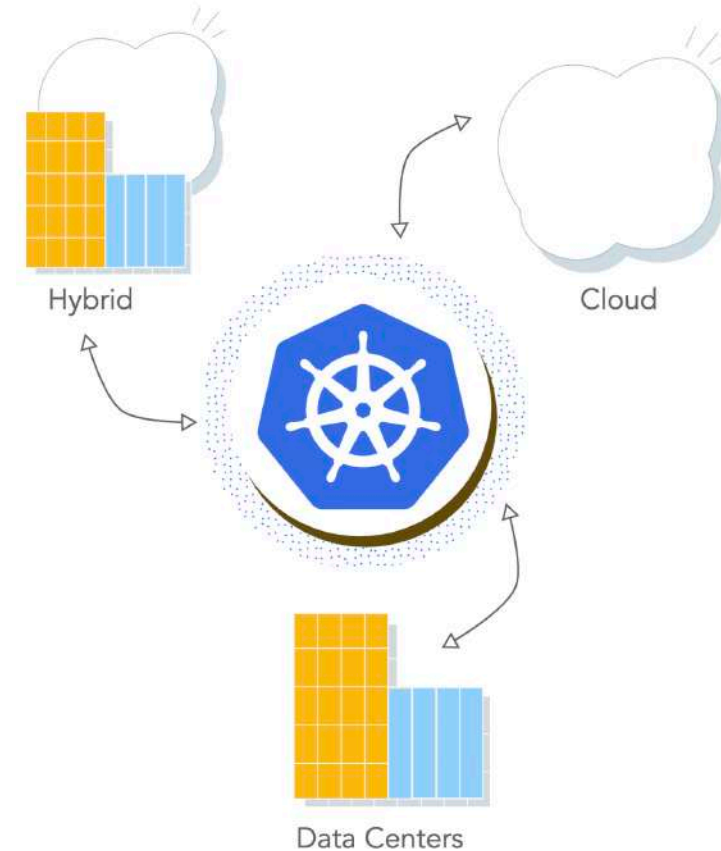




Advanced Features Enabled by Fleets

Fleets: Allow you to unlock advanced multi-cluster capabilities

- **Config Sync:** Deploy and monitor declarative configuration packages
 - **Source of Truth:** Typically stored in a **Git repository**
 - **Big Idea:** Manage config centrally using GitOps-style practices
- **Binary Authorization:** Control what can be deployed
- **Policy Controller:** Enforces declarative policies across clusters (Policies act as **guardrails**)
- **Cloud Service Mesh:** Centralized service mesh
- **Multi-cluster Gateway:** Route traffic across clusters
- **Multi Cluster Ingress:** Simplify ingress across clusters





Scenarios: Managing Hybrid Kubernetes Clusters

Scenario	Solution
You want to manage multiple related clusters together	Use a fleet to logically group GKE and external Kubernetes clusters
You need to connect a cluster outside Google Cloud	Install the Connect Agent to establish connectivity with Google Cloud
You want to apply production-wide configuration consistently	Group production clusters into the same fleet and use Config Sync
You want traffic management across clusters	Use Cloud Service Mesh , Multi-cluster Gateway , or Multi Cluster Ingress



GKE - Cluster Management - Command Line

Description	Command
Create Cluster	<code>gcloud container clusters create my-cluster --zone us-central1-a --node-locations us-central1-c,us-central1-b</code>
Resize Cluster	<code>gcloud container clusters resize my-cluster --node-pool my-node-pool --num-nodes 10</code>
Autoscale Cluster	<code>gcloud container clusters update cluster-name --enable-autoscaling --min-nodes=1 --max-nodes=10</code>
Delete Cluster	<code>gcloud container clusters delete my-cluster</code>
Adding Node Pool	<code>gcloud container node-pools create new-node-pool-name --cluster my-cluster</code>
List Images	<code>gcloud artifacts docker images list</code>



GKE - Workload Management - Command Line

Description	Command
List Pods/Service/Replica Sets	<code>kubectl get pods/services/replicasets</code>
Create Deployment	<code>kubectl apply -f deployment.yaml</code> or <code>kubectl create deployment</code>
Create Service	<code>kubectl expose deployment hello-world --type=LoadBalancer --port=8080</code>
Scale Deployment	<code>kubectl scale deployment hello-world --replicas 5</code>
Autoscale Deployment	<code>kubectl autoscale deployment hello-world --max --min --cpu</code>
Delete Deployment	<code>kubectl delete deployment hello-world</code>
Update Deployment	<code>kubectl apply -f deployment.yaml</code>
Rollback Deployment	<code>kubectl rollout undo deployment hello-world --to-revision=1</code>



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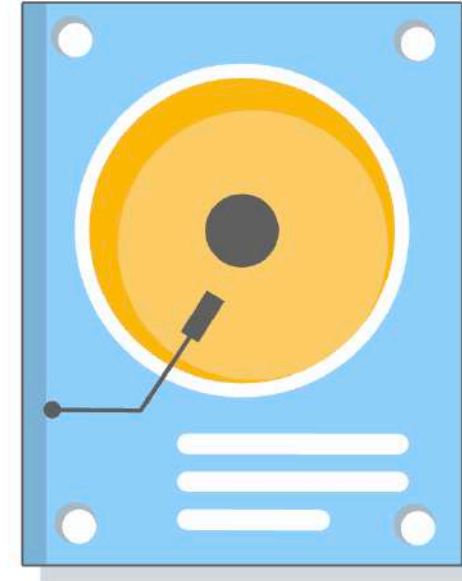
Managing Block Storage From Command Line



Managing Disks and Snapshots with Gcloud

gcloud compute disks:

- *gcloud compute disks list*
 - List Disk Types: `gcloud compute disk-types list`
- *gcloud compute disks create my-disk*
 - `--zone=us-east1-a`, `--size=SIZE` (1GB or 2TB), `--type=TYPE`
 - `--image`, `--image-family`, `--source-disk`, `--source-snapshot`
 - `--kms-key`, `--kms-project`
- *gcloud compute disks resize my-disk --size=1TB* (Only increasing disk size is supported)
- *gcloud compute disks snapshot my-disk --zone=us-central1-a --snapshot-names=my-snapshot*
 - You can also play with the snapshots which are created (`gcloud compute snapshots list/describe/delete`)

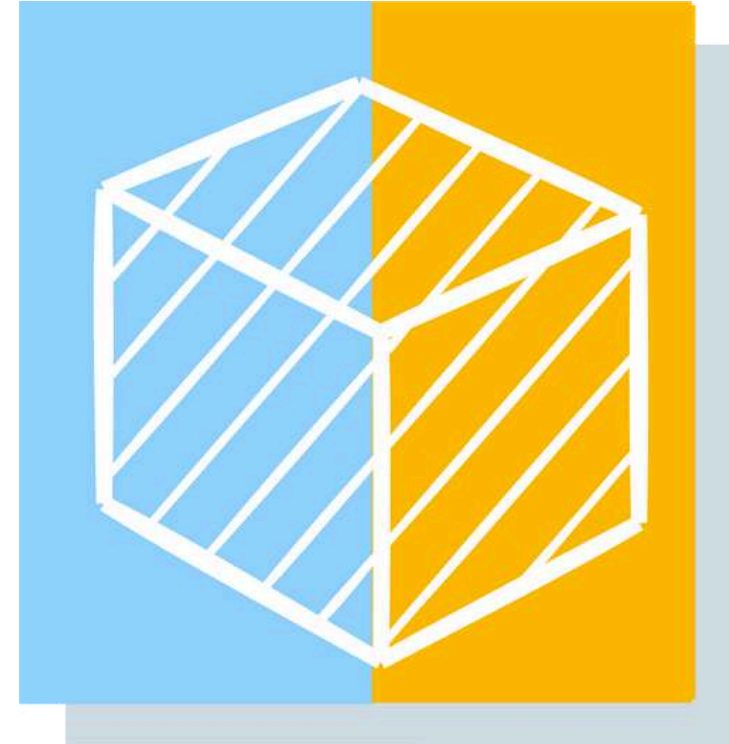




Managing Images with Gcloud [1/2]

gcloud compute images : *list, create, delete, deprecate, describe, export, import, ..*

- **List Images:** *gcloud compute images list --no-standard-images --format="value(NAME)"*
- **Create an Image:** *gcloud compute images create my-image*
 - From a Disk - `--source-disk=my-disk --source-disk-zone=us-east1-a`
 - From a Snapshot - `--source-snapshot=source-snapshot`
 - From another image - `--source-image=source-image --source-image-project=source-image-project`
 - Latest non deprecated in family - `--source-image-family=source-image-family --source-image-project=source-image-project`



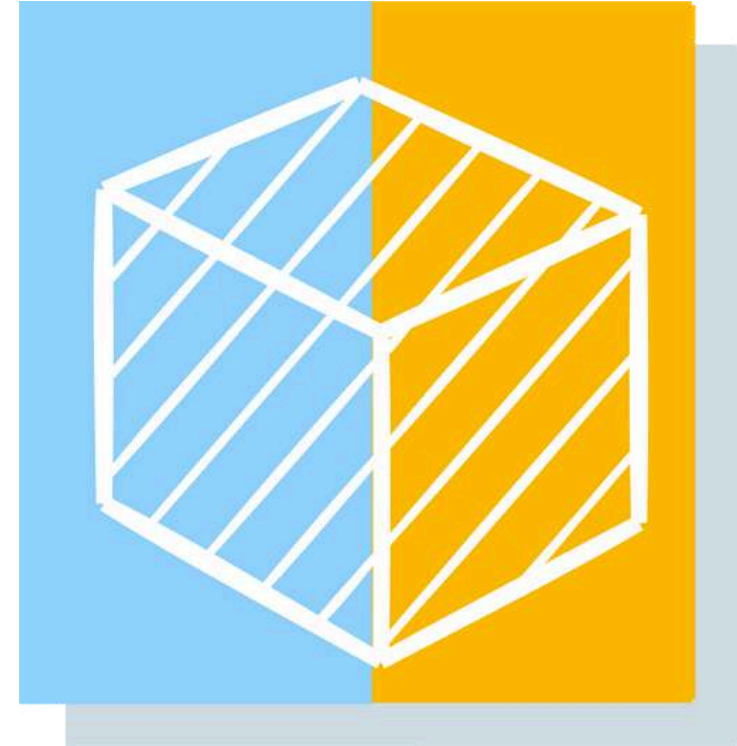


Managing Images with Gcloud [2/2]

Deprecate Image: *gcloud compute images deprecate IMAGE --state=DEPRECATED*

Export image to virtual disk: *gcloud compute images export --image=my-image --destination-uri=gs://my-bucket/my-image.vmdk --export-format=vmdk --project=my-project*

Delete Image: *gcloud compute images delete my-image1 my-image2*





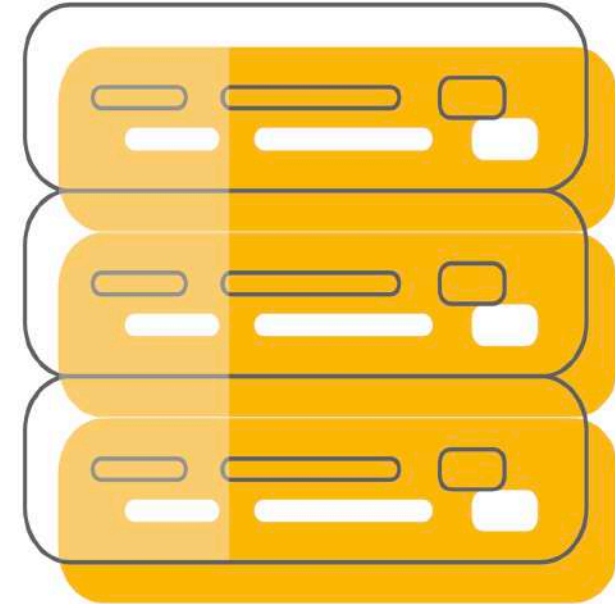
Managing Machine Images with Gcloud

Create a Machine Image: *gcloud compute machine-images create MACHINE_IMG_NAME -source-instance VM_NAME*

- **Create an Instance from the Machine Image:** *gcloud compute instances create VM_NAME --zone ZONE --source-machine-image MACHINE_IMG_NAME*

List Machine Images: *gcloud compute machine-images list*

Describe a Machine Image: *gcloud compute machine-images describe MACHINE_IMG_NAME*





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Managing Cloud Storage From Command Line

Cloud Storage From Command Line – gcloud storage



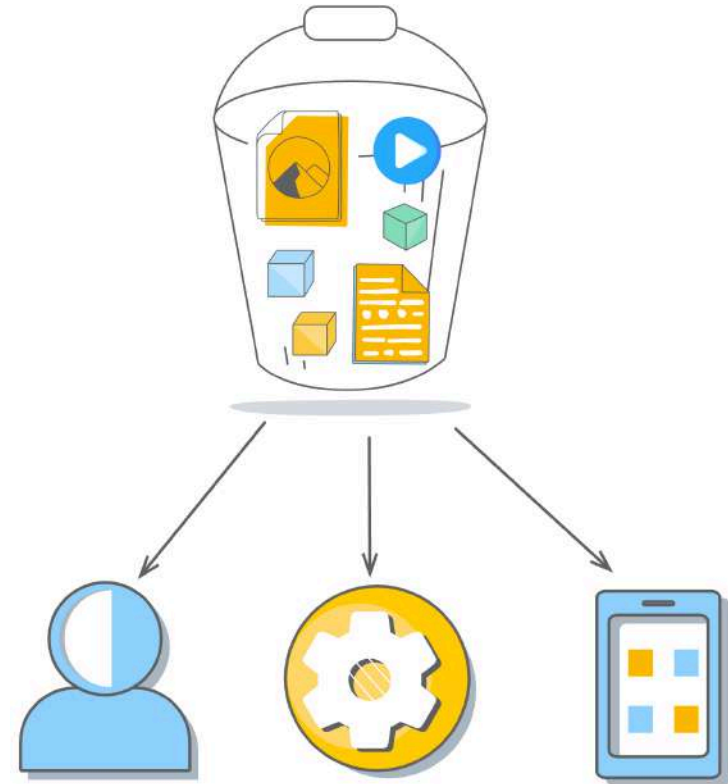
buckets: Manage Cloud Storage buckets

objects: Manage Cloud Storage objects

ls: List Cloud Storage buckets and objects

cp: Upload, download, and copy Cloud Storage objects

rm: Delete objects and buckets





Cloud Storage From Command Line – gcloud storage buckets

Description	Command
List	<code>gcloud storage buckets list</code>
Describe	<code>gcloud storage buckets describe gs://BUCKET_NAME</code>
Relocate	<code>gcloud storage buckets relocate gs://BUCKET_NAME --location=us-central1</code>
Update	<code>gcloud storage buckets update gs://BUCKET_NAME --default-storage-class=NEARLINE</code> (Key Options: <code>--[no-]uniform-bucket-level-access</code> , <code>--[no-]versioning</code> , <code>--default-encryption-key</code> , <code>--retention-period</code>)
Create	<code>gcloud storage buckets create gs://BUCKET_NAME --default-storage-class=nearline --location=asia</code>
Delete	<code>gcloud storage buckets delete gs://BUCKET_NAME</code>



Cloud Storage From Command Line – gcloud storage objects

Description	Command
List objects in a bucket recursively	<code>gcloud storage objects list gs://BUCKET_NAME/**</code>
Describe a Google Cloud Storage object	<code>gcloud storage objects describe gs://BUCKET_NAME/my-object</code>
Create a new object target.txt by concatenating a.txt and b.txt	<code>gcloud storage objects compose gs://BUCKET_NAME/a.txt gs://BUCKET_NAME/b.txt gs://BUCKET_NAME/target.txt</code>
Update a Google Cloud Storage object's storage class to NEARLINE	<code>gcloud storage objects update gs://BUCKET_NAME/my-object --storage-class=NEARLINE</code> (Other options include <code>--encryption-key</code> , <code>--content-type</code> , <code>--custom-metadata</code>)



Cloud Storage From Command Line – Miscellaneous [1/2]

Description	Command
List buckets in current active project	<code>gcloud storage ls</code>
List contents of a bucket	<code>gcloud storage ls gs://BUCKET_NAME</code>
List all .txt objects in a bucket	<code>gcloud storage ls gs://BUCKET_NAME/**/*.txt</code>
Delete an object	<code>gcloud storage rm gs://BUCKET_NAME/my-object</code>
Delete objects within a directory but not objects within subdirectories	<code>gcloud storage rm gs://BUCKET_NAME/my-dir/*</code>
Deletes all objects and subdirectories within the directory (affects ONLY live versions)	<code>gcloud storage rm gs://BUCKET_NAME/my-dir/**</code>
Deletes all versions of all resources and then deletes the bucket	<code>gcloud storage rm --recursive gs://BUCKET_NAME/</code>



Cloud Storage From Command Line – Miscellaneous [2/2]

Description	Command
Upload files from local	<code>gcloud storage cp *.txt gs://BUCKET_NAME</code>
Download text files from a bucket	<code>gcloud storage cp gs://BUCKET_NAME/*.txt .</code>
Transfer text files from a bucket to a AWS S3	<code>gcloud storage cp gs://BUCKET_NAME/*.txt s3://BUCKET_NAME</code>
Create a signed url to download an object - valid for 10 min	<code>gcloud storage sign-url gs://BUCKET_NAME/file.txt --duration=10m --impersonate-service-account=SERVICE_ACCOUNT</code>
Create a signed url, valid for one hour, for uploading a plain text file via HTTP PUT	<code>gcloud storage sign-url gs://BUCKET_NAME/file.txt --http-verb=PUT --duration=1h --headers=content-type=text/plain --impersonate-service-account=SERVICE_ACCOUNT</code>



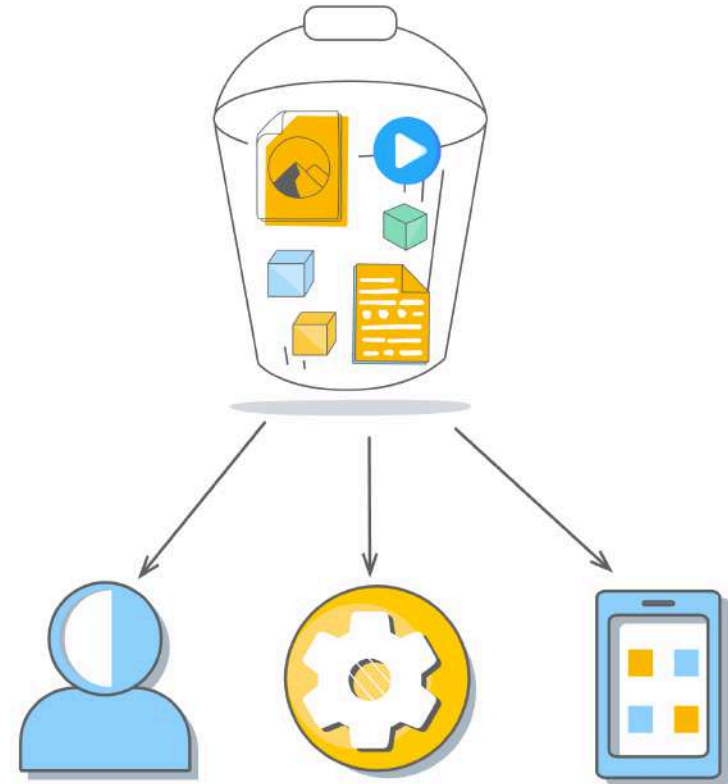
Cloud Storage From Command Line - A Peak into History

Then: Cloud Storage was managed mainly through the **gsutil** CLI

Now: The recommended interface is **gcloud**

Why the shift? Faster performance, simpler commands, and smarter defaults

- **Speed Advantage:** Reduced latency with improved parallelism
- **Ease of Use:** Automatically chooses the best transfer method, reducing operational effort





Playing with Cloud Storage using gsutil [1/2] [OLD]

Description	Command
Create Cloud Storage bucket	<code>gsutil mb gs://BUCKET_NAME</code>
List Objects	<code>gsutil ls -a gs://BUCKET_NAME</code>
Rename/Move objects	<code>gsutil mv</code>
Move object across buckets	<code>gsutil mv gs://OLD_BUCKET_NAME/OLD_OBJECT_NAME gs://NEW_BUCKET_NAME/NEW_OBJ_NAME</code>
Change storage class for objects	<code>gsutil rewrite -s STORAGE_CLASS gs://BUCKET_NAME/OBJ_PATH</code>



Playing with Cloud Storage using gsutil [2/2] [OLD]

Description	Command
Upload objects	<code>gsutil cp LOCAL_LOCATION gs://DESTINATION_BUCKET_NAME/</code>
Download objects	<code>gsutil cp gs://BUCKET_NAME/OBJ_PATH LOCAL_LOCATION</code>
Enable/Disable versioning	<code>gsutil versioning set on/off gs://BUCKET_NAME</code>
Enable/Disable uniform bucket-level access	<code>gsutil uniformbucketlevelaccess set on/off gs://BUCKET_NAME</code>
Generate signed URL for temporary access	<code>gsutil signurl -d 10m YOUR_KEY gs://BUCKET_NAME/OBJECT_PATH</code>

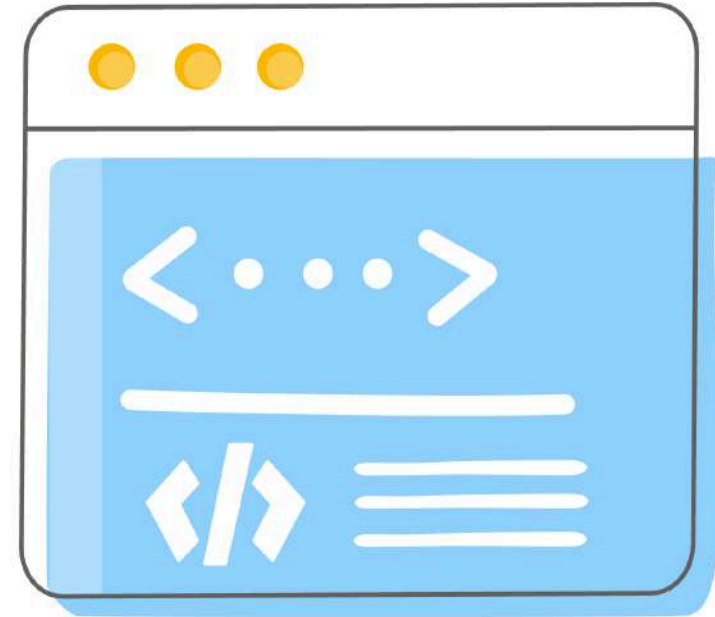


gcloud storage – Shim Option

Challenge: gsutil is older but many teams still rely on gsutil commands in automation

Shim: Shim provides **gcloud storage performance and features** while keeping gsutil syntax

- **How it works:** gsutil internally routes operations through **gcloud storage**
- In your **boto config file**, enable:
`use_gcloud_storage=True` under the **GSUtil** section
- **Best for:** Teams who cannot update old scripts





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Managing Databases From Command Line



Managing Cloud SQL using Gcloud

Action	Command
Create a Cloud SQL instance	<code>gcloud sql instances create INSTANCE</code>
Actions on Cloud SQL instance	<code>gcloud sql instances clone/delete/describe/export/import</code>
Create a database inside an instance	<code>gcloud sql databases create DATABASE --instance=INSTANCE</code>
Manage databases	<code>gcloud sql databases (delete/describe/list/patch)</code>
Connect to an instance	<code>gcloud sql connect INSTANCE [--database=DATABASE --user=root]</code>
Create a one-time backup	<code>gcloud sql backups create --async --instance [INSTANCE]</code>
Describe or list backups	<code>gcloud sql backups describe/list/delete</code>



Managing AlloyDB using Gcloud

Action	Command
Create a cluster	<code>gcloud alloydb clusters create CLUSTER --region=REGION --password=VALUE</code>
Actions on a cluster	<code>gcloud alloydb clusters create-secondary/describe/delete/export/import</code>
Create an instance	<code>gcloud alloydb instances create INSTANCE --cluster=CLUSTER</code>
Actions on an instance	<code>gcloud alloydb instances create-secondary/describe/delete/failover</code>
Connect to an instance	<code>gcloud alloydb connect INSTANCE --cluster=CLUSTER</code>
Create a one-time backup	<code>gcloud alloydb backups create --async --cluster=CLUSTER</code>
Describe or list backups	<code>gcloud alloydb backups describe/list/delete</code>



Managing BigQuery using bq

Action	Command
Show table metadata	<code>bq show bigquery-public-data:samples.shakespeare</code>
Run a query	<code>bq query 'QUERY-STRING'</code>
Estimate bytes scanned (--dry-run)	<code>bq query --dry-run 'QUERY-STRING'</code>
Export data	<code>bq extract</code>
Load data	<code>bq load</code>
(Remember) Use the standard way to set the project	<code>gcloud config set project my-project</code>



Managing Cloud Bigtable using cbt

cbt: CLI for Cloud Bigtable (NOT gcloud)

Install: gcloud components install cbt

Configuration: Create a .cbtrc file

- `echo project = project-id > ~/.cbtrc`
- `cat ~/.cbtrc`
- **Verify install:** `cbt listinstances`





Managing Cloud Bigtable using cbt – Example Commands

Action	Command
Create a Bigtable instance	<code>cbt createinstance</code>
Create a cluster inside the instance	<code>cbt createcluster</code>
Create or delete a table	<code>cbt createtable / cbt deletetable</code>
Delete an instance or cluster	<code>cbt deleteinstance / cbt deletecluster</code>
List all instances or clusters	<code>cbt listinstances / cbt listclusters</code>
List tables and column families	<code>cbt ls</code>
Reminder	Configure project and instance in <code>.cbtrc</code> file



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Managing Cloud Identity From Command Line



Exploring IAM using Gcloud [1/2]

Action	Command
Describe current project	<code>gcloud compute project-info describe</code>
Access the Cloud Platform with Google user credentials	<code>gcloud auth login</code>
Revoke access credentials	<code>gcloud auth revoke</code>
List active accounts	<code>gcloud auth list</code>
Add IAM policy binding	<code>gcloud projects add-iam-policy-binding</code>
Get IAM policy for a project	<code>gcloud projects get-iam-policy</code>
Remove IAM policy binding	<code>gcloud projects remove-iam-policy-binding</code>



Exploring IAM using Gcloud [2/2]

Action	Command
Set IAM policy	<code>gcloud projects set-iam-policy</code>
Delete a project	<code>gcloud projects delete</code>
Describe an IAM role	<code>gcloud iam roles describe</code>
Create an IAM role	<code>gcloud iam roles create</code> (options: --project, --permissions, --stage)
Copy IAM roles	<code>gcloud iam roles copy</code>



Managing Projects using Gcloud - gcloud projects

Action	Command
List all projects	<code>gcloud projects list</code>
Get project details	<code>gcloud projects describe</code>
Create or delete	<code>gcloud projects create/delete/undelete</code>
Get IAM policy with all bindings	<code>gcloud projects get-iam-policy</code>
Add IAM policy binding	<code>gcloud projects add-iam-policy-binding PROJECT_ID --member='user:test@gmail.com' --role='roles/storage.admin'</code>
Remove IAM policy binding	<code>gcloud projects remove-iam-policy-binding</code>
Move project to another organization	<code>gcloud projects move PROJECT_ID --organization=ORGANIZATION_ID</code>



Managing Projects and IAM using Gcloud - Scenarios

Scenario	Command
How can I authenticate to Google Cloud from my local machine with Google user credentials?	<code>gcloud auth login</code>
How can I copy a role from one organization to another?	<code>gcloud iam roles copy</code>
How can I get a list of project-level IAM permissions?	<code>gcloud projects get-iam-policy</code> - Lists all policy bindings
How can I give permissions to a specific user?	<code>gcloud projects add-iam-policy-binding PROJECT_ID --member='user:test@gmail.com' --role='roles/storage.admin'</code>
How can I move a project to another organization?	<code>gcloud projects move PROJECT_ID --organization=ORGANIZATION_ID ..</code>



Managing Active Services using Gcloud - gcloud services

Action	Command
List the services that are enabled for a project	<code>gcloud services list --enabled --project=PROJECT-ID</code>
List services that are available	<code>gcloud services list --available</code>
Enable a service for a specific project	<code>gcloud services enable compute.googleapis.com --project=PROJECT-ID</code>
Disable a service for a specific project	<code>gcloud services disable compute.googleapis.com --project=PROJECT-ID</code>



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Managing Cloud Networking From Command Line



Managing Cloud DNS From Gcloud

Create a managed zone: `gcloud dns managed-zones create ZONE_NAME`

- `--description`, `--dns-name`, `--visibility` (private/public), `--networks` (VPC for private zones)

Add records in three steps

- Start transaction for Zone: `gcloud dns record-sets transaction start --zone=ZONE_NAME`
- Make Changes: `gcloud dns record-sets transaction add --name=REC_NAME --ttl=TTL_VALUE --type=A/CNAME --zone=ZONE_NAME`
- End Transaction for Zone: `gcloud dns record-sets transaction execute --zone=ZONE_NAME`





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Managing Pub Sub

From Command Line



Managing Pub/Sub From Command Line

Action	Command
Create a topic	<code>gcloud pubsub topics create my-topic</code>
Create a subscription	<code>gcloud pubsub subscriptions create my-subscription --topic=my-topic</code> Options: <code>--enable-message-ordering</code> , <code>--ack-deadline</code> (how long to wait for acknowledgment?), <code>--message-filter</code> (criteria to filter messages)
Pull messages	<code>gcloud pubsub subscriptions pull my-subscription --auto-ack --limit</code>
Acknowledgement	<code>gcloud pubsub subscriptions ack my-subscription --ack-ids=[ACK_ID,...]</code>
Delete a topic	<code>gcloud pubsub topics delete my-topic</code>
List topics	<code>gcloud pubsub topics list</code>
List subscriptions for a topic	<code>gcloud pubsub topics list-subscriptions my-topic</code>



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Getting Started with Kubernetes Security



Securing GKE Clusters – Different Elements [1/2]

Focus Area	Security Element Question
Cluster Security	How do we restrict access to the Kubernetes control plane? Who is allowed to administer the cluster? What operations are allowed by whom?
Network Security	How do we restrict network access to the cluster?
Container Runtime Security	How do we protect the container runtime? How do we detect suspicious container activity?
Image Supply Chain Security	How do we ensure container images are trusted and secure? How do we prevent deploying vulnerable or unverified images?



Securing GKE Clusters – Different Elements [2/2]

Focus Area	Security Element Question
Application Security	How do we secure applications running inside containers?
	How do we prevent vulnerabilities in application code?
	How do you provide the applications minimum access to cloud resources?
	How to implement user authentication and authorization?
Workload Isolation	How do we isolate workloads from each other?
	How do we control communication between workloads?
	How do we prevent one compromised container from affecting others?
Secrets Management	How do we securely store credentials, API keys, and certificates?



Securing GKE Nodes from the Internet

Scenario: Your Kubernetes nodes run production workloads

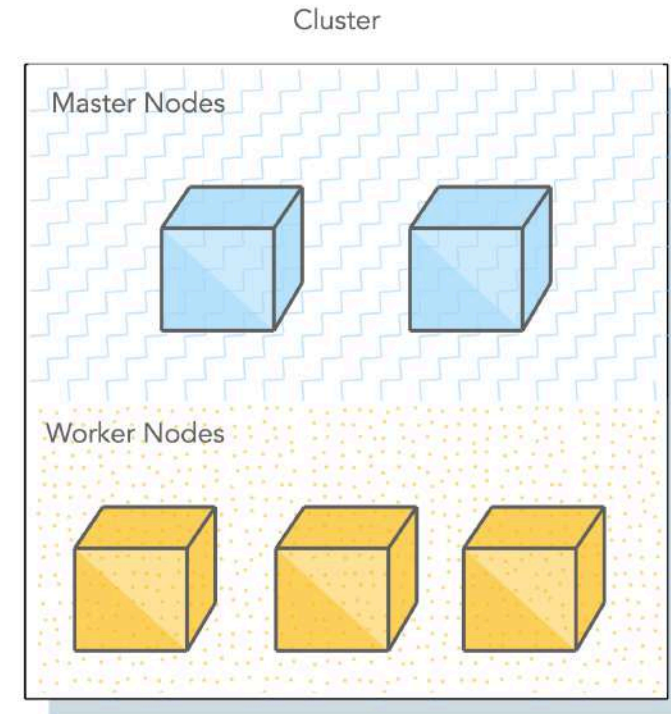
- Your Nodes have Public IP Addresses

Risk:

- Public node IPs increase the attack surface
- Direct public access to nodes is not recommended

Goal: Reduce exposure of nodes and cluster infrastructure

- **Solution:** Use private nodes





Enable Private Nodes

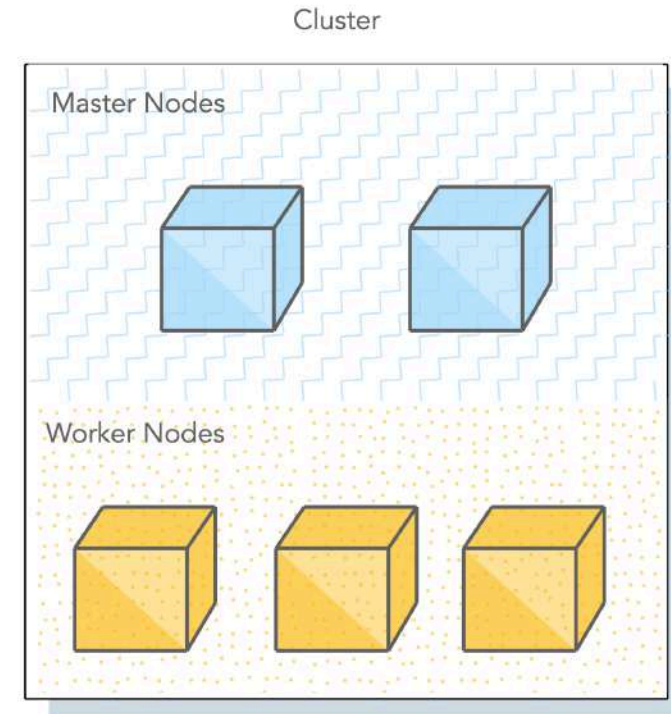
Private Nodes: Nodes are provisioned with **internal IP addresses only**

Effect: Nodes cannot be accessed directly from the public internet

- External clients cannot connect to node IPs

Result: Reduces exposure of the infrastructure layer

Important: Nodes can still communicate **within the VPC**



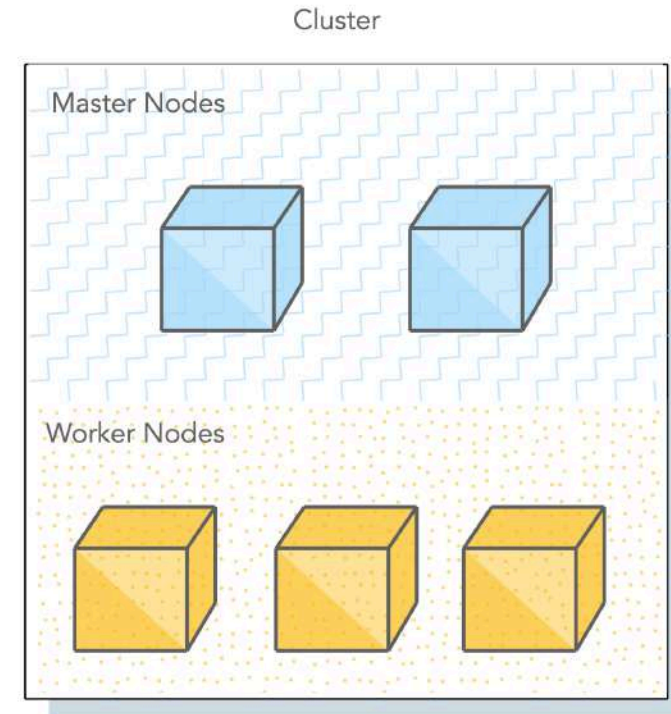


What is a Private GKE Cluster?

Private GKE Cluster: A cluster where nodes only have internal IP addresses

Isolated Nodes and Pods: Nodes and Pods are isolated from the internet by default

Control Access to Control Plane: You may choose to have no client access, limited access, or unrestricted access to the control plane.





Control Access to Control Plane - Options

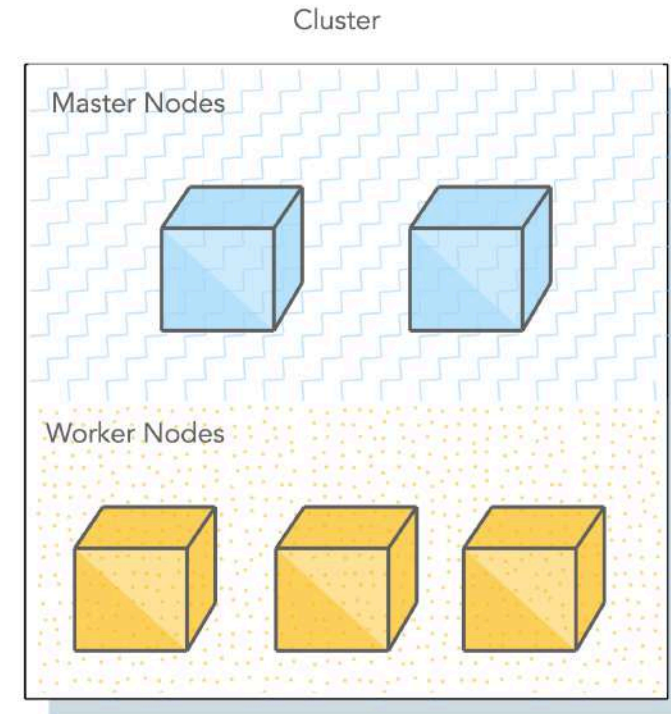
1: No client access to the public endpoint

- Access only from within the VPC

2: Limited access to the public endpoint

- Enable Master Authorized Networks
 - `gcloud container clusters update CLUSTER_NAME --enable-master-authorized-networks --master-authorized-networks 213.0.143.0/29`

3: Unrestricted access to the public endpoint (NOT RECOMMENDED)





Internet Access from Private Nodes

Private nodes still need outbound internet access

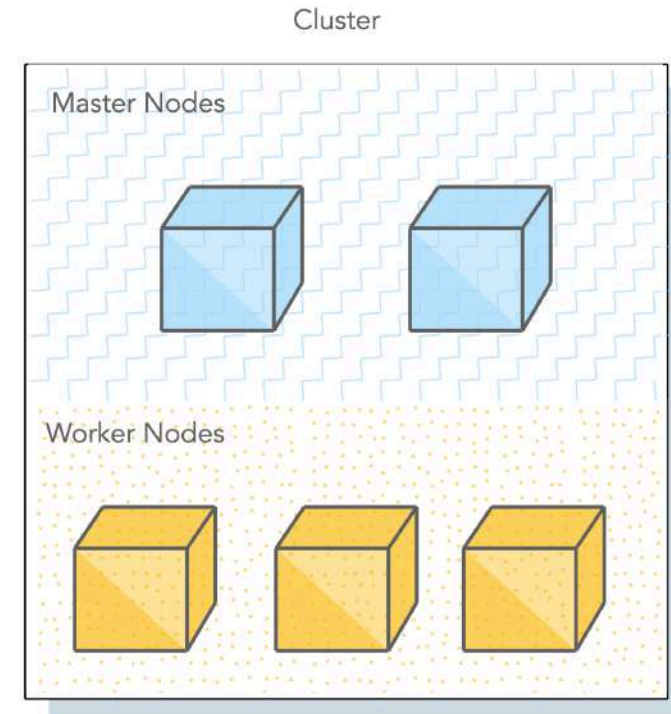
- Pull container images
- Access external APIs

Solution

- Use Cloud NAT with Cloud Router

How it works

- Nodes send traffic through NAT
- Outbound internet works
- Nodes still do not have public IPs



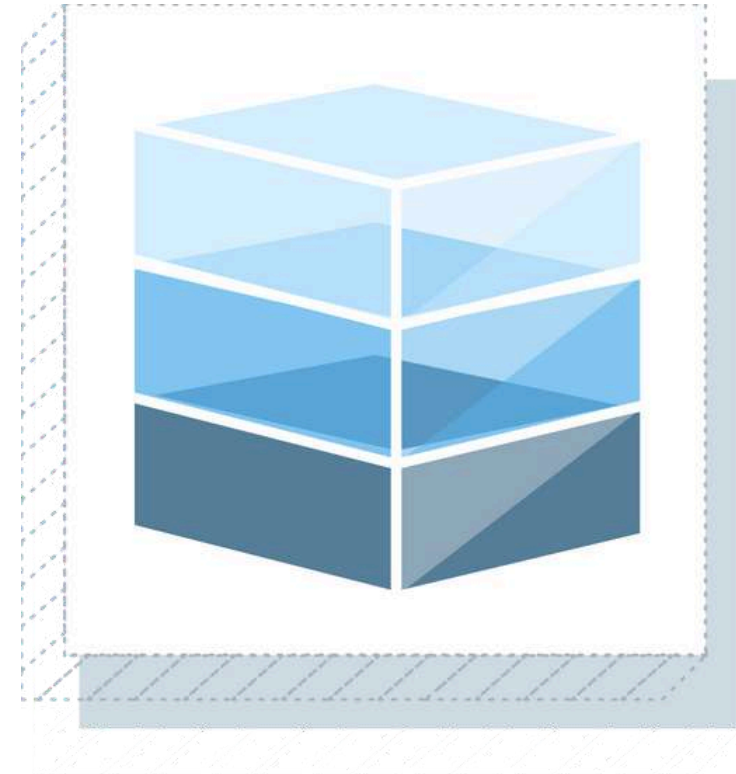


What is the need for a Network Policy?

Limit Pod Communication: Block all default Pod-to-Pod traffic; allow only what's required

- **Enforce Service Boundaries:** Permit expected flows (e.g., frontend → backend) while blocking others
- **Protect Sensitive Data Paths:** Restrict access to critical services like databases to approved Pods only
- **Control Ingress & Egress:** Define which Pods can receive external traffic or call external systems
- **Contain Compromises:** Prevent a breached Pod from reaching admin, or other sensitive workloads

Isolate by Labels/Namespace: Build logical boundaries - `role=backend` → `role=db` only





NetworkPolicy - Example 1

```
# Blocks all incoming traffic unless explicitly allowed
# WHEN: Use as a baseline before applying any allows
# WHY: Ensures "default deny" security posture
# HOW: Apply once for namespace-wide deny
```

```
apiVersion: networking.k8s.io/v1
```

```
kind: NetworkPolicy
```

```
metadata:
```

```
  name: default-deny-all
```

```
  namespace: my-microservices
```

```
spec:
```

```
  # Matches all Pods in the namespace
```

```
  podSelector: {}
```

```
  policyTypes:
```

```
    # This policy is about incoming traffic.
```

```
    - Ingress
```

```
    # No ingress section defined =>
```

```
    # no Pods are allowed to receive any traffic
```



NetworkPolicy - Example 2

```
# WHAT: Allow only 'frontend' to talk to 'backend'  
# WHY:  Enforce service-level isolation  
# WHEN: Frontend calls APIs on backend  
# WHERE: All in 'my-microservices' namespace  
# HOW:  Match Pod labels
```

```
apiVersion: networking.k8s.io/v1  
kind: NetworkPolicy  
metadata:  
  name: allow-backend-from-frontend  
  namespace: my-microservices  
spec:  
  podSelector:  
    matchLabels:  
      role: backend  
  policyTypes:  
    - Ingress  
  ingress:  
    - from:  
      - podSelector:  
          matchLabels:  
            role: frontend
```



NetworkPolicy - Example 3

```
# WHAT: Allow only 'backend' to talk to 'database'  
# WHY:  Protect DB from direct external access  
# WHEN: Backend persists data to database  
# WHERE: All in 'my-microservices' namespace  
# HOW:  Match Pod labels
```

```
apiVersion: networking.k8s.io/v1  
kind: NetworkPolicy  
metadata:  
  name: allow-database-from-backend  
  namespace: my-microservices  
spec:  
  podSelector:  
    matchLabels:  
      role: database  
  policyTypes:  
    - Ingress  
  ingress:  
    - from:  
      - podSelector:  
          matchLabels:  
            role: backend
```



Network Security for Kubernetes - Scenarios [1/2]

Scenario	Solution
How can you restrict access to control plane to specific IP addresses?	Master authorized networks allow only specific IPs to access the control plane
You are trying to connect to a GKE cluster using kubectl. You are getting timed out. What are things you can check?	Verify that the API server endpoint (DNS/IP) is correct. Check whether Master Authorized Networks is enabled and ensure your client IP (for example Cloud Shell) is included. Also verify that your kubectl credentials/context are correct (gcloud container clusters get-credentials).



Network Security for Kubernetes - Scenarios [2/2]

Scenario	Solution
A database Pod should only be accessible by backend Pods and not by frontend or other services. How can this be enforced?	Apply a NetworkPolicy that allows ingress to Pods labeled <code>role=database</code> only from Pods labeled <code>role=backend</code> , and deny all other traffic
Your GKE nodes should not be accessible from the public internet, but workloads still need internet access to pull container images. What should you configure?	Use Private Nodes with Cloud NAT + Cloud Router so nodes only have private IPs while still allowing outbound internet connectivity.
You want to implement a zero-trust network posture inside a namespace so Pods cannot talk to each other unless explicitly allowed. What should you do first?	Apply a default deny NetworkPolicy (<code>podSelector: {}</code> with no ingress rules) to block all Pod-to-Pod traffic, then create specific allow policies.



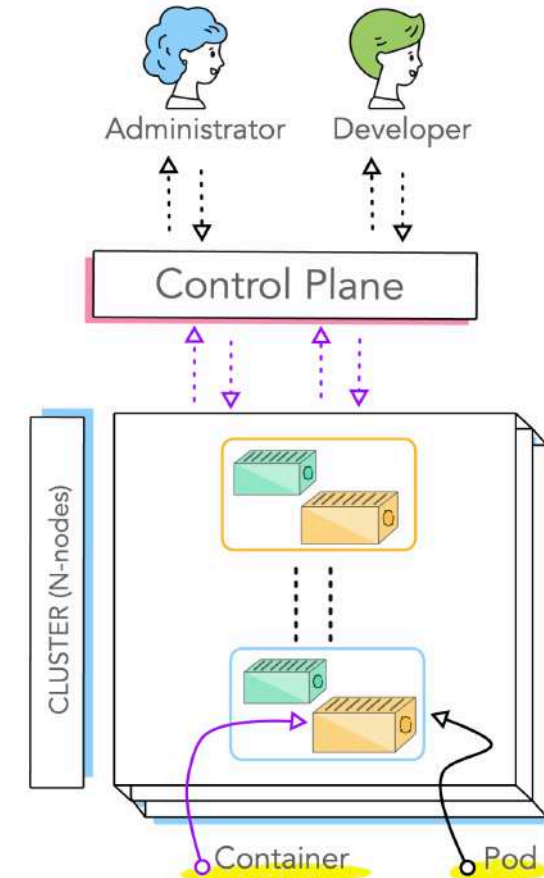
Authorization for Kubernetes Workloads

Scenario: Multiple microservices run in the same **GKE cluster**

- **Challenge:** Each microservice needs access to different Google Cloud resources (Examples – Cloud Storage, Databases, Pub/Sub)
- **Goal:** Grant the **right permissions** to each workload securely

Options:

- **Option 1** – Assign permissions at **Node** (VM) level
- **Option 2** – Assign permissions at **Pod** (individual microservice) level





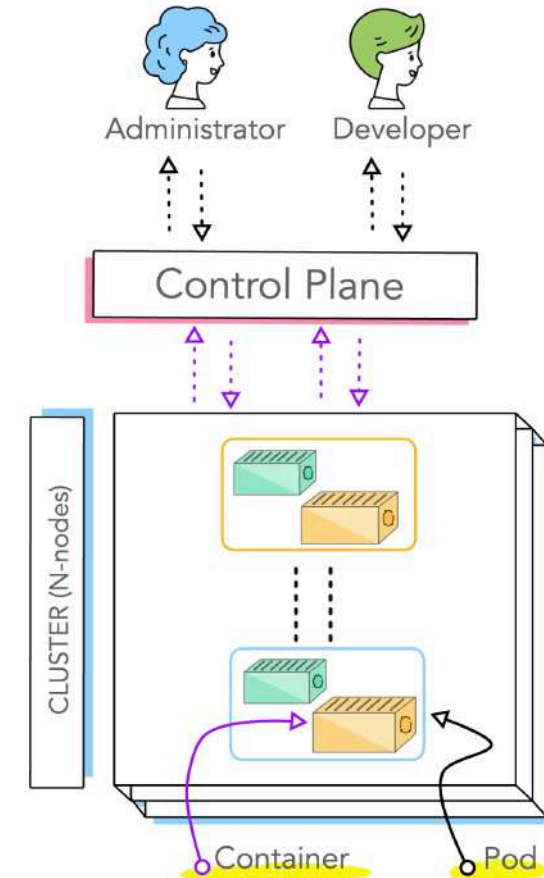
Exploring Node-Level Permissions

Each **Node** in a GKE cluster is a **GCE VM**

- **Default service account:** Compute Engine default service account
 - Has a broad range of permissions (⚠ NOT recommended)
- **Better Option:** Create a **custom service account** with limited permissions (maybe just the common permissions)

Conclusion: Node-level authorization is **not ideal**

- Violates the **principle of least privilege**





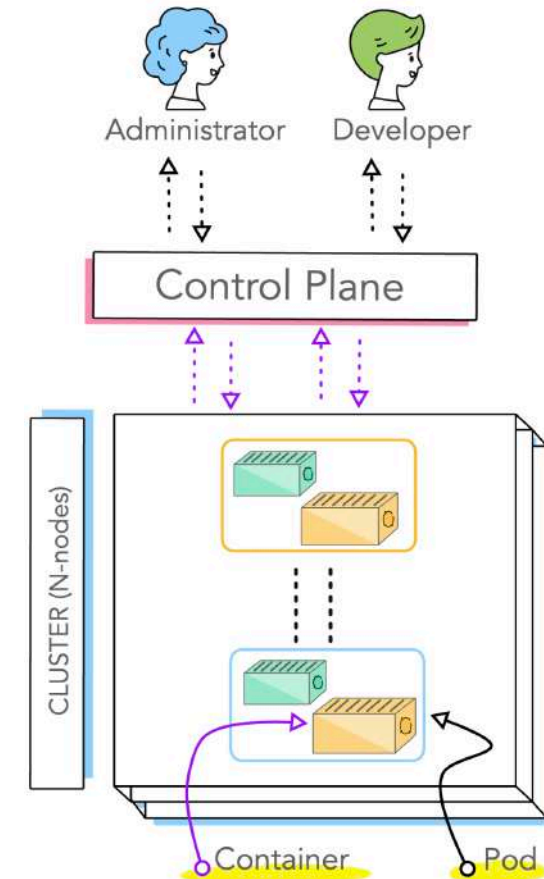
Exploring Pod-Level Permissions

Best Practice: Configure permissions per microservice (Pod)

Each microservice can use its **own service account** with custom permissions

Ways to assign service accounts:

- Option 1 – Using Kubernetes Secrets
- Option 2 – Using Workload Identity





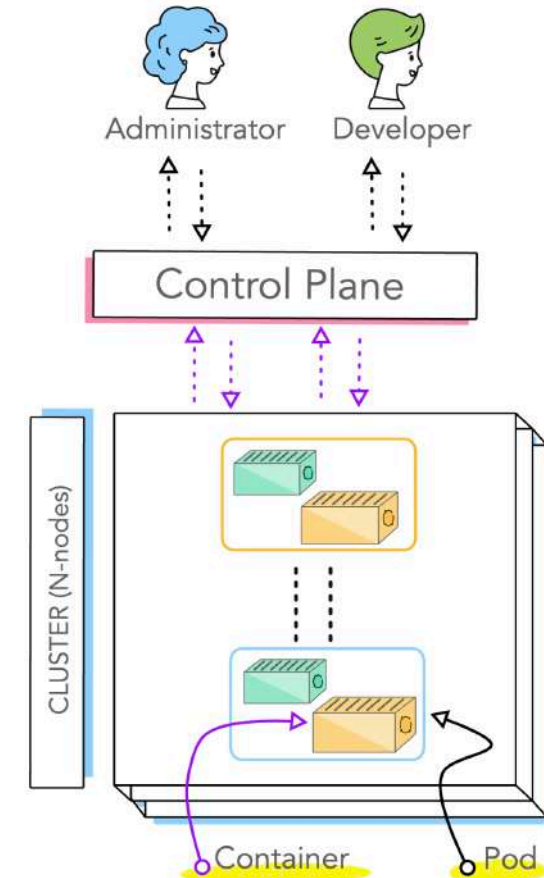
Option 1 – Using Kubernetes Secrets

Steps:

- 1 – Create a Google service account with permissions
- 2 – Generate a **JSON key** for that account
- 3 – Import it as a **Kubernetes Secret**
 - `kubectl create secret generic my-service-account-key --from-file=key.json=my-service-account-key.json`
- 4 – Update your **Deployment** to mount and use this

Advantage: Adheres to **principle of least privilege**

Limitation: Manual key management (Key rotation is difficult)





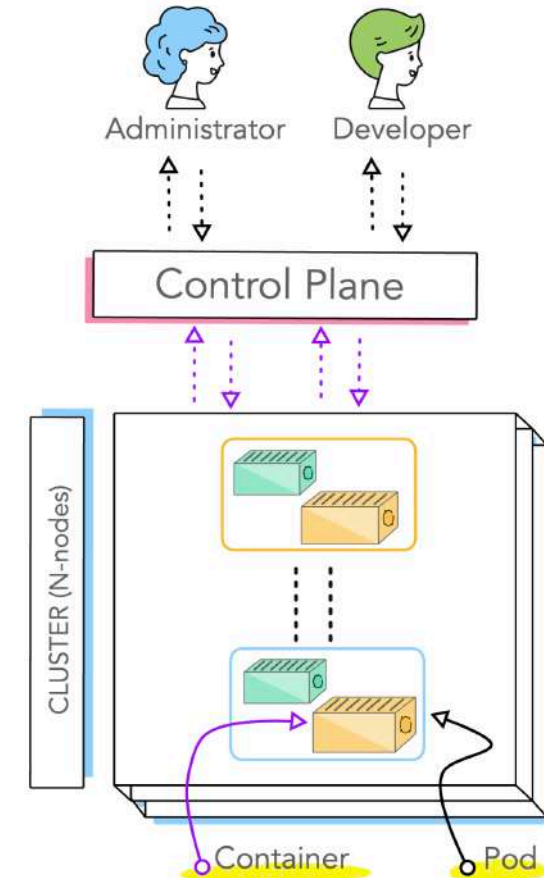
Option 2 – Using Workload Identity (Recommended)

Workload Identity: Links a Kubernetes SA (KSA) to a Google Service Account (GSA)

- Eliminates the need for secrets or key files
- **Fine-grained authorization:** For each microservice
- **Steps:**
 - Enable **Workload Identity** on your GKE cluster
 - Configure the **KSA ↔ GSA** mapping

Benefits: Strong security and simplified management

- **Recommended approach** for workloads accessing Google Cloud services





Workload Identity - Steps

1: Enable Workload Identity on GKE cluster

- At creation: `gcloud container clusters create CLUSTER_NAME --workload-pool=PROJECT_ID.svc.id.goog` OR
- At a later point: `gcloud container clusters update CLUSTER_NAME --workload-pool=PROJECT_ID.svc.id.goog`
 - Enable Workload Identity on Older Node Pools (NOT NEEDED if you created cluster with workload identity enabled)
 - `gcloud container node-pools update NODEPOOL_NAME --cluster=CLUSTER_NAME --workload-metadata=GKE_METADATA`

2: Create a Google service account `IAM_SA` with the permissions that your microservice needs

3: Create a Kubernetes service account

- `kubectl create serviceaccount K8S_SA`



Workload Identity - Steps - 2

4: Create an IAM policy binding between K8S_SA and IAM_SA

- `gcloud iam service-accounts add-iam-policy-binding --role roles/iam.workloadIdentityUser --member "serviceAccount:PROJECT_ID.svc.id.goog[K8S_SA]" IAM_SA@PROJECT_ID.iam.gserviceaccount.com`

5: Add annotation to K8S_SA: `kubectl annotate serviceaccount K8S_SA iam.gke.io/gcp-service-account=IAM_SA@PROJECT_ID.iam.gserviceaccount.com`

6: Configure container to use K8S_SA

```
spec:
  containers:
  - image: in28min/rest-api:0.0.1.RELEASE
    name: hello-world-rest-api
    serviceAccountName: K8S_SA
```



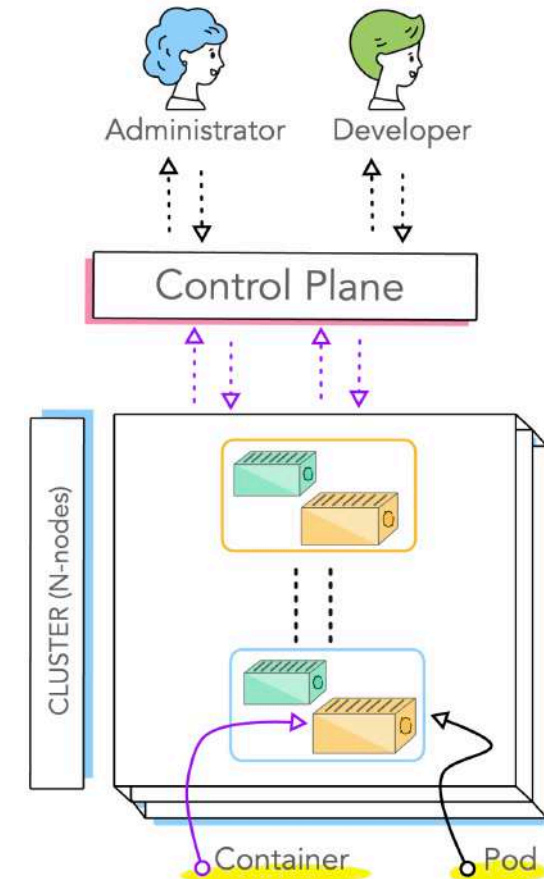

Authorization for Kubernetes Workloads - Summary

Authorization: For Kubernetes Workloads

- 1: Node (Compute Engine VM) level
- 2: Pod (individual microservice) level
 - **Solutions:** Kubernetes Secrets, Workload Identity
- Recommended option is to use **Workload Identity**
 - Follows principle of least privilege
 - Does NOT need any secrets management!

Next Focus Area: Authorization for Kubernetes cluster operators, Kubernetes workload deployers etc

- **Level 1:** Google Cloud IAM
- **Level 2:** Kubernetes RBAC



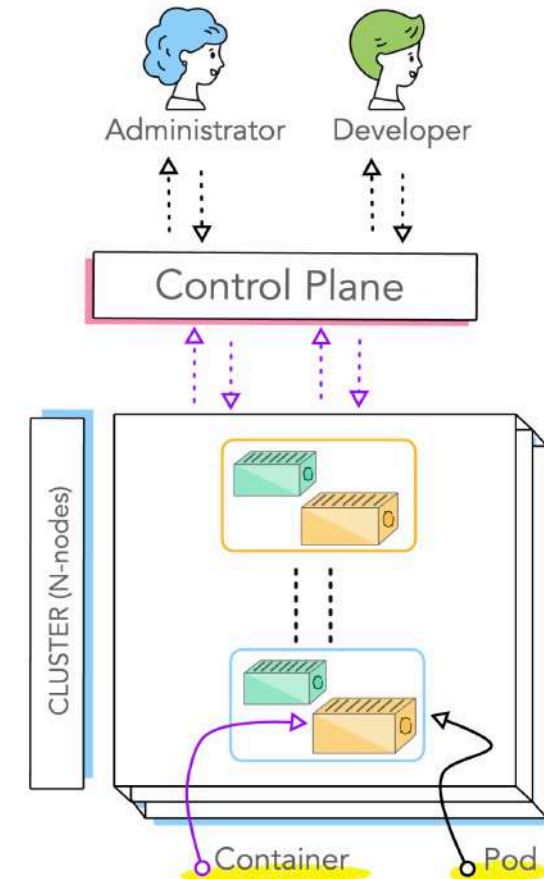


Kubernetes RBAC and Google Cloud IAM

Scenario: You want to manage access to your Kubernetes clusters

- **Two Layers of Access Control:**
 - **Google Cloud IAM** – Controls what users can do **across clusters** in a project, folder, or organization
 - **Kubernetes RBAC** – Controls what users can do **inside clusters**
- **Examples:**
 - **Cluster Access** – Create, update, or delete clusters and nodes
 - **Kubernetes API Access** – Manage deployments, services, pods, and more

Goal: Combine IAM and RBAC to enforce fine-grained, secure access control



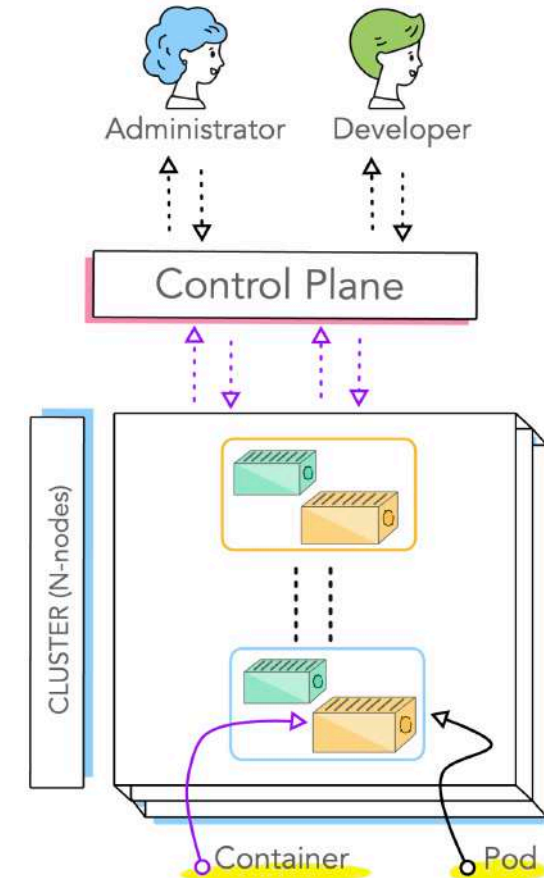


IAM vs RBAC – Who Controls What? [1/2]

Goal: Understand how Google Cloud IAM and Kubernetes RBAC work together

Google Cloud IAM – Access at infrastructure and project level

- Controls who can create, update, or delete **GKE clusters**
- Applies to **all clusters** within a project, folder, or organization





IAM vs RBAC – Who Controls What? [2/2]

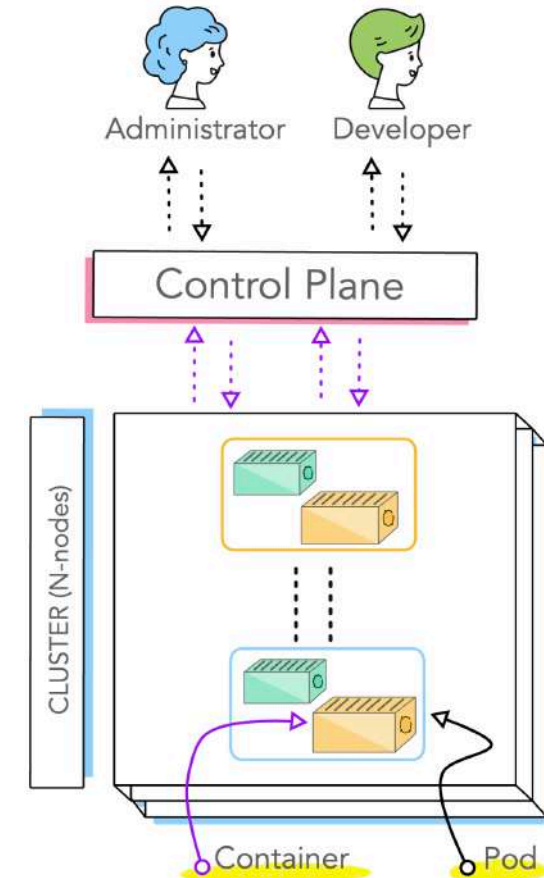
Kubernetes RBAC – Access inside the cluster

- Controls who can manage **namespaces, pods, services, and deployments**
- Applies at **cluster or namespace level**

Together:

- IAM defines **who can access the cluster**
- RBAC defines **what they can do inside the cluster**

Result: Secure and layered access management for GKE environments

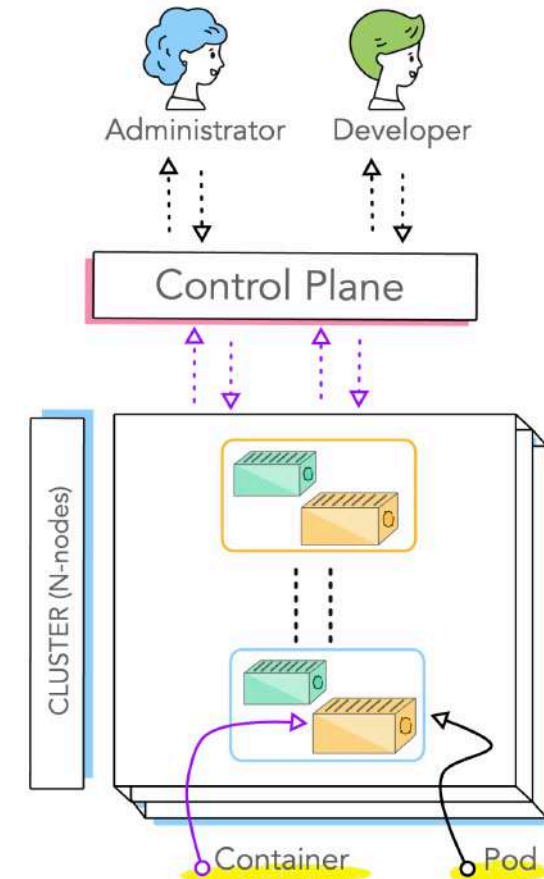




How kubectl Access Works in GKE

When a user runs `kubectl`, two authorization systems are checked

- **Step 1: Google Cloud IAM Authentication**
 - User runs: `kubectl get pods`
 - `kubectl` first retrieves cluster credentials using: `gcloud container clusters get-credentials CLUSTER_NAME`.
Required IAM Permission - `container.clusters.get` (Without this permission, the user **cannot even connect to the cluster**)
- **Step 2: Kubernetes RBAC Authorization**
 - Once authenticated, the request reaches the **Kubernetes API Server**.
 - Example request: `kubectl get pods`
 - Now **Kubernetes RBAC checks**: Is the user allowed to perform `get` on `pods` in this `namespace`? If not → request denied.

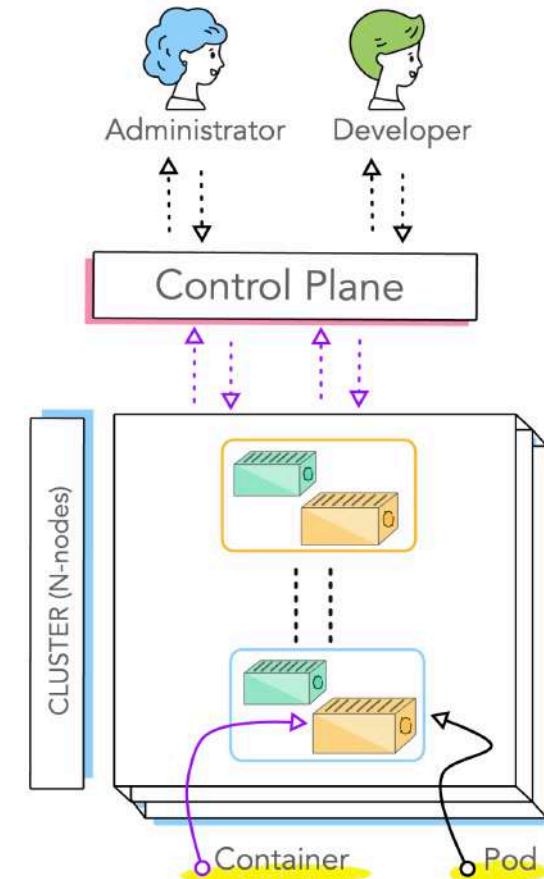




Exploring Google Cloud IAM Roles for GKE

Apply For A: Project, Folder, or Organization

- Affect **all GKE clusters** within that scope
- **Basic Roles:** Owner, Editor, Viewer
 - NOT recommended
- **Predefined GKE Roles:**
 - **Kubernetes Engine Admin** – Full access to clusters and Kubernetes API objects
 - **Kubernetes Engine Cluster Admin** – Manage clusters ONLY
 - **Kubernetes Engine Cluster Viewer** – Read access to clusters
 - **Kubernetes Engine Developer** – Manage API objects and view cluster info
 - **Kubernetes Engine Viewer** – Read-only for both clusters and API objects

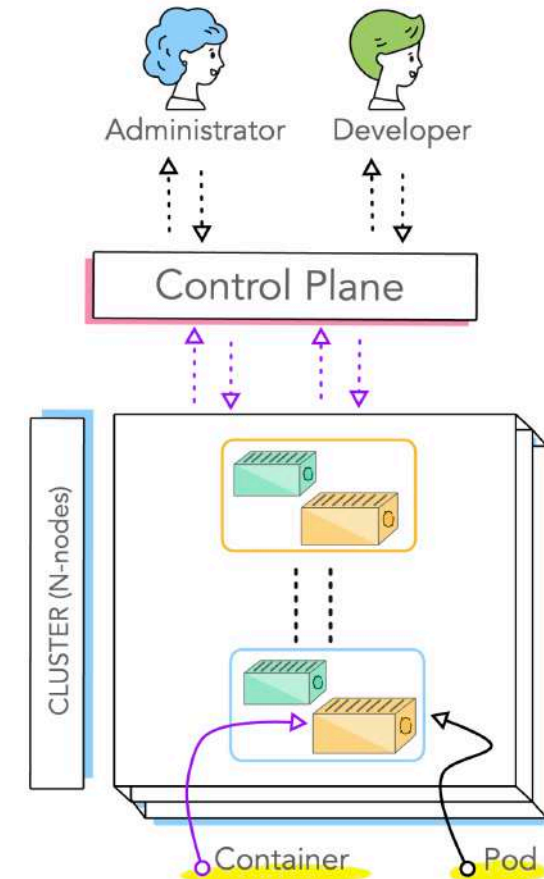




Exploring Kubernetes Role-Based Access Control (RBAC)

Kubernetes RBAC: Cluster-level feature

- **Fine-grained permissions:** In a cluster or namespace
- **Controls Access To:** Deployments, Services, Pods, Secrets, ConfigMaps, and more
- **Key RBAC Objects:**
 - **Role** – Defines allowed permissions in a **namespace**
 - **ClusterRole** – Defines permissions across the **entire cluster**
 - **RoleBinding** – Maps users or service accounts to a Role or ClusterRole **within a namespace**
 - **ClusterRoleBinding** – Maps users or service accounts to a ClusterRole **across all namespaces**
- **Remember:** These are **Kubernetes objects**, just like Service or Deployment





Understanding Kubernetes RBAC - Role

Represents a set of allowed permissions in a namespace

Define:

- **namespace**
- **name**
- **resources:** Which resource?
 - can be any namespaced resource - "services", "pods" etc
- **verbs:** What do you want to allow?
 - examples: "get", "list", "create", "update", "delete" etc

```
apiVersion: rbac.authorization.k8s.io/v1
kind: Role
metadata:
  # namespace is mandatory
  namespace: default
  name: all-configmaps-editor-role

rules:
  - apiGroups: ["" ]
    resources: ["configmaps"]
    # resourceName: ["my-configmap"]
    verbs: ["update", "get"]
```




Understanding Kubernetes RBAC - ClusterRole

Represents a set of allowed permissions
in a cluster

- **name**
- Does NOT need a namespace
- **resources:** Which resource?
 - can be any namespaced resource - "services", "pods" etc (applies to all namespaces in cluster)
 - can be cluster-scoped resources - "nodes"
- **verbs:** What do you want to allow?
 - examples: "get", "list", "create", "update", "delete" etc

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  # "namespace" is not configured
  name: all-secrets-viewer-role
rules:
  - apiGroups: ["" ]
    resources: ["secrets"]
    verbs: ["get", "watch", "list"]
```



Kubernetes RBAC - RoleBinding and ClusterRoleBinding

Map Subjects to a Role

- **Subjects:** Google Cloud user account (email address) or
 - Kubernetes service account (name) or
 - IAM service account (service account email address) or
 - Google Group address
- **Role:** a Role or a ClusterRole

```
apiVersion: rbac.authorization.k8s.io/v1
kind: RoleBinding
#kind: ClusterRoleBinding
metadata:
  name: edit-configmaps
  # NOT needed for ClusterRoleBinding
  namespace: default

subjects:
- kind: User # Group or ServiceAccount
  name: jane@in28minutes.com
- kind: ServiceAccount # Kubernetes SA
  name: kubernetes-service-account-name
- kind: User # IAM service account
  name: iam-service-account@PROJECT-
ID.iam.gserviceaccount.com

roleRef:
  kind: Role # Role or ClusterRole
  name: all-configmaps-editor-role
```



Understanding Kubernetes Security Best Practices

Best Practice	What It Ensures
Enable Node Auto-Upgrade	Keeps GKE nodes patched with latest security fixes
Use Shielded GKE Nodes	Provides secure boot and protects against rootkit attacks
Use Workload Identity	Maps Kubernetes SA to IAM for least-privilege access
Avoid Default Service Account	Prevents nodes from inheriting broad IAM permissions
Use Namespaces and RBAC	Controls who can access which cluster resources
Restrict Pod-to-Pod Traffic	Limits lateral movement using Network Policies
Use Kubernetes Secrets or Secrets Manager	Provides secure storage for sensitive values
Consider HashiCorp Vault for Multi-Cloud	Centralized, secure secret management across clouds



Securing GKE Clusters – Different Elements [1/2]

Focus Area	Things You Can Do
Cluster Security	Restrict control plane access using Private Clusters & Master Authorized Networks. Manage who administers the cluster using Google Cloud IAM & K8S RBAC. Control allowed operations using IAM and RBAC permissions.
Network Security	Restrict network access using Private Nodes, and Kubernetes Network Policies to control Pod-to-Pod traffic.
Container Runtime Security	Protect runtime using GKE Sandbox. Harden your nodes (Container-Optimized OS, Shielded GKE Nodes, Auto updates) Monitor important metrics and logs, use runtime threat detection tools.
Image Supply Chain Security	Use Artifact Registry to store trusted images. Scan images with Container/Artifact Analysis (vulnerability scanning). Enforce trusted deployments using Binary Authorization.



Securing GKE Clusters – Different Elements [2/2]

Focus Area	Security Element Question
Application Security	Secure applications through secure coding practices , dependency scanning , and application vulnerability scanning . Provide minimal cloud access using Workload Identity . Implement authentication and authorization using application-level auth mechanisms (OAuth, OIDC) .
Workload Isolation	Isolate workloads using Namespaces , Kubernetes RBAC , and Network Policies .
Secrets Management	Store secrets using Kubernetes Secrets or Google Secret Manager . (Consider HashiCorp Vault for Multi-Cloud)



Authorization for Kubernetes - Scenarios

Scenario	Solution
Your microservices is deployed to GKE. You want to give each microservice specific access to Google Cloud resources.	Workload Identity
Your microservices are going to be deployed to GKE. You want to give common permissions to Cloud Logging and Cloud Monitoring for all microservices.	Node Pool Service Account
You are sharing GKE cluster between different teams. You want to ensure that each team has access to deploy only to their specific namespaces.	Kubernetes RBAC
Which role gives you complete access to GKE clusters and the Kubernetes API objects?	Kubernetes Engine Admin (<code>roles/container.admin</code>)
You are assigned permissions using Kubernetes RBAC. But you are unable to access the cluster.	Check if IAM member has the permission - <code>container.clusters.get</code>



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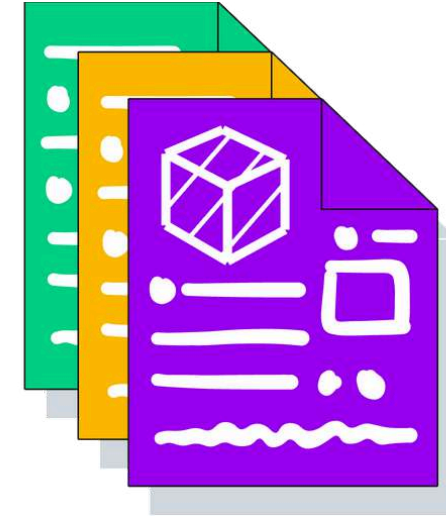
Going Deeper with Compute Engine



GCE VMs - Project and Instance Custom Metadata

Metadata: Additional key-value pairs to attach with VM instances

- **Examples:** Startup script, shutdown script, SSH keys, etc.
- **Levels:** Project or VM Instance (Metadata at project level **applies to all VM instances in a project**)
- **Metadata Server:** Stores all metadata
(<http://metadata.google.internal/computeMetadata/v1>)



Optional: Use **custom metadata** to store application configuration

- **Create custom attributes** - `gcloud compute project-info/instances add-metadata --metadata name=value`
- **Examples:** env=prod, service_url=*<http://external.service.com>*



Executing Shutdown Script on a GCE VM

Shutdown Script: Execute commands before a GCE VM is stopped, terminated or restarted

- Perform cleanup or tasks like export of logs
- **Very similar to startup script**
 - Run as: `root` in Linux VMs, `System` account in Windows VMs
 - **Stored as metadata:** `--metadata-from-file shutdown-script=script.sh`
 - You can store startup and shutdown scripts in cloud storage - `--metadata shutdown-script-url=gs://bucket-in-cloud-storage/file`

Run on: Best-effort basis

- No guarantee that it will run always





Troubleshooting VM startup and ...

Check 1: Are there **Quota** errors?

Check 2: Is **boot disk** full?

Check 3: Check **serial port output**

- Each VM instance has 4 virtual serial ports that capture system-level logs like BIOS, bootloader, and kernel outputs
 - Useful for troubleshooting crashes, startup/shutdown issues
 - Send serial port output to Cloud Logging: `gcloud compute project-info add-metadata --metadata serial-port-logging-enable=true`
 - Interactive access to serial console allows login and debugging of boot issues - `gcloud compute instances get-serial-port-output`

Check 4: Does disk have a **valid file system**? (Attach boot disk as data disk to another VM & check)





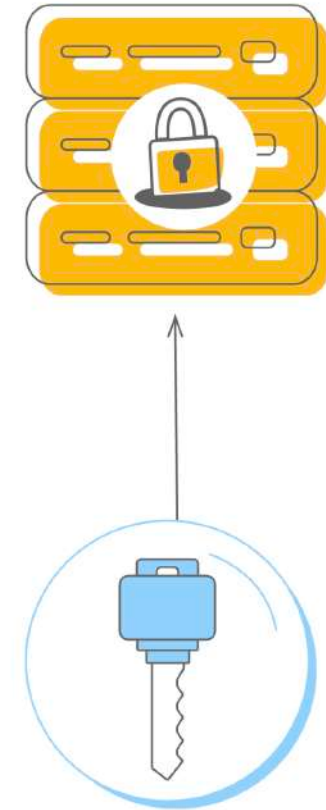
Why Do You Need to Connect to a GCE VM?

You want to troubleshoot a problem or install software: Inspect logs, fix issues, install a package ..

Windows VMs: Use username + password

Linux VMs: Typically SSH keys are recommended for secure authentication

- **Challenge grows with team size:** Manually managing SSH keys becomes messy
- *How do you safely, cleanly, and scalably SSH into Linux VMs?*





SSH Access Options for Linux VMs

Option	Details
Console SSH	Click SSH in the Cloud Console. GCE creates a temporary SSH key pair, adds the public key to instance metadata for a limited time, and logs you in.
gcloud SSH	Run <code>gcloud compute ssh VM_NAME</code> . GCE generates a persistent SSH key pair and stores it locally. Public key is added to VM/project metadata & reused for future logins.
Metadata-Managed Keys	You manually upload your public SSH key to project or instance metadata - <code>gcloud compute project-info add-metadata</code> . VMs configured to use metadata-based SSH keys allow login using these keys. <i>Simple, but becomes messy as teams grow.</i>
OS Login	Enable OS Login (<code>gcloud compute project-info/instances add-metadata --metadata enable-oslogin=TRUE</code>) Your Google identity becomes your Linux identity. Recommended for large organizations because access is controlled centrally using IAM. User should have IAM roles (<code>roles/compute.osLogin</code> , <code>roles/compute.osAdminLogin</code>)



Quick Reference: When to Use Each Method

Console SSH – Quick browser-based troubleshooting

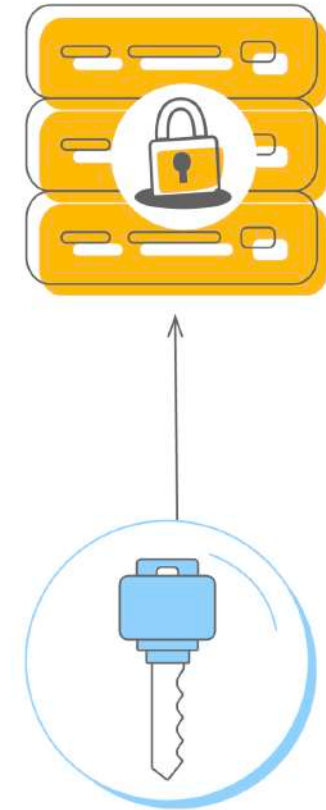
gcloud SSH – Frequent CLI-based access

Metadata-Managed Keys – Small teams with simple access needs

OS Login – Best for large organizations

Windows VMs Reminder

- Use username + password
- Generate using: `gcloud compute reset-windows-password`





Why OS Login Simplifies SSH Access

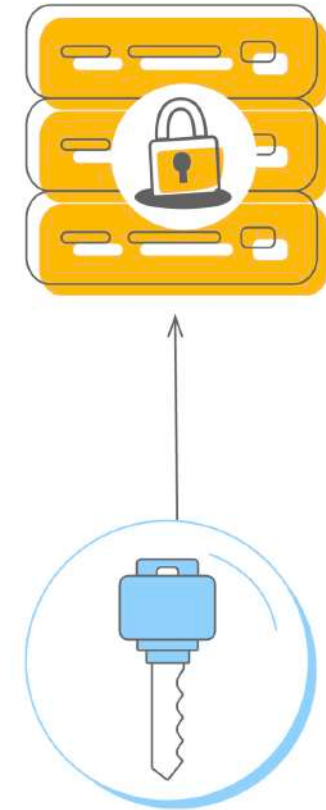
Identity-based access: Your Google identity is mapped to a Linux user account

- Disable a user in IAM → SSH access instantly revoked
- **Enterprise Identity Federation:** Use enterprise identities from on-prem AD or LDAP

Easy enablement: `gcloud compute project-info add-metadata --metadata enable-oslogin=TRUE`

No per-user key management: No uploading keys to metadata

- No outdated or forgotten SSH keys





Connecting to Compute Engine VMs - Scenarios

Scenario	Solution
How is authentication to Windows VMs different from Linux VMs?	Windows uses username/password. Linux typically uses SSH keys.
What is the problem with Metadata-Managed Keys?	You manually upload your public SSH key to project or instance metadata. Becomes messy as teams grow. ONLY for very small teams with simple access needs.
Which approach allows automatic lifecycle management as members join or leave?	OS Login. Removing IAM access immediately revokes SSH access.
What is recommended for managing SSH access in a large organization?	Use OS Login with IAM Group-based access control. No need to manage SSH keys in metadata. Grant access to IAM Groups instead of individual users.



Compute Engine - Other Key Features

Feature	Details
VM Manager	Suite of tools to manage OS for large VM fleets running Windows and Linux - Apply on-demand and scheduled patches, collect and review OS information, install, remove, and auto-update software packages
VM instance insights (Recommender)	Understand the CPU, memory, and network usage of VMs (<i>In the last 12 days for 90% of the time, CPU usage was greater than or equal to 83%</i>) Get insights - machine type recommendations (<i>If your instance has had low CPU utilization most of the time, Compute Engine recommends a machine type with fewer virtual CPUs.</i>) and idle VM recommendations
Delete Protection	Prevent accidental VM deletion. Constraints apply.
Confidential VM	Uses a specific machine type and keeps your sensitive code and other data encrypted in memory during processing (encryption-in-use)



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Getting Started with Artificial Intelligence

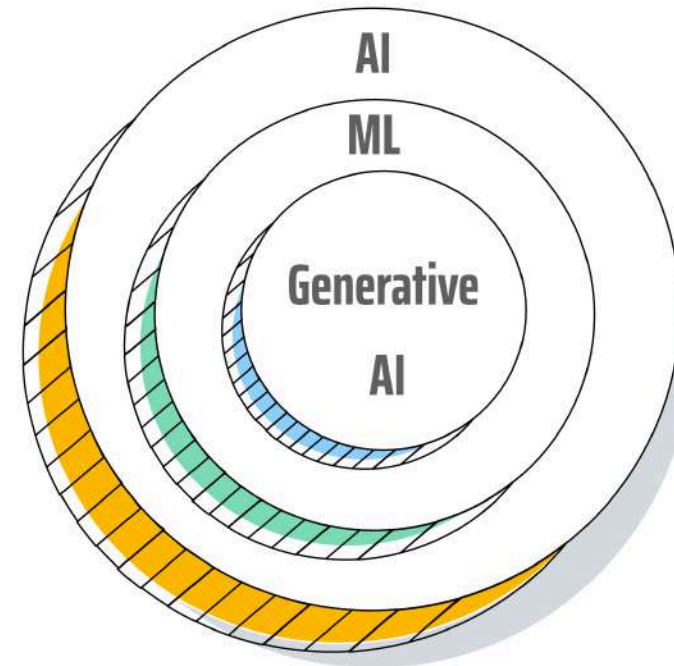
Section Overview - Introduction to AI, ML and Generative AI



Focus: What will we learn?

- What is Artificial Intelligence?
- What is Machine Learning?
- How does Machine Learning work?
- Machine Learning vs Traditional Programming
- What is Generative AI?
- How is Generative AI different from Traditional AI and ML?

Foundation: Build the core concepts needed for the rest of the course





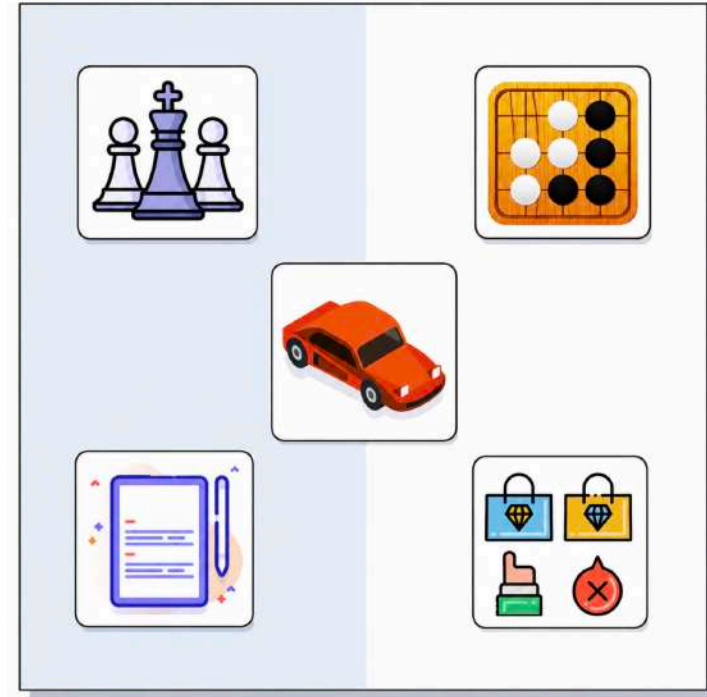
What is Artificial Intelligence?

Humans are great at:

- **Decision-Making:** Choose from multiple options
- **Visual Perception:** Recognize faces and objects
- **Language Understanding:** Understand written text
- **Speech Recognition:** Understand spoken language
- And a lot more..

Artificial Intelligence: Theory & development of computer systems able to perform tasks normally requiring human intelligence

- **Simplified Goal:** Make computers do things that normally require human intelligence





Are All AI Systems Equally Intelligent?

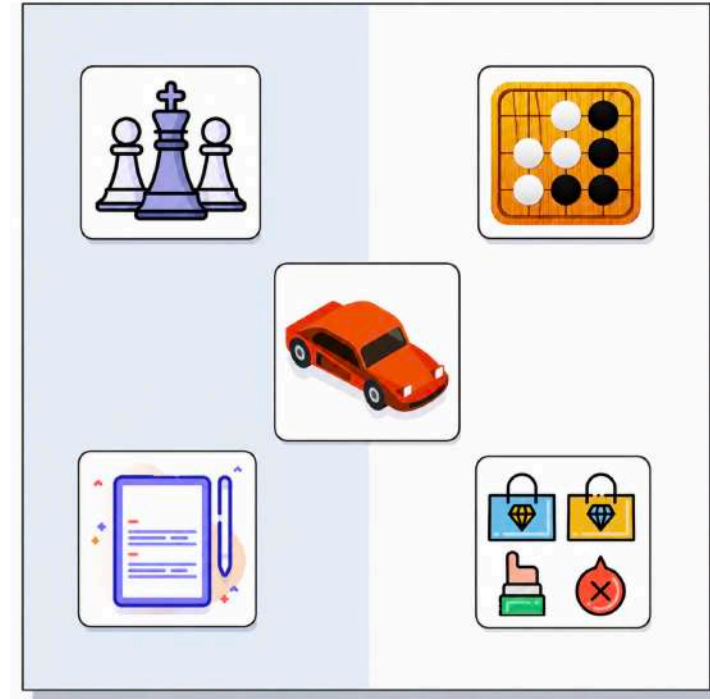
AI can have different levels of capability

Some AI systems are designed to:

- Perform **one specific task**
- Perform **many different tasks**

This leads to two broad types of AI:

- Strong AI
- Narrow AI





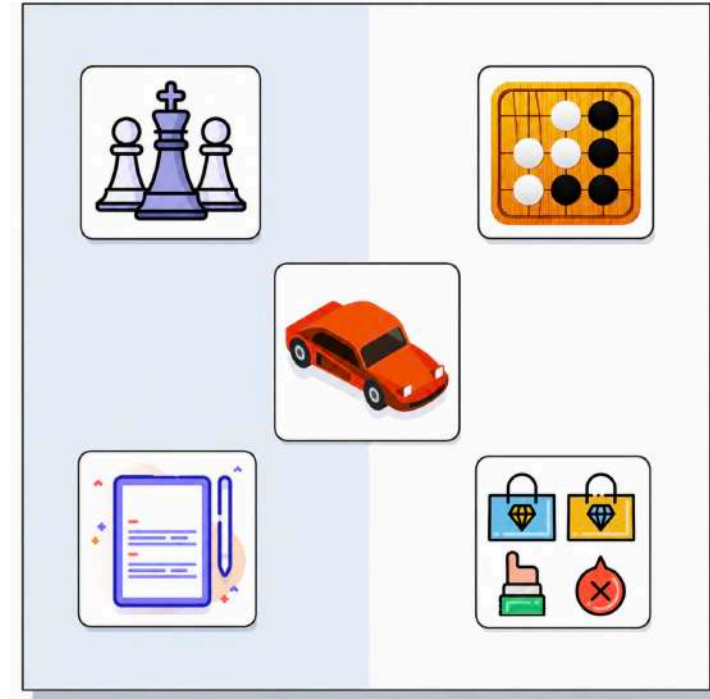
Understanding Types of AI - Strong AI

Strong AI: Goal is to create machines that are as intelligent as humans

Capabilities: Solve problems, learn, and plan for the future

- Expert at everything
- Make complex decisions
- Learns like a child

Also Known As: General AI or Artificial General Intelligence (AGI)





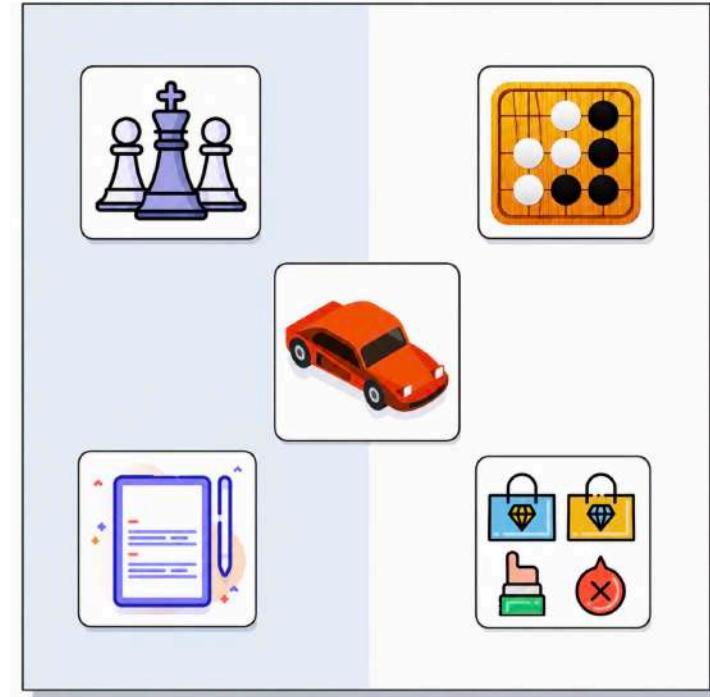
Understanding Types of AI - Narrow AI

Narrow AI: AI focused on a specific task

Examples:

- **Self-driving Car:** Can drive but cannot play chess
- **Chess App:** Can beat champions but cannot drive
- **Weather Predictor:** Knows weather but cannot write poetry

Also Known As: Weak AI





Getting Started with AI - Scenarios

Scenario	Solution
What is the Goal of AI?	Make computers do things that normally require human intelligence
Strong or Narrow AI: Build a system that can learn any game, write poetry, and drive a car like a human	Strong AI (General AI)
Build a computer system that focuses on a specific task such as self-driving, virtual assistants, or object detection	Narrow AI (or Weak AI)
The AI used in your smartphone to recognize your face	Narrow AI



Why are we talking a lot more about AI now?

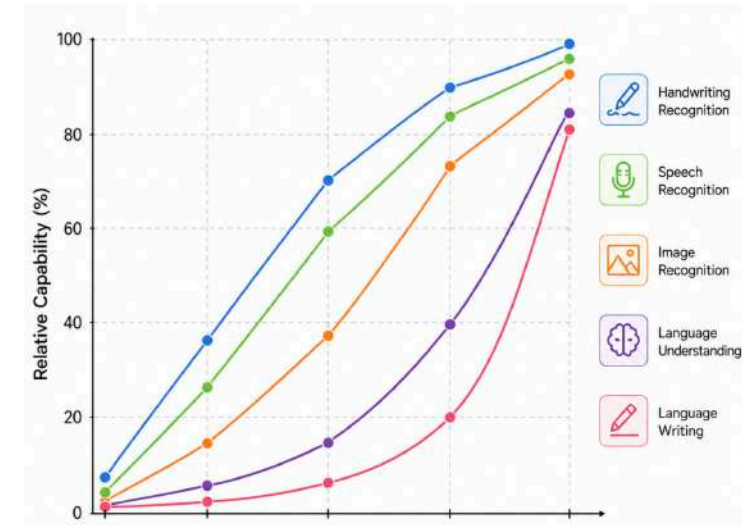
AI is Not New: Researchers have been working on AI since 1950s

Question: Why are we talking a lot more about AI now?

- **Answer:** AI capabilities improved dramatically in recent decades

Question: How can we measure progress in Artificial Intelligence?

- **Early Answer:** The **Turing Test**





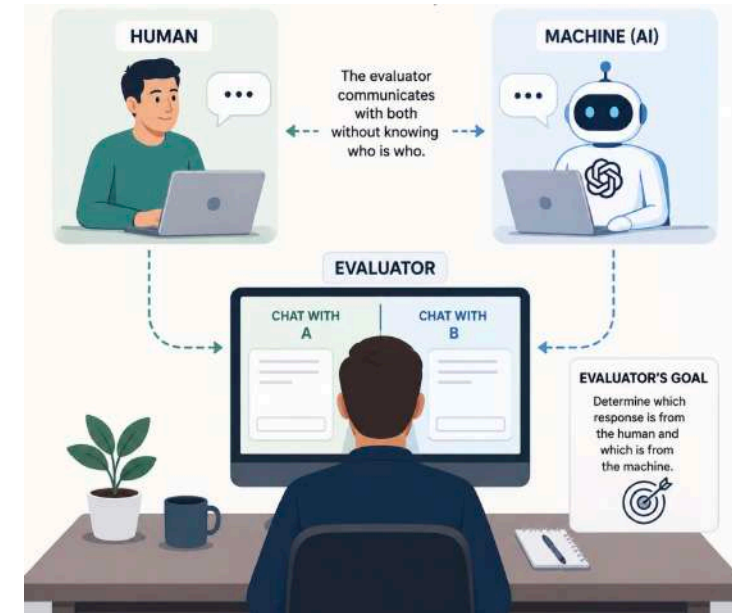
What is Turing Test?

Goal: Check if a AI system can communicate like a human

- 1: An evaluator chats with a human and a AI system
- 2: The evaluator asks questions through text
- 3: Both respond to questions
- 4: Evaluator tries to identify the machine

Early AI: AI systems struggled to imitate human conversation

Today: Advanced AI systems can often produce human-like responses





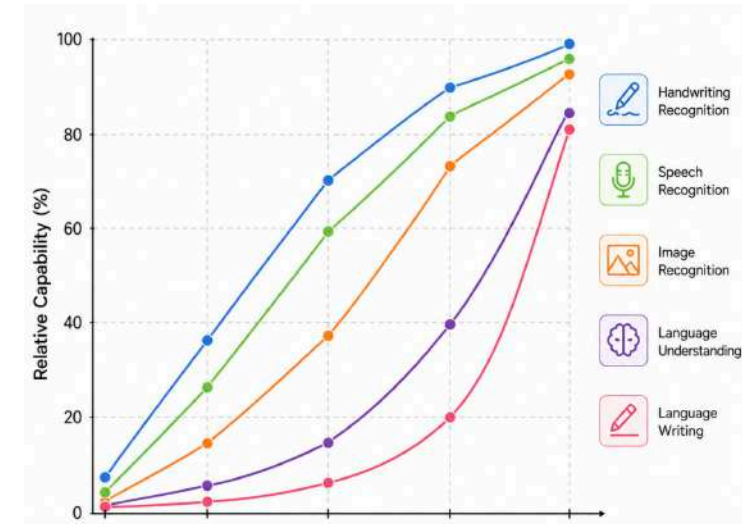
AI Has Made a Lot of Progress

AI Improved Rapidly: Across many areas

- **Image Recognition:** Recognizing object in images
- **Speech Recognition:** Convert spoken words into written text
- **Language Generation:** Large language models are capable of fluent writing
- And a number of others

How: How did this progress happen?

- Coming up!





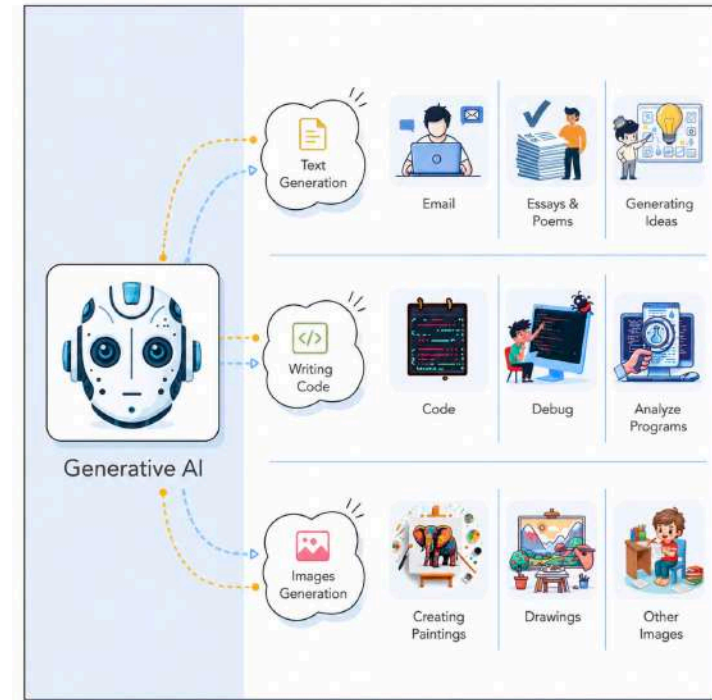
Playing with Generative AI

Checklist: Generate a list of items I need for a 15-day Everest Base Camp trek

- Update the list assuming I will stay at a tea house as part of a guided group

Multimodal: Let's create an image showing the atmosphere at a cricket match

- A vibrant, high-energy night scene at an Indian Premier League (IPL) cricket match. Focus on the electric atmosphere. Show fans wearing team jerseys, waving flags, and cheering loudly. Include digital scoreboards and a sense of celebration.



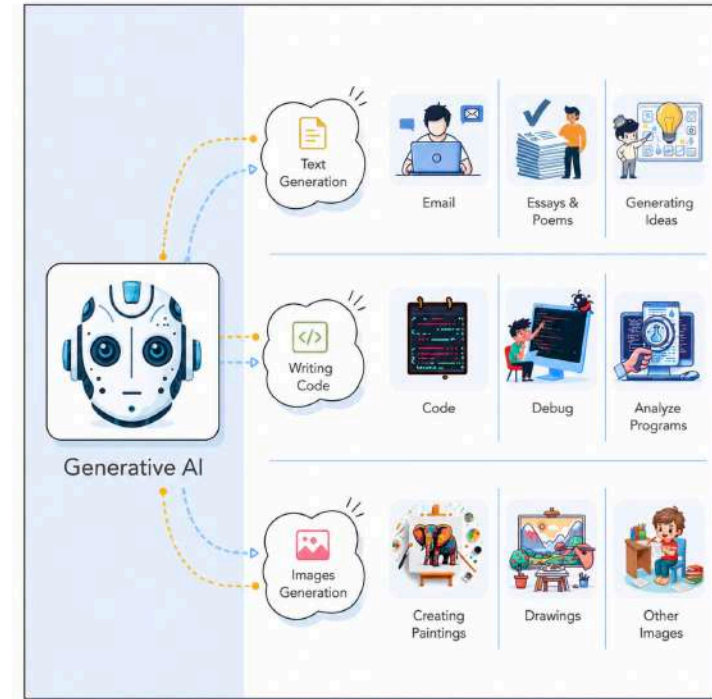


Playing with Generative AI - Observations

Powerful, Not Perfect: Can produce incorrect, biased, or inappropriate responses

- **Use with Judgment:** Outputs should be reviewed and verified
- **Still Valuable:** When used with the right expectations, Generative AI can be extremely useful

How does it work?: Let's start with Machine Learning and build our way to Generative AI





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Getting Started with Machine Learning



How is Traditional Programming Done? [1/2]

Approach: Rule-based logic

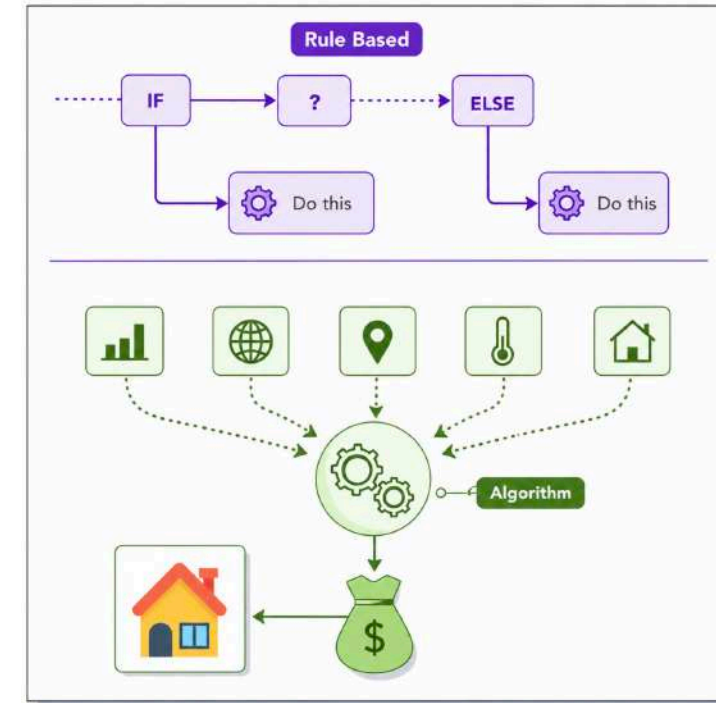
Mechanism: You explicitly tell the computer what to do

Flow:

- **Input:** Data
- **Program:** Rules such as IF this DO that
- **Output:** Result

Example: Calculating tax

- **Rule:** If income > 50k, tax is 20%





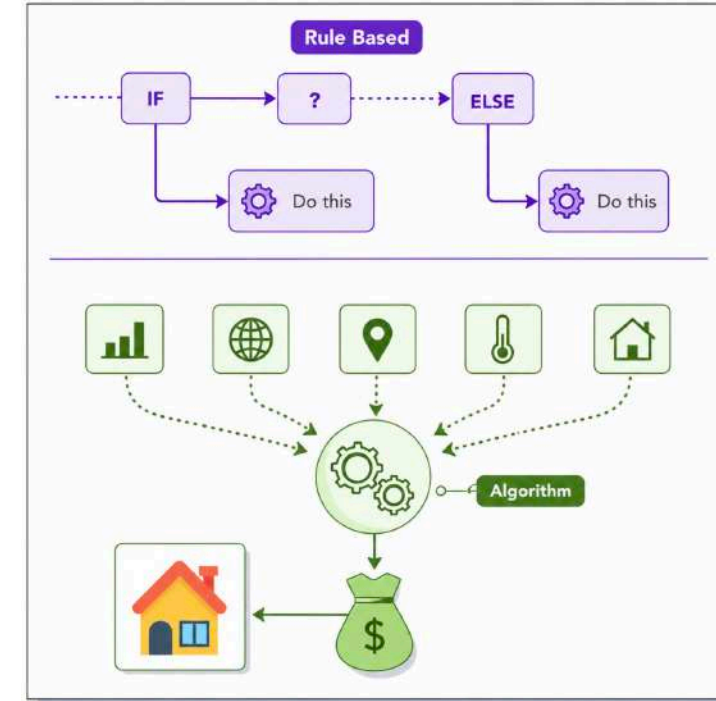
How is Traditional Programming Done? [2/2]

Example: Predicting house price

Approach: You design an algorithm that considers all key factors

- Location
- Home size
- Age
- Condition
- Market

Result: The computer does exactly what you tell it, nothing more and nothing less





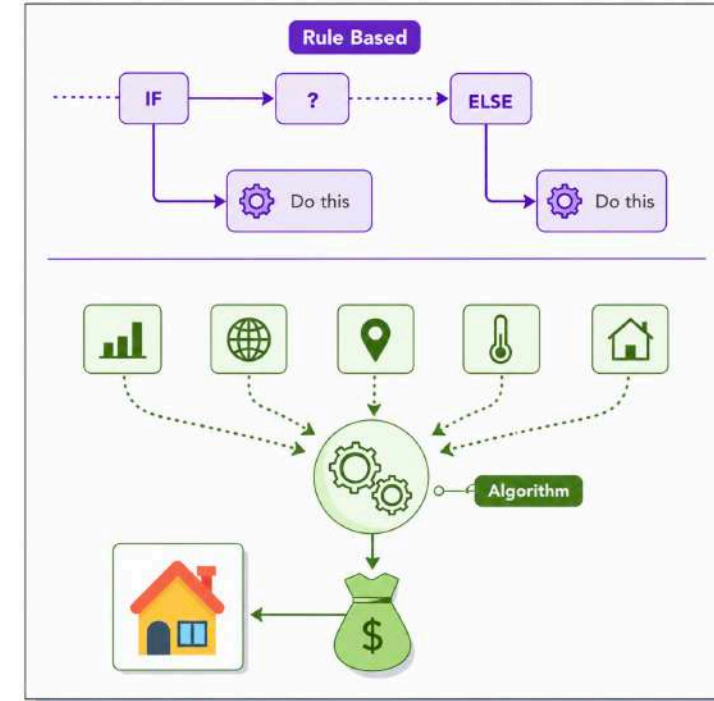
What is the Limitation of Traditional Programming?

Traditional Programming: Input --> Program (Rules or Algorithm) --> Output

Think About This: Can you write the rules to recognize a cat?

- **Example:** If pixel 1 is light and pixel 2 is dark, and so on
- **Reality:** Almost impossible to hand-code such kind of rules

Result: We hit a wall with traditional coding





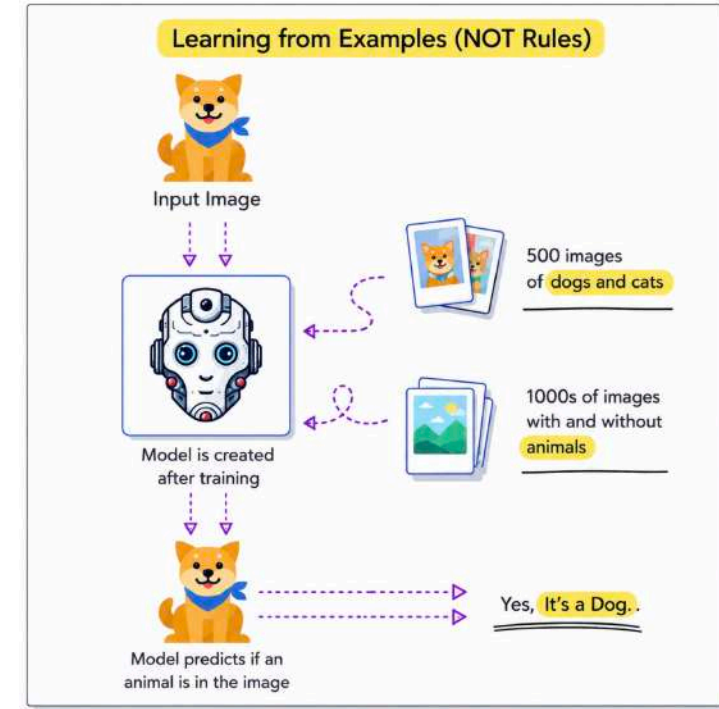
How Does Machine Learning Work?

What Happens?: ML learns from data

Machine Learning: Give examples, ML finds patterns, then predicts the outcome

Example:

- **House Price Prediction:** Learn from past sales data





Machine Learning - Simplified Workflow

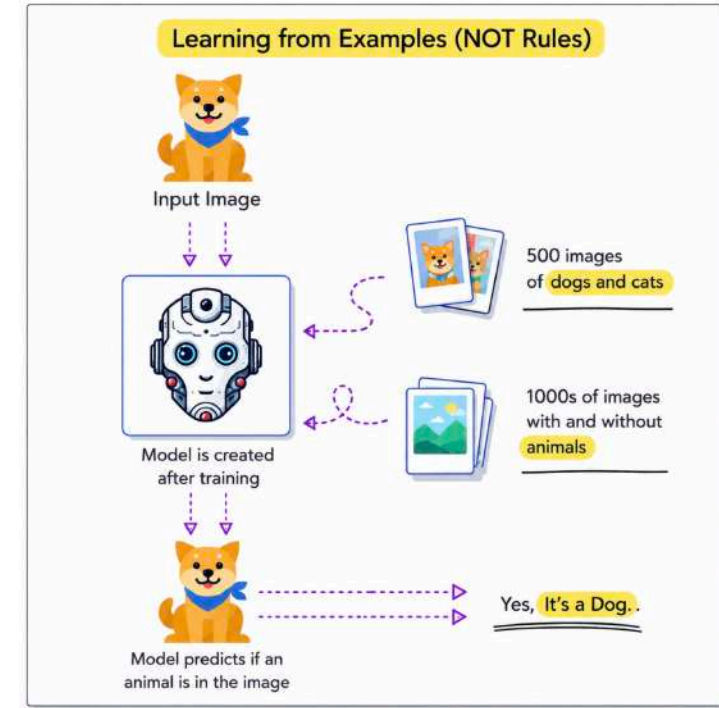
Step 1 - Obtain Data: Provide examples

- **Example:** Photos of cats, dogs and other animals with labels

Step 2 - Train Model: The system analyzes images to find patterns

- **Learned Patterns:** Shapes, edges, and textures

Step 3 - Predict: Give it a new photo and get a prediction





Traditional Programming vs Machine Learning

Traditional Programming

Programmer writes the rules (program)

Input → Program (Rules or Algorithm) → Output

Works well when rules are known

Example: Tax calculation

Example: Payroll processing

Machine Learning

Model learns the rules from data

Input + Expected Outputs → Training → Model

Works well when rules are difficult to define

Example: House price prediction

Example: Spam email detection



Machine Learning Fundamentals - Scenarios

Scenario	Solution
Developer writes logic to tell the computer what to do	Traditional Programming
System finds patterns from data that humans cannot explicitly define	Machine Learning
Category of AI that focuses on learning from data	Machine Learning
How is ML different from traditional programming?	Traditional Programming uses rules, ML uses examples



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Getting Started with

Machine Learning Stages



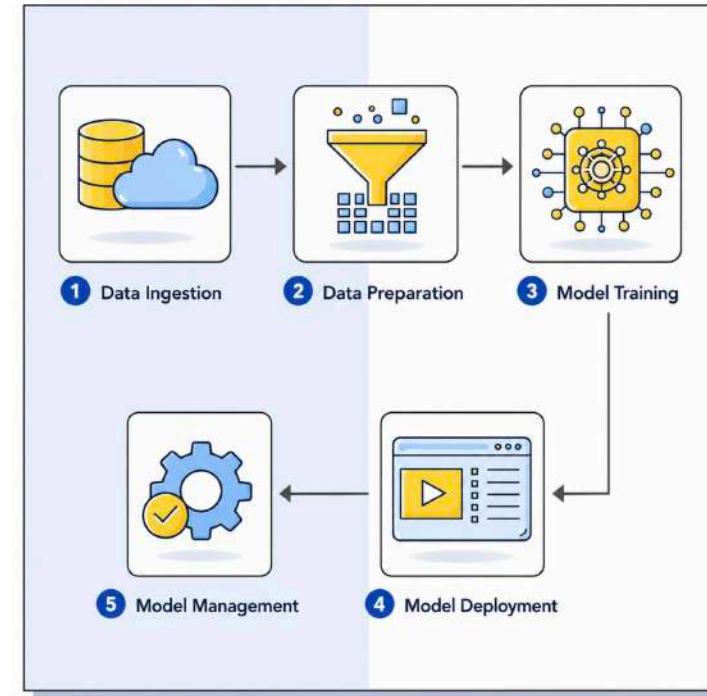
ML Lifecycle: Data Related Stages

Data Ingestion: Collect raw data from different sources

- **Example:** Streaming user activity from a mobile app
- **Example:** Pulling transaction logs from a sales system

Data Preparation: Clean and format data for training

- **Example:** Remove duplicates
- **Example:** Handle missing values in patient health data





ML Lifecycle: Model Related Stages

Model Training: Use data to train the model

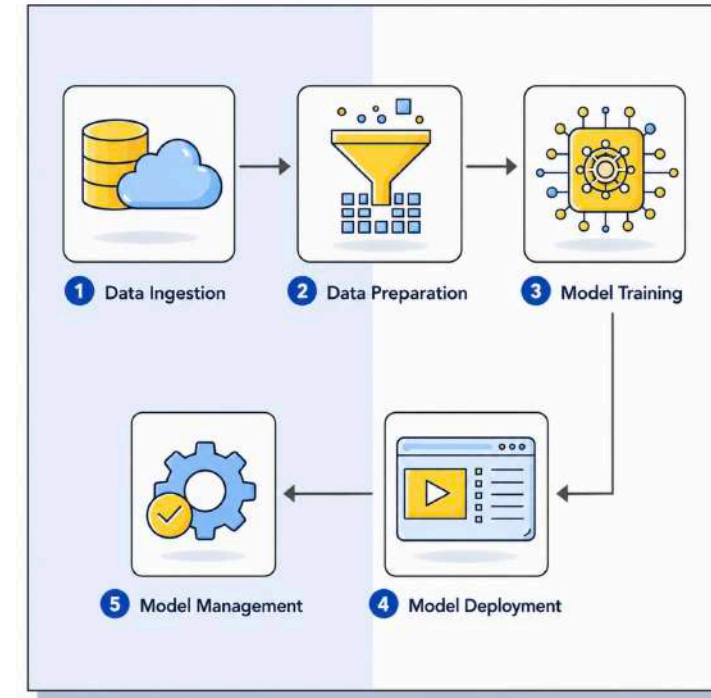
- **Example:** Train a fraud detection model with transactions
- **Example:** Train a price prediction model for used cars

Model Deployment: Make the trained model available to users

- **Example:** Deploy a chatbot in production (Inference)

Model Management: Monitor and maintain the model after deployment

- **Example:** Update the model as new financial data comes in (Re-Training)





ML Lifecycle Recap: End-to-End Example

Use Case: ML model to detect spam in email

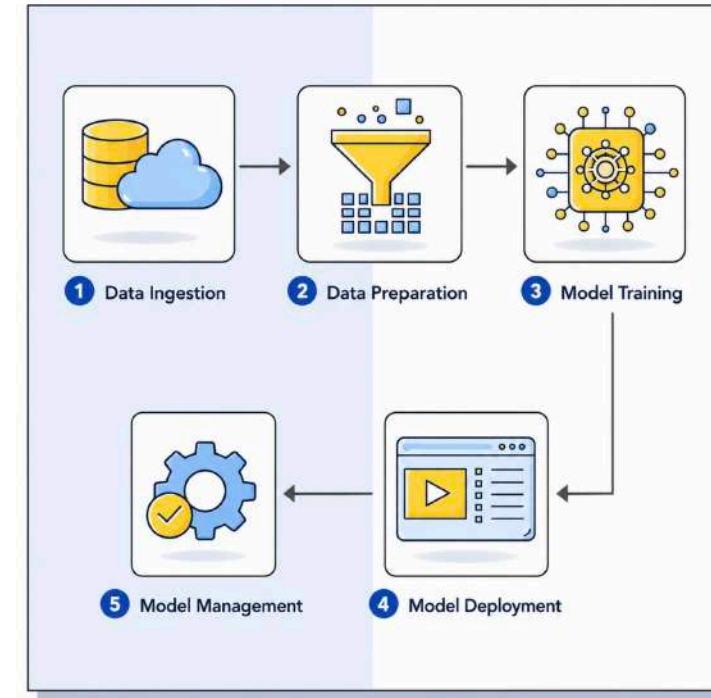
1: Collect email history

2: Prepare labels

3: Train with examples

4: Deploy into email platform

5: Manage performance





Identify the Stage in ML Lifecycle - 1

Activity	Stage
Pulling product transaction logs from a retail store's log store	Data Ingestion
Streaming real-time user activity from a mobile app	Data Ingestion
Removing duplicate customer records from the training dataset	Data Preparation
Filling in missing values in patient data using averages	Data Preparation
Labeling emails as spam or not spam for training	Data Preparation
Using thousands of labeled housing records to teach a model to predict price	Model Training
Training a model to detect credit card fraud using past transaction data	Model Training



Identify the Stage in ML Lifecycle - 2

Activity	Stage
Making your trained model available via an API to a web application	Model Deployment
Deploying an image classification model into a mobile app	Model Deployment
Monitoring how accurate your chatbot is after deployment	Model Management
Updating your model with recent data after noticing performance drop	Model Management
Creating a dashboard that displays model inference accuracy trends	Model Management



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Getting Started with Generative AI



Generative AI - Generates New Content

Goal: Generate New Content

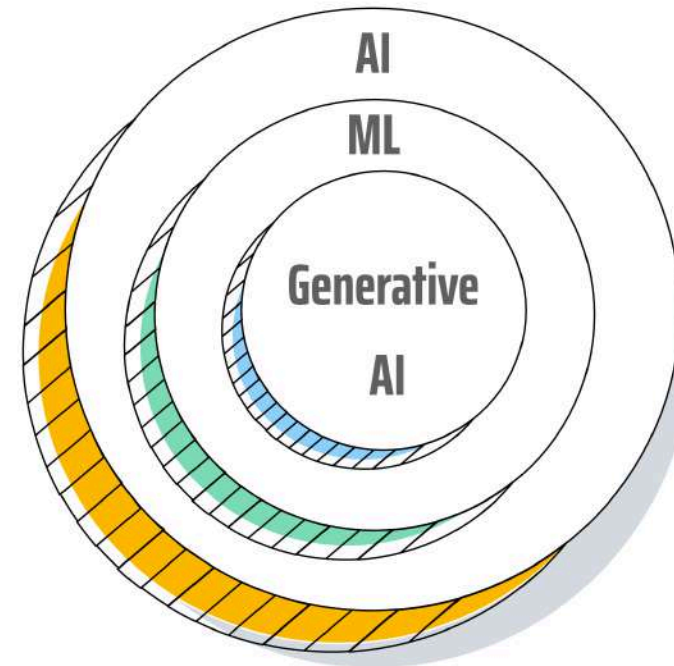
- Instead of making predictions, Generative AI focuses on creating new content

Text Generation: Writing e-mails, essays & poems. Generating ideas.

Code Generation: Generate code snippets. Create unit tests.

Media Creation: Design images. Create instructional videos.

And a lot more features!



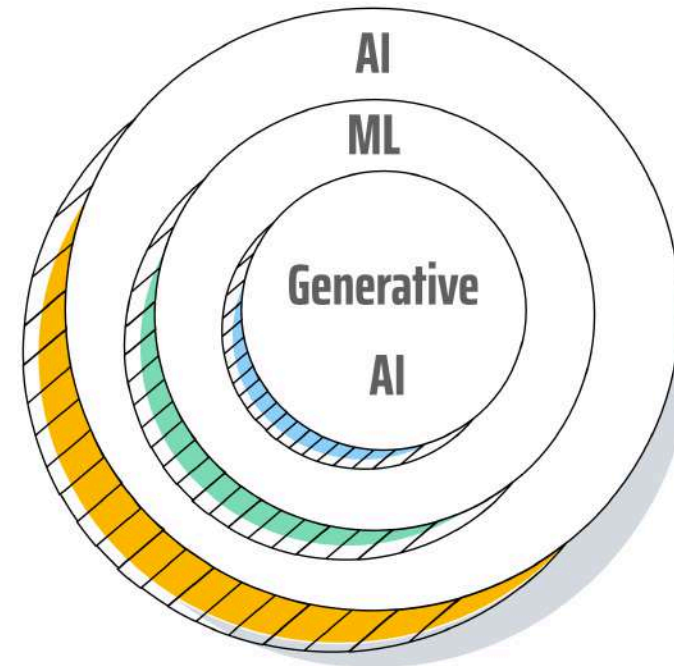


Generative AI - How is it different?

Artificial intelligence (AI): Create machines that simulate human-like intelligence and behavior

Machine learning (ML): A subset of AI where machines learn from data

Generative AI: AI that learns patterns from data to **generate new content**





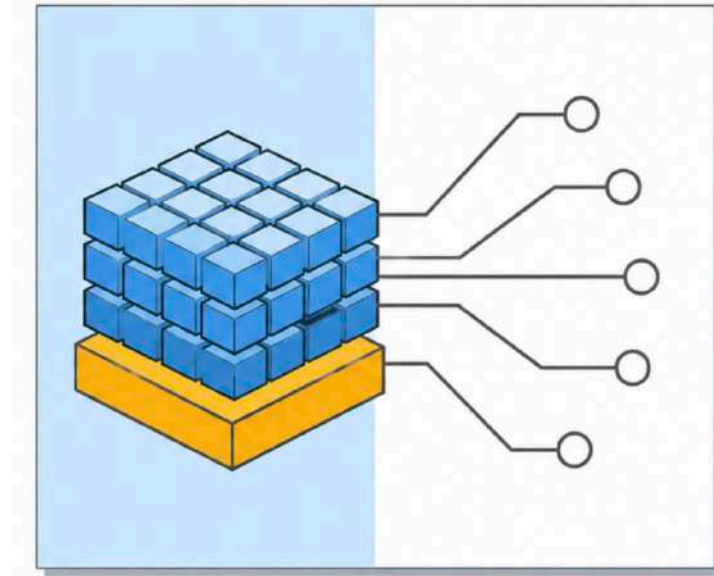
Foundation of Generative AI - The Foundation Models

Foundation Model: The core model powering many generative AI applications

- **One Model, Many Tasks:** Answer questions, summarize, generate code, create content
- **Examples:** GPT, Gemini, Claude, Llama

Pre-trained: Trained on massive amounts of text, code, images, and other data

- **Expensive to Train:** Requires significant compute, time, and data
- **Easy to Reuse:** Build applications without training a model from scratch



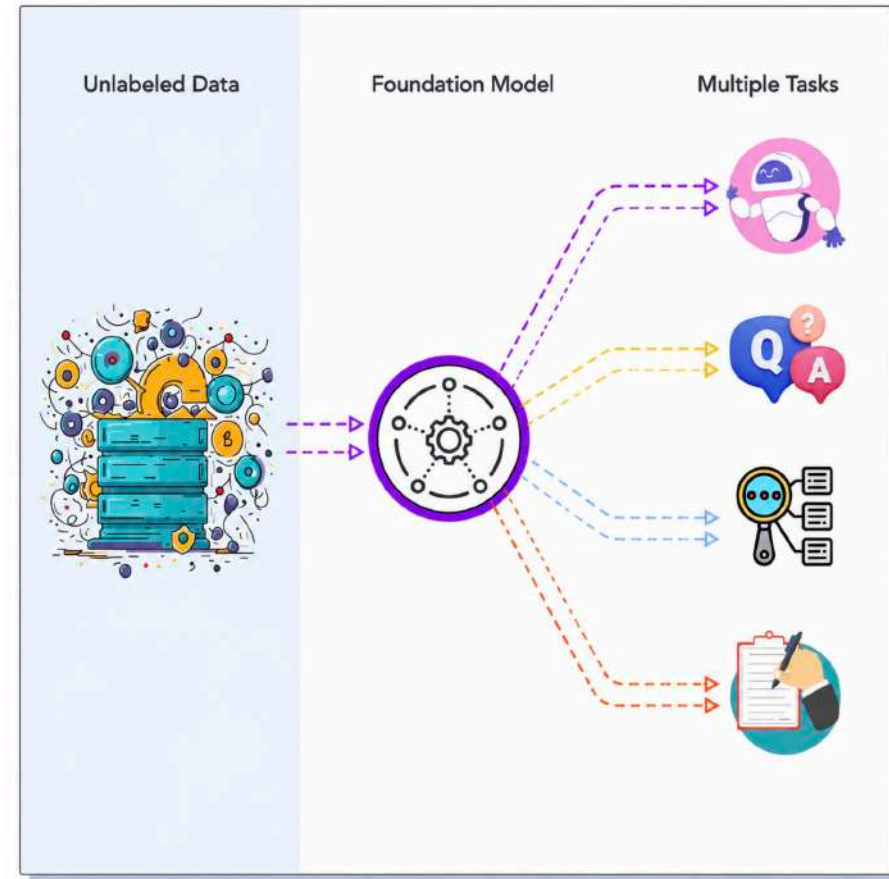


Training Foundation Models is Complex

Complex Training Process: Training is computationally intensive and complex

- **Massive Datasets:** Books, code, web pages, images, captions, and more
- **Specialized Hardware:** GPUs, TPUs, distributed storage, and parallel processing
- **Takes Time:** Training runs for weeks or months

Outcome: Broad general knowledge - understands how humans write, speak and code





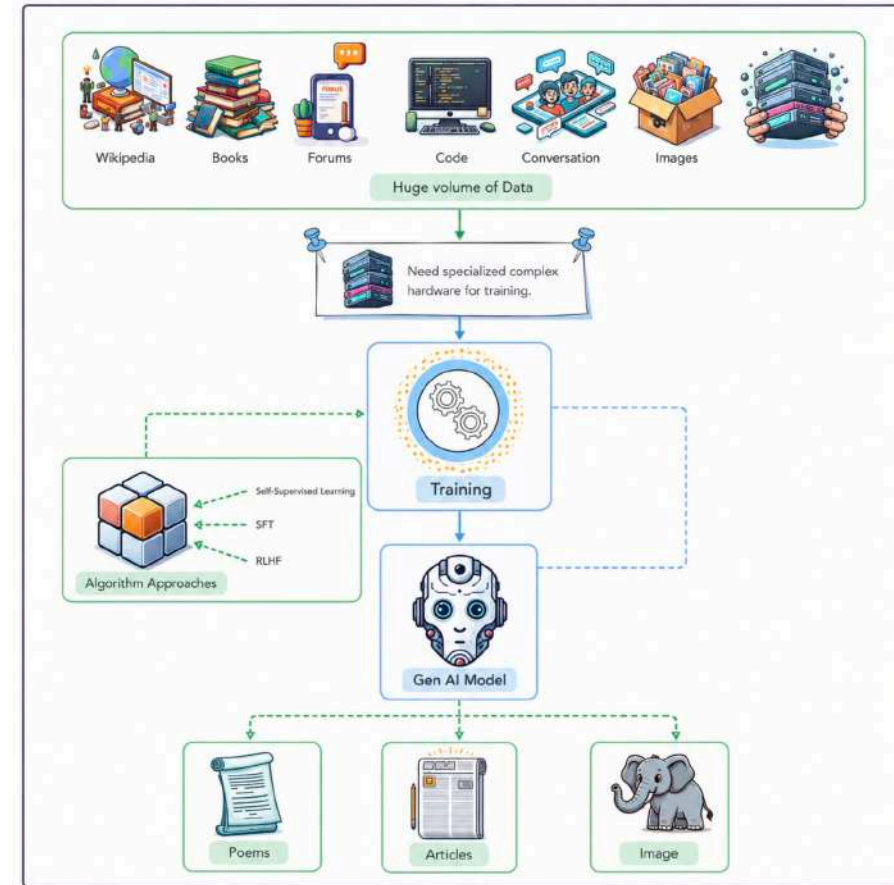
Foundation Models Are Multi Task Models

Traditional ML Models:

- Needed task specific training
- Multiple tasks => Multiple trainings => Multiple models

Foundation Models: Multi Task Models

- Trained once (called pre-training)
- Same model can be used for multiple tasks
 - Chatbot, Classification, Summarization, etc.
- Some models are multi modal as well: Text, video, audio, image...





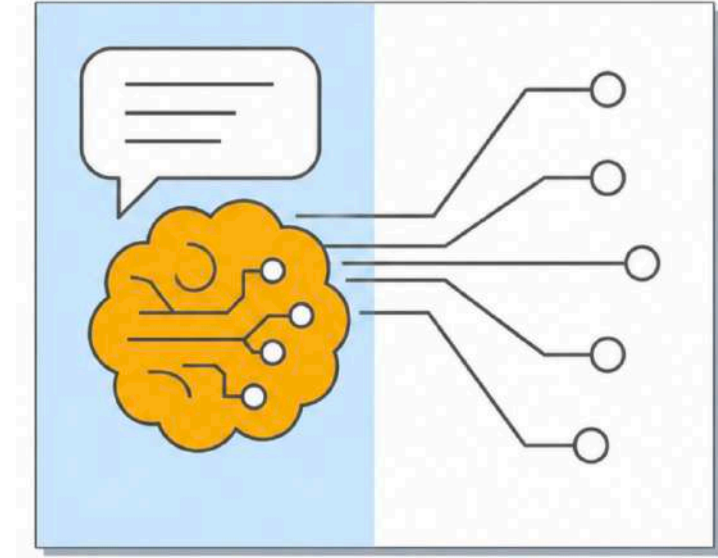
Text Foundation Models are called Large Language Models

Trained on **huge volumes** of diverse text data

- Books, articles, websites, code, and more

Learn to **predict the next token** in a sequence

- Example: "The cat sat on the ..."
- Model gives probabilities:
 - "mat" (40%), "table" (20%), "chair" (20%), "moon" (10%)
- It usually picks the most likely token — but we can tune settings to make it more creative
 - e.g., using the temperature parameter



Use cases: Chat, summarization, translation, Q&A, ..



Diffusion Models: Turning Noise into Art

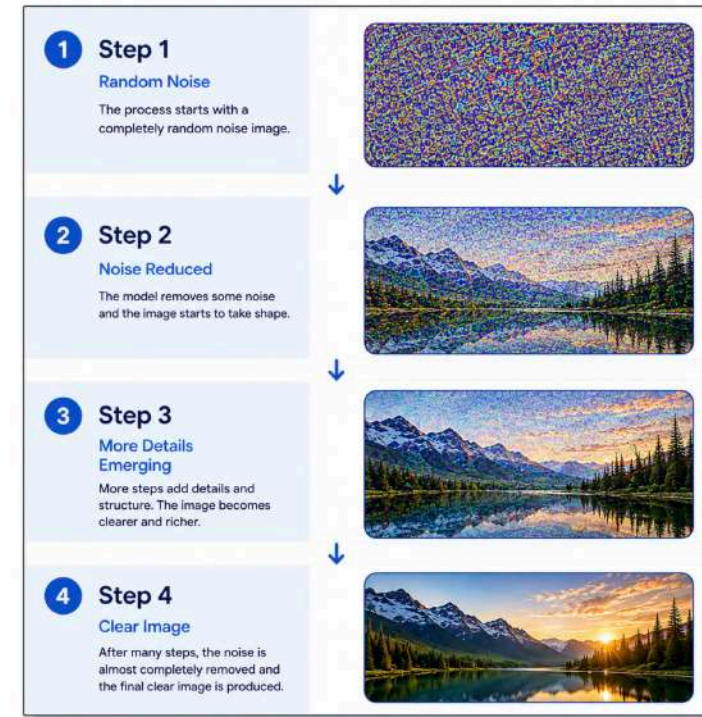
Diffusion models: Great at generating images, audio, and video

How they work:

- Start with **random noise**
- Iteratively **refine it** step-by-step
- Result: **Structured, realistic output**

Used for:

- Creating AI art and realistic pictures from text prompts
- Generating life-like human voices or music
- Producing short AI-generated videos





Foundation Models - Scenarios

Scenario	Solution
Models trained on large, diverse datasets across domains. Adaptable to a wide variety of tasks.	Foundation Models
Text foundation models trained on huge volumes of diverse text data.	Large Language Models
Models that generate images, audio, or video starting with random noise and refining step-by-step	Diffusion Models



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Getting Started with AI Agents



Why Do We Need AI Agents?

Models are great at understanding, reasoning, and generating outputs

- **Real-world tasks** often require more than generating a response:
 - **Decide** what needs to happen next
 - **Interact** with external systems
 - **Take Actions** based on results
 - **Complete Tasks** involving multiple steps

Question: How do we go one level above models and focus on achieving a specific goal?

- **Answer:** AI Agents





What Are AI Agents?

AI Agent: A system that uses AI models to pursue a goal

- **Observe:** Understand user input and context
- **Reason:** Determine what needs to be done
- **Use Tools:** Access data, invoke API and interact with other applications

Agentic AI: Moves beyond one-off prompts

- Uses models to pursue goals (Can adapt across multiple steps)

Key Idea: Move beyond generating responses to getting things done





AI Agents - An Example

User Request: "Summarize report and send to team"

A model can:

- **Generate:** Summarize the report

An agent can:

- **Understand:** Identify the user's goal
- **Generate:** Create the summary using a model
- **Find Information:** Identify the team members
- **Use Tools:** Connect to email
 - **Take Action:** Send the summary
- **Result:** The task is completed





Use Cases for AI Agents [1/2]

Customer Agents: Assist customers and perform customer-facing tasks

- **Example:** Answer account questions and help resolve issues

Employee Agents: Help employees complete internal tasks

- **Example:** Answer HR questions and submit leave requests

Creative Agents: Assist with content creation workflows

- **Example:** Create campaign copy and visuals





Use Cases for AI Agents [2/2]

Data Agents: Analyze data and generate insights

- **Example:** Analyze sales data and summarize important trends

Coding Agents: Assist with software development tasks

- **Example:** Generate code, review, and fix issues

Security Agents: Assist with detecting and responding to threats

- **Example:** Investigate suspicious activity and initiate remediation





Agent Pattern: Conversational Agents

Goal: Interact with users through natural language

Flow:

- **1: Receive Request:** User requests an action
- **2: Understand:** Agent understands the request
- **3: Use Tools:** Fetch information or perform actions
- **4: Respond:** Return the result to the user

Examples:

- **Healthcare:** "Book an appointment with Dr. XYZ tomorrow"
- **Retail:** "Where is my order?"





Agent Pattern: Workflow Agents

Goal: Execute multi-step processes to achieve a business goal

- **1: Trigger:** User or system initiates a task
- **2: Determine Steps:** Determine what to do
- **3: Execute:** Use tools and external systems
- **4: Adapt:** Determine subsequent steps
- **5: Complete:** Deliver the final result

Examples:

- **HR On-boarding Agent:** Create account → assign equipment → schedule orientation
- **IT Operations Agent:** Diagnose issue → restart service → create incident → notify team





Getting Started with AI Agent Architecture [1/2]

Persona: Defines the agent's role, behavior, and communication style

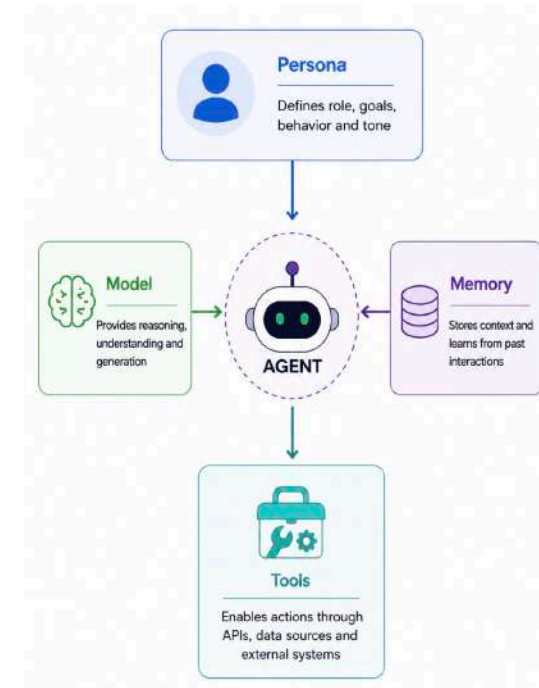
- **Defined Using:** Agent instructions or system prompts

Model (LLM): Provides the core intelligence for the agent

- **Understand:** Comprehend user requests and instructions
- **Reason:** Determine what needs to be done
- **Generate:** Create responses

Memory: Provides context across interactions

- **Short-Term:** Current conversation and task context
- **Long-Term:** Historical information and past interactions





Getting Started with AI Agent Architecture [2/2]

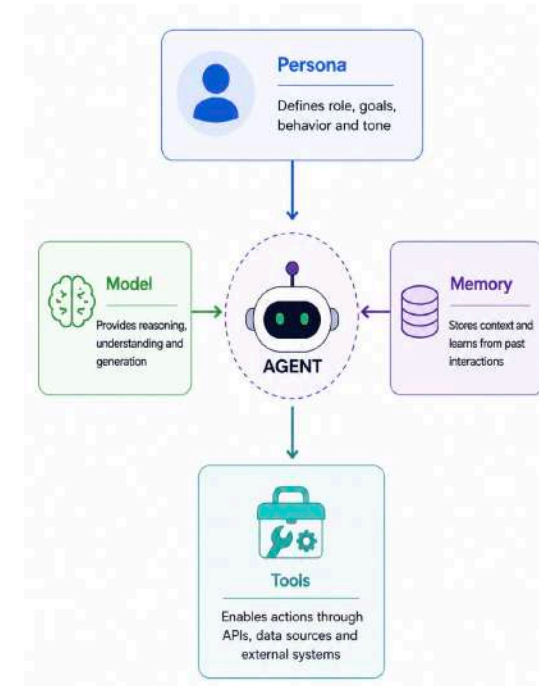
Tools: Extend what an agent can do

Knowledge Tools: Access external information

- **Examples:** Search engines, documents, databases, etc.

Action Tools: Perform tasks in external systems

- **APIs:** Connect to services such as translation or travel booking
- **Functions:** Execute application logic such as `calculate_xyz()`
- **Integrations:** Connect to calendars, email, payment systems, etc





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Getting Started with Model Deployment

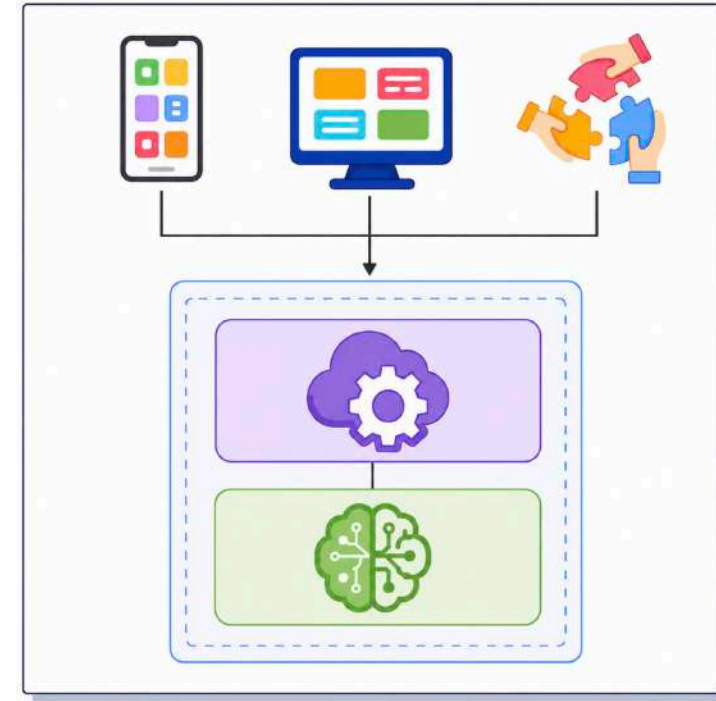


Goal of Model Deployment

Goal: Make an AI model easy to use in production

Endpoint: Typically expose the model through an API

- **Request:** Application/user sends a prompt
- **Inference:** Run the input through the model
 - **Processing:** Apply system instructions and other configuration
- **Response:** Return the generated output
 - **Examples:** Text, image, audio, or structured JSON



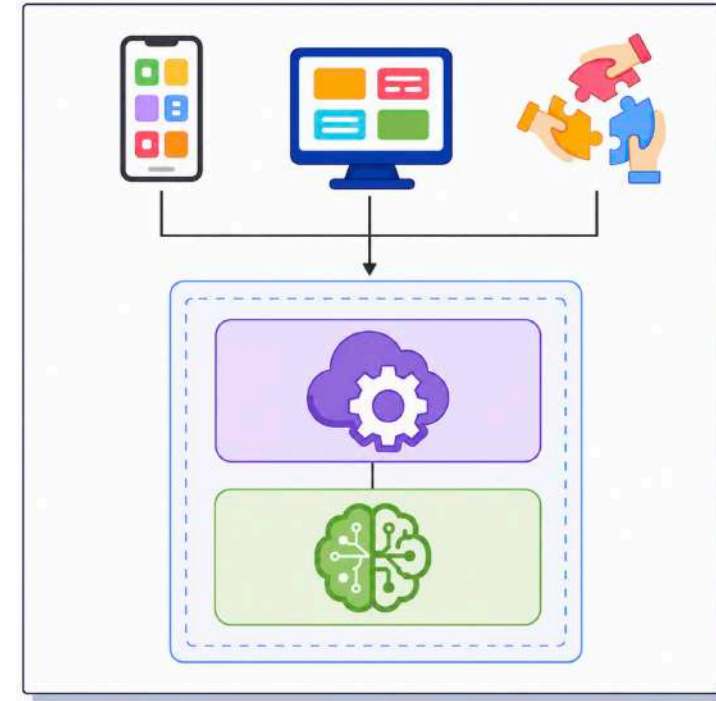


Model Deployment is Challenging

Goal: Make the **generative AI model** available through a stable, scalable, and secure endpoint

You need to handle a lot yourself:

- **Hardware:** Provision Servers - CPUs, GPUs, etc.
- **Deployment:** Install dependencies and deploy the model
- **Availability:** Keep the model endpoint running
- **Scaling:** Add or remove capacity based on traffic
- **Security:** Control access to the model
- **Operations:** Monitor performance, failures, and resource usage



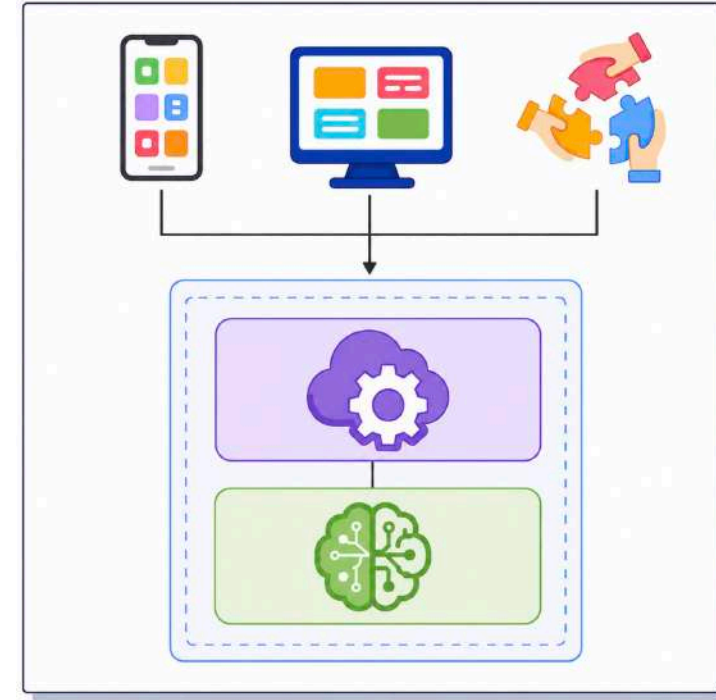


How Does Cloud Make Model Deployment Easy?

Managed AI Services: Reduce complexity

- **1: Pick a Model:** Choose a supported foundation model or your own model
- **2: Configure Deployment:** Select compute and deployment configuration
- **3: Deploy:** Cloud platform creates the infrastructure and endpoint
- **4: Use the API:** Send requests to the endpoint
- **5: Built-in Ops:** Platform helps with scaling, availability, monitoring, and security

Focus: Build AI applications instead of managing model infrastructure





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Getting Started with

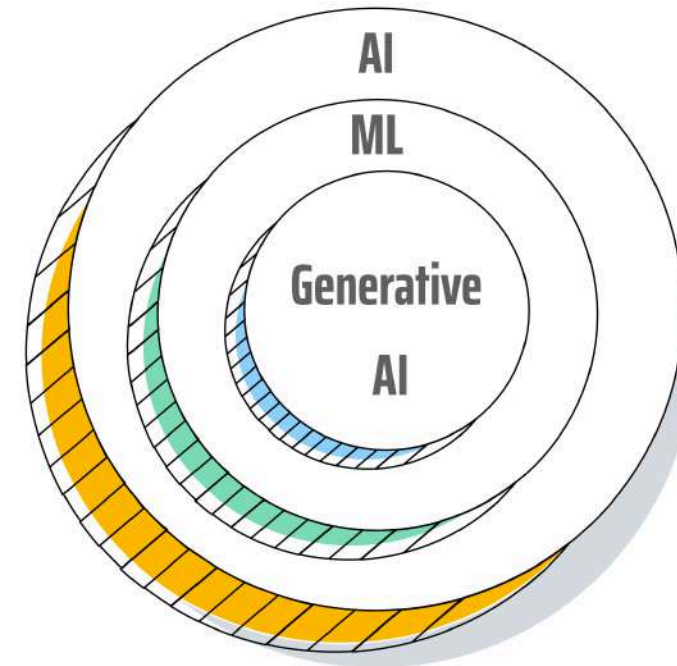
ML Landscape in Cloud



The ML Landscape - Before vs After GenAI

Before GenAI: The 3 Options

- **Pre-built API:** Specific tasks (Vision, Speech, Language, etc.)
- **AutoML:** Custom models without code
- **Training ML Models Using ML Frameworks:** Use frameworks like PyTorch, TensorFlow, Scikit-learn to build models



After GenAI: 2 More Options

- **Foundation Models (API):** Create new content
- **Tune Foundation Models:** Customize Foundation Models for your data



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ML Landscape

Before Generative AI



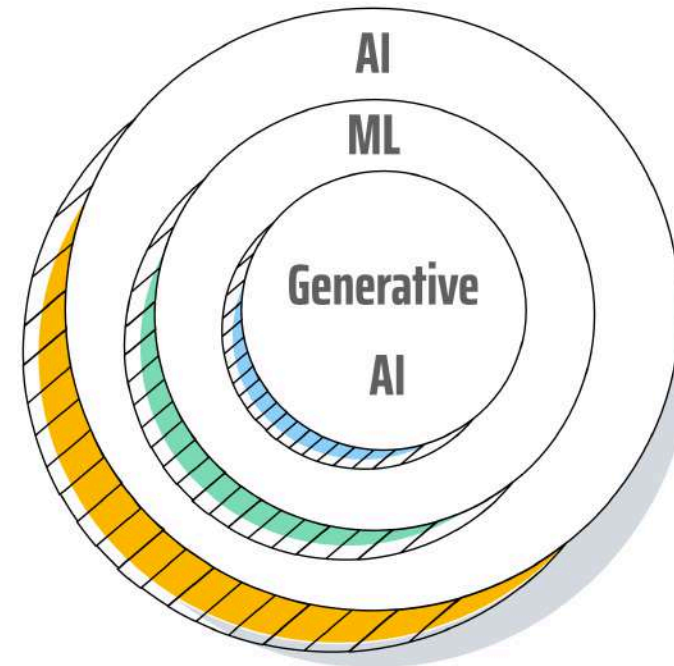
ML Landscape Before Generative AI – Overview

Before Generative AI: 3 Options

- Choose based on your team's expertise and business needs

Three Options:

- **Prebuilt ML API:** Ready-made API for speech, vision, or text
- **AutoML tools:** Train custom models without deep ML expertise
- **Training ML Models Using ML Frameworks:** Use frameworks like PyTorch, TensorFlow, Scikit-learn to build models





ML Approach 1 – Pre-Built ML API

ML API: Ready-made API provided by cloud platforms

Speech API: Convert calls and meetings into transcripts

Vision API: Detect objects, labels, or text inside images

Language API: Classify sentiment or extract keywords

And several others





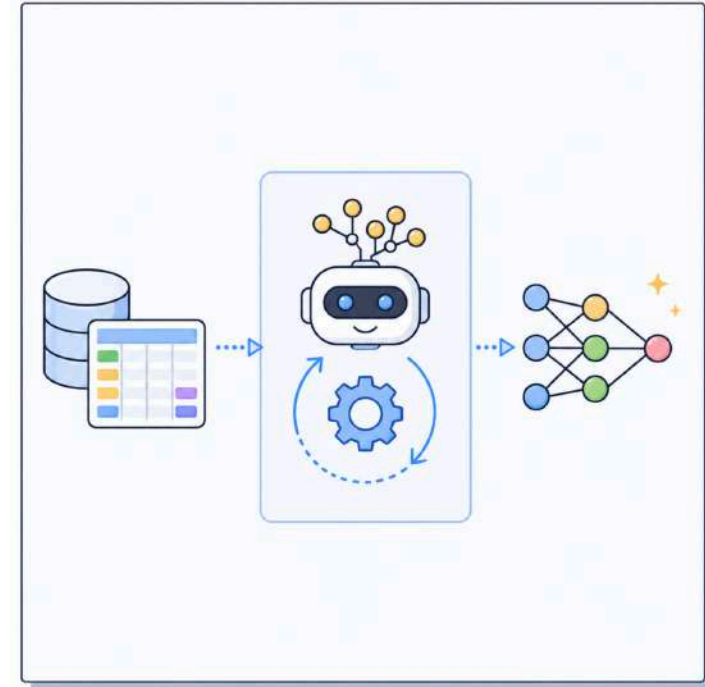
ML Approach 2 – Custom Models with AutoML

AutoML: Train custom models without ML expertise

- Upload labeled data → AutoML handles training and deployment
- **Use Case:** Classify support tickets into categories

Do You Know?: AutoML opened ML to everyone

- No notebooks, no code, no ML expertise needed
- Just data





ML Approach 3 – Training ML Models Using ML Frameworks

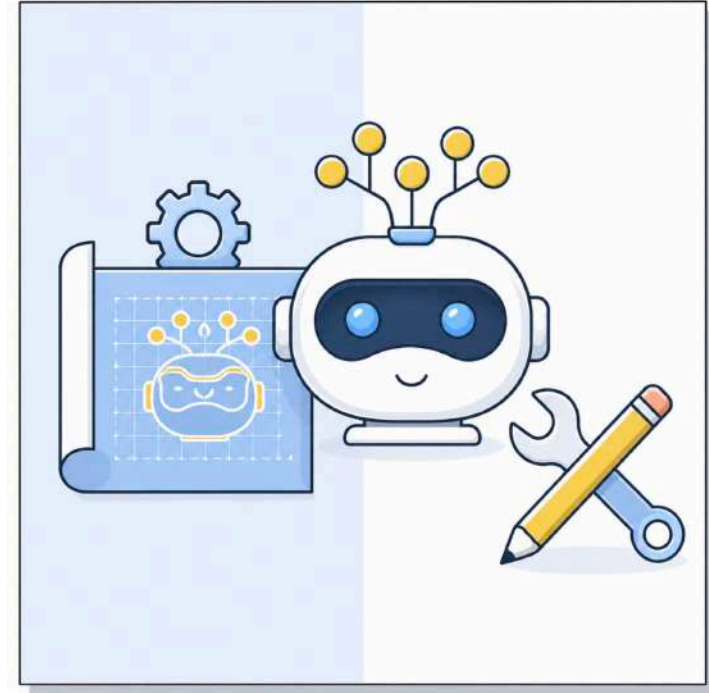
Training ML Models Using ML Frameworks:

Write your ML code to build your models

- **Use Frameworks to Build Models:** Like PyTorch, TensorFlow, Scikit-learn
- Bring your own data
- **Needs:** Deep AI/ML expertise
- **Use Case:** Train models for anomaly detection

Do You Know?: Maximum flexibility BUT maximum skill and maximum effort

- Requires ML engineers with expertise





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Generative AI

New Capabilities



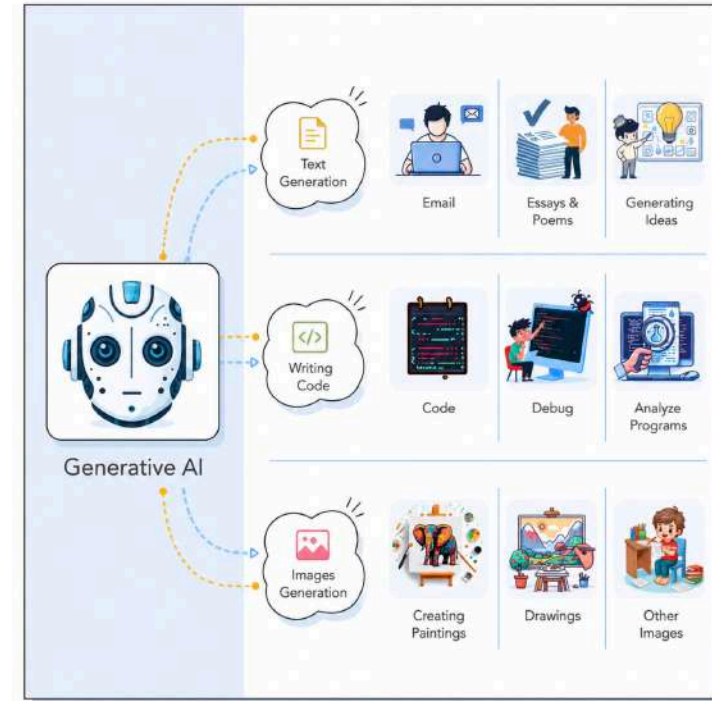
Generative AI Brought in New Capabilities

Foundation Models: Pre-trained on massive datasets - create text, code, images

- *Use Case:* Generate product descriptions automatically for e-commerce listings
- *Use Case:* Write code snippets based on natural language prompts

New Capabilities: Summarization, chat, translation, etc.

- *Use Case:* Summarize feedback
- *Use Case:* Translate product manuals

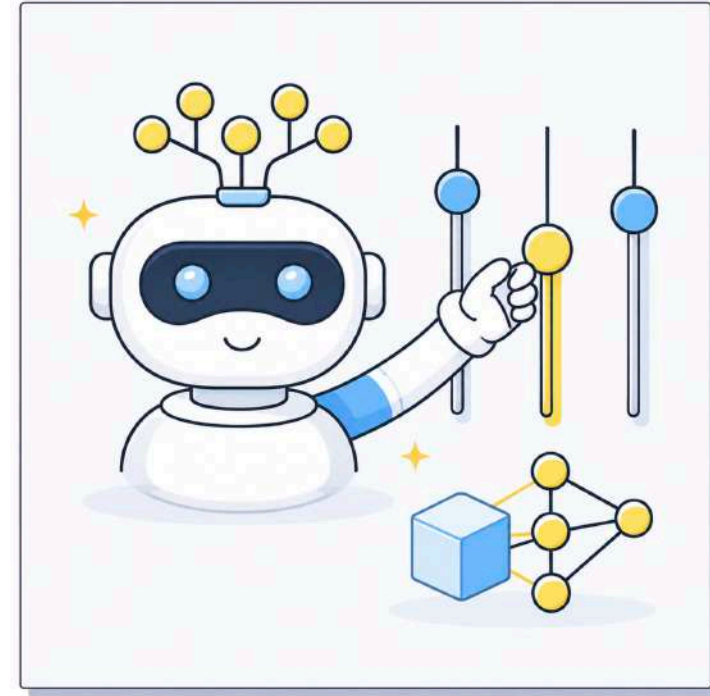




Generative AI Brought in New Capabilities - Tuning

Models Can Be Tuned Easily: Foundation models can be fine-tuned with your own data to adjust tone, context, or behavior

- **Use Case:** Fine-tune a chatbot to use your company's tone and answer product-specific questions
- **Use Case:** Customize a model with your brand guidelines to generate on-brand marketing copy
- **Use Case:** Provide domain-specific examples (medical, legal, financial) for more accurate, compliant responses
- **Use Case:** Tune the coding model using your codebase to improve code suggestions





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ML Landscape

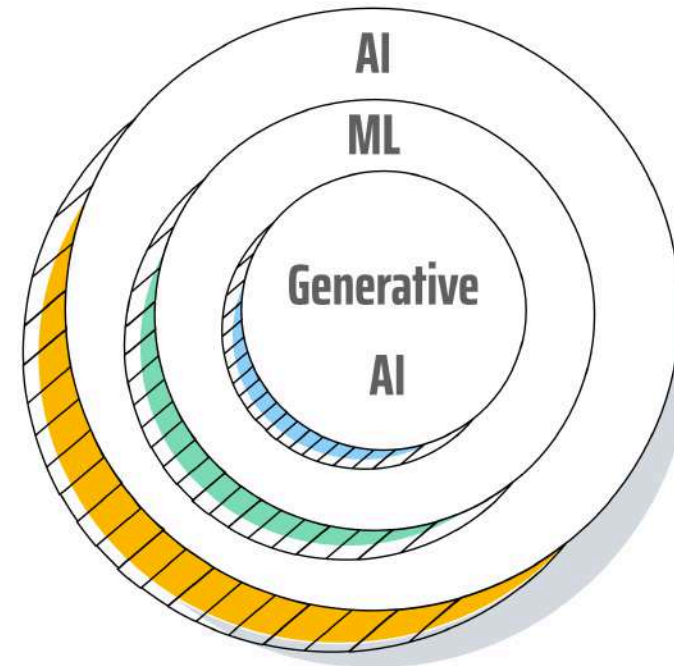
After Generative AI



The New ML Landscape After Generative AI

The ML world expanded from 3 options to 5:

- **Pre-built API** - no change
- **AutoML** - no change
- **Training ML Models Using ML Frameworks** - no change
- **Foundation Models via API** - *brand new*
- **Tuning Foundation Models** - *brand new*





The New ML Landscape After Generative AI

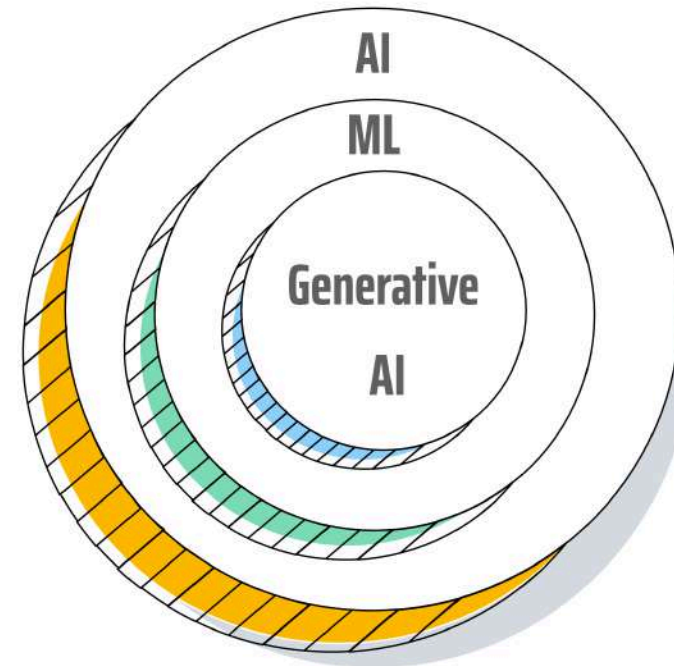
Use Pretrained API (NO CHANGE) - Use ready-made API: text, speech, image,...

Use Foundation Models via API (NEW) - New capabilities: chat, summarize, translate, generate code

- *Use Case:* Summarize long customer feedback into short highlights

Tuning Foundation Models (NEW) - fine-tune on your own data

- *Use Case:* Fine-tune a chatbot to use your company's tone and answer product-specific questions





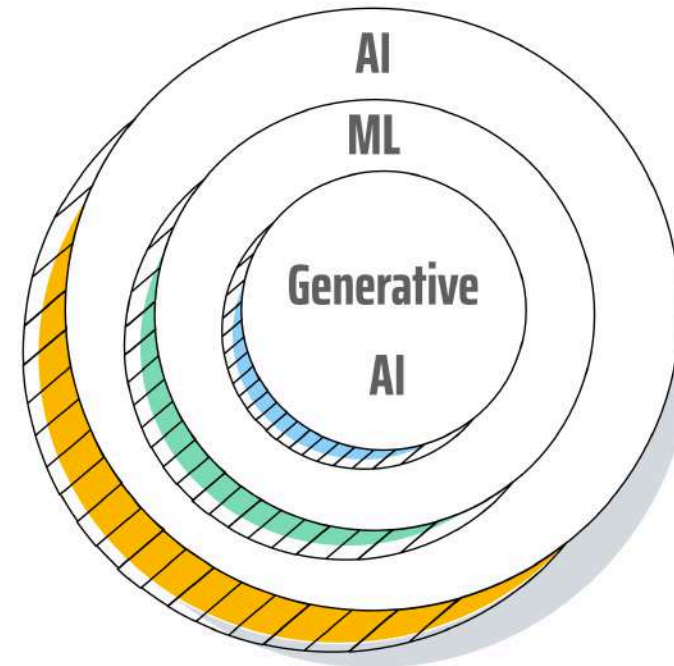
The New ML Landscape After Generative AI - 2

AutoML (NO CHANGE) - Build models without writing code

- **Use Case:** Classify support tickets into categories

Training ML Models Using ML Frameworks - Use frameworks like PyTorch, TensorFlow, Scikit-learn to build models

- *Use Case:* Train a deep learning model for detecting anomalies in financial transactions





When Should You Use Each Approach?

Scenario	Best Choice
You need quick results for speech, vision, or text tasks without ML expertise	Pretrained ML API
You want chat, summarization, translation, reasoning, or code generation	Foundation Models via API
You need AI that speaks in your brand voice or answers domain-specific questions	Tune Foundation Models
You have labeled data and want to create a model without writing any ML code	AutoML
You need full control over model architecture, training, and infrastructure	Training ML Models Using ML Frameworks
You want to add quick AI features to an app (summaries, Q&A)	Foundation Models via API



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Getting Started with ML in Google Cloud



1 - Pre-built AI API [1/2]

Concept: Pre-trained models ready for you!

Examples of AI API:

- **Vision:** "Is there a cat in this photo?"
- **Speech-to-Text:** "Transcribe this audio file"
- **Natural Language:** "Identify sentiment of a user review"

Best For: Standard tasks where you don't need customization





1 - Pre-built AI API [2/2]

Usecase: Derive insights from unstructured text

- Natural Language - <https://cloud.google.com/natural-language>

Usecase: Speech to Text

- Speech to Text - <https://cloud.google.com/speech-to-text>

Usecase: Convert text into speech

- Text to Speech - <https://cloud.google.com/text-to-speech>

Usecase: Extract insights from images, videos and documents

- Vision - <https://cloud.google.com/vision>



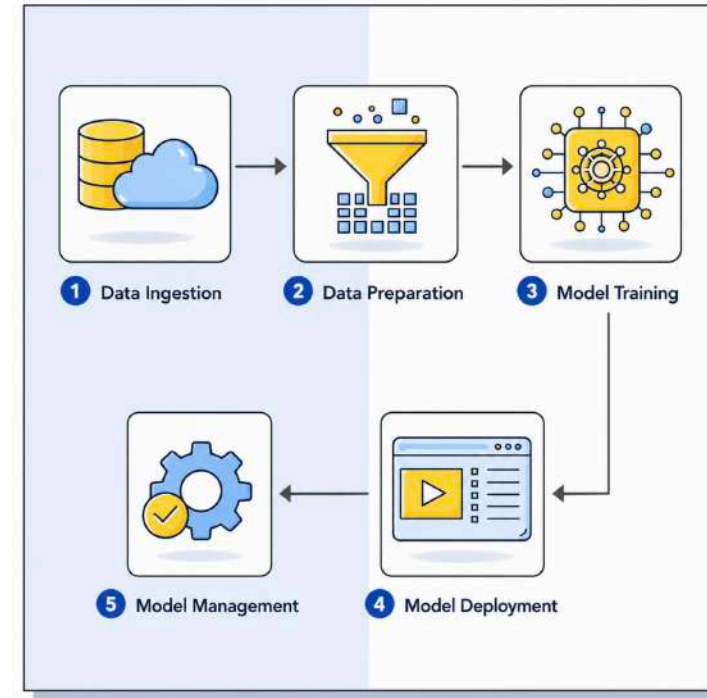


Agent Platform - The Unified AI Platform [1/2]

Agent Platform: One Platform for Machine Learning, Generative AI and Agentic AI

Train and Deploy Your Models: Automate your complete ML workflow (AutoML and Custom Training)

- **Infrastructure for Training and Inference:** Servers, GPUs, TPUs
- **Datasets:** Manage your training data
- **Model Monitoring:** Monitor your model; auto-trigger retraining if needed
- **Model Registry:** Maintain model versions



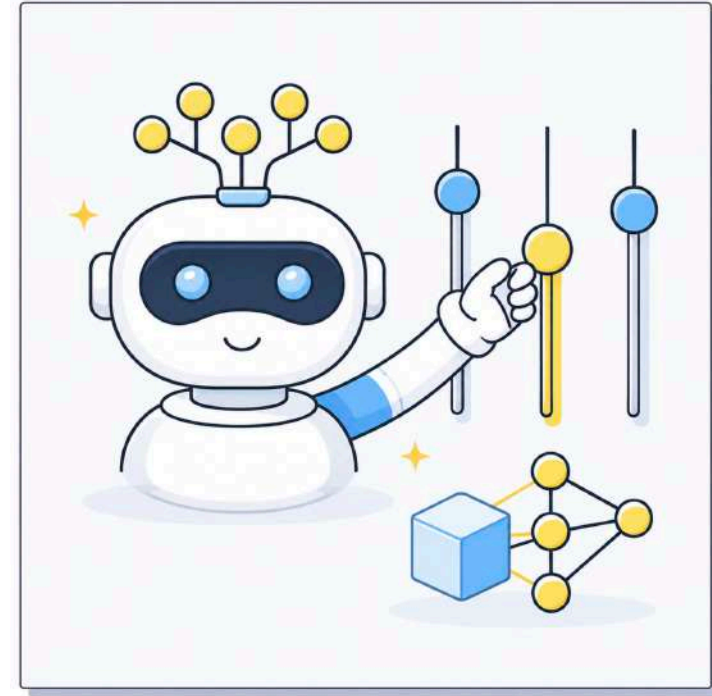


Agent Platform - The Unified AI Platform [2/2]

Model Garden: Models from Google, partners, and open-source

- *First Party Google Models:* Gemini, Imagen, Veo, ..
- *Open Models:* Gemma (Lightweight open models from Google), CodeGemma, PaliGemma, Llama, Mistral
- *Partner Models:* Claude (Anthropic), and more
- **Prompt Gallery:** Ready-to-use prompts
- **Fine-Tuning:** Fine-tune foundation models
- **Agent Platform Studio:** Rapid prototyping and testing of generative AI models

Build and Deploy Agents: Build enterprise-grade agents using generative AI models



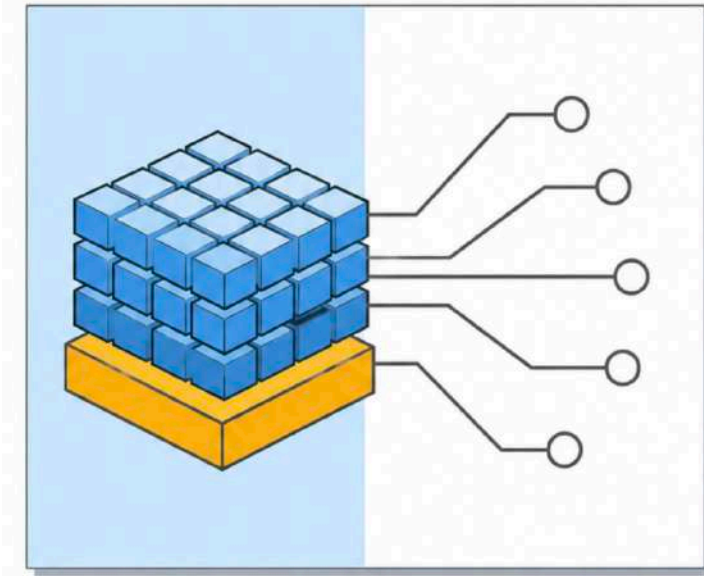


2 - Use Foundation Models via API

What: Access powerful Foundation Models via API

Agent Platform - Model Garden: A library of models (Gemini, Llama, Mistral, etc.) you can deploy

Best For: Chatbots, Content Creation, Coding help, etc.





3 - Fine-Tune Foundation Models

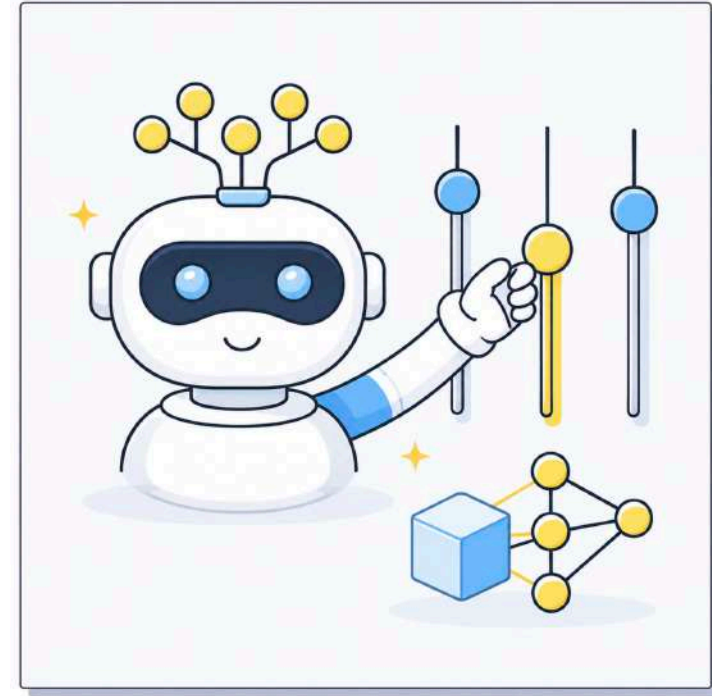
Scenario: Foundation Models do NOT know your business communication style

Solution: Agent Platform - Model Garden - Fine Tuning

- Take a foundation model and "teach" it your style
- Feed it your own data: Support emails or legal docs

Result: A version of Foundation Model that speaks "Your Language"

Best For: Domain-specific or Enterprise-specific customization





4 - AutoML - Train Without Code

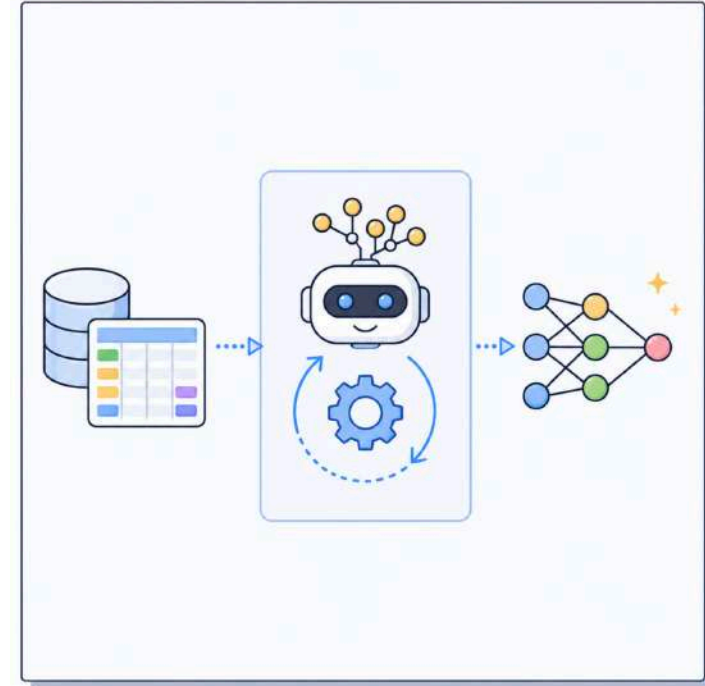
Scenario: The APIs aren't enough

- *Ex:* Vision API sees a "Dog", but you need to know if it's "Your Company's Mascot"

Solution: AutoML on Agent Platform

- **1. Upload Data:** Photos of your mascot
- **2. Click Train:** Auto ML automates the training
- **3. Deploy:** You get a custom API

Best For: Custom model without needing ML expertise





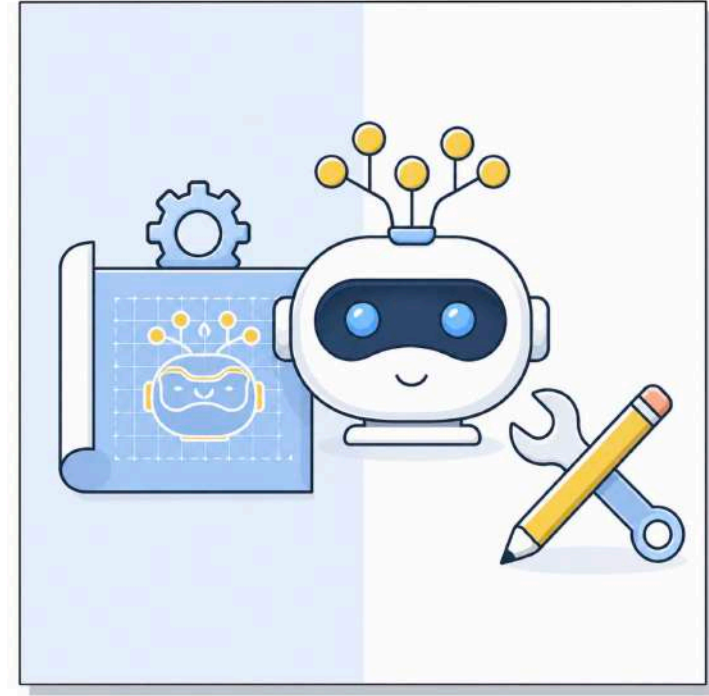
5 - Training ML Models Using ML Frameworks

Scenario: You have a complex ML problem to solve

- You have a team of Data Scientists
- You want to use specific ML Frameworks (**TensorFlow**, **PyTorch**, **scikit-learn**, etc)

Solution: Agent Platform - Custom Training

- Bring your own code
- Custom Training provides the infrastructure needed
- Deploy the model and take complete control of the entire ML cycle





Summary - ML Managed Services in Google Cloud

ML Approach	Description	Google Cloud
Use Pre-built APIs	Use ready-made APIs for text, speech, image, or video tasks	Cloud Natural Language API, Speech-to-Text API, Cloud Vision API, Video Intelligence API, etc.
Use Foundation Models via API	Use models for chat, code generation, multimodal tasks, etc	Gemini Enterprise Agent Platform - Model Garden
Fine-Tune Foundation Models	Take a foundation model and "teach" it your style	Gemini Enterprise Agent Platform - Model Garden
AutoML	Build models from zero using your data without writing code	Gemini Enterprise Agent Platform - AutoML
Training ML Models Using Frameworks	Build and train custom ML models using frameworks such as TensorFlow and PyTorch	Gemini Enterprise Agent Platform - Custom Training



AI and ML in Google Cloud – Scenarios

Scenario	Solution
A company wants to build a chatbot using a foundation model without any fine-tuning	Use Foundation Models from Model Garden
A healthcare company wants their AI assistant to respond in a medical tone with accurate terminology	Fine Tune Foundation Model
An e-commerce platform wants an image generator that matches their brand aesthetic	Fine Tune a Foundation Model (with brand-specific data)
A retail team wants to build a custom model to categorize images but doesn't have ML engineers	Agent Platform - AutoML
A telecom firm has a team of data scientists building a complex churn prediction model using TensorFlow	Build Complex Custom Models (Agent Platform Custom Training)
A logistics company wants to train a route optimization model using PyTorch	Build Complex Custom Models (Agent Platform Custom Training)



Google AI Studio vs Agent Studio

Feature	Google AI Studio	Agent Studio
Audience	Beginners and Experimenters	Enterprise teams
Access	Google account	Google Cloud
Use Case	Try prompts and models quickly	Build, tune, and deploy production-grade models
Governance & Security	Basic	Enterprise-grade access control and compliance



Agent Search (Gemini Enterprise Agent Platform)

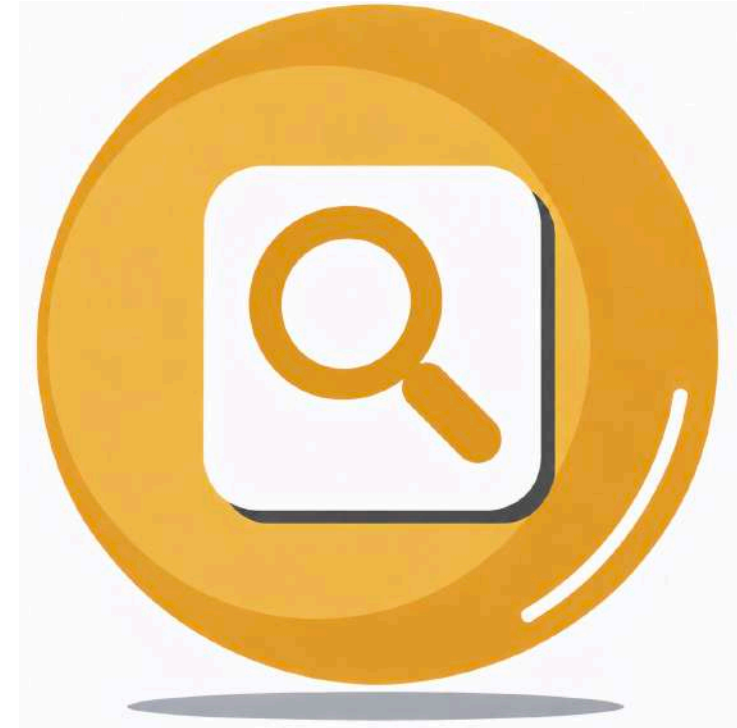
Agent Search: Fully managed service for adding AI-powered search to your websites and applications

- Understands user intent and finds relevant information across your content

AI Answers: Generates answers grounded in your data, with source citations

- **Supported Content:** Websites, documents such as PDFs, and structured data from BigQuery

Example: Let customers ask questions and find answers across your product manuals





RAG and Grounding Options in Google Cloud

Option	How it works	Best for
Agent Search (Agent Platform)	Index your content and provide search results or grounded AI answers	Building search and question-answering experiences quickly
RAG Engine API (Agent Platform)	Ingest and process documents, then retrieve relevant content to ground model responses	Building custom RAG applications
Grounding with Google Search	Use public Google Search results to ground Gemini responses	Questions that need current public web information



Getting Started with AI Hypercomputer

Problem: Training and running AI models requires substantial infrastructure

- Compute, n/w, and storage can become bottlenecks

AI Hypercomputer: Google's integrated infrastructure for AI workloads

- **Compute:** Choose TPUs or NVIDIA GPUs (GKE or GCE)
- **Networking:** High-speed network connections
- **Storage:** Efficient access to training data & model files
- **Software:** Support for frameworks such as PyTorch

Key Benefit: Optimal performance for AI training and inference



Data Center



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Getting Started with

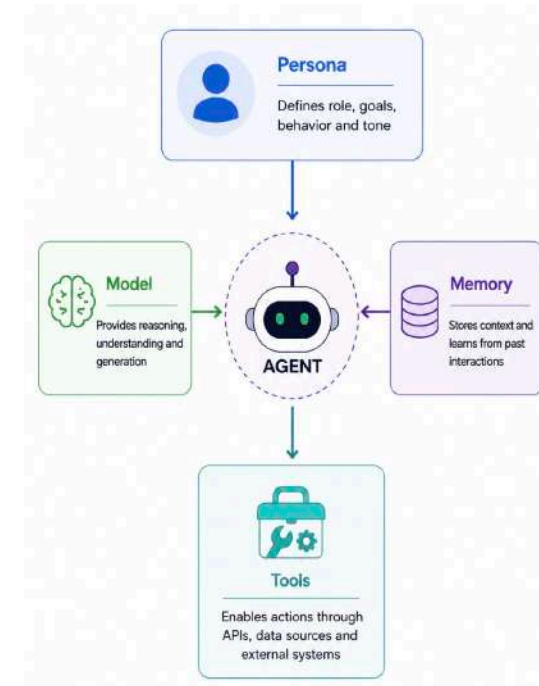
AI Agents - Google Cloud



Getting Started with AI Agents in Google Cloud

Gemini Enterprise Agent Platform: Google's unified platform to build, deploy, and optimize AI agents

- **Agent Studio:** Build agents with a low-code experience
- **Agent Development Kit (ADK):** Open-source framework to build and orchestrate sophisticated agents
- **Agent Runtime:** Managed runtime to deploy, scale, and operate agents in production
- **Models:** Gemini & 200+ supported models - Model Garden
- **Tools & Integrations:** Connect agents to search, RAG, APIs, enterprise systems, and other services
- **Protocols:** Supports MCP for tools and A2A for agent-to-agent collaboration





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Getting Started with Gemini Notebook



What Is Gemini Notebook (NotebookLM)

Gemini Notebook = AI-first research notebook using Gemini models

Grounded in documents **you provide**

Great for:

- Research
- Summarizing PDFs or presentations
- Creating outlines, Q&A, speaker notes, or training materials

Output always based on your uploaded sources - not random web data





How Does Gemini Notebook Work

Upload variety of docs (PDFs, slides, audio)

Features:

- Create one-click summaries, slide decks, FAQs, and briefing docs
- Create quizzes and infographics
- Ask questions for deeper insights and get answers with citations
- Create a Podcast Episode (Audio Overview) and listen on-the-go!





Gemini Notebook - Scenarios

Scenario	How Gemini Notebook Helps
A new hire needs to understand company policy documents	Upload the employee handbook and generate a summarized FAQ
A product manager needs to brief executives on a recent market report	Upload the report and generate a concise one-click briefing doc
A professor wants to prepare study guide for a long lecture slide deck	Upload the slide deck and generate study guide
A finance team wants a voice summary of quarterly results	Generate an audio overview from uploaded earnings reports



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Getting Started with Gemini Enterprise



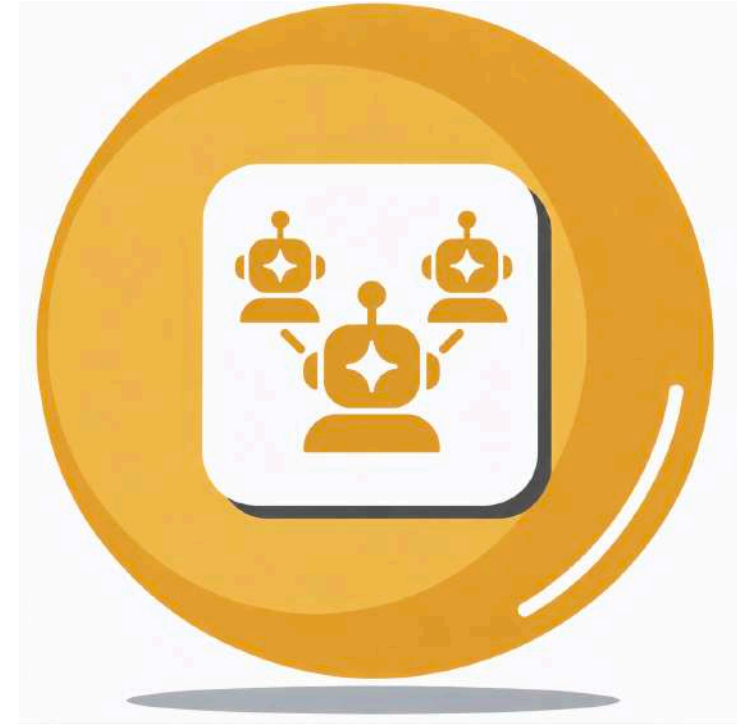
Why Use Gemini Enterprise (Agentspace)

Want to make your employees more **productive**?

Want to make **enterprise information easier to find**?

- Even when spread across Google Drive, Jira, Confluence, and SharePoint?

Gemini Enterprise: Helps employees find information, get answers, and complete tasks using AI agents





What are the Key Features of Gemini Enterprise

Feature	What it helps employees do
Enterprise Search	Find information across Google Drive, Jira, Confluence, and SharePoint
AI Assistant	Ask questions, get answers using company information, and summarize documents
Ready-to-Use Agents	Perform tasks such as research using prebuilt AI agents
Custom Agents	Create agents for specific business tasks and workflows
Gemini Notebook Enterprise	AI-powered research tool to summarize and extract information from your knowledge sources
Security and Access Controls	Access company information based on user permissions



Why Modernize Customer Support?

Customers expect: Fast, and personalized support

Common challenges:

- Long wait times
- Inconsistent answers
- Limited support outside business hours
- Repeating information when transferred between agents

Opportunity: Use AI to help customers and support teams





Gemini Enterprise for CX (Customer Support Solutions)

Component	What it helps you do
Conversational Agents	Provide 24/7 text and voice support, answer routine questions like checking order status
Agent Assist	Help human agents with suggested responses, relevant knowledge articles, live transcription, and conversation summaries
Customer Experience Insights	Analyze customer conversations to identify common issues, track sentiment, evaluate service quality, and uncover coaching needs
Contact Center as a Service (CCaaS)	Run a cloud contact center with voice and chat, request routing, and AI tools for support teams
Remember	Earlier offered under Google Customer Engagement Suite brand



Getting Started with Dialogflow CX

Dialogflow CX: Build conversational agents for customer questions and tasks

- Example: Order a pizza or book an appointment
- **Natural Conversations:** Understand requests and maintain context across multiple turns
- **Visual Builder:** Design and test conversation flows
- **Text and Voice:** Connect agents to websites, mobile apps, and phone systems

Goal: Resolve routine requests quickly and hand off to human agents when needed





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Getting Started with Gemini Integrations



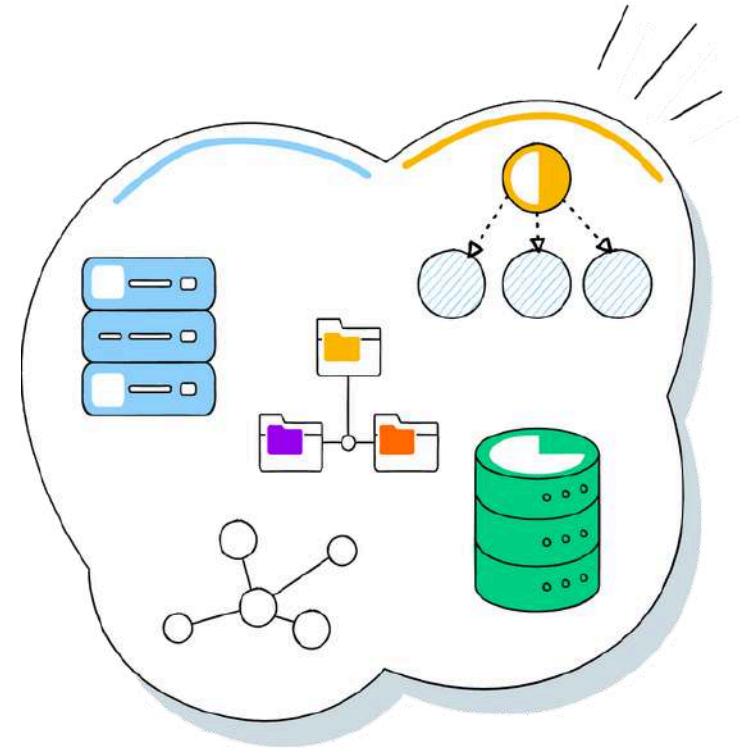
Gemini for Google Cloud - Your Cloud AI Companion

Gemini is integrated into the Google Cloud ecosystem

- Use **natural language** to interact with different services

Available in:

- **Cloud Console:** Create or manage resources
- **BigQuery:** Write, fix, and explain SQL queries
- **Looker:** Explore and visualize data through plain English
- **Security Command Center:** Understand threats and get mitigation guidance
- ...





What is Google Workspace?

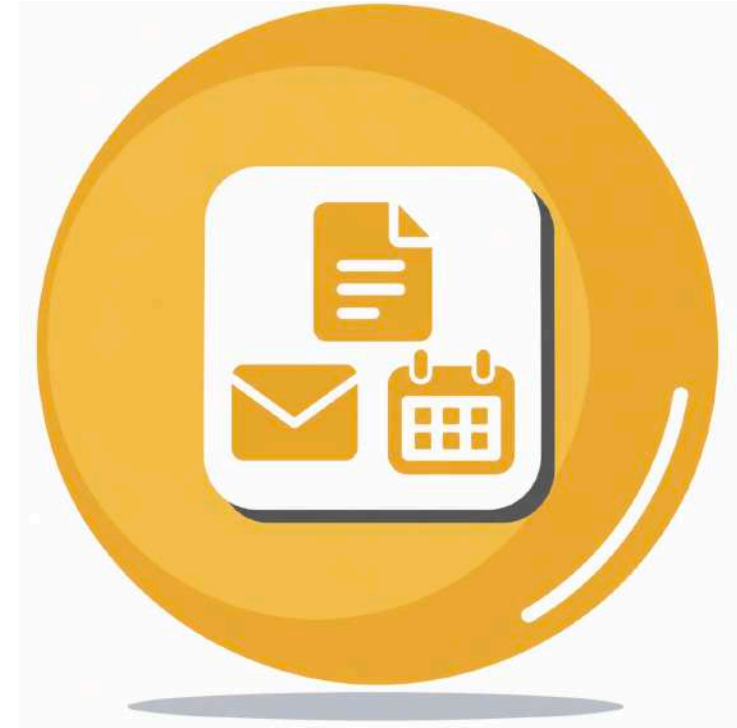
A cloud-based productivity suite by Google

Combines communication, collaboration, and productivity tools

Includes apps like:

- Gmail, Google Docs, Google Sheets, Google Slides
- Google Meet, Google Drive, Google Calendar, and more

Used by millions of businesses and individuals worldwide



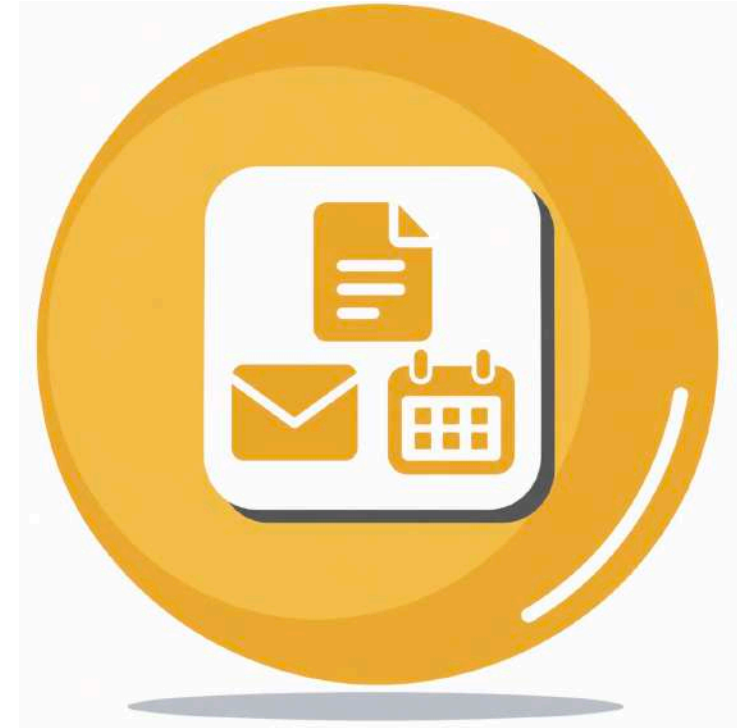


Gemini for Google Workspace

Gemini is built into your favorite apps:

- **Gmail:** Write, summarize, and reply to emails
- **Docs:** Help writing content
- **Sheets:** Generate formulas
- **Slides:** Create presentations
- **Meet:** Capture notes or summarize meetings

Boost productivity across teams with smart, contextual AI help





Other Gemini Integrations

Google Vids: AI-powered video creator/editor

- Automatically generate storyboards, scripts, & visuals

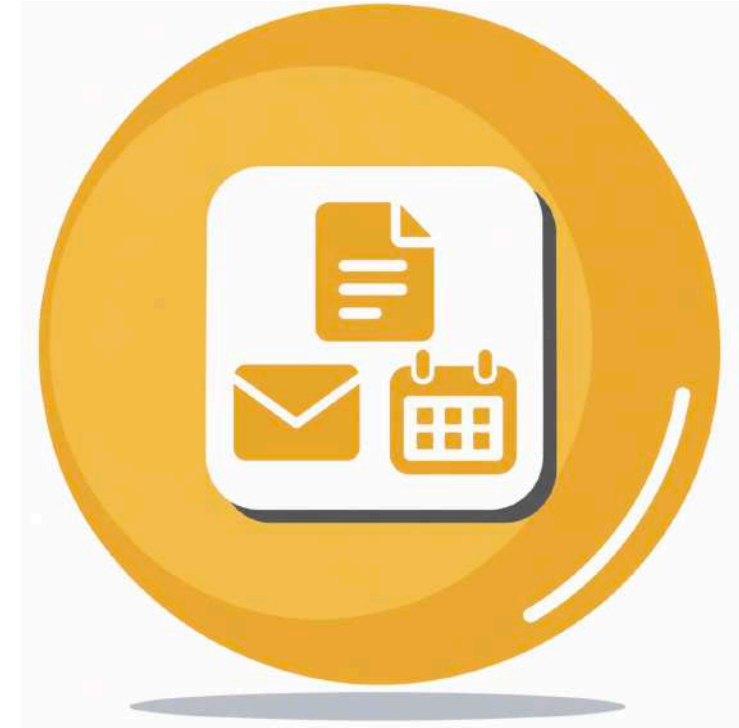
AppSheet: no-code platform (build apps without writing a line of code)

- Use natural language to define workflows and logic
- Turn spreadsheets and data into fully functional apps

Gemini Code Assist: An AI coding assistant

We already talked about:

- **Gemini for Google Workspace:** AI in Gmail, Docs, ..
- **Gemini for Google Cloud:** Assist with code, architecture, and cloud operations





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Getting Started with Case Studies

Case Studies - Importance



Key Part - Of certification exam

Download - From *Official Guide*

Key Resources

- **Google Cloud Well-Architected Framework** - Best practices for designing, building, and operating secure, efficient, and reliable cloud solutions on Google Cloud
- **Cloud Architecture Center** - *Discover reference architectures, design guidance, and best practices*





Case Studies - Mindset & Exam Best Practices

Best Attitude - Test yourself as architect

- **Understand Domains** - ML, Gen AI, Healthcare, Retail, Media & Streaming
- **Analyze & Reflect** - Form your own opinions
- **Challenge Solutions** - Do not accept recommendations blindly

During Exam - You can read case study but do NOT depend on it

- **Before Exam** - Get clear overview of each case
- **Exam Tip** - Group questions per case study and answer together





Altostrat Media Case Study - Overview

Company - Leading media provider with vast audio and video library

- Podcasts, interviews, news broadcasts, and documentaries
- **Priorities:** Reliability and cost management

Goal - Modernize content management and user engagement strategies using AI

- Drive revenue growth through dynamic pricing, targeted marketing, and personalized product recommendations
- Seamless self-service support

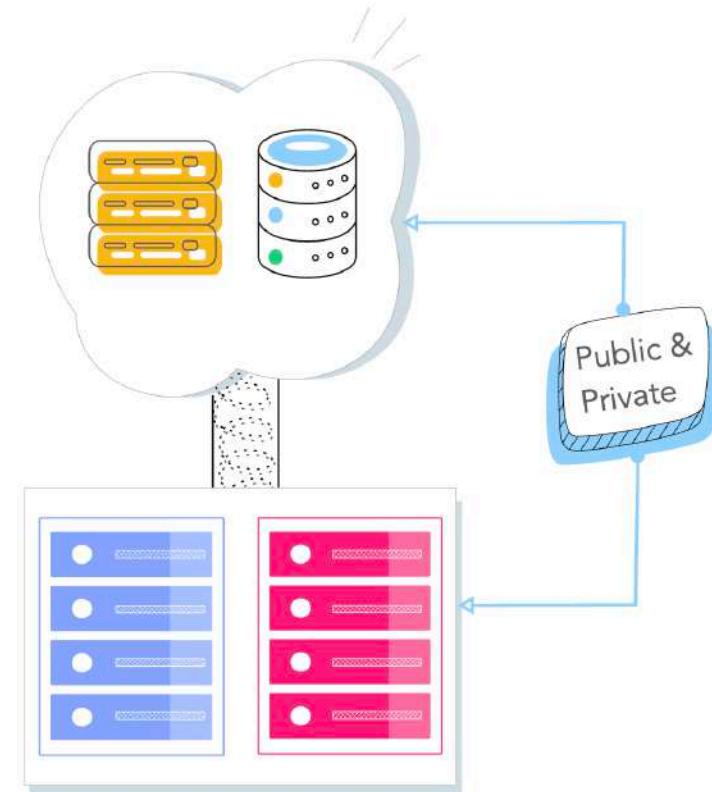




Altostrat Media - Existing Technical Environment

Hybrid Setup - Some on-prem legacy systems for content ingestion and archival

- Slated for modernization & migration to Google Cloud
- **GKE** - Content management and delivery platform
- **Cloud Storage** - Hosts their vast media library
- **BigQuery** - Data Warehouse (Analyze user behavior, content consumption, and audience demographics)
- **Cloud Run Functions** - Event driven tasks (video transcoding, metadata extraction, and personalized content recommendations)
- **Identity** - Google Identity + third-party providers
- **Observability** - Cloud Monitoring + Prometheus (email alerts)



Altostrat Media - Business Requirements



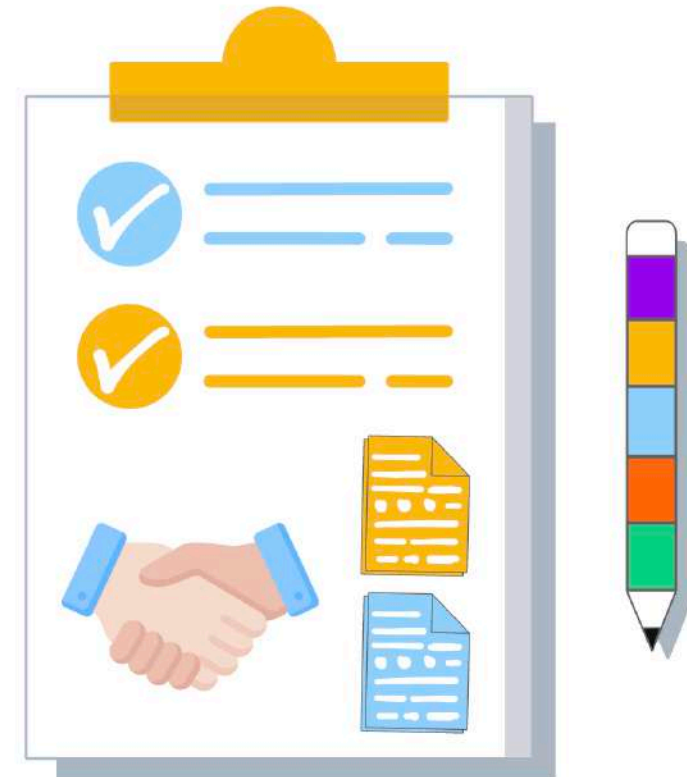
Improve Reliability - For Operational Workflows across hybrid environments

- **Simplify Infra Management** - Faster CI/CD

Cost Optimization - For cloud storage

AI: Better Content Mgmt, User Engagement

- **Data Driven Decisions** - Content strategy
 - Identify trends and extract insights
- **Summarization** - Auto-summarize media content
- **Metadata Extraction** - Directly from content
- **Content Filtering** - Detect inappropriate content
- **Customer Support** - Enable chat and support (24 X 7)





Altostrat Media - Technical Requirements

Secure performant hybrid cloud connectivity - For data ingestion

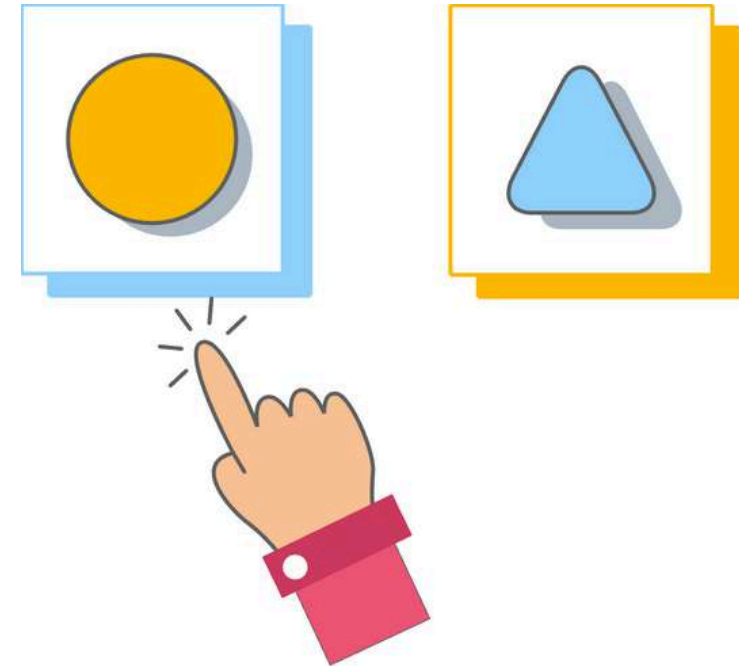
Scalable GKE Clusters - Across Hybrid Cloud

- **Modernize CI/CD** - Centralized deployment

Optimize Costs - For Cloud Storage

Make Best Use of AI - Detect harmful content, ensure auditability and explainability

- **Personalize Experiences** - AI recommendations
- **Advanced Chatbots** - Customer Assistance
- **Automated Summarization** - Of media content





Altostrat Media - Discussion [1/2]

Service	Discussion
GKE Enterprise	Scalable container platform for hybrid workloads
BigQuery	Data Driven Decisions - Analyze user behavior, content trends
Pub/Sub & Dataflow	Real-time ingestion and transformation pipelines
Cloud Storage	Tiered storage for cost optimization (Classes - Standard, Nearline etc)
Cloud Build + Cloud Deploy + Artifact Registry	Modernize CI/CD
Cloud Observability	Unified observability and alerting
Cloud Interconnect	Secure, high-performance hybrid connectivity
Cloud Run Jobs	Functions have execution time limits - Cloud Run Jobs can be used for heavy, long-running jobs (video transcoding etc)



Altostrat Media - Discussion [2/2]

Service	Discussion
Agent Platform	Train, tune, and deploy ML models (Cloud Audit Logs Integration for Auditability, Explainable AI for Explainability)
BigQuery ML	Create ML models using familiar SQL
Vision AI, Video AI	Video and Image content filtering
Cloud Speech-to-Text API + Generative AI models on Agent Platform	Auto-summarize media content and metadata extraction
Conversational AI Platforms (CAIP) with Dialogflow	Build virtual agents (chatbots)
Customer Engagement Suite with Google AI (Conversational Agents + Agent Assist + Conversational Insights + Contact Center as a Service)	Enable chat and support (24 X 7)



Cymbal Retail Case Study - Overview

Cymbal - Online retailer experiencing significant growth with diverse product portfolio

Challenge - Complex catalog management and manual workflows

Solution Concept: 3 Focus Areas

- **1:** Catalog and Content Enrichment
- **2:** Conversational Commerce with Product Discovery
- **3:** Technical Stack Modernization



Cymbal Retail Case Study - 3 Focus Areas



Solution Concept: 3 Focus Areas

- **Catalog and Content Enrichment** - Use Gen AI to improve accuracy of product catalog
 - Generate product attributes, descriptions, and images from supplier-provided information
 - **Goal:** Streamline catalog mgmt, reduce manual effort and errors, and ensure consistent info across sales channels
- **Conversational Commerce with Product Discovery** - Integrate AI-powered agents to provide personalized experience (using Google Cloud's Discovery AI)
- **Technical Stack Modernization:** Streamline operations and reduce manual costs (Cloud-based infrastructure, Secure and efficient data handling, Proactive monitoring and security)





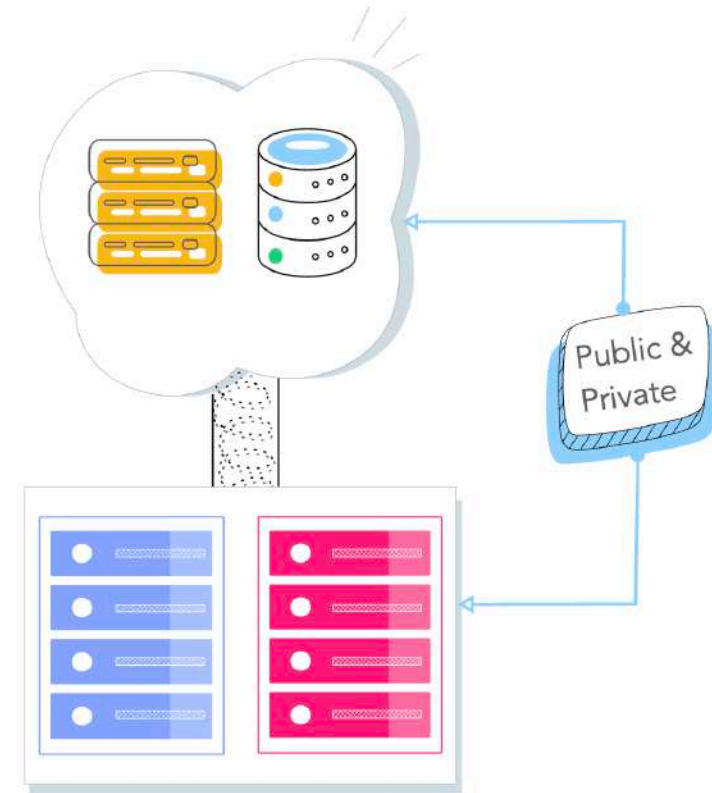
Cymbal Retail - Existing Technical Environment

Infrastructure - Mix of on-prem and cloud

- **Databases** - MySQL, SQL Server, Redis, MongoDB
- **Apps** - Kubernetes clusters running workloads
- **Hybrid Cloud** - Legacy file transfers + batch ETL jobs

Applications: Web App + IVR

- **Web App** - Browse product catalog (integrated with relational databases)
- **IVR (Interactive Voice Response)** - Handle customer calls and route to appropriate agents
- **Call center agents** - Handle transfers from IVR and manually enter orders when customers face issues
- **Monitoring** - Various tools - Grafana, Nagios, Elastic



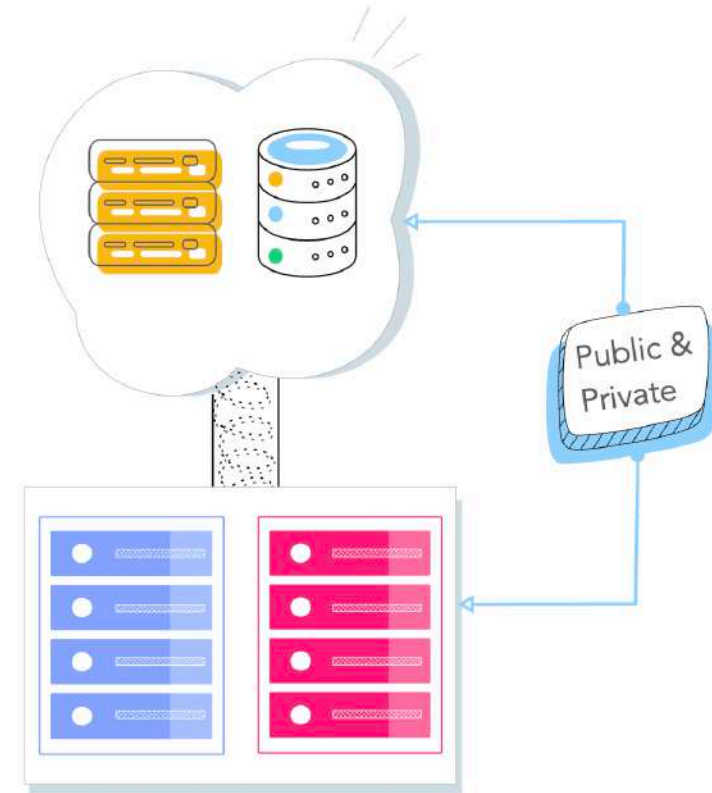
Cymbal Retail - Existing Technical Environment - Challenges



Manual Processes: Make things time-consuming and error-prone

Data Silos: Provides limits unified view of the customer journey

Integrating New Technologies: Is difficult





Cymbal Retail - Business Requirements

Automate Product Catalog Enrichment -

Reduce manual effort, minimize errors

Improve Product Discoverability -

Smarter search and recommendations

Increase Customer Engagement -

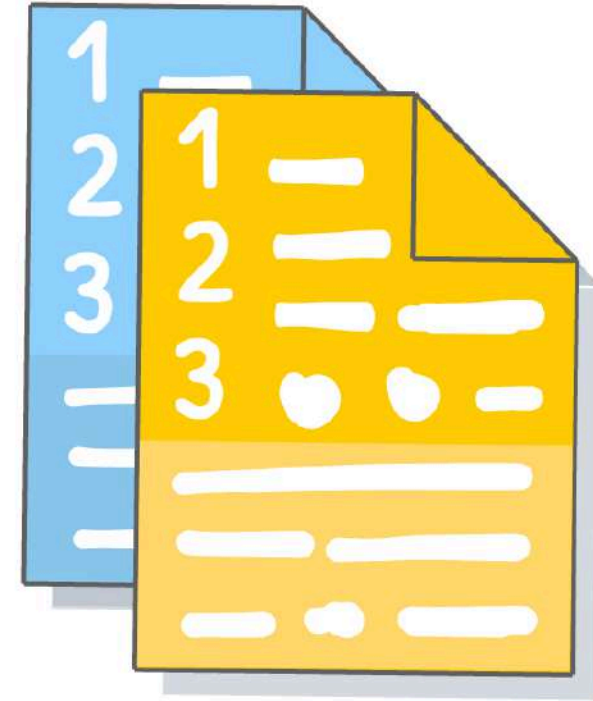
Interactive and personalized shopping experience

Drive Sales Conversions -

Intuitive and helpful shopping experience

Cut Costs -

Reduce call center staffing costs and data-center hosting costs





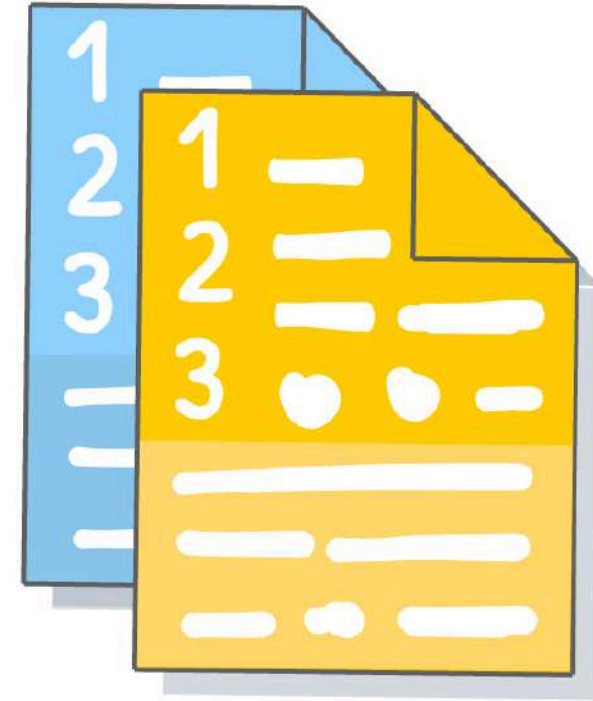
Cymbal Retail - Technical Requirements

Automate Product Discovery - Natural language search for customers

NFRs - Securely handle growing catalog with good performance

Human-in-the-Loop (HITL) Review - UI for team to review/reject AI-generated content

- **Attribute Generation** - Derive product attributes from supplier data - titles, descriptions, and images
- **Image Generation** - Create variations from base image (support background changes, color adjustments, text overlays)





Cymbal Retail - Discussion [1/2]

Service	Discussion
Google Cloud's Discovery AI	Retail Product Recommendations
Agent Search	Enable natural language product search and discovery
Agent Platform Studio, Generative AI Models, Vision AI, Natural Language AI	Derive product attributes from titles, descriptions and images
Agent Platform Studio, Generative AI Image Models	Create and enhance product images (color, background, text overlays)
Conversational AI Platforms (CAIP) with Dialogflow	Build virtual agents (chatbots)
Customer Engagement Suite with Google AI (Conversational Agents + Agent Assist + Conversational Insights + Contact Center as a Service)	Enable chat and support (24 X 7)



Cymbal Retail - Discussion [2/2]

Service	Discussion
Cloud Run (custom UI)	Enable Human-in-the-Loop content review
Google Kubernetes Engine (GKE), Cloud Run	Modernize application hosting for scalability and performance
Cloud Spanner, Cloud SQL, AlloyDB, Firestore, MemoryStore (Redis)	Support scalable, low-latency databases for catalog and transactions
Cloud Build/Deploy + Artifact Registry	Modernize CI/CD
Dataflow, Pub/Sub, BigQuery	Efficient data handling
Google Cloud Observability	Proactive monitoring
Security Command Center, VPC Service Controls, Cloud KMS, IAM, Cloud DLP	Ensure data security, privacy, and compliance

EHR Healthcare Case Study - Overview



Company - Leading provider of electronic health record (EHR) software

- **Model** - SaaS for hospitals, clinics, insurance providers

Challenge - Handling exponential growth

- **Solution** - Cloud migration for scalability, resilience, and agility

Goal - Modernize and scale - Google Cloud

- Adapt existing disaster recovery plan
- Update software at a fast pace (roll out new continuous deployment capabilities)



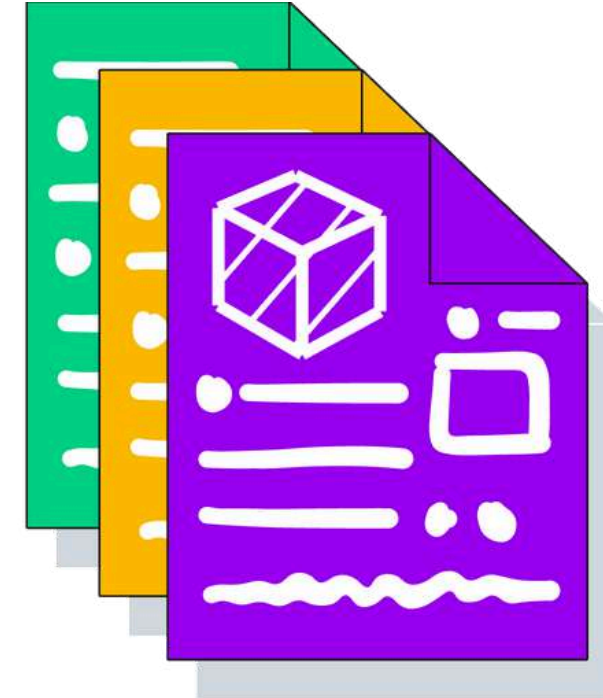
EHR Healthcare - Executive Statement



Current Approach: On-premises strategy worked successfully for years

- **High Effort:** Major investment for team training
- **Complex Operations:** Teams managing similar but separate environments
- **Frequent Outages:** Caused by misconfigured systems and capacity limits
- **Monitoring Gaps:** Inconsistent practices led to delayed responses

Future Direction: Use Google Cloud to build scalable, resilient, multi-environment platform





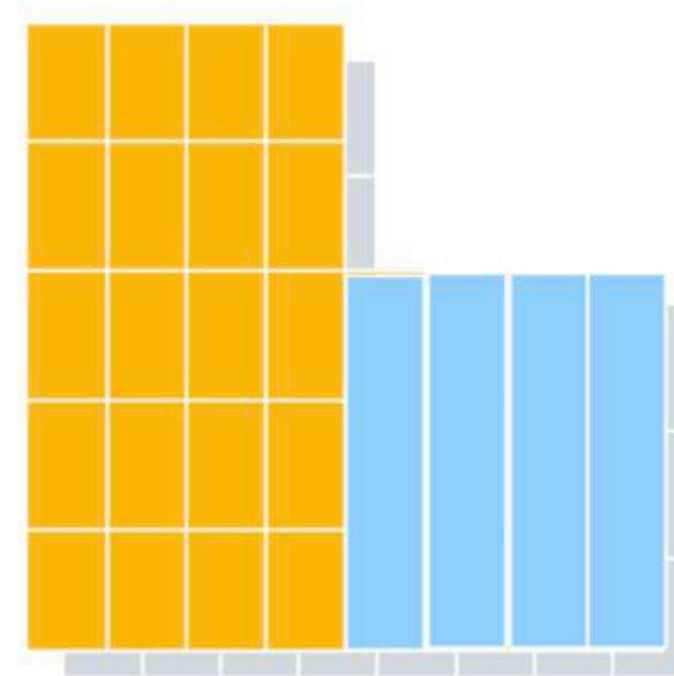
EHR Healthcare - Existing Technical Environment

Hosting - Multiple colocation facilities, one lease expiring soon

- **Apps** - Web-based, containerized on Kubernetes clusters
- **Data** - MySQL, MS SQL Server, Redis, MongoDB mix
- **Integrations** - Legacy file/API-based insurance systems on-prem (scheduled to be replaced)

Identity - Microsoft Active Directory

Monitoring - Open source tools (BUT alerts ignored often)





EHR Healthcare - Business Requirements

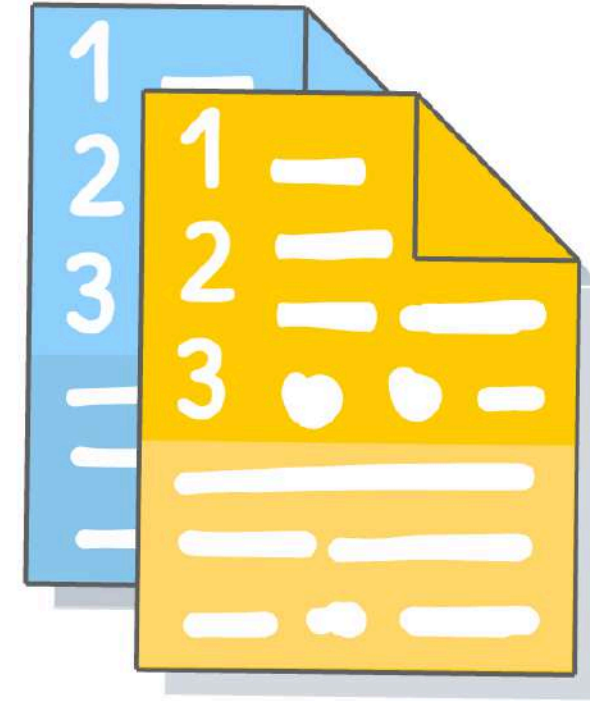
Faster Partner Onboarding: Add new insurance providers quickly

Data Insights: Generate healthcare trend analytics

High Availability: Maintain min 99.9% uptime

- **Global Reach:** Reduce latency - worldwide customers
- **Centralized Visibility:** Monitor system performance proactively

Regulatory Compliance: Maintain security and privacy standards





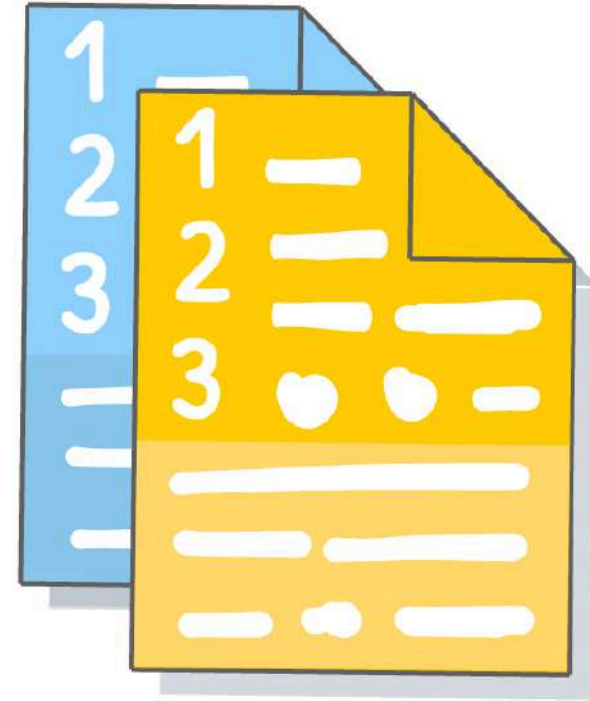
EHR Healthcare - Technical Requirements

Hybrid Connectivity - Secure high speed link between on-prem and cloud (for legacy interfaces to insurance providers)

Container Management - Multiple environments with consistency

- **Scalability** - Auto scale environments
- **Observability** - Consistent logging, log retention, monitoring, and alerting capabilities

Create New Provider Interfaces - To ingest and process data





EHR Healthcare - A Few More Details

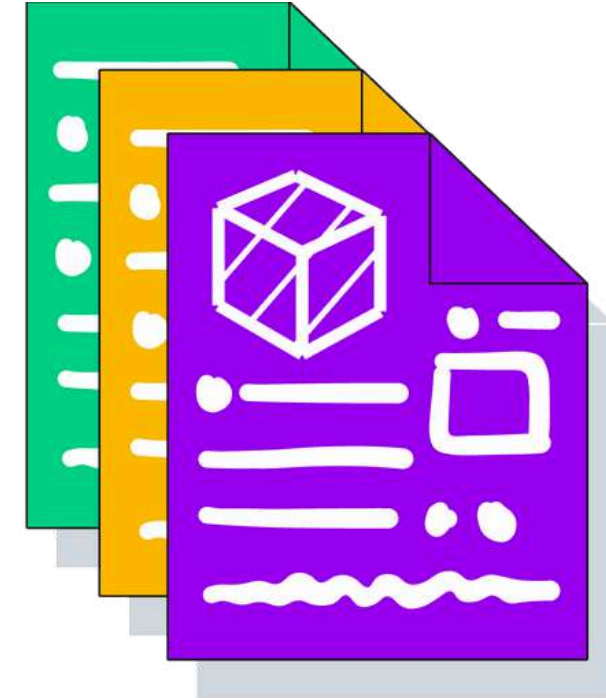
Web-based customer-facing applications running on Kubernetes clusters

- Requirement: Consistent way to manage container apps
- Requirement: Manage multiple container-based environments

Legacy file- and API-based integrations with insurance providers

- Scheduled for replacement over next few years - Will NOT be moved to Google Cloud

Facing outages: With mis-configured systems





Case Study - EHR Healthcare - Discussion [1/2]

Service	Discussion
GKE Enterprise	Run Kubernetes clusters anywhere (cloud and on-premises) Config Mgmt - Central policies - Service Mesh, Access control Service Mesh (logging, monitoring, distributed tracing, dashboards) CI/CD - Watch for updates in the Git repository and applies changes to all relevant clusters automatically
Databases	MySQL, MS SQL Server => Cloud SQL (or Cloud Spanner or AlloyDB) Redis => Memorystore MongoDB => MongoDB Atlas
BigQuery	Make predictions and generate reports on industry trends Batch: Cloud Storage > Dataflow > (BigQuery,) Stream: Cloud Pub/Sub > Dataflow > (BigQuery,)



Case Study - EHR Healthcare - Discussion [2/2]

Service	Discussion
Active Directory Federation Services (AD FS)	For single sign-on Google Cloud Directory Sync (synchronize users and groups from Active Directory to Cloud Identity)
Cloud Dedicated Interconnect	Secure and high-performance connection
Cloud CDN	Reduced latency
Cloud Observability	Alerting Policies for Notifications
Cloud Logging > (Cloud Storage, BigQuery)	Log Retention

KnightMotives Automotive Case Study - Overview



KnightMotives: Global car manufacturer specializing in self-driving vehicles

- **Product Range:** Battery Electric Vehicles (BEVs), Hybrids, & Internal Combustion Engine (ICE) cars
 - **Current Status:** BEVs feature modern in-vehicle systems, but hybrid and ICE models lag behind
 - **Challenge:** Outdated technology causing declining sales
- **5 Year Vision:** Modernize consumer experience
 - **Opportunity:** Use AI to transform in-vehicle, shopping, and service experiences
- **Strained Dealer Relationships:** Because of unreliable ordering system
- **Goal:** Enhance technology & dealer tools to enable built-to-order vehicles & better customer satisfaction





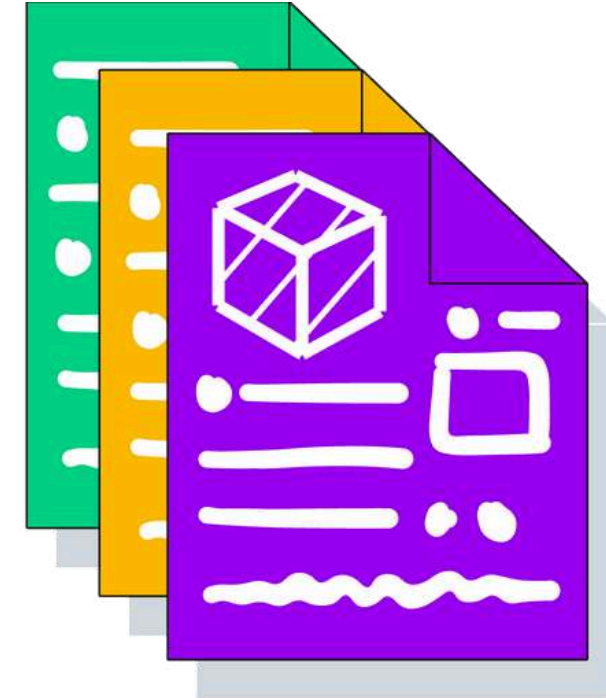
KnightMotives Automotive - Executive Statement

Commitment: Enhance safety and save lives through intelligent data use

- **Data Foundation:** Leverage driving behavior, road conditions, and crash statistics
- **AI Excellence:** Consistently outperform national safety benchmarks

Unified Experience: Deliver the trusted KnightMotives experience across all models

Vision: Create compelling, data-driven digital experiences for every driver



KnightMotives Automotive - Existing Technical Environment

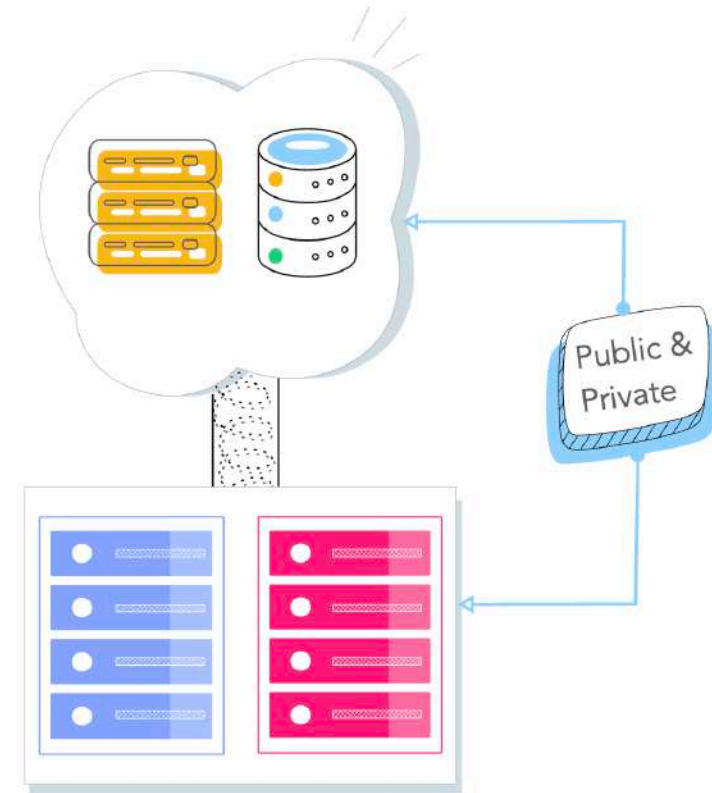


Infrastructure: Mostly on-premises with few workloads on major cloud platforms

- **Core Systems:** Supply chain runs on outdated mainframes, Enterprise Resource Planning outdated
 - **Problem:** Hard to launch new promotions & dealer discounts
- **Dealer Constraint:** Limited budget for h/w upgrades
- **Connectivity Gaps:** Unreliable links to manufacturing plants and rural vehicles

Code Fragmentation: Multiple code bases across vehicle models

- **Technical Debt:** Heavy backward compatibility requirements slowing innovation





KnightMotives Automotive - Business Requirements

Personalized Consistent Experience: Across models (build deeper relationships)

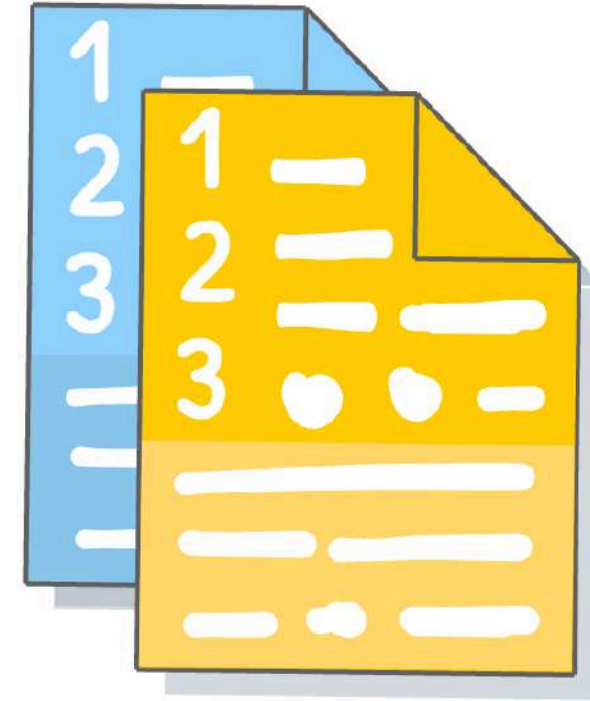
- **Goal:** Better build-to-order model
- **Expand:** Fully autonomous driving in favorable regions

Data Monetization: Leverage corporate data

- Fund projects to fix AI infra. and siloed corporate data

Security & Compliance: With EU data laws

Workforce Enablement: Upskill employees, attract top technical talent



KnightMotives Automotive - Technical Requirements

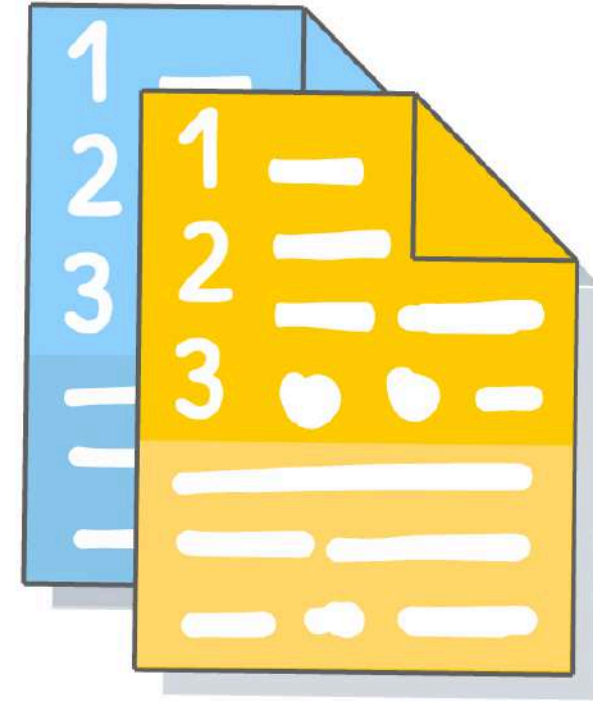


In-Vehicle Experience: Unified AI-powered UX

- **Reliable Connectivity:** For rural vehicles
- **Autonomous Vehicle Development:** Invest in AI and simulation for testing and compliance

Cloud Modernization: Hybrid strategy to modernize IT (& phase out legacy platforms)

- **Secure Data Platform:** For data management, monetization, and AI-driven analytics
- **Security & Risk Management:** Establish a framework
- **Dealer & Customer Experience:** Enhance build-to-order system, dealer tools, and CRM for seamless engagement





KnightMotives Automotive - Discussion [1/2]

Service	Discussion
Android Automotive OS	Modern, connected in-vehicle infotainment & UX systems
Edge TPUs	Deploy ML models on vehicles for real-time decisions
Dialogflow	AI-powered voice assistants - in-vehicle interactions
Agent Platform	Train, tune, and deploy ML models
HPC - Powerful GPUs & CPUs on GCE VMs	Simulate road conditions and driving behavior
Cloud Interconnect	Reliable connectivity for dealers, and vehicles
Pub/Sub, Dataflow	Handle streaming data from connected vehicles
BigQuery	Fix siloed corporate data and base for secure data platform
Sensitive Data Protection (Cloud DLP)	Anonymize data before storing it in BigQuery



KnightMotives Automotive - Discussion [2/2]

Service	Discussion
Mainframe Connector	Integrate your mainframe data and apps with Google Cloud
Replatform Mainframe Applications	Mainframe apps can run "as-is" in Google Cloud on Compute Engine
Mainframe Assessment Tool	Assesses and reverse-engineers your mainframe apps and data - detailed code explanations, dependency insights, etc.
Google Gemini Code Assist	Mainframe Code Rewrite - Transform legacy mainframe code into modern languages
GKE (or Cloud Run)	Managed compute services for refactored mainframe code
API Gateway	Data Monetization. Unify legacy systems through secure APIs.
Security Command Center	Centralized multi-cloud security risk dashboard



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Getting Ready

For the Certification Exam



What to Expect in the Exam

Exam Format: 50-60 multiple choice/select questions

No Penalty: Wrong answers do not reduce your score

Instant Result: Result shown immediately after submitting

Email Confirmation: Sent within a few days





Certification Exam - My Recommendations

Read the Entire Question

- Identify the **key requirements and constraints**
- Read **all answer options** before choosing

Eliminate Wrong Answers

- Remove clearly incorrect options first

Mark and Review

- Mark difficult questions for review
- Revisit them before final submission

Manage Your Time

- Do not spend too much time on one question
- Move on and return later if needed





Get Ready For Your Exam

Great Effort: You made strong progress throughout the course

Strengthen Your Prep: Reinforce your understanding before taking the exam

- **Refresh Concepts:** Review the presentation end-to-end
- **Practice Tests:** Attempt at least one practice test to understand exam patterns
- **Close Your Gaps:** Revisit areas where you did not do well in practice tests and re-watch videos as needed





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WOW!

Congratulations



Let's Clap for You!

Congratulations: Congratulations on your patience and persistence

You Have Put Your Best Foot Forward toward getting certified

Final Step: Review the key concepts and complete your exam preparation

Good Luck!





Do Not Forget!

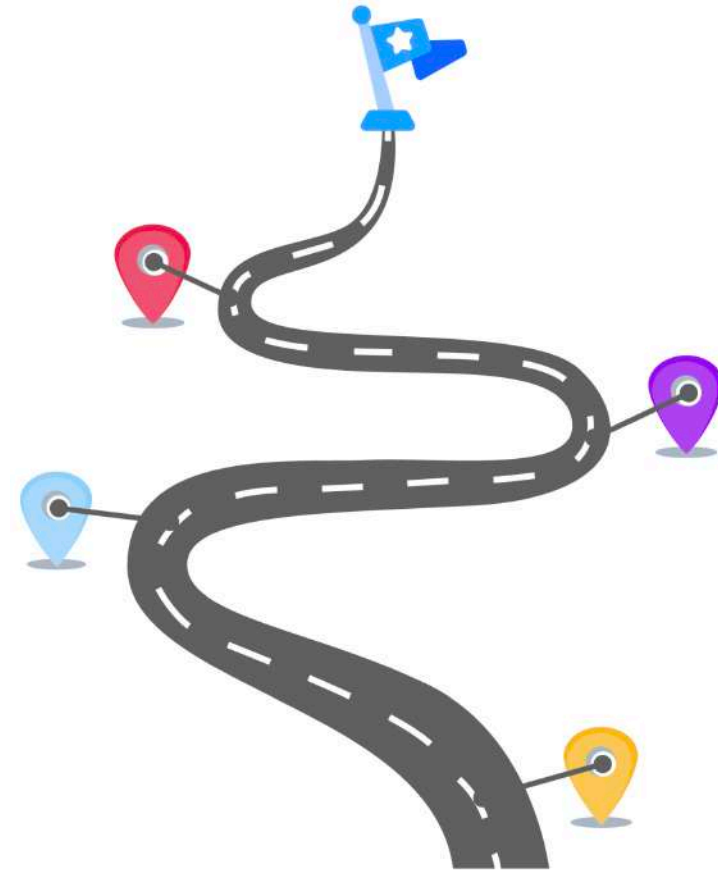
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Leave a Review: Your feedback helps future learners

Your Success = Our Success

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- Share your journey and lessons learned in Q&A to inspire others

Continue Your Learning Journey: Checkout our Roadmaps!





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